

(vi) The path of D will intersect the projector of D at d'' . Join $c'd''$. It is front view of CD .

(vii) With c' as centre and radius equal to $c'd''$, draw an arc cutting a line drawn parallel to xy from c' at d''' . Project d''' to cut a line drawn with c as centre and $c'd'$ (true length) as radius, at d_3 . From d_3 , draw a line parallel to xy , representing path of D in top view.

(viii) Project d'' to cut path of D in top view at d_4 . Join cd_4 . It is top view of CD .

(ix) The results are shown in fig. 10-67.

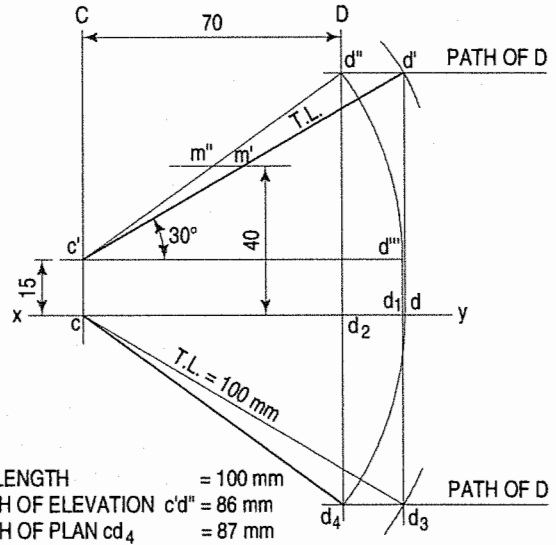


FIG. 10-67

EXERCISES 10(b)

1. A line AB , 75 mm long, is inclined at 45° to the H.P. and 30° to the V.P. Its end B is in the H.P. and 40 mm in front of the V.P. Draw its projections and determine its traces.
2. Draw the projections of a line AB , 90 mm long, its mid-point M being 50 mm above the H.P. and 40 mm in front of the V.P. The end A is 20 mm above the H.P. and 10 mm in front of the V.P. Show the traces and the inclinations of the line with the H.P. and the V.P.
3. The front view of a 125 mm long line PQ measures 75 mm and its top view measures 100 mm. Its end Q and the mid-point M are in the first quadrant, M being 20 mm from both the planes. Draw the projections of the line PQ .
4. A line AB , 75 mm long is in the second quadrant with the end A in the H.P. and the end B in the V.P. The line is inclined at 30° to the H.P. and at 45° to the V.P. Draw the projections of AB and determine its traces.
5. The end A of a line AB is in the H.P. and 25 mm behind the V.P. The end B is in the V.P. and 50 mm above the H.P. The distance between the end projectors is 75 mm. Draw the projections of AB and determine its true length, traces and inclinations with the two planes.
6. The top view of a 75 mm long line CD measures 50 mm. C is 50 mm in front of the V.P. and 15 mm below the H.P. D is 15 mm in front of the V.P. and is above the H.P. Draw the front view of CD and find its inclinations with the H.P. and the V.P. Show also its traces.
7. A line PQ , 100 mm long, is inclined at 45° to the H.P. and at 30° to the V.P. Its end P is in the second quadrant and Q is in the fourth quadrant. A point R on PQ , 40 mm from P is in both the planes. Draw the projections of PQ .
8. A line AB , 65 mm long, has its end A in the H.P. and 15 mm in front of the V.P. The end B is in the third quadrant. The line is inclined at 30° to the H.P. and at 60° to the V.P. Draw its projections.

9. The front view of a line AB measures 65 mm and makes an angle of 45° with xy . A is in the H.P. and the V.T. of the line is 15 mm below the H.P. The line is inclined at 30° to the V.P. Draw the projections of AB and find its true length and inclination with the H.P. Also locate its H.T.
10. A room is 4.8 m $\times$ 4.2 m $\times$ 3.6 m high. Determine graphically the distance between a top corner and the bottom corner diagonally opposite to it.
11. A line AB is in the first quadrant. Its end A and B are 20 mm and 60 mm in front of the V.P. respectively. The distance between the end projectors is 75 mm. The line is inclined at 30° to the H.P. and its H.T. is 10 mm above xy . Draw the projections of AB and determine its true length and the V.T.
12. Two oranges on a tree are respectively 1.8 m and 3 m above the ground, and 1.2 m and 2.1 m from a 0.3 m thick wall, but on the opposite sides of it. The distance between the oranges, measured along the ground and parallel to the wall is 2.7 m. Determine the real distance between the oranges.
13. Draw an isosceles triangle abc of base ab 40 mm and altitude 75 mm with a in xy and ab inclined at 45° to xy . The figure is the top view of a triangle whose corners A , B and C are respectively 75 mm, 25 mm and 50 mm above the H.P. Determine the true shape of the triangle and the inclination of the side AB with the two planes.
14. Three points A , B and C are 7.5 m above the ground level, on the ground level and 9 m below the ground level respectively. They are connected by roads with each other and are seen at angles of depression of 10° , 15° and 30° respectively from a point O on a hill 30 m above the ground level. A is due north-east, B is due north and C is due south-east of O . Find the lengths of the connecting roads.
15. A pipe-line from a point A , running due north-east has a downward gradient of 1 in 5. Another point B is 12 m away from and due east of A and on the same level. Find the length and slope of a pipe-line from B which runs due 15° east of north and meets the pipe-line from A .
16. The guy ropes of two poles 12 m apart, are attached to a point 15 m above the ground on the corner of a building. The points of attachment on the poles are 7.5 m and 4.5 m above the ground and the ropes make 45° and 30° respectively with the ground. Draw the projections and find the distances of the poles from the building and the lengths of the guy ropes.
17. A plate chimney, 18 m high 0.9 m diameter is supported by two sets of three guy wires each, as shown in fig. 10-68.
One set is attached at 3 m from the top and anchored 6 m above the ground level. The other set is fixed to the chimney at its mid-height and anchored on the ground. Determine the length and slope with the ground, of one of the wires from each set.
18. The projectors drawn from the H.T. and the V.T. of a straight line AB are 80 mm apart while those drawn from its ends are 50 mm apart. The H.T. is 35 mm in front of the V.P., the V.T. is 55 mm above the H.P. and the end A is 10 mm above the H.P. Draw the projections of AB and determine its length and inclinations with the reference planes.

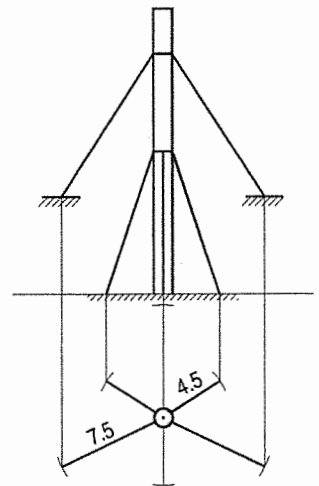


FIG. 10-68

19. Three guy ropes AB , CD and EF are tied at points A , C and E on a vertical post 15 m long at heights of 14 m, 12 m and 10 m respectively from the ground. The lower ends of the ropes are tied to hooks at points B , D and F on the ground level. If the points B , D and F lie at the corners of an equilateral triangle of 9 m long sides and if the post is situated at the centre of this triangle, determine graphically the length of each rope and its inclination with the ground. Assume the thickness of the post and the ropes to be equal to that of a line.
20. A line AB , 80 mm long, makes an angle of 60° with the H.P. and lies in an auxiliary vertical plane (A.V.P.), which makes an angle of 45° with the V.P. Its end A is 10 mm away from both the H.P. and the V.P.
Draw the projections of AB and determine (i) its true inclination with the V.P. and (ii) its traces.
21. A line PQ is 75 mm long and lies in an auxiliary inclined plane (A.I.P.) which makes an angle of 45° with the H.P. The front view of the line measures 55 mm and the end P is in the V.P. and 20 mm above the H.P.
Draw the projections of PQ and find (i) its inclinations with both the planes and (ii) its traces.
22. A line AB , 80 mm long, makes an angle of 30° with the V.P. and lies in a plane perpendicular to both the H.P. and the V.P. Its end A is in the H.P. and the end B is in the V.P. Draw its projections and show its traces.
23. The front view of a line makes an angle of 30° with xy . The H.T. of the line is 45 mm in front of the V.P., while its V.T. is 30 mm below the H.P. One end of the line is 10 mm above the H.P. and the other end is 100 mm in front of the V.P.
Draw the projections of the line and determine (i) its true length, and (ii) its inclinations with the H.P. and the V.P.
24. A room is 6 m $\times$ 5 m $\times$ 3.5 m high. An electric bracket light is above the centre of the longer wall and 1 m below the ceiling. The bulb is 0.3 m away from the wall. The switch for the light is on an adjacent wall, 1.5 m above the floor and 1 m away from the other longer wall. Find graphically the shortest distance between the bulb and the switch.
25. Three lines oa , ob and oc are respectively 25 mm, 45 mm and 65 mm long, each making 120° angles with the other two and the shortest line being vertical. The figure is the top view of the three rods OA , OB and OC whose ends A , B and C are on the ground, while O is 100 mm above it. Draw the front view and determine the length of each rod and its inclination with the ground.
26. The projectors of the ends of a line PQ are 90 mm apart. P is 20 mm above the H.P. while Q is 45 mm behind the V.P. The H.T. and the V.T. of the line coincide with each other on xy , between the two end projectors and 35 mm away from the projector of the end P . Draw the projections of PQ and determine its true length and inclinations with the two planes.
27. A person on the top of a tower 30 m high, which rises from a horizontal plane, observes the angles of depression (below the horizon) of two objects H and K on the plane to be 15° and 25° , the direction of H and K from the tower being due north and due west respectively. Draw the top view to a scale of 1 mm = 0.5 m showing the relative positions of the person and the two objects. Measure and state in metres the distance between H and K .

28. Two pegs A and B are fixed in each of the two adjacent side walls (of a rectangular room) which meet in a corner. Peg A is 1.5 m above the floor, 1.2 m from the side wall and is protruding 0.3 m from the wall. Peg B is 2 m above the floor, 1 m from the other side wall and is also protruding 0.3 m from the wall. Find the distance between the ends of the pegs.
29. Two objects A and B , 10 m above and 7 m below the ground level respectively, are observed from the top of a tower 35 m high from the ground. Both the objects make an angle of depression of 45° with the horizon. The horizontal distance between A and B is 20 m. Draw to scale 1:250, the projections of the objects and the tower and find (a) the true distance between A and B , and (b) the angle of depression of another object C situated on the ground midway between A and B .
30. A room measures 8 m long, 5 m wide and 4 m high. An electric point hangs in the centre of the ceiling and 1 m below it. A thin straight wire connects the point to a switch kept in one of the corners of the room and 2 m above the floor. Draw the projections of the wire, and find the length of the wire and its slope-angle with the floor.
31. A rectangular tank 4 m high is strengthened by four stay rods one at each corner, connecting the top corner to a point in the bottom 0.7 m and 1.2 m from the sides of the tank. Find graphically the length of the rod required and the angle it makes with the surface of the tank.
32. Three vertical poles AB , CD and EF are respectively 5, 8 and 12 metres long. Their ends B , D and F are on the ground and lie at the corners of an equilateral triangle of 10 metres long sides. Determine graphically the distance between the top ends of the poles, viz. AC , CE and EA .
33. The front view of a line AB measures 70 mm and makes an angle of 45° with xy . A is in the H.P. and the V.T. of the line is 15 mm below the H.P. The line is inclined at 30° to the V.P. Draw the projections of AB , and find its true length, inclination with the H.P. and its H.T.
34. A line AB measures 100 mm. The projectors through its V.T. and the end A are 40 mm apart. The point A is 30 mm below the H.P. and 20 mm behind the V.P. The V.T. is 10 mm above the H.P. Draw the projections of the line and determine its H.T. and inclinations with the H.P. and the V.P.
35. A horizontal wooden platform is 3.5 m long and 2 m wide. It is suspended from a hook by means of chains attached at its four corners. The hook is situated vertically above the centre of the platform and at a distance of 5 m above it. Determine graphically the length of each chain and the angle which it makes with the platform. Assume the thickness of the platform and the chain to be equal to that of a line. Scale: 10 mm = 0.5 m.
36. A picture frame 2 m wide and 1 m high is to be fixed on a wall railing by two straight wires attached to the top corners. The frame is to make an angle of 40° with the wall and the wires are to be fixed to a hook on the wall on the centre line of the frame and 1.5 m above the railing. Find the length of the wires and the angle between them.
37. The top view of line AB measures 60 mm and inclined to reference line at 60° . The end point A is 15 mm above the H.P. and 30 mm in front of the V.P. Draw the projections of the line when it is inclined at 45° to the H.P. and is situated in the first quadrant. Find true length and inclination of the line with the V.P. and traces.

Chapter 11



PROJECTIONS ON AUXILIARY PLANES

11-0. INTRODUCTION

Two views of an object, viz. the front view and the top view (projected on the principal planes of projection), are sometimes not sufficient to convey all the information regarding the object. Additional views, called *auxiliary views*, are therefore, projected on other planes known as *auxiliary planes*. These views are often found necessary in technical drawings. Auxiliary views may also be used for determining

- (i) the true length of a line,
- (ii) the point-view of a line,
- (iii) the edge-view of a plane,
- (iv) the true size and form of a plane etc.

They are thus very useful in finding solutions of problems in practical solid geometry.

This chapter deals with the following topics:

1. Types of auxiliary planes and views.
2. Projection of a point on an auxiliary plane.
3. Projections of lines and planes by the use of auxiliary planes.
4. To determine true length of a line.
5. To obtain point-view of a line and edge-view of a plane.
6. To determine true shape of a plane figure.

11-1. TYPES OF AUXILIARY PLANES AND VIEWS

Auxiliary planes are of two types:

- (i) *auxiliary vertical plane* or A.V.P., and
 - (ii) *auxiliary inclined plane* or A.I.P.
- (i) Auxiliary vertical plane is perpendicular to the H.P. and inclined to the V.P. Projection on an A.V.P. is called *auxiliary front view*.
- (ii) Auxiliary inclined plane is perpendicular to the V.P. and inclined to the H.P. Projection on an A.I.P. is called *auxiliary top view*.

The orthographic views of the auxiliary projections are drawn by rotating the auxiliary plane about that principal plane to which it is perpendicular.

11-2. PROJECTION OF A POINT ON AN AUXILIARY PLANE



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 25 for the projection of a point on an auxiliary plane.

(1) Projection of a point on an auxiliary vertical plane:

A point A [fig. 11-1(i)] is situated in front of the V.P. and above the H.P. A.V.P. is the auxiliary vertical plane inclined at an angle α to the V.P. The H.P. and the A.V.P. meet at right angles in the line x_1y_1 .

a' and a are respectively, the front view and the top view of the point A. a'_1 is the auxiliary front view obtained by drawing a projector Aa'_1 perpendicular to the A.V.P. It can be clearly seen that a'_1o_1 (the distance of the auxiliary front view a'_1 from x_1y_1) = $a'o$ (the distance of the front view a' from xy) = Aa (the distance of the point A from the H.P.).

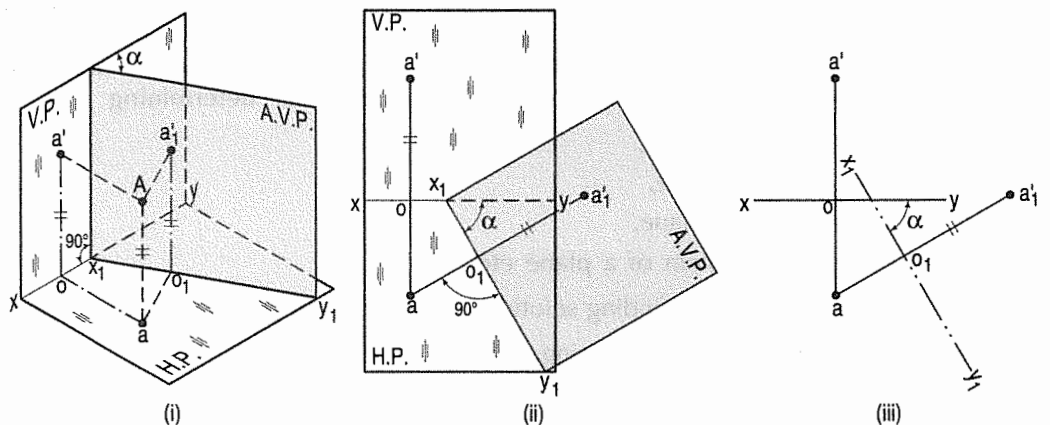


FIG. 11-1

Fig. 11-1(ii) shows the V.P. and the A.V.P. rotated about the H.P. to which they are perpendicular. The line of intersection x_1y_1 between the A.V.P. and the H.P., is inclined at the angles α to xy . The line joining the top view a with the auxiliary front view a'_1 , is at right angles to x_1y_1 and intersects it at o_1 . Note that $a'_1o_1 = a'o$.

To draw the orthographic views [fig. 11-1(iii)], start with the reference line xy and mark the front view a' and the top view a .

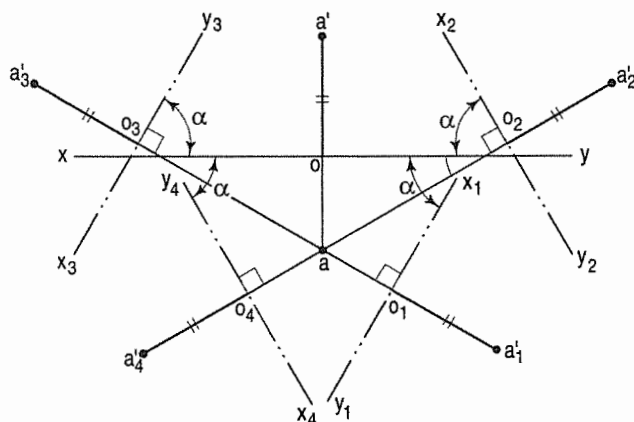


FIG. 11-2

Draw a new reference line x_1y_1 , making the angle α with xy . Through the top view a , draw a projector aa'_1 perpendicular to and intersecting x_1y_1 at o_1 and such that $a'_1o_1 = a'o$. a'_1 is the required auxiliary front view.

The new reference line making the angle α with xy , can be drawn in four different positions, as shown in fig. 11-2 by lines x_1y_1 , x_2y_2 etc. All the front views are projected from the top view a and their distances from their respective reference lines are equal, i.e. $a'_1o_1 = a'_2o_2 \dots = a'o$.

(2) Projection of a point on an auxiliary inclined plane:

A point P [fig. 11-3(i)] is situated above the H.P. and in front of the V.P. A.I.P. is an auxiliary inclined plane making an angle β with the H.P. It meets the V.P. at right angles and in a line x_1y_1 .

p' and p are respectively the front view and the top view of the point P . p_1 is the auxiliary top view obtained by drawing the projector Pp_1 perpendicular to the A.I.P. It can be seen that p_1o_1 (the distance of the auxiliary top view p_1 from x_1y_1) = po (the distance of the top view p from xy) = Pp' (the distance of the point P from the V.P.).

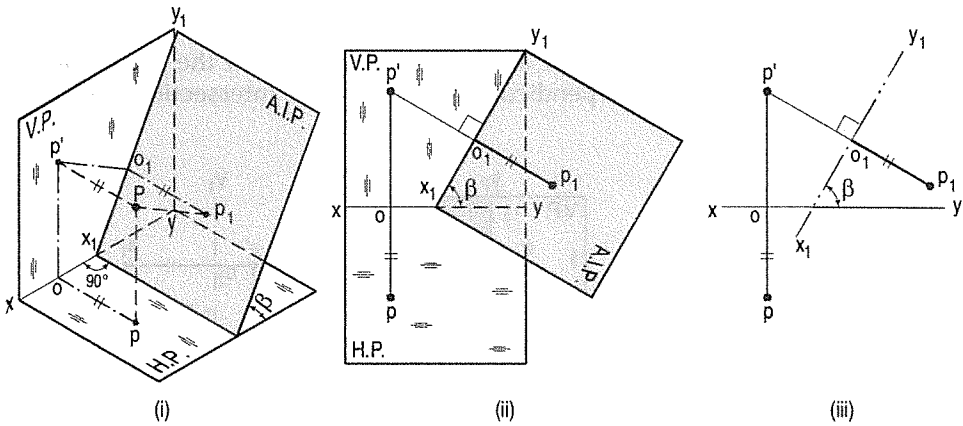


FIG. 11-3

The H.P. and the A.I.P. are then rotated about the V.P. to which they are perpendicular [fig. 11-3(ii)]. x_1y_1 , the line of intersection between the V.P. and the A.I.P. makes the angle β with xy . The line joining the front view p' and the auxiliary top view p_1 is at right angles to x_1y_1 and intersects it at o_1 . Note that $p_1o_1 = po$.

To draw the orthographic views [fig. 11-3(iii)], draw xy and mark p' and p . Draw x_1y_1 making the angle β with xy .

Through the front view p' , draw a projector $p'p_1$ perpendicular to and intersecting x_1y_1 at o_1 and such that $p_1o_1 = po$. p_1 is the required auxiliary front view.

In this case also, the new reference line can be drawn in four different positions as shown in fig. 11-4 by lines x_1y_1 , x_2y_2 etc., each inclined at β to xy . All the top views are projected from the front view p' and their distances from their respective reference lines are equal, i.e. $p_1o_1 = p_2o_2 \dots = po$.

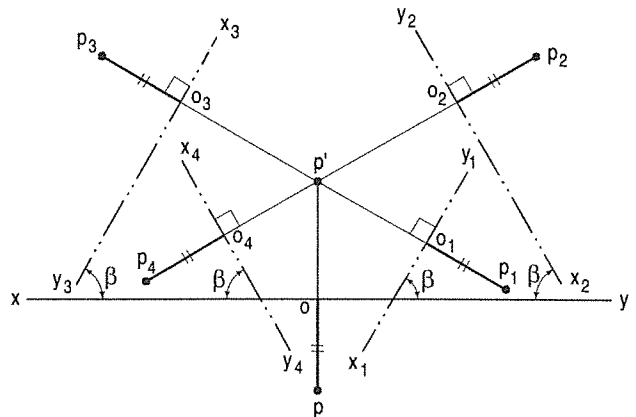


FIG. 11-4

If the inclination of the A.V.P. with the V.P. is increased so that $\alpha = 90^\circ$, the A.V.P. will be perpendicular to both the planes. Similarly, if the inclination of the A.I.P. with the H.P. is increased so that $\beta = 90^\circ$, it will also be perpendicular to both the H.P. and the V.P. This plane is called the *profile plane* (P.P.). It may be rotated about any one of the two principal planes. The view on this plane can, therefore, be projected from either the top view or the front view and named accordingly.

(3) Projection of a point on an auxiliary plane perpendicular to both the principal planes:

In fig. 11-5(i), A is a point. P.P. is an auxiliary plane perpendicular to the H.P. and the V.P. a_1 is the auxiliary view projected on the P.P. It can be seen that $a_1o_2 = a'o = Aa$ (the distance of A from the H.P.). Also $a_1o_1 = ao = Aa'$ (the distance of A from the V.P.).

Fig. 11-5(ii) shows the P.P. rotated about the V.P. a_1 lies on the projector drawn through the front view a' and perpendicular to the line of intersection between the V.P. and the P.P.

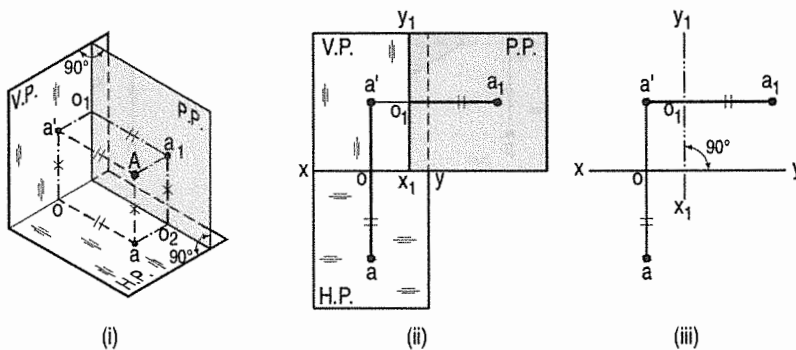


FIG. 11-5

It is thus projected from the front view and hence, called the auxiliary top view. In technical drawings, this view is generally termed as the *side view*, *end view* or *end front view*. Note that $a_1o_1 = ao$.

Note that, when seen from the left, the new reference line and the side view are placed to the right of the front view. When seen from the right, they would be placed to the left of the front view. Thus, the view seen from any side of the front view is placed on its other side.

Fig. 11-6 shows the P.P. rotated about the H.P. The view on the P.P. now lies on the projector drawn through the top view a . Hence, it is called the auxiliary front view. In this case, $a'_1o_2 = a'o$.

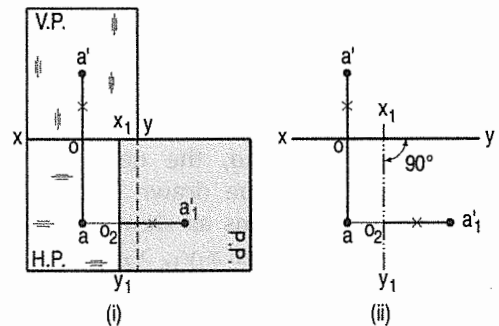


FIG. 11-6

The orthographic views [fig. 11-5(iii) and fig. 11-6(ii)] in both cases are self-explanatory.

General conclusions:

- (i) The auxiliary top view of a point lies on a line drawn through the front view, perpendicular to the new reference line (x_1y_1) and at a distance from it, equal to the distance of the first top view from its own reference line (xy).
- (ii) The auxiliary front view of a point lies on a line drawn through the top view, perpendicular to the new reference line (x_1y_1) and at a distance from it, equal to the distance of the first front view from its own reference line (xy).
- (iii) The distances of all the front views of the same point (projected from the same top view) from their respective reference lines are equal.
- (iv) The distances of all the top views of the same point (projected from the same front view) from their respective reference lines are equal.

Problem 11-1. (fig. 11-7 and fig. 11-8): The projections of a line AB are given. Draw (i) an auxiliary front view of the line on an A.V.P. inclined at 60° to the V.P. and (ii) an auxiliary top view on an A.I.P. making an angle of 75° with the H.P.

Let ab and $a'b'$ be the given projections.

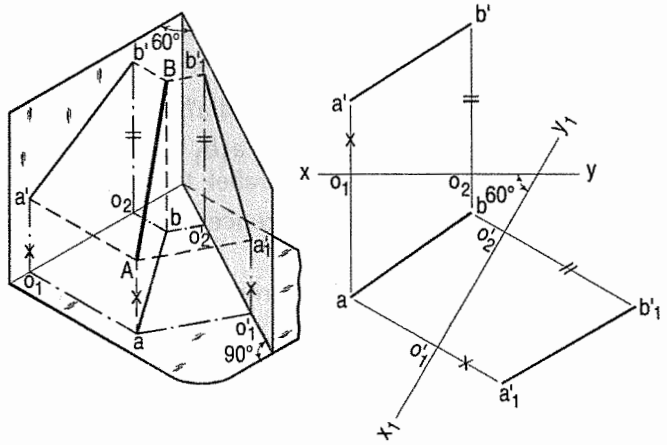


FIG. 11-7

- (i) Draw a new reference line x_1y_1 , inclined at 60° to xy to represent the A.V.P. (fig. 11-7).
- (ii) Project the auxiliary front view $a_1'b_1$ from the top view ab , by making $a_1'o_1$ equal to $a'o_1$, and $b_1'o_2$ equal to $b'o_2$.
- (iii) Similarly, draw x_2y_2 for the A.I.P. inclined at 75° to xy (fig. 11-8).
- (iv) Project the auxiliary top view a_2b_2 from the front view $a'b'$, making $a_2o''_1$ equal to ao_1 and $b_2o''_2$ equal to bo_2 .

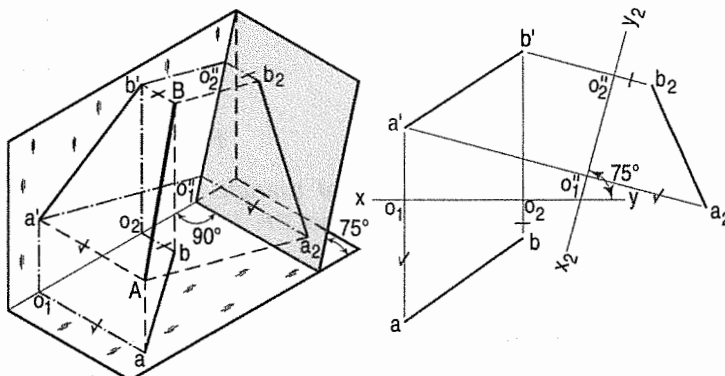


FIG. 11-8

11-3. PROJECTIONS OF LINES AND PLANES BY THE USE OF AUXILIARY PLANES

Projections of lines and planes at given inclinations to one or both the planes may also be obtained by the use of auxiliary planes. The method adopted is called the *alteration or change-of-reference-line method*.

The line, in its initial position, is assumed to be parallel to both the planes of projection. Then, instead of making the line inclined to one of the planes, an auxiliary plane inclined to the line is assumed, i.e. a new reference line is drawn and the view is projected on it.

In case of a plane, it is kept parallel to one of the planes of projection in the initial stage, the required views being obtained by projecting it on new reference lines.

Problem 11-2. (fig. 11-9): A line AB , 50 mm long, is inclined at 30° to the H.P. and its top view makes an angle of 60° with the V.P. Draw its projections.

- Draw the projections ab and $a'b'$, assuming AB to be parallel to both the planes.
- Project a new top view a_1b_1 on a new reference line x_1y_1 inclined at 30° to $a'b'$. a_1b_1 is still parallel to x_1y_1 .
- Draw the another reference line x_2y_2 to represent an A.V.P. inclined at 60° to the top view a_1b_1 .
- Project the required front view $a'_1b'_1$ on x_2y_2 . Note that $a_1o_1 = a'o$, $a'_1o_2 = a'o_1$ etc.

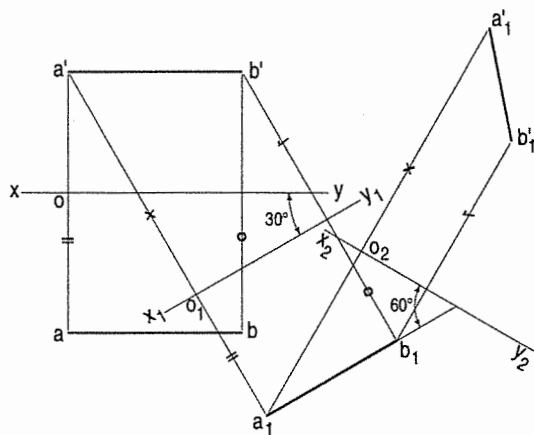


FIG. 11-9

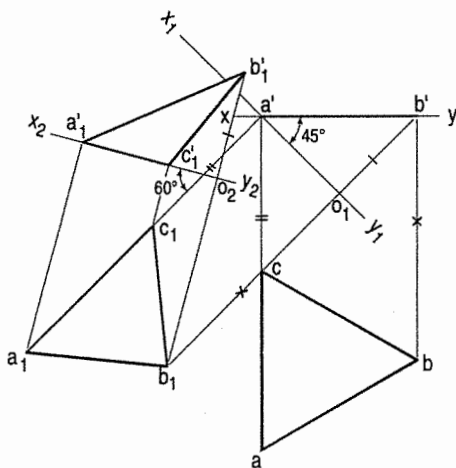


FIG. 11-10

Problem 11-3. (fig. 11-10): An equilateral triangle of 40 mm long sides has an edge on the ground and inclined at 60° to the V.P. Its plane makes an angle of 45° with the H.P. Draw its projections.

- Draw the top view abc and the front view $a'b'c'$, assuming the triangle to be lying on the ground and one side perpendicular to the V.P.
- Draw x_1y_1 inclined at 45° to $a'b'c'$ (the front view) and passing through a' (because the triangle has its edge on the ground).

- (iii) Project the new top view $a_1b_1c_1$. Again draw a new reference line x_2y_2 inclined at 60° to the line a_1c_1 (which is to make 60° angle with the V.P.).
- (iv) Project the final front view $a'_1b'_1c'_1$ on x_2y_2 . Note that $a'_1c'_1$ is in x_2y_2 .

11-4. TO DETERMINE TRUE LENGTH OF A LINE

We have seen that the true length of a line and its inclinations with the planes of projection can be determined by making each of its projections parallel to xy (chapter 10).

Instead of changing the position of the projection, that of the plane may be altered, i.e. a new reference line representing an auxiliary plane may be drawn parallel to the projection. The auxiliary projection on that reference line will show the true length and true inclination of the line with the other plane.

Problem 11-4. (fig. 11-11): *The projections of a line AB are given. To determine its true length and true inclinations with the reference planes.*

Let ab and $a'b'$ be the given projections of AB .

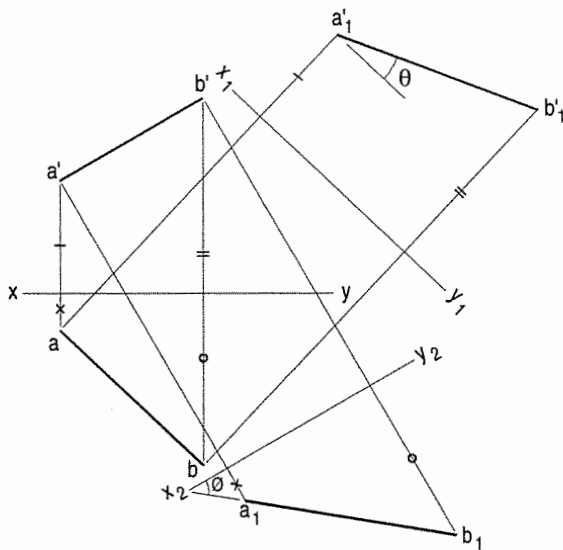


FIG. 11-11

- (i) Draw a reference line x_1y_1 to represent an A.V.P. parallel to ab , the top view.
- (ii) Project the auxiliary front view $a'_1b'_1$ which is the true length of AB . θ is its inclination with the H.P.
- (iii) Similarly, draw x_2y_2 parallel to the front view $a'b'$ and project the auxiliary top view a_1b_1 . It is the true length of AB and ϕ is its true inclination with the V.P.

Problem 11-5. (fig. 11-12): *The projections of a line AB viz. ab and $a'b'$ are on the same projector as shown in fig. 11-12(ii). Find the true length, inclinations with the H.P. and the V.P. and the traces of AB.*

- (i) Draw a reference line x_1y_1 parallel to the projections. It will be perpendicular to xy and will represent an auxiliary plane at right angles to both the H.P. and the V.P. as shown in fig. 11-12(i).
- (ii) Project the side view $a'_1b'_1$, which will be the true length of AB . θ and ϕ are its true inclinations with the H.P. and the V.P. respectively.

- (iii) Produce $a'_1b'_1$ to meet xy at v and x_1y_1 at h . The H.T. will be on the top view ab -produced, so that $oH.T. = o_1h$. The V.T. is on the front view $a'b'$ -produced, so that $oV.T. = o_1v$.

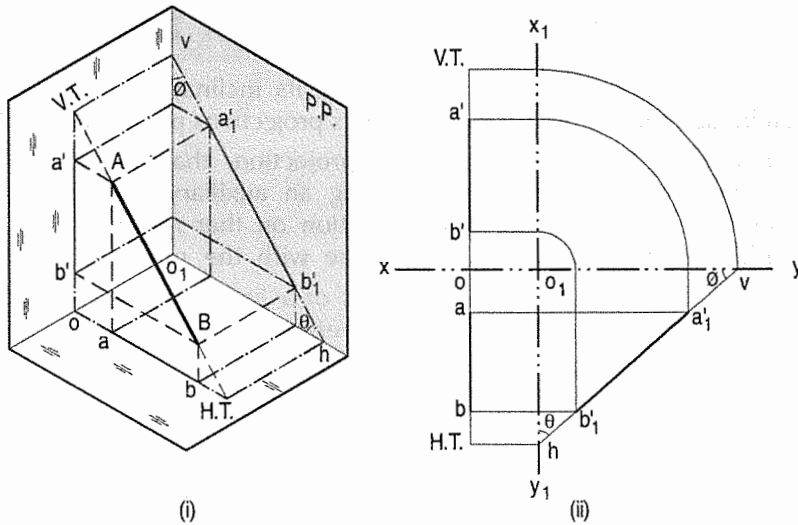


FIG. 11-12

The distances of the ends of the line from x_1y_1 are kept equal to their respective distances above xy by drawing lines inclined at 45° as shown.

11-5. TO OBTAIN POINT-VIEW OF A LINE AND EDGE-VIEW OF A PLANE

We have seen in chapter 10 that when a line is perpendicular to a reference plane, its projection on that plane is a point; while its projection on the other reference plane shows its true length. In other words, the projection of the view of a line showing its true length, on an auxiliary plane perpendicular to that view will be a point.

Similarly, a plane will be seen as a line, when it is projected on a plane, perpendicular to the true length of any one of its elements. The projection of the line-view or edge-view of a plane on an auxiliary plane parallel to it, will show the true shape and size of the plane.

Uses of the point-view of a line and edge-view of a plane on auxiliary planes are illustrated in problems 11-6 and 11-7 respectively.

Problem 11-6. (fig. 11-13): The projections of a line PQ are given. Determine

- the distance of its mid-point from xy , and
- the shortest distance of the line from xy .

Let pq and $p'q'$ be the given projections of PQ .

- Draw a new reference line x_1y_1 perpendicular to xy and project the side view $p'_1q'_1$ on it.

When considering the side view, x_1y_1 is the edge-view of the V.P. and xy is that of the H.P. The point o is the side view or the point-view of the line of intersection of the V.P. and the H.P., i.e. x_1y_1 and xy . The line joining any point on $p'_1q'_1$, with o will show the shortest distance of that point from xy .

- (ii) Find the mid-point a'_1 of $p'_1q'_1$ and join it with o . oa'_1 is the distance of the mid-point of PQ from xy .
- (iii) From o , draw a line ob perpendicular to $p'_1q'_1$. ob is the shortest distance of PQ from xy . It will be perpendicular to both PQ and xy .

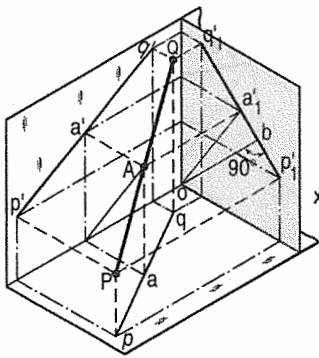


FIG. 11-13

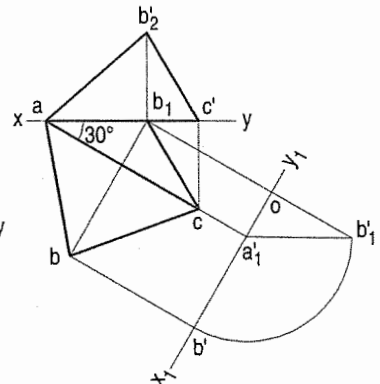
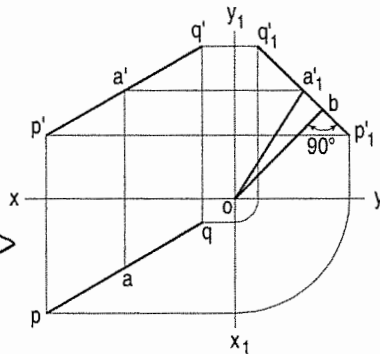


FIG. 11-14

Problem 11-7. (fig. 11-14): An isosceles triangle ABC , base 60 mm and altitude 40 mm has its base AC in the H.P. and inclined at 30° to the V.P. The corners A and B are in the V.P. Draw its projections.

Assume the triangle to be lying in the H.P. with the base AC inclined at 30° to the V.P. and A in the V.P. Its true shape will be seen in the top view.

- (i) Therefore, in the top view, draw ac 60 mm long and inclined at 30° to xy and complete the triangle abc .
- (ii) Project c to c' on xy . When the triangle is tilted about the edge AC , so that the corner B is in the V.P., in the top view the point b will move along a line perpendicular to ac , to a point b_1 in xy . The distance of the front view of B below xy may now be determined by means of an auxiliary plane.
- (iii) Draw a reference line x_1y_1 perpendicular to ac .
- (iv) Project an auxiliary front view of the triangle abc . It will be a line $b'a'_1$, showing the edge-view of the triangle. a'_1 is the point-view of the line ac . When b moves to b_1 , b' will move along the arc drawn with a'_1 as centre and radius equal to a'_1b' to b'_1 on the line through b_1 drawn perpendicular to x_1y_1 . ob'_1 is the required distance.
- (v) Therefore, through b_1 , draw a line perpendicular to xy and on it, mark a point b'_2 such that $b_1b'_2 = ob'_1$.
- (vi) Join a and c' with b'_2 . ab'_2c' and ab_1c are the required projections.

11-6. TO DETERMINE TRUE SHAPE OF A PLANE FIGURE

The true shape of any plane figure may be determined by means of its projections on auxiliary planes, as illustrated in problems 11-8 and 11-9.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 26 for the following problem.

Problem 11-8. (fig. 11-15):

Projections of a pentagon resting on the ground on one of its sides are given. Determine the true shape of the pentagon.

ae is the true length of the side because it is parallel to the H.P.

- (i) Draw a reference line x_1y_1 perpendicular to ae . Project an auxiliary front view on it. It will give a point-view a'_1 and e'_1 of ae and an edge-view $a'_1c'_1$ of the pentagon.
- (ii) Draw another reference line x_2y_2 parallel to $a'_1c'_1$ and project an auxiliary top view $a_1b_1c_1d_1e_1$, which will be the true shape of the pentagon.

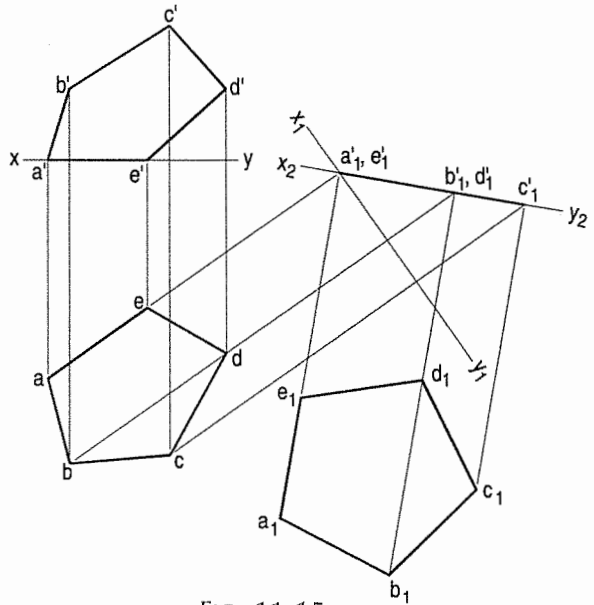


FIG. 11-15

Problem 11-9. Fig. 11-16 shows the top view $abcd$ and front view $a'b'c'd'$ of a quadrilateral. Determine its true shape.

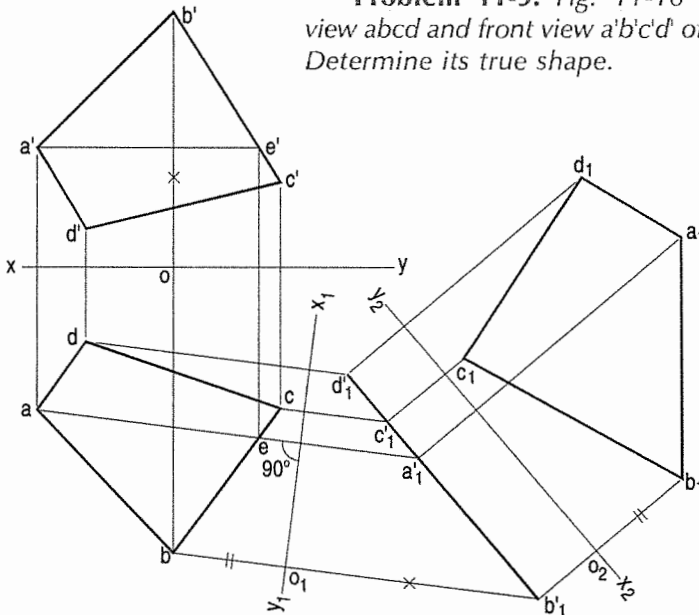


FIG. 11-16

- (i) Through any corner, say a' , draw a line parallel to xy and meeting $b'c'$ at e' .
- (ii) Project e' to e on the line bc in the top view. ae is the true length of the element AE .
- (iii) Draw a reference line x_1y_1 perpendicular to ae . Project a new front view from the top view. It is a line $d'_1b'_1$.
- (iv) Again, draw another reference line x_2y_2 parallel to the line-view $d'_1b'_1$ and project on it a new top view $a_1b_1c_1d_1$ which will show the true shape of the quadrilateral.

Note that $b'_1o_1 = b'o$, $b'_1o_2 = bo_1$ etc.

Problem 11-10. (fig. 11-17): The projections of a triangular plate PQR appear as under:

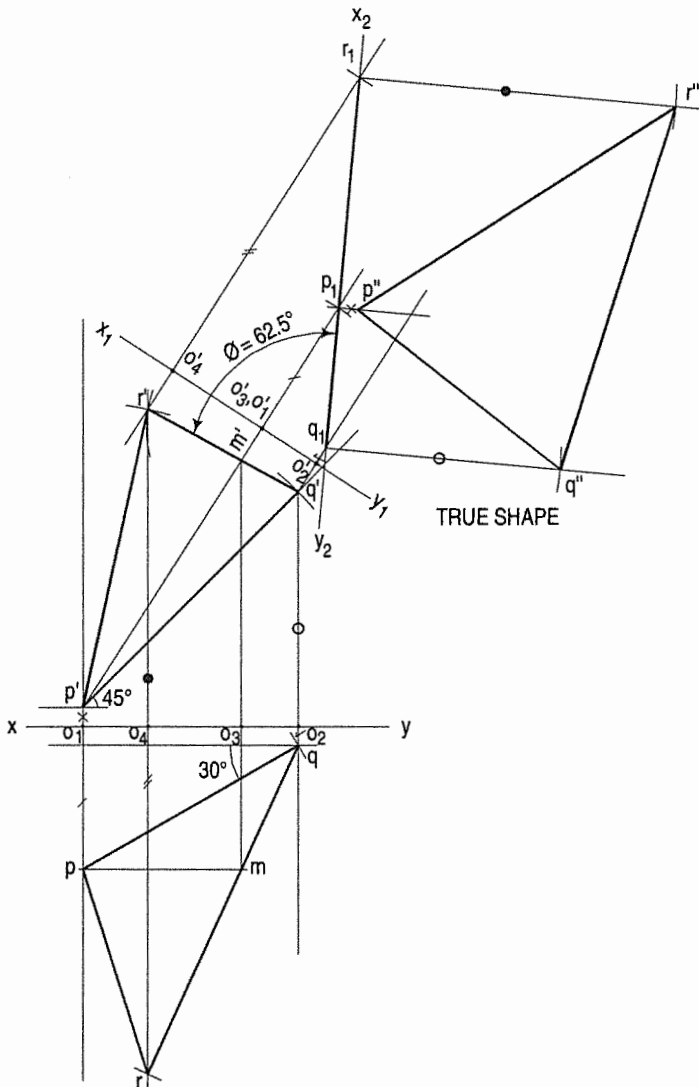


FIG. 11-17

(a) Top view : pq makes 30° with xy .

$qr = 95$ mm. The corner q is 5 mm in front of the V.P.

(b) Front view : $p'q' = 80$ mm; $q'r' = 45$ mm and $r'p' = 80$ mm.

$p'q'$ makes an angle of 45° with xy . The corner P of the plate is 5 mm above the H.P.

Draw the projections of the plate and determine true shape of the triangular plater

- (i) Mark xy . Draw top view and front view from the given dimensions.
- (ii) Mark a line pm in the top view parallel to xy . Draw the projection of pm in the front view $p'm'$ as shown.
- (iii) Select x_1y_1 perpendicular to $p'm'$ and project new top view taking the distances o_1p , o_4r and o_2q from the top view.
- (iv) Select x_2y_2 and draw new front view as shown in fig. 11-17.
- (v) Measure angle ϕ .

Problem 11-11. A regular pentagon of 50 mm side is resting on one of its sides on the H.P. having that side parallel to and 25 mm in front of V.P. It is tilted about that side so that its highest corner rests in the V.P. Draw the projections of the pentagon.

- (i) Keep the plane of pentagon parallel to V.P. Draw the front view and top view as shown in fig. 11-18.
- (ii) Project $a'b'c'd'e'$ on the line parallel to and 25 mm behind x_1y_1 . Taking a'' as a center, tilt $a''d''$ such that d'' touches x_1y_1 at d''' . Complete the projections as shown in fig. 11-18.

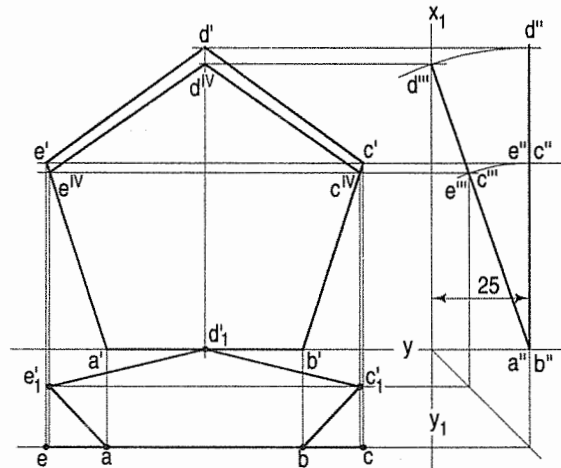


FIG. 11-18

Note that the points e'' and c'' should be transferred in the top view correctly as shown.

Problem 11-12. A thin square plate with 60 mm side stands on one of its corners in H.P. and the opposite corner is raised so that one of its diagonals is twice that of the others. If, one of the diagonal is parallel to both the planes, draw its projection and determine an inclination of the plane of the plate with H.P.

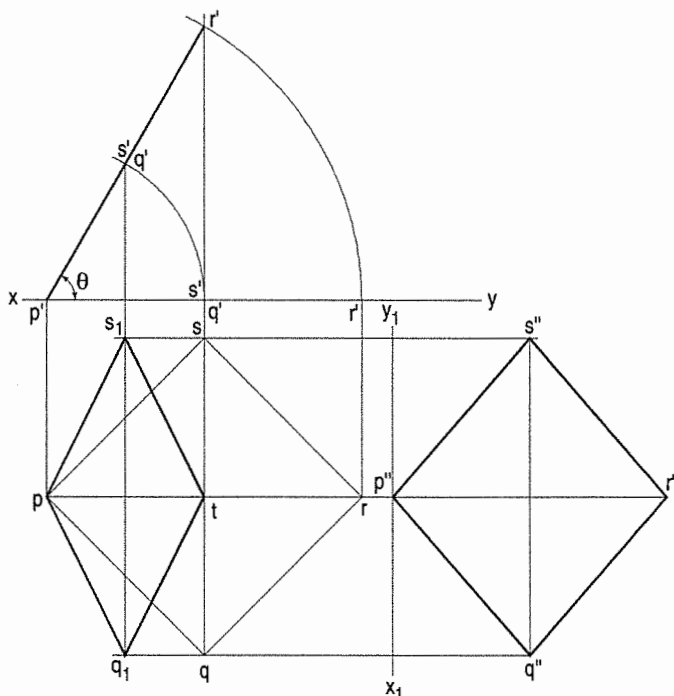


FIG. 11-19

- (i) Assuming that the square plate is equally inclined with a vertical plane, draw its top view and front view as shown in fig. 11-19.
- (ii) Raise r' so that the diagonal qs becomes twice the diagonal pt . Draw the top view p, q_1, t and s_1 . Measure angle θ .
- (iii) Draw new reference line x_1y_1 parallel to s_1q_1 . Project new front view, keeping the same distances of points p', q', r' and s' from xy .
- (iv) Measure angle θ .

EXERCISES 11



Solve the following exercises by applying the method of projections on auxiliary planes:

1. Determine the true length, inclinations with the H.P. and the V.P., and the traces of the line PQ in problem 10-15 of chapter 10.
2. Find the true length and the distance of the mid-point from xy of the line whose projections are given in problem 10-13 of chapter 10.
3. Draw the projections of the various plane figures as required in Exercises 1, 2 and 3 of chapter 12.
4. abc is an equilateral triangle of altitude 50 mm with ab in xy and c below it. abc' is an isosceles triangle of altitude 75 mm and c' is above xy . Determine the true shape of the triangle ABC , of which abc is the top view and abc' is the front view.

5. Determine the true shape of the figure, the top view of which is a regular pentagon of 35 mm sides, having one side inclined at 30° to xy and whose front view is a straight line making an angle of 45° to xy .
6. An equilateral triangle ABC of sides 75 mm long has its side AB in the V.P. and inclined at 60° to the H.P. Its plane makes an angle of 45° with the V.P. Draw its projections.
7. An isosceles triangle PQR having the base PQ 50 mm long and altitude 75 mm has its corners P , Q , and R 25 mm, 50 mm and 75 mm respectively above the ground. Draw its projections.
8. A thin regular pentagonal plate of 60 mm long edges has one of its edges in the H.P. and perpendicular to the V.P. while its farthest corner is 60 mm above the H.P. Draw the projections of the plate. Project another front view on an A.V.P. making an angle of 45° with the V.P.
9. A thin composite plate consists of a square of 70 mm long sides with an additional semi-circle constructed on CD as diameter. The side AB is vertical and the surface of the plate makes an angle of 45° with the V.P. Draw its projections. Project another top view on an A.I.P. making an angle of 30° with the side AB .
10. A 60° set-square of 125 mm longest side is so kept that the longest side is in the H.P. making an angle of 30° with the V.P. and the set-square itself inclined at 45° to the H.P. Draw the projections of the set-square.
11. A plane figure is composed of an equilateral triangle ABC and a semi-circle on AC as diameter. The length of the side AB is 50 mm and is parallel to the V.P. The corner B is 20 mm behind the V.P. and 15 mm below the H.P. The plane of the figure is inclined at 45° to the H.P. Draw the projections of the plane figure.
12. An equilateral triangle ABC having side length as 50 mm is suspended from a point O on the side AB 15 mm from A in such a way that the plane of the triangle makes an angle of 60° with the V.P. The point O is 20 mm below the H.P. and 40 mm behind the V.P. Draw the projections of the triangle.
13. A hexagonal plate of side 40 mm, is resting on a corner in V.P. with its surface making an angle of 30° with the V.P. The front view of the diagonal passing through that corner is inclined at 45° to the line xy . Draw the projections of the hexagonal plate.
14. A rectangular plate 50 mm $\times$ 70 mm stands on one of its shorter edges in H.P. and is raised about this edge so that the top view becomes a square of 50 mm. Determine an inclination of the plate with the horizontal plane.

Chapter 12



PROJECTIONS OF PLANES

12-0. INTRODUCTION

Plane figures or surfaces have only two dimensions, viz. length and breadth. They do not have thickness. A plane figure may be assumed to be contained by a plane, and its projections can be drawn, if the position of that plane with respect to the principal planes of projection is known.

In this chapter, we shall discuss the following topics:

1. Types of planes and their projections.
2. Traces of planes.

12-1. TYPES OF PLANES

Planes may be divided into two main types:

- (1) Perpendicular planes.
- (2) Oblique planes.

(1) **Perpendicular planes:** These planes can be divided into the following sub-types:

- (i) Perpendicular to both the reference planes.
- (ii) Perpendicular to one plane and parallel to the other.
- (iii) Perpendicular to one plane and inclined to the other.

(i) **Perpendicular to both the reference planes** (fig. 12-1): A square $ABCD$ is perpendicular to both the planes. Its H.T. and V.T. are in a straight line perpendicular to xy .

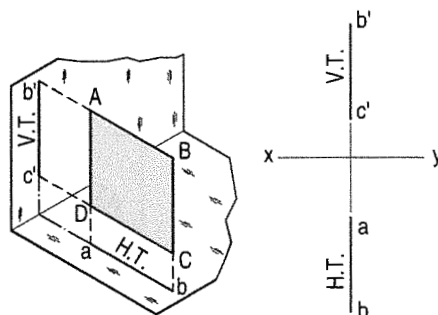


FIG. 12-1

The front view $b'c'$ and the top view ab of the square are both lines coinciding with the V.T. and the H.T. respectively.

(ii) Perpendicular to one plane and parallel to the other plane:

- (a) Plane, perpendicular to the H.P. and parallel to the V.P. [fig. 12-2(i)].

A triangle PQR is perpendicular to the H.P. and is parallel to the V.P. Its H.T. is parallel to xy . It has no V.T.

The front view $p'q'r'$ shows the exact shape and size of the triangle. The top view pqr is a line parallel to xy . It coincides with the H.T.

- (b) Plane, perpendicular to the V.P. and parallel to the H.P. [fig. 12-2(ii)].

A square $ABCD$ is perpendicular to the V.P. and parallel to the H.P. Its V.T. is parallel to xy . It has no H.T.

The top view $abcd$ shows the true shape and true size of the square. The front view $a'b'$ is a line, parallel to xy . It coincides with the V.T.

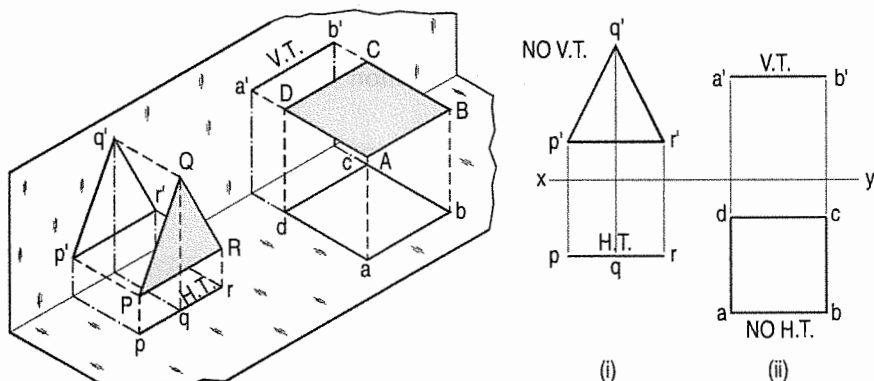


FIG. 12-2

(iii) Perpendicular to one plane and inclined to the other plane:

- (a) Plane, perpendicular to the H.P. and inclined to the V.P. (fig. 12-3).

A square $ABCD$ is perpendicular to the H.P. and inclined at an angle θ to the V.P. Its V.T. is perpendicular to xy . Its H.T. is inclined at θ to xy .

Its top view ab is a line inclined at θ to xy . The front view $a'b'c'd'$ is smaller than $ABCD$.

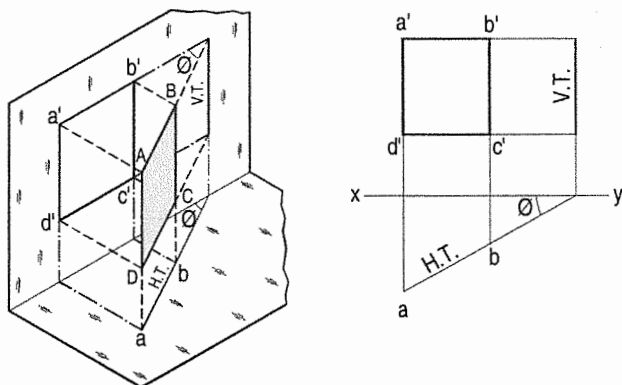


FIG. 12-3

- (b) Plane, perpendicular to the V.P. and inclined to the H.P. (fig. 12-4).

A square $ABCD$ is perpendicular to the V.P. and inclined at an angle θ to the H.P. Its H.T. is perpendicular to xy . Its V.T. makes the angle θ with xy . Its front view $a'b'$ is a line inclined at θ to xy . The top view $abcd$ is a rectangle which is smaller than the square $ABCD$.

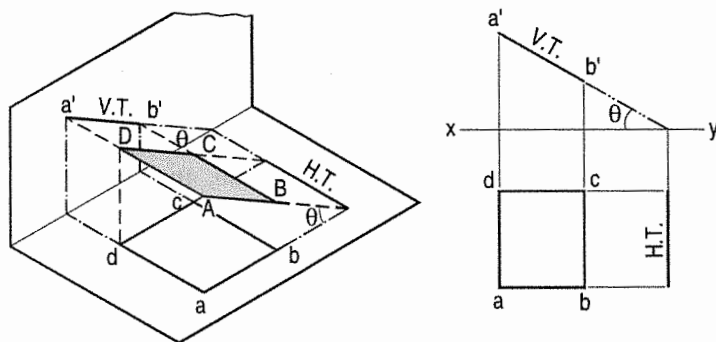


FIG. 12-4

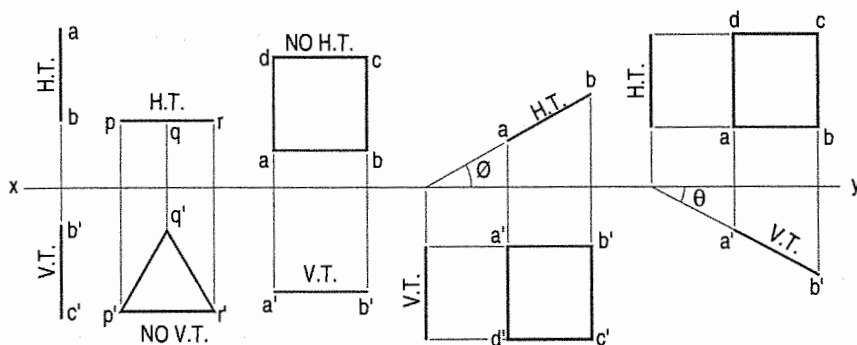


FIG. 12-5

Fig. 12-5 shows the projections and the traces of all these perpendicular planes by third-angle projection method.

(2) **Oblique planes:** Planes which are inclined to both the reference planes are called *oblique planes*. Representation of oblique planes by their traces is too advanced to be included in this book.

A few problems on the projections of plane figures inclined to both the reference planes are however, illustrated at the end of the chapter. They will prove to be of great use in dealing with the projections of solids.

12-2. TRACES OF PLANES

A plane, extended if necessary, will meet the reference planes in lines, unless it is parallel to any one of them.

These lines are called the *traces* of the plane. The line in which the plane meets the H.P. is called the *horizontal trace* or the H.T. of the plane. The line in which it meets the V.P. is called its *vertical trace* or the V.T. A plane is usually represented by its traces.

12-3. GENERAL CONCLUSIONS



(1) Traces:

- (a) When a plane is perpendicular to both the reference planes, its traces lie on a straight line perpendicular to xy .
- (b) When a plane is perpendicular to one of the reference planes, its trace upon the other plane is perpendicular to xy (except when it is parallel to the other plane).
- (c) When a plane is parallel to a reference plane, it has no trace on that plane. Its trace on the other reference plane, to which it is perpendicular, is parallel to xy .
- (d) When a plane is inclined to the H.P. and perpendicular to the V.P., its inclination is shown by the angle which its V.T. makes with xy . When it is inclined to the V.P. and perpendicular to the H.P., its inclination is shown by the angle which its H.T. makes with xy .
- (e) When a plane has two traces, they, produced if necessary, intersect in xy (except when both are parallel to xy as in case of some oblique planes).

(2) Projections:

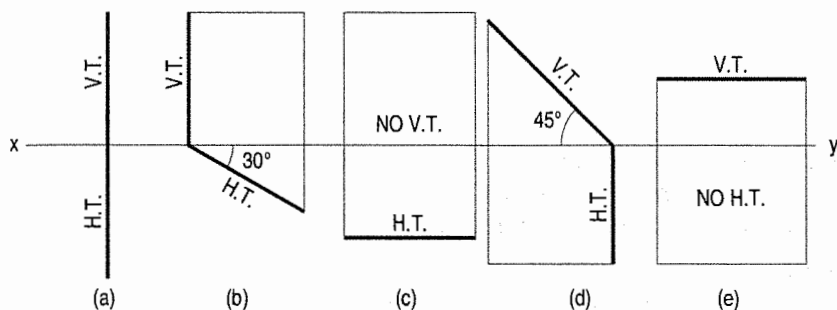
- (a) When a plane is perpendicular to a reference plane, its projection on that plane is a straight line.
- (b) When a plane is parallel to a reference plane, its projection on that plane shows its true shape and size.
- (c) When a plane is perpendicular to one of the reference planes and inclined to the other, its inclination is shown by the angle which its projection on the plane to which it is perpendicular, makes with xy . Its projection on the plane to which it is inclined, is smaller than the plane itself.

Problem 12-1. Show by means of traces, each of the following planes:

- (a) Perpendicular to the H.P. and the V.P.
- (b) Perpendicular to the H.P. and inclined at 30° to the V.P.
- (c) Parallel to and 40 mm away from the V.P.
- (d) Inclined at 45° to the H.P. and perpendicular to the V.P.
- (e) Parallel to the H.P. and 25 mm away from it.

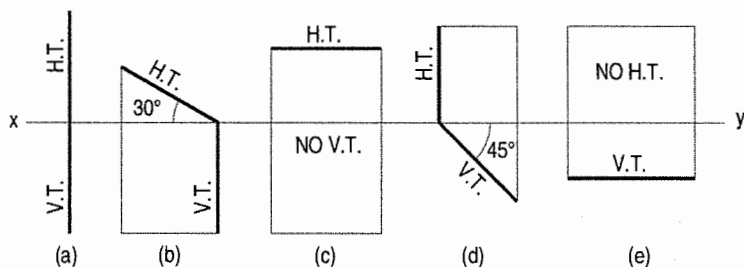
Fig. 12-6 and fig. 12-7 show the various traces.

- (a) The H.T. and the V.T. are in a line perpendicular to xy .
- (b) The H.T. is inclined at 30° to xy ; the V.T. is normal to xy ; both the traces intersect in xy .
- (c) The H.T. is parallel to and 40 mm away from xy . It has no V.T.
- (d) The H.T. is perpendicular to xy ; the V.T. makes 45° angle with xy ; both intersect in xy .
- (e) The V.T. is parallel to and 25 mm away from xy . It has no H.T.



(First-angle projection)

FIG. 12-6



(Third-angle projection)

FIG. 12-7

12-4. PROJECTIONS OF PLANES PARALLEL TO ONE OF THE REFERENCE PLANES

The projection of a plane on the reference plane parallel to it will show its true shape. Hence, beginning should be made by drawing that view. The other view which will be a line, should then be projected from it.

(1) When the plane is parallel to the H.P.: The top view should be drawn first and the front view projected from it.

Problem 12-2. (fig. 12-8): An equilateral triangle of 50 mm side has its V.T. parallel to and 25 mm above xy . It has no H.T. Draw its projections when one of its sides is inclined at 45° to the V.P.

As the V.T. is parallel to xy and as there is no H.T. the triangle is parallel to the H.P. Therefore, begin with the top view.

- (i) Draw an equilateral triangle abc of 50 mm side, keeping one side, say ac , inclined at 45° to xy .
- (ii) Project the front view, parallel to and 25 mm above xy , as shown.

(2) When the plane is parallel to the V.P.: Beginning should be made with the front view and the top view projected from it.

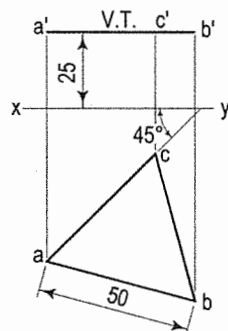


FIG. 12-8

Problem 12-3. (fig. 12-9): A square $ABCD$ of 40 mm side has a corner on the H.P. and 20 mm in front of the V.P. All the sides of the square are equally inclined to the H.P. and parallel to the V.P. Draw its projections and show its traces.

As all the sides are parallel to the V.P., the surface of the square also is parallel to it. The front view will show the true shape and position of the square.

- (i) Draw a square $a'b'c'd'$ in the front view with one corner in xy and all its sides inclined at 45° to xy .
- (ii) Project the top view keeping the line ac parallel to xy and 30 mm below it. The top view is its H.T. It has no V.T.

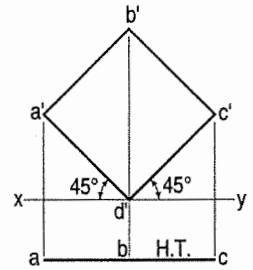


FIG. 12-9

12-5. PROJECTIONS OF PLANES INCLINED TO ONE REFERENCE PLANE AND PERPENDICULAR TO THE OTHER

When a plane is inclined to a reference plane, its projections may be obtained in two stages. In the initial stage, the plane is assumed to be parallel to that reference plane to which it has to be made inclined. It is then tilted to the required inclination in the second stage.

(1) **Plane, inclined to the H.P. and perpendicular to the V.P.:** When the plane is inclined to the H.P. and perpendicular to the V.P., in the initial stage, it is assumed to be parallel to the H.P. Its top view will show the true shape. The front view will be a line parallel to xy . The plane is then tilted so that it is inclined to the H.P. The new front view will be inclined to xy at the true inclination. In the top view the corners will move along their respective paths (parallel to xy).

Problem 12-4. (fig. 12-10): A regular pentagon of 25 mm side has one side on the ground. Its plane is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections and show its traces.

Assuming it to be parallel to the H.P.

- (i) Draw the pentagon in the top view with one side perpendicular to xy [fig. 12-10(i)]. Project the front view. It will be the line $a'c'$ contained by xy .
- (ii) Tilt the front view about the point a' , so that it makes 45° angle with xy .
- (iii) Project the new top view $ab_1c_1d_1e$ upwards from this front view and horizontally from the first top view. It will be more convenient if the front view is reproduced in the new position separately and the top view projected from it, as shown in fig. 12-10(ii). The V.T. coincides with the front view and the H.T. is perpendicular to xy , through the point of intersection between xy and the front view-produced.

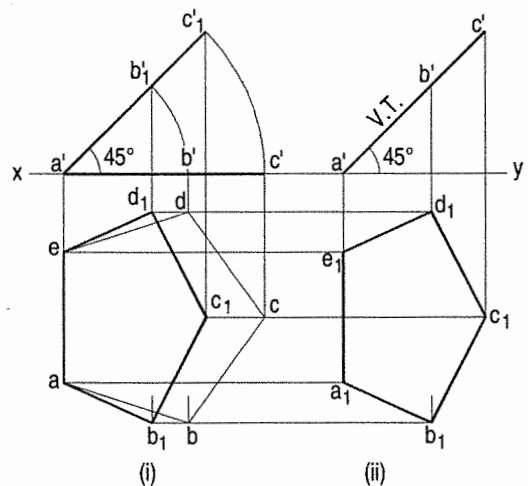


FIG. 12-10

The V.T. coincides with the front view and the H.T. is perpendicular to xy , through the point of intersection between xy and the front view-produced.

(2) **Plane, inclined to the V.P. and perpendicular to the H.P.:** In the initial stage, the plane may be assumed to be parallel to the V.P. and then tilted to the required position in the next stage. The projections are drawn as illustrated in the next problem.

Problem 12-5. (fig. 12-11): Draw the projections of a circle of 50 mm diameter, having its plane vertical and inclined at 30° to the V.P. Its centre is 30 mm above the H.P. and 20 mm in front of the V.P. Show also its traces.

A circle has no corners to project one view from another. However, a number of points, say twelve, equal distances apart, may be marked on its circumference.

- (i) Assuming the circle to be parallel to the V.P., draw its projections. The front view will be a circle [fig. 12-11(i)], having its centre 30 mm above xy . The top view will be a line, parallel to and 20 mm below xy .
- (ii) Divide the circumference into twelve equal parts (with a 30° - 60° set-square) and mark the points as shown. Project these points in the top view. The centre O will coincide with the point 4.

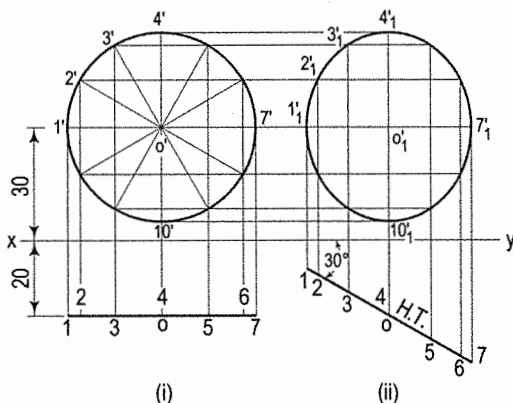


FIG. 12-11

- (iii) When the circle is tilted, so as to make 30° angle with the V.P., its top view will become inclined at 30° to xy . In the front view all the points will move along their respective paths (parallel to xy). Reproduce the top view keeping the centre o at the same distance, viz. 20 mm from xy and inclined at 30° to xy [fig. 12-11(ii)].
- (iv) For the final front view, project all the points upwards from this top view and horizontally from the first front view. Draw a freehand curve through the twelve points $1'_1, 2'_1$ etc. This curve will be an ellipse.

12-6. PROJECTIONS OF OBLIQUE PLANES

When a plane has its surface inclined to one plane and an edge or a diameter or a diagonal parallel to that plane and inclined to the other plane, its projections are drawn in three stages.

- (1) If the surface of the plane is inclined to the H.P. and an edge (or a diameter or a diagonal) is parallel to the H.P. and inclined to the V.P.,
 - (i) in the initial position the plane is assumed to be parallel to the H.P. and an edge perpendicular to the V.P.
 - (ii) It is then tilted so as to make the required angle with the H.P. As already explained, its front view in this position will be a line, while its top view will be smaller in size.
 - (iii) In the final position, when the plane is turned to the required inclination with the V.P., only the position of the top view will change. Its shape and size will not be affected. In the final front view, the corresponding distances of all the corners from xy will remain the same as in the second front view.

If an edge is in the H.P. or on the ground, in the initial position, the plane is assumed to be lying in the H.P. or on the ground, with the edge perpendicular to the V.P. If a corner is in the H.P. or on the ground, the line joining that corner with the centre of the plane is kept parallel to the V.P.

- (2) Similarly, if the surface of the plane is inclined to the V.P. and an edge (or a diameter or a diagonal) is parallel to the V.P. and inclined to the H.P.,
 - (i) in the initial position, the plane is assumed to be parallel to the V.P. and an edge perpendicular to the H.P.
 - (ii) It is then tilted so as to make the required angle with the V.P. Its top view in this position will be a line, while its front view will be smaller in size.
 - (iii) When the plane is turned to the required inclination with the H.P., only the position of the front view will change. Its shape and size will not be affected. In the final top view, the corresponding distances of all the corners from xy will remain the same as in the second top view.

If an edge is in the V.P., in the initial position, the plane is assumed to be lying in the V.P. with an edge perpendicular to the H.P. If a corner is in the V.P., the line joining that corner with centre of the plane is kept parallel to the H.P.

Problem 12-6. (fig. 12-12): A square $ABCD$ of 50 mm side has its corner A in the H.P., its diagonal AC inclined at 30° to the H.P. and the diagonal BD inclined at 45° to the V.P. and parallel to the H.P. Draw its projections.

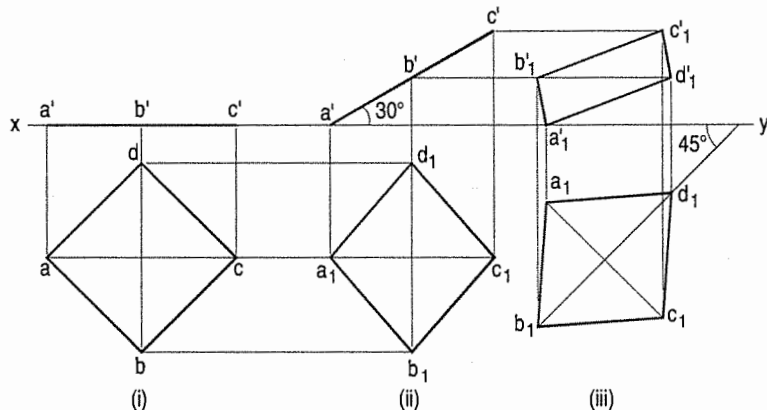


FIG. 12-12

In the initial stage, assume the square to be lying in the H.P. with AC parallel to the V.P.

- (i) Draw the top view and the front view. When the square is tilted about the corner A so that AC makes 30° angle with the H.P., BD remains perpendicular to the V.P. and parallel to the H.P.
- (ii) Draw the second front view with $a'c'$ inclined at 30° to xy , keeping a' or c' in xy . Project the second top view. The square may now be turned so that BD makes 45° angle with the V.P. and remains parallel to the H.P. Only the position of the top view will change. Its shape and size will remain the same.

- (iii) Reproduce the top view so that b_1d_1 is inclined at 45° to xy . Project the final front view upwards from this top view and horizontally from the second front view.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 27 for the following problem.

Problem 12-7. (Fig. 12-13): A rectangular plane surface of size $L \times W$ is positioned in the first quadrant and is inclined at an angle of 60° with the H.P. and 30° with the V.P. Draw its projections.

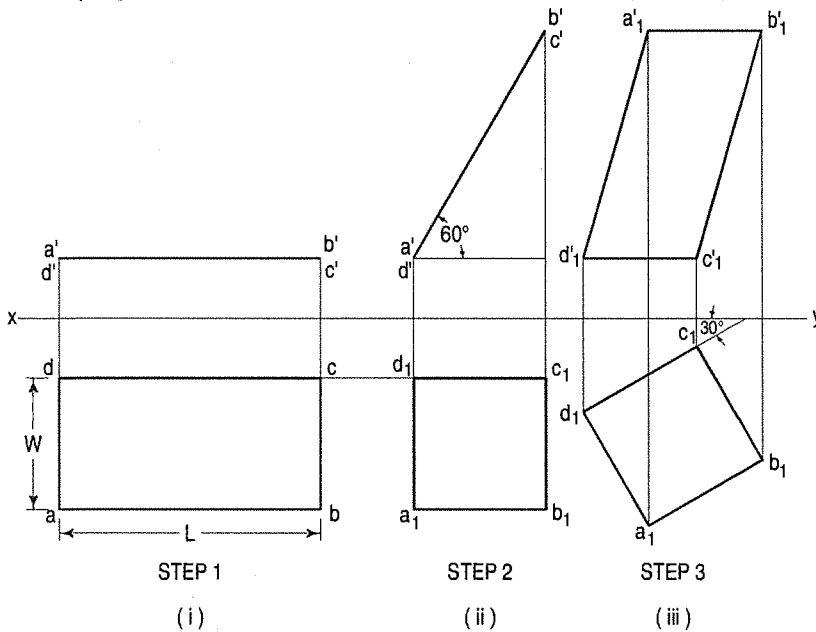


FIG. 12-13

- The plane is first assumed to be parallel to H.P. with its shorter edge perpendicular to V.P. In this position, true shape and size of the plane is given by its projection on H.P. The front view will be a true line parallel to the reference line xy .
- Rotate the front view projection by 60° (the angle of inclination of plane with H.P.) as shown in Step 2 of fig. 12-13(ii). Draw vertical lines from the ends of line $a'd'$ and $b'c'$ to intersect horizontal lines drawn from the top view $abcd$ (step 1) at points b_1, c_1, d_1 and a_1 . Join $a_1b_1c_1d_1$ to obtain the top view of the plane in this inclined position.
- Now rotate the edge d_1c_1 of the top view (step 2) by 30° (the angle of inclination of plane with V.P.) and reproduce it as shown in step 3 of the fig. 12-13(iii). Draw projections from a_1, b_1, c_1 and d_1 to intersect the horizontal projections from $a'd'$ and $b'c'$ to get the points a'_1, b'_1, c'_1 and d'_1 . Join the lines $a'_1b'_1c'_1d'_1$ to obtain the final front view of the given plane surface.

Problem 12-8. (fig. 12-14): Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the H.P. and inclined at 60° to the V.P., and its surface making an angle of 45° with the H.P.

- (i) Draw the hexagon in the top view with one side perpendicular to xy . Project the front view $a'c'$ in xy .
- (ii) Draw $a'c'$ inclined at 45° to xy keeping a' or c' in xy and project the second top view.
- (iii) Reproduce this top view making a_1f_1 inclined at 60° to xy and project the final front view.

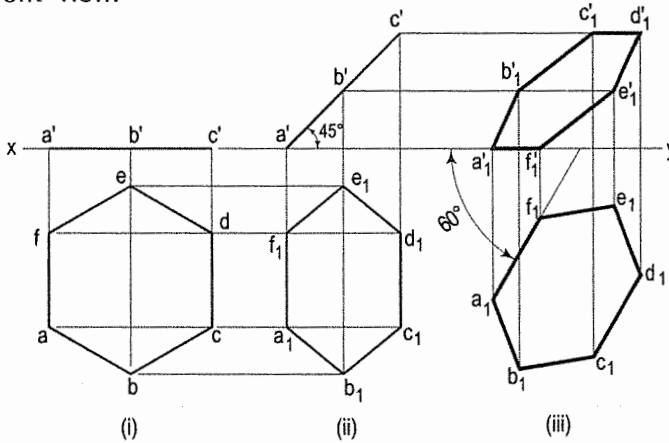


FIG. 12-14

Problem 12-9. (fig. 12-15): Draw the projections of a circle of 50 mm diameter resting in the H.P. on a point A on the circumference, its plane inclined at 45° to the H.P. and

- (a) the top view of the diameter AB making 30° angle with the V.P.;
- (b) the diameter AB making 30° angle with the V.P.

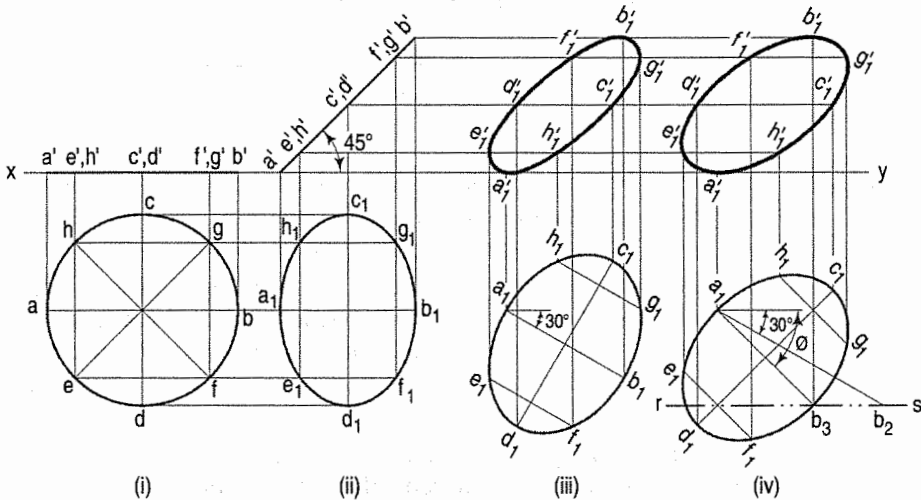


FIG. 12-15

Draw the projections of the circle with A in the H.P. and its plane inclined at 45° to the H.P. and perpendicular to the V.P. [fig. 12-15(i) and fig. 12-15(ii)].

- (a) In the second top view, the line a_1b_1 is the top view of the diameter AB. Reproduce this top view so that a_1b_1 makes 30° angle with xy [fig. 12-15(iii)]. Project the required front view.

- (b) If the diameter AB , which makes 45° angle with the H.P., is inclined at 30° to the V.P. also, its top view a_1b_1 will make an angle greater than 30° with xy . This apparent angle of inclination is determined as described below.

Draw any line a_1b_2 equal to AB and inclined at 30° to xy [fig. 12-15(iv)]. With a_1 as centre and radius equal to the top view of AB , viz. a_1b_1 , draw an arc cutting rs (the path of B in the top view) at b_3 . Draw the line joining a_1 with b_3 , and around it, reproduce the second top view. Project the final front view. It is evident that a_1b_3 is inclined to xy at an angle ϕ which is greater than 30° .

Problem 12-10. (fig. 12-16): A thin 30° - 60° set-square has its longest edge in the V.P. and inclined at 30° to the H.P. Its surface makes an angle of 45° with the V.P. Draw its projections.

In the initial stage, assume the set-square to be in the V.P. with its hypotenuse perpendicular to the H.P.

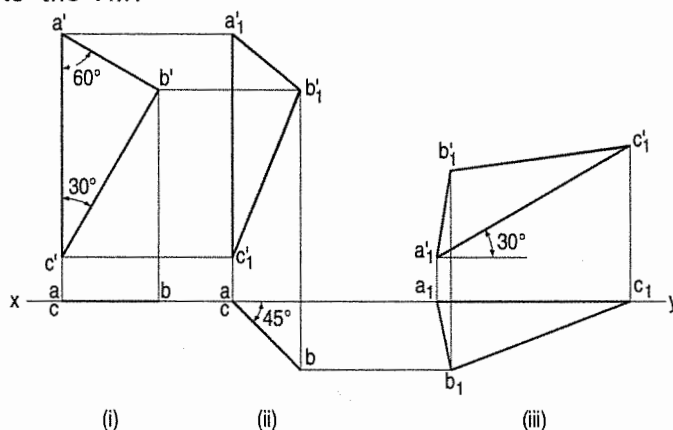


FIG. 12-16

- (i) Draw the front view $a'b'c'$ and project the top view ac in xy .
- (ii) Tilt ac around the end a so that it makes 45° angle with xy and project the front view $a'1b'1c'1$.
- (iii) Reproduce the second front view $a'1b'1c'1$ so that $a'1b'1$ makes an angle of 30° with xy . Project the final top view $a_1b_1c_1$.

Problem 12-11. (fig. 12-17): A thin rectangular plate of sides $60 \text{ mm} \times 30 \text{ mm}$ has its shorter side in the V.P. and inclined at 30° to the H.P. Project its top view if its front view is a square of 30 mm long sides.

As the front view of the plate is a square, its surface must be inclined to the V.P. Hence, assume the plate to be in the V.P. with its shorter edge perpendicular to the H.P.

- (i) Draw the front view $a'b'c'd'$ and project the top view ab in xy [fig. 12-17(i)].
- (ii) The line ab should be so inclined to xy that the front view becomes a square. Therefore, draw the square $a'1b'1c'1d'1$ of side equal to $a'd'$. With a as centre and radius equal to ab draw an arc cutting the projector through $b'1$ at b . Then ab is the new top view.
- (iii) Reproduce the second front view in such a way that $a'1d'1$ makes 30° angle with xy . Project the final top view as shown.

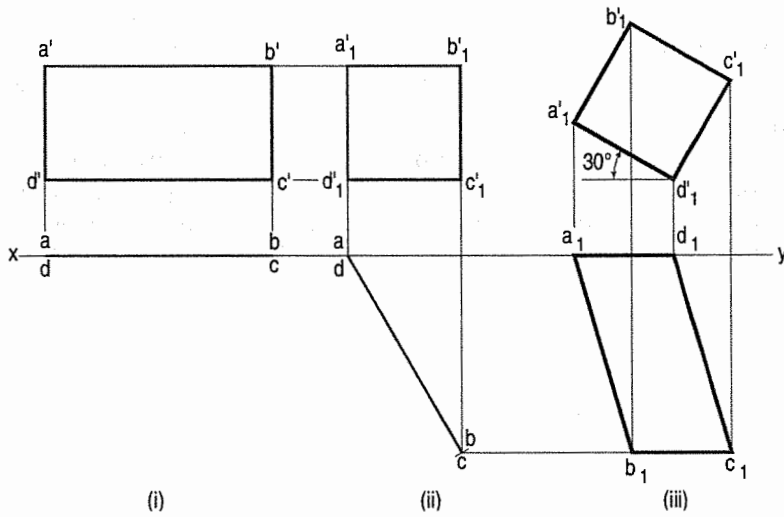


FIG. 12-17

Problem 12-12. (fig. 12-18): A circular plate of negligible thickness and 50 mm diameter appears as an ellipse in the front view, having its major axis 50 mm long and minor axis 30 mm long. Draw its top view when the major axis of the ellipse is horizontal.

As the plate is seen as an ellipse in the front view, its surface must be inclined to the V.P.

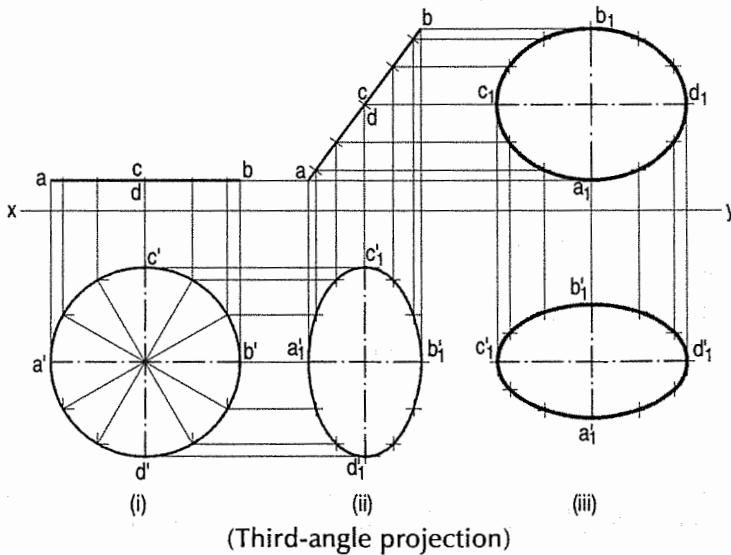


FIG. 12-18

- (i) Therefore, assume it to be parallel to the V.P. and draw its front view and the top view.
- (ii) Turn the line ab so that its length in the front view becomes 30 mm, and project the front view. It will be an ellipse whose major axis is vertical.
- (iii) Reproduce this view so that the major axis c_1d_1 is horizontal, and project the required top view.

Problem 12-13. Fig. 12-19 shows a thin plate of negligible thickness. It rests on its PQ edge with its plane perpendicular to V.P. and inclined 40° to the H.P. Draw its projections.

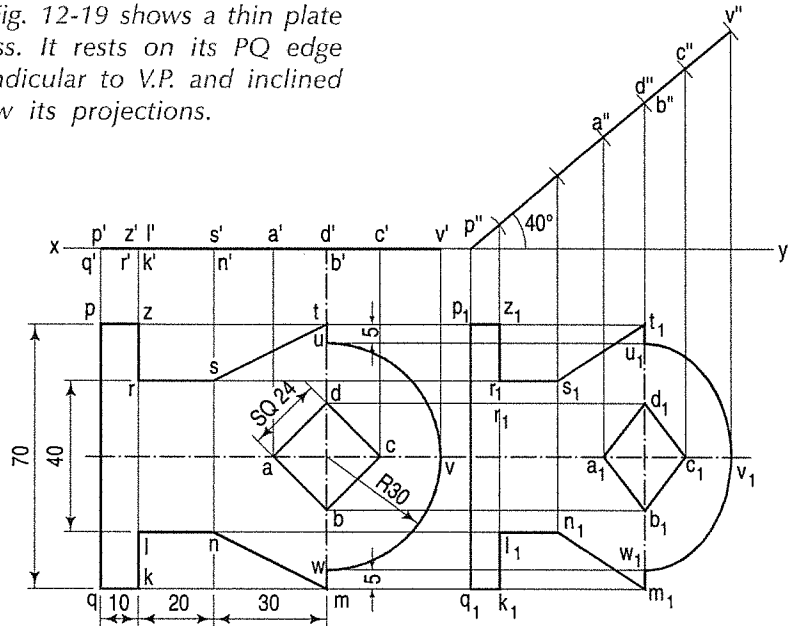


FIG. 12-19

- (i) Keep the plane of plate in the H.P. and draw the projections as shown.
- (ii) Tilt front view at p'' making an angle of 40° .
- (iii) Project p'' , a'' , d'' , c'' , v'' etc. in the top view. Draw horizontal projectors intersecting previously drawn projectors from the front view. Join by smooth curve to complete the top view.

Problem 12-14. (fig. 12-20): A pentagonal plate of 45 mm side has a circular hole of 40 mm diameter in its centre. The plane stands on one of its sides on the H.P. with its plane perpendicular to V.P. and 45° inclined to the H.P. Draw the projections.

- (i) Keep the plane of plate in the horizontal plane.
- (ii) Draw top view and front view as shown.
- (iii) Tilt the front view a'' d'' at a'' making an angle of 45° . Draw the projectors from various points a'' d'' .
- (iv) Draw horizontal projectors from the top view $abcd$ as shown. Join the intersection points and complete new top view a_1 , b_1 , c_1 , d_1 , e_1 , as shown.

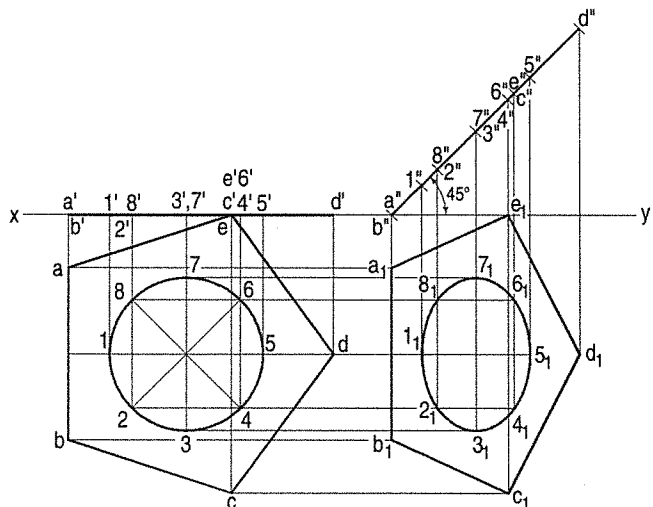


FIG. 12-20

Problem 12-15. (fig. 12-21): A thin circular plate of 70 mm diameter is resting on its circumference such that its plane is inclined 60° to the H.P. and 30° to the V.P. Draw the projections of the plate.

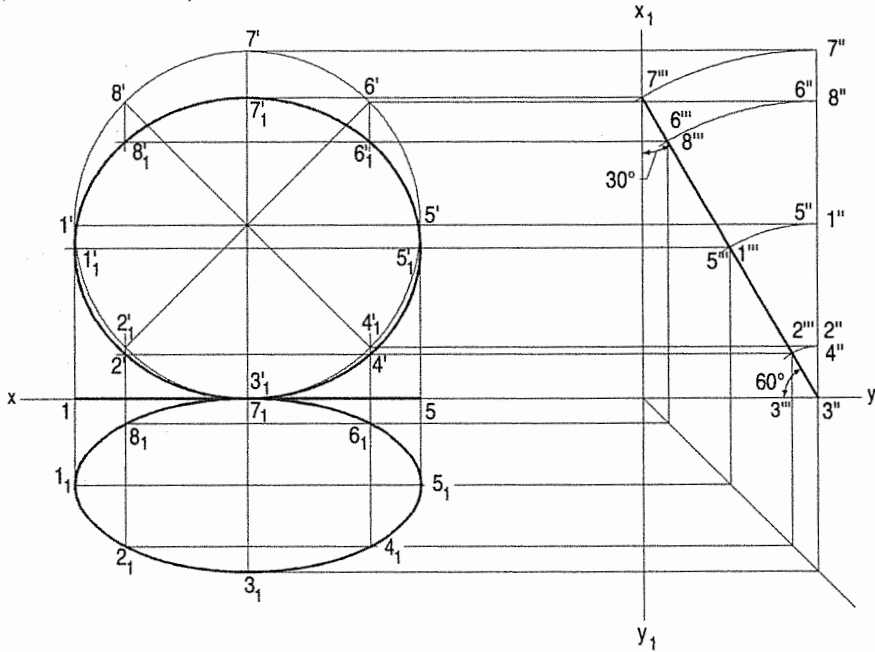


FIG. 12-21

- (i) Draw the projection of the plate keeping its plane parallel to the V.P. as shown in fig. 12-21.
- (ii) Mark a reference line x_1y_1 perpendicular to xy line to represent the auxiliary plane which is at right angle to both the H.P. and the V.P.
- (iii) Divide the front view in eight parts and mark the points $1', 2' \dots 8'$. Project these points on the side view as $1'', 2'' \dots 8''$.
- (iv) Tilt the side view $3'' 7''$ such that it touches the x_1y_1 line and also makes 60° with the xy line.
- (v) Complete the projection as shown in fig. 12-21.

Problem 12-16. (fig. 12-22): PQRS is a rhombus having diagonal $PR = 60$ mm and $QS = 40$ mm and they are perpendicular to each other. The plane of the rhombus is inclined with H.P. such that its top view appears to be square. The top view of PR makes 30° with the V.P. Draw its projections and determine inclination of the plane with the H.P.

- (i) Assume that the rhombus is lying in H.P. with its longest diagonal parallel to xy line.
- (ii) Draw the plans of diagonals $PR = 60$ mm and $QS = 40$ mm (true length) perpendicular each other as shown.
- (iii) Join points p, q, r and s . It is top view of the rhombus. Project the points p, q, r and s in the xy line. It is front view of the rhombus points p', q', r' and s' in the xy line as the plane of rhombus is perpendicular to the V.P.

- (iv) PR and QS are lying in H.P. pr and qs are true length. As the plane of the rhombus is inclined to H.P., the top view of the rhombus is going to be a square. But diagonal qs does not change in the length as it is perpendicular to V.P.
- (v) Draw the projectors from the points p, q and r, s parallel to the xy . From q_1 and s_1 draw square $p_1 q_1 r_1 s_1$ such that $s_1 q_1 = p_1 r_1$ as shown in fig. 12-22.
- (vi) Draw vertical projectors from p_1, q_1, r_1 and s_1 .
- (vii) Projector of p_1 intersects at p'' in xy line. Taking p'' as centre and the radius equal to 60 mm ($p' r'$), draw the arc to intersect the vertical projectors of r_1 at r'' . Join $p'' r''$. Measure angle of $p'' r''$ with xy line.
- (viii) Tilt diagonal $p_1 r_1$ at 30° with xy and reproduce square $p_2 q_2 r_2 s_2$. Draw vertical projectors from p_2, q_2, r_2 and s_2 to intersect the horizontal projectors from p'', q'', r'' and s'' at p''', q''', r''' and s''' . Join the points p''', q''', r''' and s''' as shown.

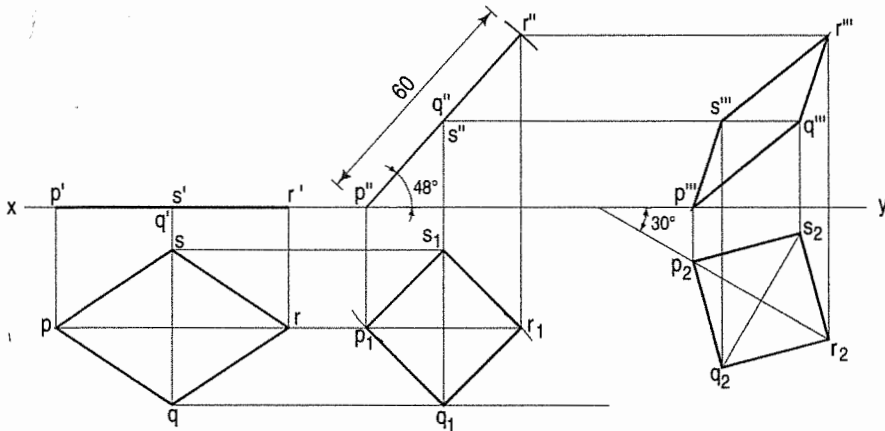


FIG. 12-22

EXERCISES 12

1. Draw an equilateral triangle of 75 mm side and inscribe a circle in it. Draw the projections of the figure, when its plane is vertical and inclined at 30° to the V.P. and one of the sides of the triangle is inclined at 45° to the H.P.
2. A regular hexagon of 40 mm side has a corner in the H.P. Its surface is inclined at 45° to the H.P. and the top view of the diagonal through the corner which is in the H.P. makes an angle of 60° with the V.P. Draw its projections.
3. Draw the projections of a regular pentagon of 40 mm side, having its surface inclined at 30° to the H.P. and a side parallel to the H.P. and inclined at an angle of 60° to the V.P.
4. Draw the projections of a rhombus having diagonals 125 mm and 50 mm long, the smaller diagonal of which is parallel to both the principal planes, while the other is inclined at 30° to the H.P.

5. Draw a regular hexagon of 40 mm side, with its two sides vertical. Draw a circle of 40 mm diameter in its centre. The figure represents a hexagonal plate with a hole in it and having its surface parallel to the V.P. Draw its projections when the surface is vertical and inclined at 30° to the V.P. Assume the thickness of the plate to be equal to that of a line.
6. Draw the projections of a circle of 75 mm diameter having the end *A* of the diameter *AB* in the H.P., the end *B* in the V.P., and the surface inclined at 30° to the H.P. and at 60° to the V.P.
7. A semi-circular plate of 80 mm diameter has its straight edge in the V.P. and inclined at 45° to the H.P. The surface of the plate makes an angle of 30° with the V.P. Draw its projections.
8. The top view of a plate, the surface of which is perpendicular to the V.P. and inclined at 60° to the H.P. is a circle of 60 mm diameter. Draw its three views.
9. A plate having shape of an isosceles triangle has base 50 mm long and altitude 70 mm. It is so placed that in the front view it is seen as an equilateral triangle of 50 mm sides and one side inclined at 45° to *xy*. Draw its top view.
10. Draw a rhombus of diagonals 100 mm and 60 mm long, with the longer diagonal horizontal. The figure is the top view of a square of 100 mm long diagonals, with a corner on the ground. Draw its front view and determine the angle which its surface makes with the ground.
11. A composite plate of negligible thickness is made-up of a rectangle 60 mm $\times$ 40 mm, and a semi-circle on its longer side. Draw its projections when the longer side is parallel to the H.P. and inclined at 45° to the V.P., the surface of the plate making 30° angle with the H.P.
12. A 60° set-square of 125 mm longest side is so kept that the longest side is in the H.P. making an angle of 30° with the V.P. and the set-square itself inclined at 45° to the H.P. Draw the projections of the set-square.
13. A plane figure is composed of an equilateral triangle *ABC* and a semi-circle on *AC* as diameter. The length of the side *AB* is 50 mm and is parallel to the V.P. The corner *B* is 20 mm behind the V.P. and 15 mm below the H.P. The plane of the figure is inclined at 45° to the H.P. Draw the projections of the plane figure.
14. An equilateral triangle *ABC* having side length as 50 mm is suspended from a point *O* on the side *AB* 15 mm from *A* in such a way that the plane of the triangle makes an angle of 60° with the V.P. The point *O* is 20 mm below the H.P. and 40 mm behind the V.P. Draw the projections of the triangle.
15. *PQRS* and *ABCD* are two square thin plates with their diagonals measuring 30 mm and 60 mm. They are touching the H.P. with their corners *P* and *A* respectively, and touching each other with their corresponding opposite corners *R* and *C*. If the plates are perpendicular to each other and perpendicular to V.P. also, draw their projections and determine the length of their sides.

Chapter 13



PROJECTIONS OF SOLIDS

13-0. INTRODUCTION



A solid has three dimensions, viz. length, breadth and thickness. To represent a solid on a flat surface having only length and breadth, at least two orthographic views are necessary. Sometimes, additional views projected on auxiliary planes become necessary to make the description of a solid complete.

This chapter deals with the following topics:

1. Types of solids.
2. Projections of solids in simple positions.
 - (a) Axis perpendicular to the H.P.
 - (b) Axis perpendicular to the V.P.
 - (c) Axis parallel to both the H.P. and the V.P.
3. Projections of solids with axes inclined to one of the reference planes and parallel to the other.
 - (a) Axis inclined to the V.P. and parallel to the H.P.
 - (b) Axis inclined to the H.P. and parallel to the V.P.
4. Projections of solids with axes inclined to both the H.P. and the V.P.
5. Projections of spheres.

13-1. TYPES OF SOLIDS



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 28 for the types of solids.

Solids may be divided into two main groups:

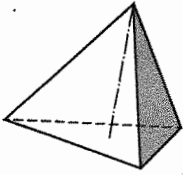
- (1) Polyhedra
- (2) Solids of revolution.

(1) **Polyhedra:** A polyhedron is defined as a solid bounded by planes called *faces*. When all the faces are equal and regular, the polyhedron is said to be regular.

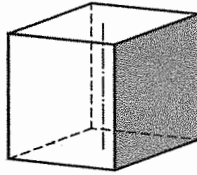
There are seven regular polyhedra which may be defined as stated below:

- (i) *Tetrahedron* (fig. 13-1): It has four equal faces, each an equilateral triangle.
- (ii) *Cube or hexahedron* (fig. 13-2): It has six faces, all equal squares.

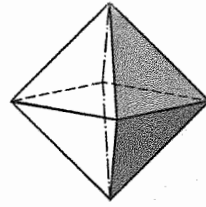
(iii) *Octahedron* (fig. 13-3): It has eight equal equilateral triangles as faces.



Tetrahedron
FIG. 13-1



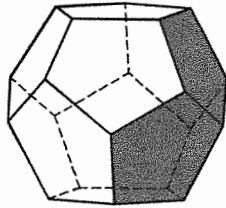
Cube
FIG. 13-2



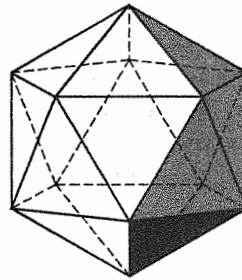
Octahedron
FIG. 13-3

(iv) *Dodecahedron* (fig. 13-4): It has twelve equal and regular pentagons as faces.

(v) *Icosahedron* (fig. 13-5): It has twenty faces, all equal equilateral triangles.



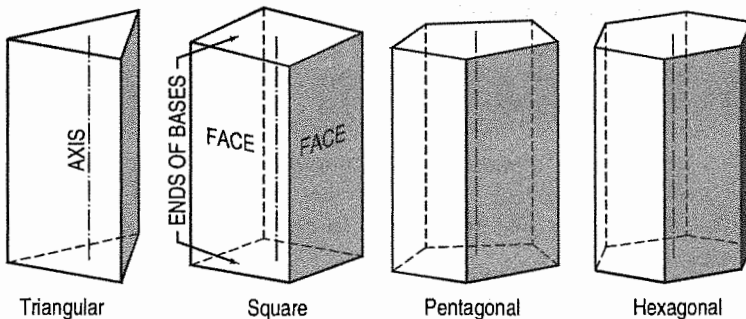
Dodecahedron
FIG. 13-4



Icosahedron
FIG. 13-5

(vi) *Prism*: This is a polyhedron having two equal and similar faces called its ends or bases, parallel to each other and joined by other faces which are parallelograms. The imaginary line joining the centres of the bases is called the axis.

A right and regular prism (fig. 13-6) has its axis perpendicular to the bases. All its faces are equal rectangles.



Triangular

Square

Pentagonal

Hexagonal

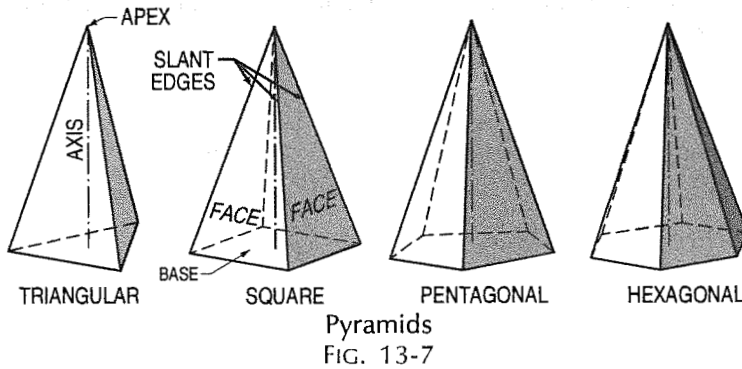
Prisms
FIG. 13-6

(vii) *Pyramid*: This is a polyhedron having a plane figure as a base and a number of triangular faces meeting at a point called the vertex or apex. The imaginary line joining the apex with the centre of the base is its axis.

A right and regular pyramid (fig. 13-7) has its axis perpendicular to the base which is a regular plane figure. Its faces are all equal isosceles triangles.

Oblique prisms and pyramids have their axes inclined to their bases.

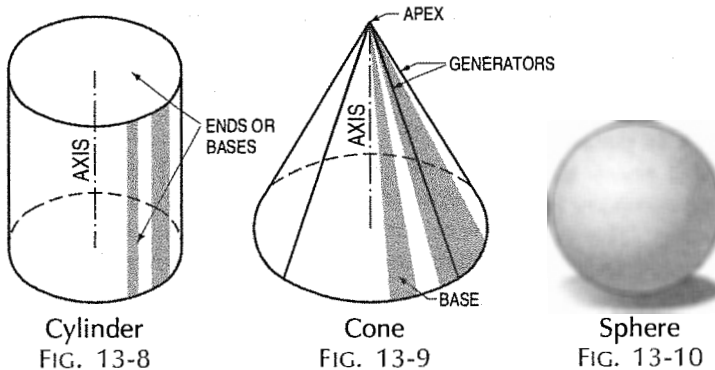
Prisms and pyramids are named according to the shape of their bases, as triangular, square, pentagonal, hexagonal etc.



(2) Solids of revolution:

- (i) *Cylinder* (fig. 13-8): A *right circular cylinder* is a solid generated by the revolution of a rectangle about one of its sides which remains fixed. It has two equal circular bases. The line joining the centres of the bases is the axis. It is perpendicular to the bases.
- (ii) *Cone* (fig. 13-9): A *right circular cone* is a solid generated by the revolution of a right-angled triangle about one of its perpendicular sides which is fixed.

It has one circular base. Its axis joins the apex with the centre of the base to which it is perpendicular. Straight lines drawn from the apex to the circumference of the base-circle are all equal and are called *generators* of the cone. The length of the generator is the slant height of the cone.



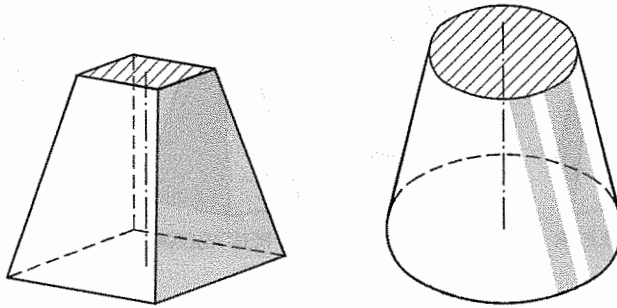
- (iii) *Sphere* (fig. 13-10): A *sphere* is a solid generated by the revolution of a semi-circle about its diameter as the axis. The mid-point of the diameter is the centre of the sphere. All points on the surface of the sphere are equidistant from its centre.

Oblique cylinders and cones have their axes inclined to their bases.

- (iv) *Frustum*: When a pyramid or a cone is cut by a plane parallel to its base, thus removing the top portion, the remaining portion is called its *frustum* (fig. 13-11).

- (v) *Truncated*: When a solid is cut by a plane inclined to the base it is said to be *truncated*.

In this book mostly right and regular solids are dealt with. Hence, when a solid is named without any qualification, it should be understood as being right and regular.



Frustums
FIG. 13-11

13-2. PROJECTIONS OF SOLIDS IN SIMPLE POSITIONS

A solid in simple position may have its axis perpendicular to one reference plane or parallel to both. When the axis is perpendicular to one reference plane, it is parallel to the other. Also, when the axis of a solid is perpendicular to a plane, its base will be parallel to that plane. We have already seen that when a plane is parallel to a reference plane, its projection on that plane shows its true shape and size.

Therefore, the projection of a solid on the plane to which its axis is perpendicular, will show the true shape and size of its base.

Hence, when the axis is perpendicular to the ground, i.e. to the H.P., the top view should be drawn first and the front view projected from it.

When the axis is perpendicular to the V.P., beginning should be made with the front view. The top view should then be projected from it.

When the axis is parallel to both the H.P. and the V.P., neither the top view nor the front view will show the actual shape of the base. In this case, the projection of the solid on an auxiliary plane perpendicular to both the planes, viz. the side view must be drawn first. The front view and the top view are then projected from the side view. The projections in such cases may also be drawn in two stages.

(1) Axis perpendicular to the H.P.:

Problem 13-1. (fig. 13-12): Draw the projections of a triangular prism, base 40 mm side and axis 50 mm long, resting on one of its bases on the H.P. with a vertical face perpendicular to the V.P.

- (i) As the axis is perpendicular to the ground i.e. the H.P. begin with the top view. It will be an equilateral triangle of sides 40 mm long, with one of its

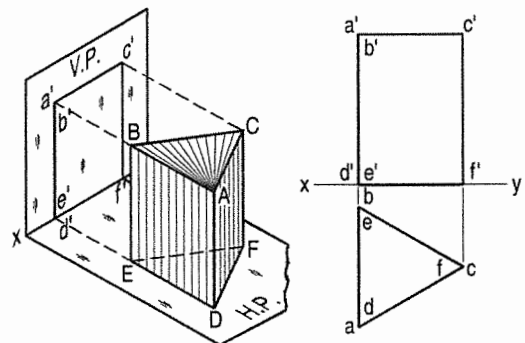


FIG. 13-12

sides perpendicular to xy . Name the corners as shown, thus completing the top view. The corners d, e and f are hidden and coincide with the top corners a, b and c respectively.

- (ii) Project the front view, which will be a rectangle. Name the corners. The line $b'e'$ coincides with $a'd'$.

Problem 13-2. (fig. 13-13): Draw the projections of a pentagonal pyramid, base 30 mm edge and axis 50 mm long, having its base on the H.P. and an edge of the base parallel to the V.P. Also draw its side view.

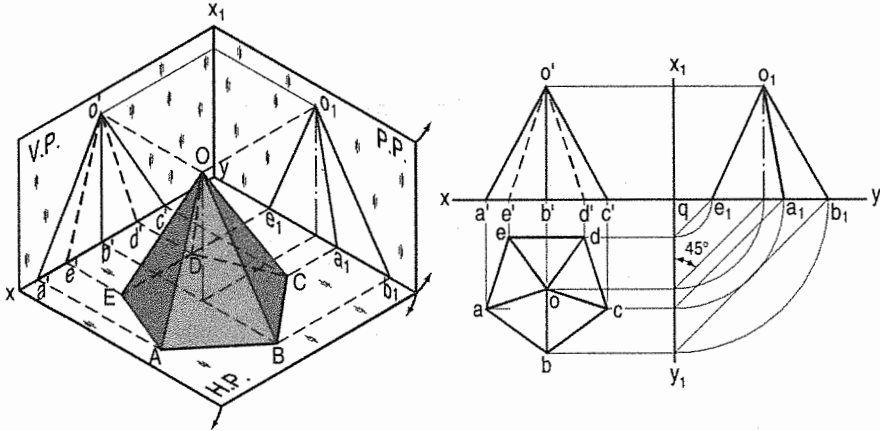


FIG. 13-13

- (i) Assume the side DE which is nearer the V.P., to be parallel to the V.P. as shown in the pictorial view.
- (ii) In the top view, draw a regular pentagon $abcde$ with ed parallel to and nearer xy . Locate its centre o and join it with the corners to indicate the slant edges.
- (iii) Through o , project the axis in the front view and mark the apex o' , 50 mm above xy . Project all the corners of the base on xy . Draw lines $o'a'$, $o'b'$ and $o'c'$ to show the visible edges. Show $o'd'$ and $o'e'$ for the hidden edges as dashed lines.
- (iv) For the side view looking from the left, draw a new reference line x_1y_1 perpendicular to xy and to the right of the front view. Project the side view on it, horizontally from the front view as shown. The respective distances of all the points in the side view from x_1y_1 , should be equal to their distances in the top view from xy . This is done systematically as explained below:
- (v) From each point in the top view, draw horizontal lines upto x_1y_1 . Then draw lines inclined at 45° to x_1y_1 (or xy) as shown. Or, with q , the point of intersection between xy and x_1y_1 as centre, draw quarter circles. Project up all the points to intersect the corresponding horizontal lines from the front view and complete the side view as shown in the figure. Lines o_1d_1 and o_1c_1 coincide with o_1e_1 and o_1a_1 respectively.

Problem 13-3. (fig. 13-14): Draw the projections of (i) a cylinder, base 40 mm diameter and axis 50 mm long, and (ii) a cone, base 40 mm diameter and axis 50 mm long, resting on the H.P. on their respective bases.

- (i) Draw a circle of 40 mm diameter in the top view and project the front view which will be a rectangle [fig. 13-14(ii)].
- (ii) Draw the top view [fig. 13-14(iii)]. Through the centre o , project the apex o' , 50 mm above xy . Complete the triangle in the front view as shown.

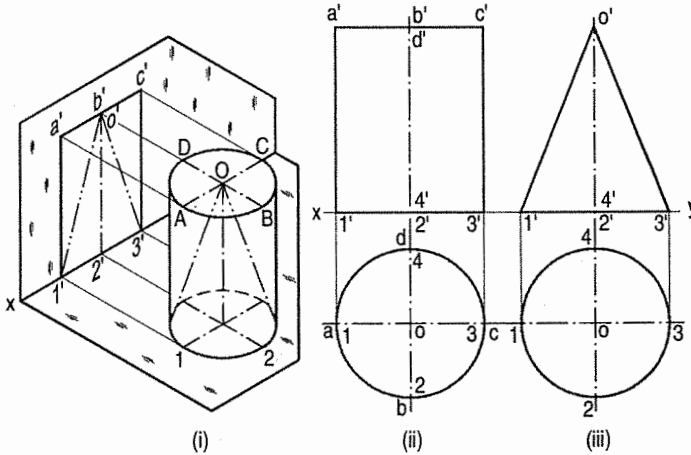


FIG. 13-14

In the pictorial view [fig. 13-14(i)], the cone is shown as contained by the cylinder.

Problem 13-4. (fig. 13-15): A cube of 50 mm long edges is resting on the H.P. with its vertical faces equally inclined to the V.P. Draw its projections.

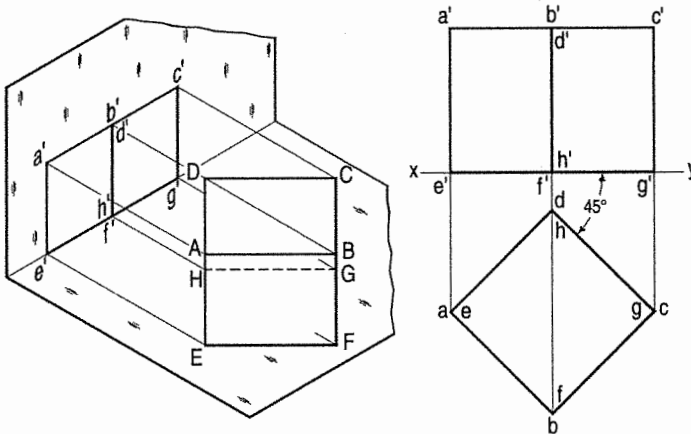


FIG. 13-15

Begin with the top view.

- (i) Draw a square $abcd$ with a side making 45° angle with xy .
- (ii) Project up the front view. The line $d' h'$ will coincide with $b' f'$.

Problem 13-5. (fig. 13-16): Draw the projections of a hexagonal pyramid, base 30 mm side and axis 60 mm long, having its base on the H.P. and one of the edges of the base inclined at 45° to the V.P.

- (i) In the top view, draw a line af 30 mm long and inclined at 45° to xy . Construct a regular hexagon on af . Mark its centre o and complete the top view by drawing lines joining it with the corners.
- (ii) Project up the front view as described in problem 13-2, showing the line $o'e'$ and $o'f'$ for hidden edges as dashed lines.

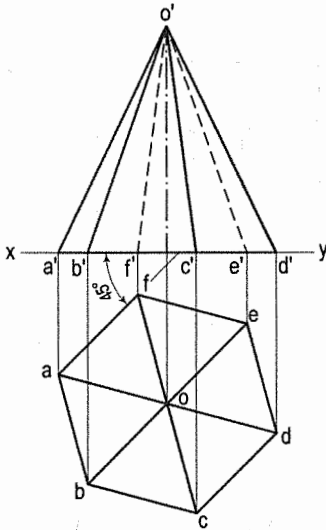


FIG. 13-16

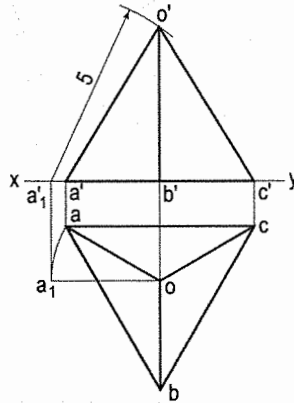


FIG. 13-17

Problem 13-6. (fig. 13-17): A tetrahedron of 5 cm long edges is resting on the H.P. on one of its faces, with an edge of that face parallel to the V.P. Draw its projections and measure the distance of its apex from the ground.

All the four faces of the tetrahedron are equal equilateral triangles of 5 cm side.

- (i) Draw an equilateral triangle abc in the top view with one side, say ac , parallel to xy . Locate its centre o and join it with the corners.
- (ii) In the front view, the corners a' , b' and c' will be in xy . The apex o' will lie on the projector through o so that its true distance from the corners of the base is equal to 5 cm.
- (iii) To locate o' , make oa (or ob or oc) parallel to xy . Project a_1 to a'_1 on xy . With a'_1 as centre and radius equal to 5 cm cut the projector through o in o' . Draw lines $o'a'$, $o'b'$ and $o'c'$ to complete the front view. $o'b'$ will be the distance of the apex from the ground.

(2) Axis perpendicular to the V.P.:

Problem 13-7. (fig. 13-18): A hexagonal prism has one of its rectangular faces parallel to the H.P. Its axis is perpendicular to the V.P. and 3.5 cm above the ground.

Draw its projections when the nearer end is 2 cm in front of the V.P. Side of base 2.5 cm long; axis 5 cm long.

- (i) Begin with the front view. Construct a regular hexagon of 2.5 cm long sides with its centre 3.5 cm above xy and one side parallel to it.
- (ii) Project down the top view, keeping the line for nearer end, viz. 1-4, 2 cm below xy .

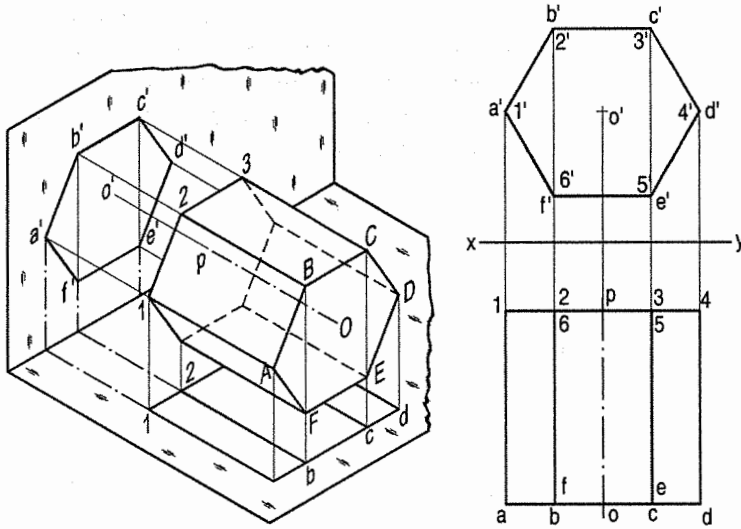


FIG. 13-18

Problem 13-8. (fig. 13-19): A square pyramid, base 40 mm side and axis 65 mm long, has its base in the V.P. One edge of the base is inclined at 30° to the H.P. and a corner contained by that edge is on the H.P. Draw its projections.

- (i) Draw a square in the front view with the corner d' in xy and the side $d'e'$ inclined at 30° to it. Locate the centre o' and join it with the corners of the square.
- (ii) Project down all the corners in xy (because the base is in the V.P.). Mark the apex o on a projector through o' . Draw lines for the slant edges and complete the top view.

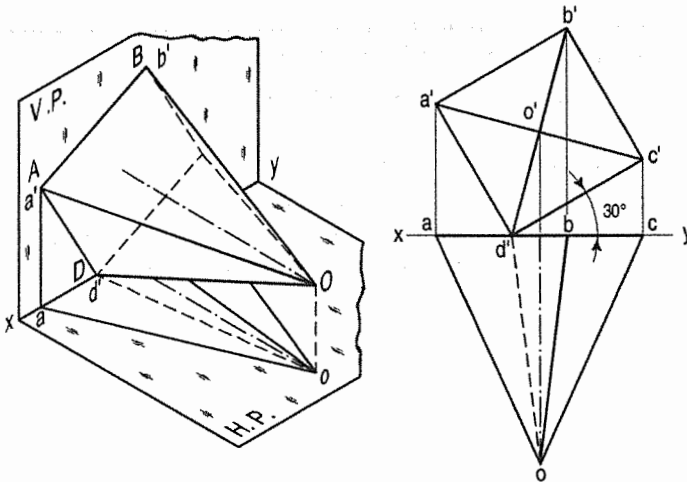


FIG. 13-19

(3) Axis parallel to both the H.P. and the V.P.:

Problem 13-9. (fig. 13-20): A triangular prism, base 40 mm side and height 65 mm is resting on the H.P. on one of its rectangular faces with the axis parallel to the V.P. Draw its projections.

As the axis is parallel to both the planes, begin with the side view.

- (i) Draw an equilateral triangle representing the side view, with one side in xy .
- (ii) Project the front view horizontally from this triangle.
- (iii) Project down the top view from the front view and the side view, as shown.

This problem can also be solved in two stages as explained in the next article.

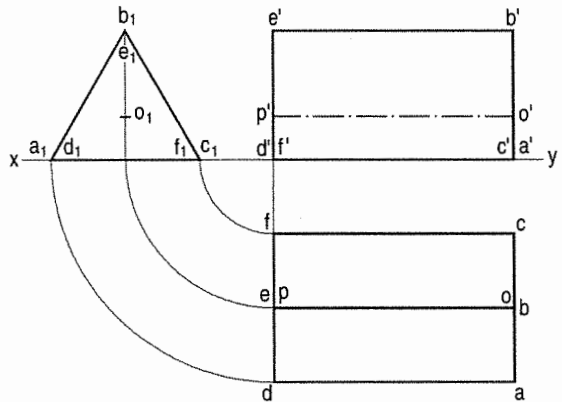


FIG. 13-20

EXERCISES 13(a)

Draw the projections of the following solids, situated in their respective positions, taking a side of the base 40 mm long or the diameter of the base 50 mm long and the axis 65 mm long.

1. A hexagonal pyramid, base on the H.P. and a side of the base parallel to and 25 mm in front of the V.P.
2. A square prism, base on the H.P., a side of the base inclined at 30° to the V.P. and the axis 50 mm in front of the V.P.
3. A triangular pyramid, base on the H.P. and an edge of the base inclined at 45° to the V.P.; the apex 40 mm in front of the V.P.
4. A cylinder, axis perpendicular to the V.P. and 40 mm above the H.P., one end 20 mm in front of the V.P.
5. A pentagonal prism, a rectangular face parallel to and 10 mm above the H.P., axis perpendicular to the V.P. and one base in the V.P.
6. A square pyramid, all edges of the base equally inclined to the H.P. and the axis parallel to and 50 mm away from both the H.P. and the V.P.
7. A cone, apex in the H.P. axis vertical and 40 mm in front of the V.P.
8. A pentagonal pyramid, base in the V.P. and an edge of the base in the H.P.

13-3. PROJECTIONS OF SOLIDS WITH AXES INCLINED TO ONE OF THE REFERENCE PLANES AND PARALLEL TO THE OTHER

When a solid has its axis inclined to one plane and parallel to the other, its projections are drawn in two stages.

- (1) In the initial stage, the solid is assumed to be in simple position, i.e. its axis perpendicular to one of the planes.

If the axis is to be inclined to the ground, i.e. the H.P., it is assumed to be perpendicular to the H.P. in the initial stage. Similarly, if the axis is to be inclined to the V.P., it is kept perpendicular to the V.P. in the initial stage.

Moreover

- (i) if the solid has an edge of its base parallel to the H.P. or in the H.P. or on the ground, that edge should be kept perpendicular to the V.P.; if the edge of the base is parallel to the V.P. or in the V.P., it should be kept perpendicular to the H.P.
- (ii) If the solid has a corner of its base in the H.P. or on the ground, the sides of the base containing that corner should be kept equally inclined to the V.P.; if the corner is in the V.P., they should be kept equally inclined to the H.P.

(2) Having drawn the projections of the solid in its simple position, the final projections may be obtained by one of the following two methods:

(i) **Alteration of position:** The position of one of the views is altered as required and the other view projected from it.

(ii) **Alteration of reference line or auxiliary plane:** A new reference line is drawn according to the required conditions, to represent an auxiliary plane and the final view projected on it.

In the first method, the reproduction of a view accurately in the altered position is likely to take considerable time, specially, when the solid has curved surfaces or too many edges and corners. In such cases, it is easier and more convenient to adopt the second method. Sufficient care must however be taken in transferring the distances of various points from their respective reference lines.

After determining the positions of all the points for the corners in the final view, difficulty is often felt in completing the view correctly. The following sequence for joining the corners may be adopted:

- (a) Draw the lines for the edges of the visible base. The base, which (compared to the other base) is further away from xy in one view, will be fully visible in the other view.
- (b) Draw the lines for the longer edges. The lines which pass through the figure of the visible base should be dashed lines.
- (c) Draw the lines for the edges of the other base.

It should always be remembered that, when two lines representing the edges cross each other, one of them must be hidden and should therefore be drawn as a dashed line.

13-3-1. AXIS INCLINED TO THE V.P. AND PARALLEL TO THE H.P.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 29 for the following problem.

Problem 13-10. (fig. 13-21): Draw the projections of a pentagonal prism, base 25 mm side and axis 50 mm long, resting on one of its rectangular faces on the H.P., with the axis inclined at 45° to the V.P.

In the simple position, assume the prism to be on one of its faces on the ground with the axis perpendicular to the V.P.

Draw the pentagon in the front view with one side in xy and project the top view [fig. 13-21(i)].

The shape and size of the figure in the top view will not change, so long as the prism has its face on the H.P. The respective distances of all the corners in the front view from xy will also remain constant.

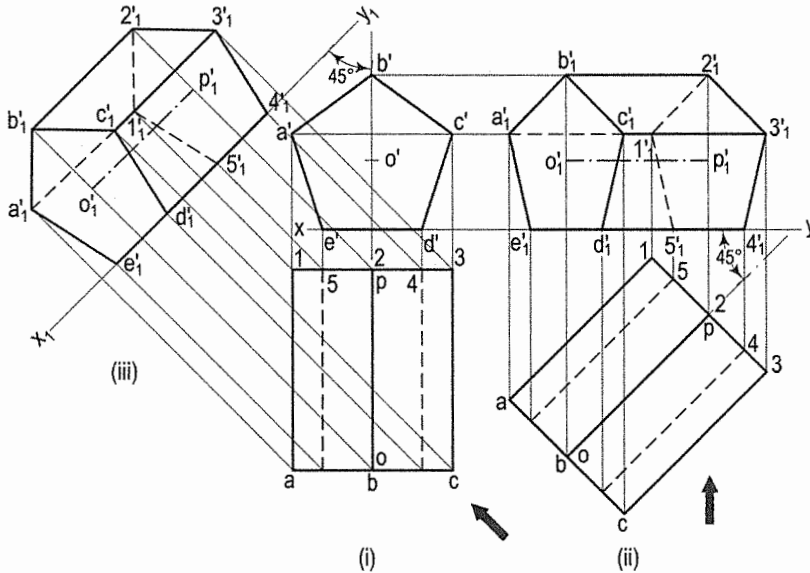


FIG. 13-21

Method I: [fig. 13-21(ii)]:

- (i) Alter the position of the top view, i.e. reproduce it so that the axis is inclined at 45° to xy . Project all the points upwards from this top view and horizontally from the first front view, e.g. a vertical from a intersecting a horizontal from a' at a point a'_1 .
- (ii) Complete the pentagon $a'_1b'_1c'_1d'_1e'_1$ for the fully visible end of the prism. Next, draw the lines for the longer edges and finally, draw the lines for the edges of the other end. Note carefully that the lines $a'_11'_1$, $1'_12'_1$ and $1'_15'_1$ are dashed lines. $e'_15'_1$ is also hidden but it coincides with other visible lines.

Method II: [fig. 13-21(iii)]:

- (i) Draw a new reference line x_1y_1 , making 45° angle with the top view of the axis, to represent an auxiliary vertical plane.
- (ii) Draw projectors from all the points in the top view perpendicular to x_1y_1 and on them, mark points keeping the distance of each point from x_1y_1 equal to its distance from xy in the front view. Join the points as already explained. The auxiliary front view and the top view are the required projections.

Problem 13-11. (fig. 13-22): Draw the projections of a cylinder 75 mm diameter and 100 mm long, lying on the ground with its axis inclined at 30° to the V.P. and parallel to the ground.

Adopt the same methods as in the previous problem. The ellipses for the ends should be joined by common tangents. Note that half of the ellipse for the hidden base will be drawn as dashed line.

Fig. 13-22(iii) shows the front view obtained by the method II.

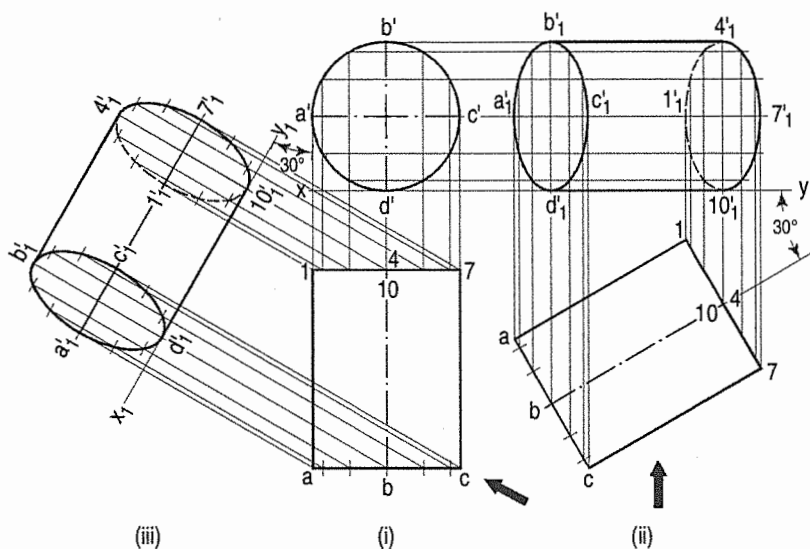


FIG. 13-22

13-3-2. AXIS INCLINED TO THE H.P. AND PARALLEL TO THE V.P.

Problem 13-12. (fig. 13-23): A hexagonal pyramid, base 25 mm side and axis 50 mm long, has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections.

In the initial position assume the axis to be perpendicular to the H.P.

Draw the projections with the base in xy and its one edge perpendicular to the V.P. [fig. 13-23(i)].

If the pyramid is now tilted about the edge AF (or CD) the axis will become inclined to the H.P. but will remain parallel to the V.P. The distances of all the corners from the V.P. will remain constant.

The front view will not be affected except in its position in relation to xy . The new top view will have its corners at same distances from xy , as before.

Method I: [fig. 13-23(ii)]:

- Reproduce the front view so that the axis makes 30° angle with xy and the point a' remains in xy .
- Project all the points vertically from this front view and horizontally from the first top view. Complete the new top view by drawing (a) lines joining the apex o'_1 with the corners of the base and (b) lines for the edges of the base.

The base will be partly hidden as shown by dashed line a_1b_1 , e_1f_1 and f_1a_1 . Similarly o_1f_1 and o_1a_1 are also dashed lines.

Method II: [fig. 13-23(iii)]:

- Through a' draw a new reference line x_1y_1 inclined at 30° to the axis, to represent an auxiliary inclined plane.
- From the front view project the required top view on x_1y_1 , keeping the distance of each point from x_1y_1 equal to the distance of its first top view from xy , viz. $e_1q = eb'$ etc.

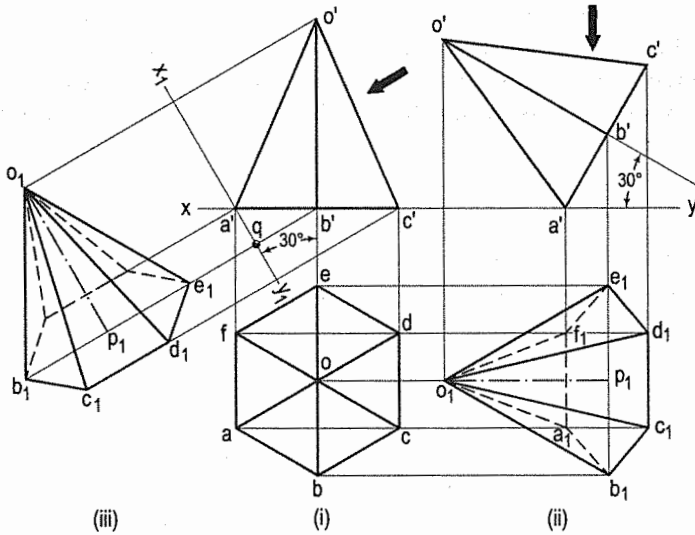


FIG. 13-23

Problem 13-13. (fig. 13-24): Draw the projections of a cone, base 75 mm diameter and axis 100 mm long, lying on the H.P. on one of its generators with the axis parallel to the V.P.

- (i) Assuming the cone to be resting on its base on the ground, draw its projections.
- (ii) Re-draw the front view so that the line $o'7'$ (or $o'1'$) is in xy . Project the required top view as shown. The lines from o_1 should be tangents to the ellipse.

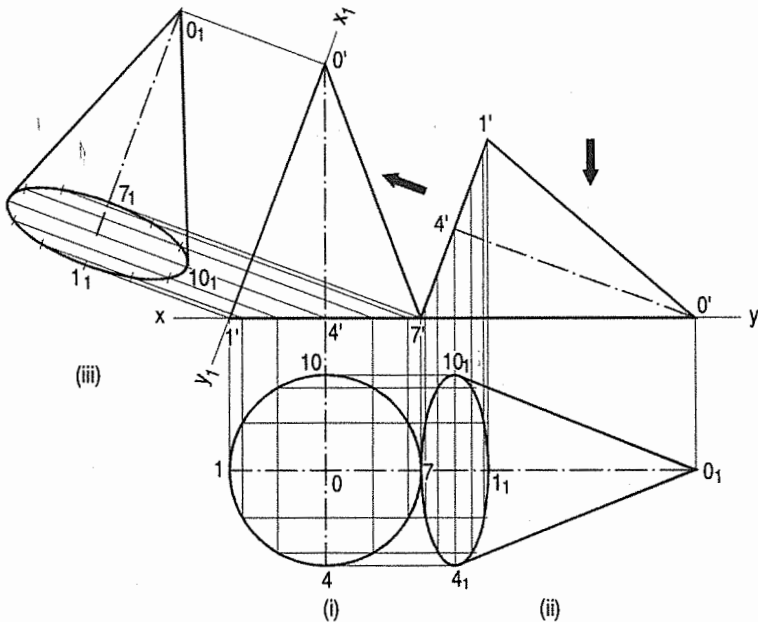


FIG. 13-24

The top view obtained by auxiliary-plane method is shown in fig. 13-24(iii). The new reference line x_1y_1 is so drawn as to contain the generator $o'1'$ instead of $o'7'$ (for sake of convenience). The cone is thus lying on the generator $o'1'$. Note that $1'1_1 = 1'1$, $o'o_1 = 4'o$ etc. Also note that the base is fully visible in both the methods.

Problem 13-14. The projections of a cylinder resting centrally on a hexagonal prism are given in fig. 13-25(i). Draw its auxiliary front view on a reference line inclined at 60° to xy .

See fig. 13-25(ii) which is self-explanatory.

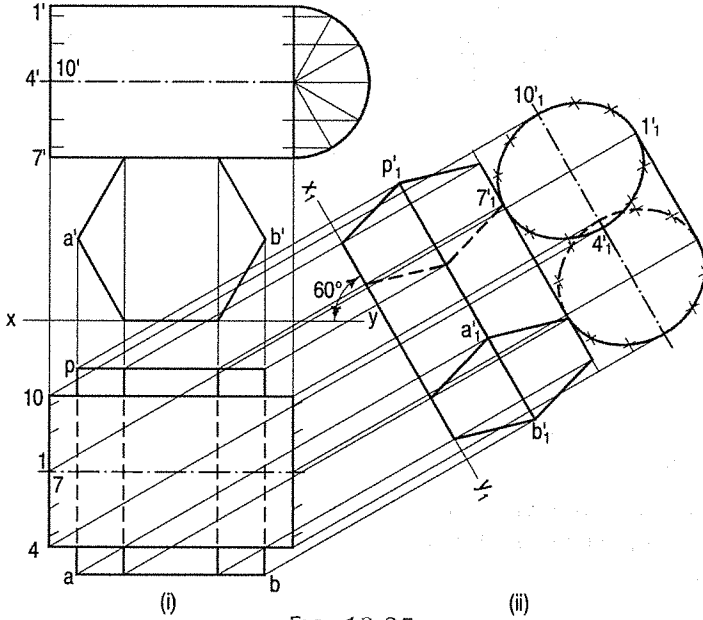


FIG. 13-25

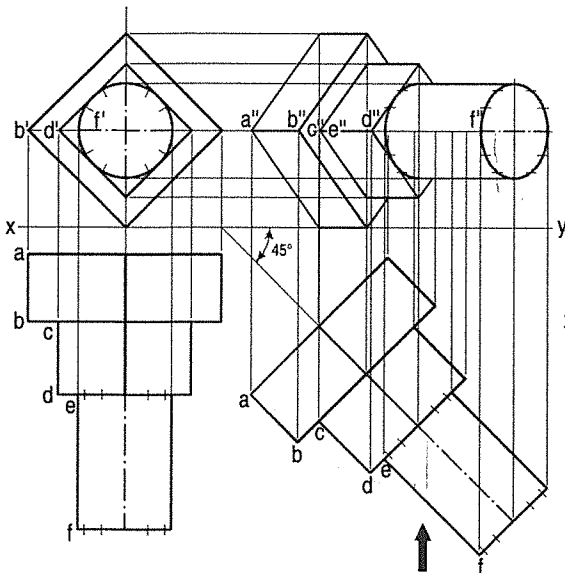


FIG. 13-26

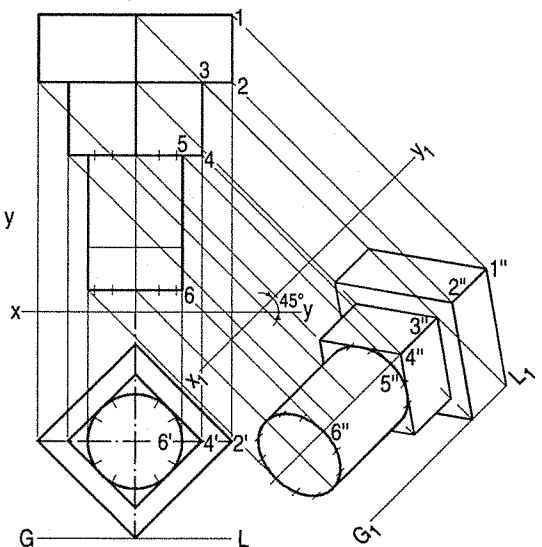


FIG. 13-27

Problem 13-15. (fig. 13-26 and fig. 13-27): A square-headed bolt 25 mm diameter, 125 mm long and having a square neck has its axis parallel to the H.P. and inclined at 45° to the V.P.

All the faces of the square head are equally inclined to the H.P. Draw its projections neglecting the threads and chamfer.

See fig. 13-26. The projections are obtained by the change-of-position method. The length of the bolt is taken shorter.

Fig. 13-27 shows the views in third-angle projection, obtained by the auxiliary-plane method.

Problem 13-16. (fig. 13-28): A hexagonal prism, base 40 mm side and height 40 mm has a hole of 40 mm diameter drilled centrally through its ends. Draw its projections when it is resting on one of its corners on the H.P. with its axis inclined at 60° to the H.P. and two of its faces parallel to the V.P.

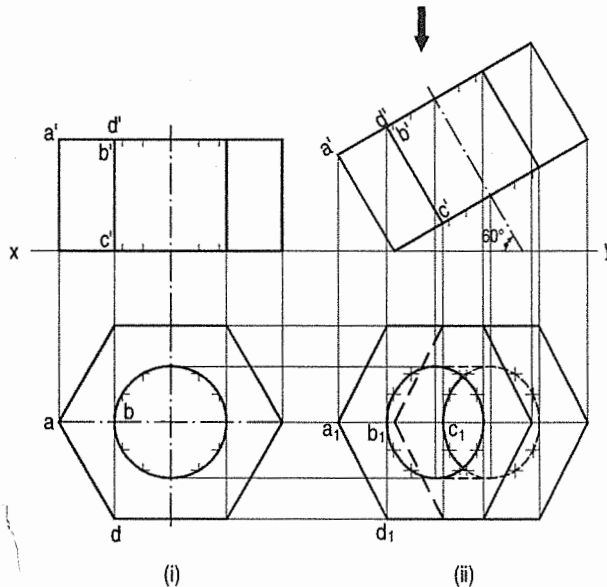


FIG. 13-28

- (i) Begin with the top view and project up the front view assuming the axis to be vertical.
- (ii) Tilt the front view, and project the required top view. Note that a part of the ellipse for the lower end of the hole will be visible.

Problem 13-17. (fig. 13-29): The projections of a hopper made of tin sheet are given. Project another top view on an auxiliary inclined plane making 45° angle with the H.P.

- (i) Draw a new reference line x_1y_1 inclined at 45° to xy and project the required top view on it, from the front view.
- (ii) Show carefully, the visible ellipses for the outer as well as the inner parts of the hopper rings.

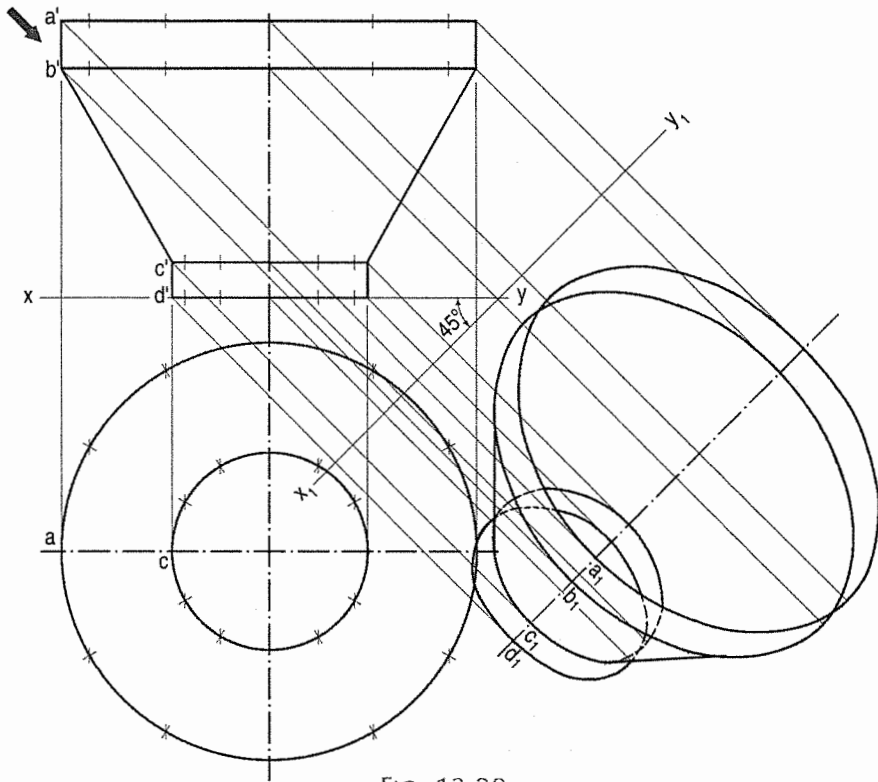


FIG. 13-29

13-4. PROJECTIONS OF SOLIDS WITH AXES INCLINED TO BOTH THE H.P. AND THE V.P.

The projections of a solid with its axis inclined to both the planes are drawn in three stages:

- (i) Simple position
- (ii) Axis inclined to one plane and parallel to the other
- (iii) Final position.

The second and final positions may be obtained either by the alteration of the positions of the solid, i.e. the views, or by the alteration of reference lines.

Problem 13-18. A square prism, base 40 mm side and height 65 mm, has its axis inclined at 45° to the H.P. and has an edge of its base, on the H.P. and inclined at 30° to the V.P. Draw its projections.

Method I: (fig. 13-30):

- (i) Assuming the prism to be resting on its base on the ground with an edge of the base perpendicular to the V.P., draw its projections.
Assume the prism to be tilted about the edge which is perpendicular to the V.P., so that the axis makes 45° angle with the H.P.
- (ii) Hence, change the position of the front view so that the axis is inclined at 45° to xy and f' (or e') is in xy . Project the second top view.

Again, assume the prism to be turned so that the edge on which it rests, makes an angle of 30° with the V.P., keeping the inclination of the axis with the ground constant. The shape and size of the second top view will remain the same; only its position will change. In the front view, the distances of all the corners from xy will remain the same as in the second front view.

- (iii) Therefore, reproduce the second top view making f_1g_1 inclined at 30° to xy . Project the final front view upwards from this top view and horizontally from the second front view, e.g. a vertical from a_1 and a horizontal from a' intersecting at a'_1 . As the top end is further away from xy in the top view it will be fully visible in the front view. Complete the front view showing the hidden edges by dashed lines.
- (iv) The second top view may be turned in the opposite direction as shown. In this position, the lower end of the prism, viz. $e'_1f'_1g'_1h'_1$ will be fully visible in the front view.

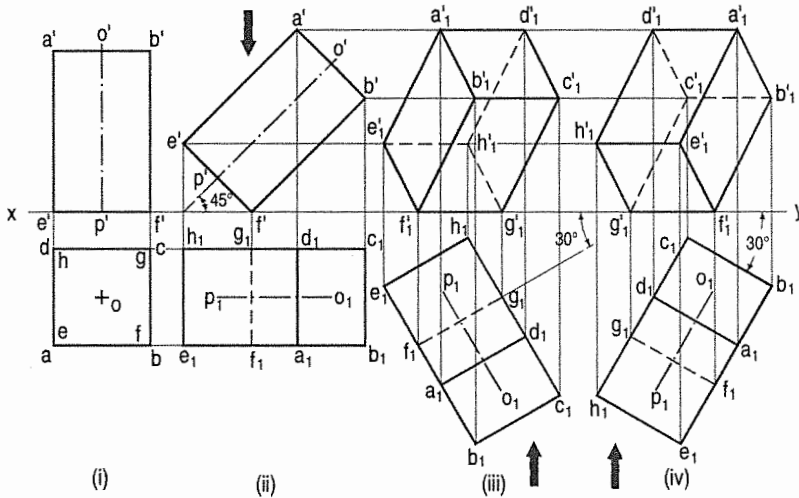


FIG. 13-30

Method II: (fig. 13-31):

- (i) Draw the top view and the front view in simple position.
- (ii) Through f' , draw a new reference line x_1y_1 making 45° angle with the axis. On it, project the auxiliary top view.
- (iii) Draw another reference line x_2y_2 inclined at 30° to the line f_1g_1 . From the auxiliary top view, project the required front view, keeping the distance of each point from x_2y_2 , equal to its distance (in the first front view) from x_1y_1 i.e. $a'_1q_1 = a'q$ etc. The problem is thus solved by change-of-reference line method only.

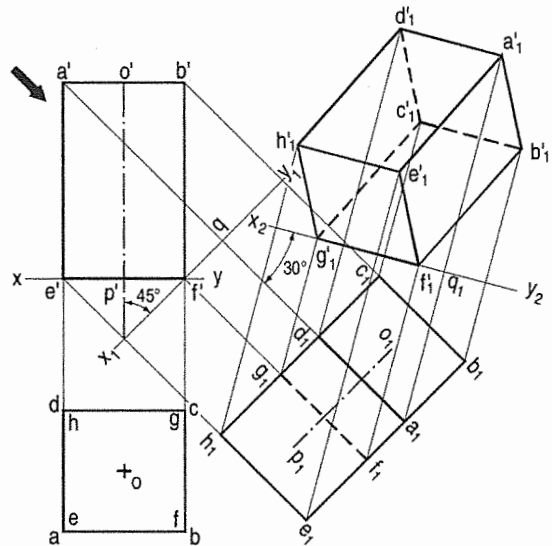


FIG. 13-31

Note: The new reference line satisfying the required conditions may be drawn in various positions, as explained in chapter 11.

Problem 13-19. (fig. 13-32): Draw the projections of a cone, base 45 mm diameter and axis 50 mm long, when it is resting on the ground on a point on its base circle with (a) the axis making an angle of 30° with the H.P. and 45° with the V.P.; (b) the axis making an angle of 30° with the H.P. and its top view making 45° with the V.P.

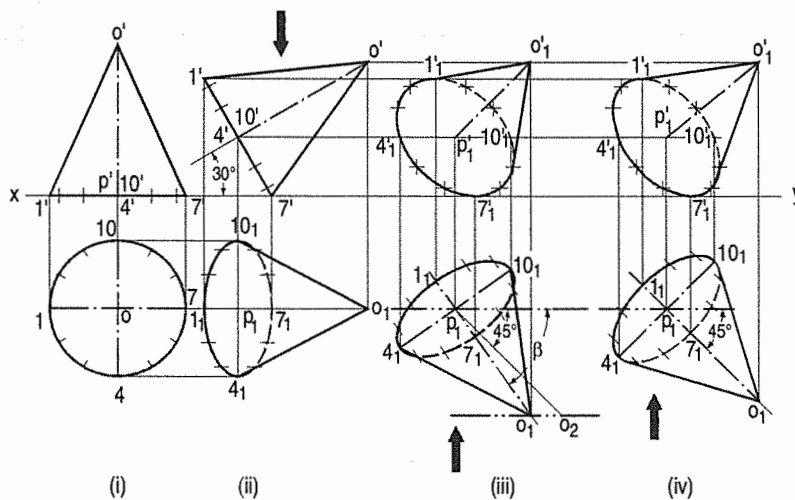


FIG. 13-32

- (i) Draw the top view and the front view of the cone with the base on the ground.
- (ii) Tilt the front view so that the axis makes 30° angle with xy . Project the second top view.
- (a) In order that the axis may make an angle of 45° with the V.P., let us determine the apparent angle of inclination which the top view of the axis, viz. o_1p_1 should make with xy and which will be greater than 45° .
- (iii) Mark any point p_1 below xy . Draw a line p_1o_2 equal to the true length of the axis, viz. $o'p'$, and inclined at 45° to xy . With p_1 as centre and radius equal to p_1o_1 (the length of the top view of the axis) draw an arc cutting the locus of o_2 at o_1 . Then β is the apparent angle of inclination and is greater than 45° . Around p_1o_1 as axis, reproduce the second top view and project the final front view as shown.

Note that the base of the cone is not visible in the front view because it is nearer xy in the top view.

- (b) When the top view of the axis is to make 45° angle with the V.P., it is evident that p_1o_1 should be inclined at 45° to xy . Hence, reproduce the top view accordingly and project the required front view [fig. 13-32(iv)].

Problem 13-20. (fig. 13-33): A pentagonal pyramid, base 25 mm side and axis 50 mm long has one of its triangular faces in the V.P. and the edge of the base contained by that face makes an angle of 30° with the H.P. Draw its projections.

- (i) In the initial position, assume the pyramid as having its base in the V.P. and an edge of the base perpendicular to the H.P. The front view will have to be drawn first and the top view projected from it.
- (ii) Change the position of the top view so that the line o_1 (for the face $o_1 5$) is in xy . Project the second front view.
- (iii) Tilt this front view so that the line $1'_1 5'_1$ makes 30° angle with xy . Project the final top view. Note that the base is not visible in the top view as it is nearer xy in the front view.

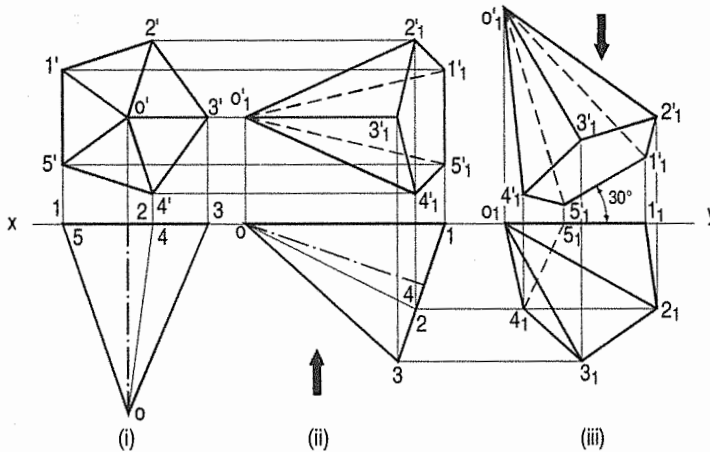


FIG. 13-33

Problem 13-21. (fig. 13-34): A square pyramid, base 38 mm side and axis 50 mm long, is freely suspended from one of the corners of its base. Draw its projections, when the axis as a vertical plane makes an angle of 45° with the V.P. When a pyramid is suspended freely from a corner of its base, the imaginary line joining that corner with the centre of gravity of the pyramid will be vertical.

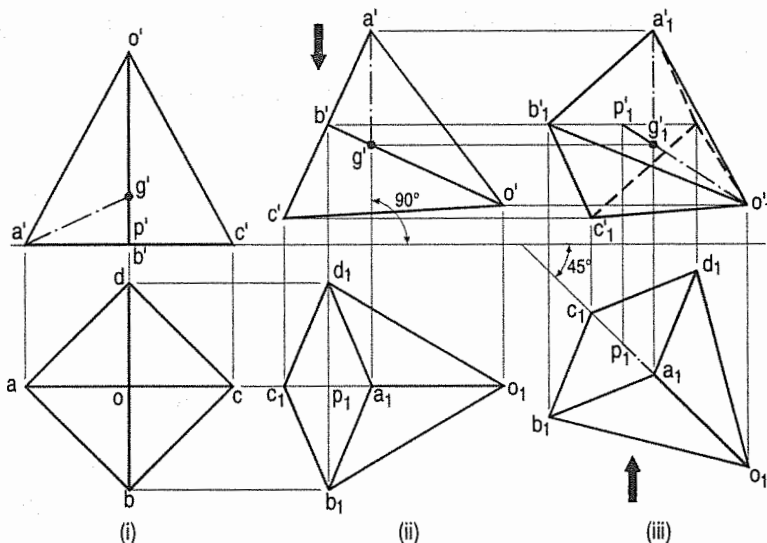


FIG. 13-34

The centre of gravity of a pyramid lies on its axis and at a distance equal to $\frac{1}{4}$ of the length of the axis from the base.

Assume the pyramid to be suspended from the corner A of the base.

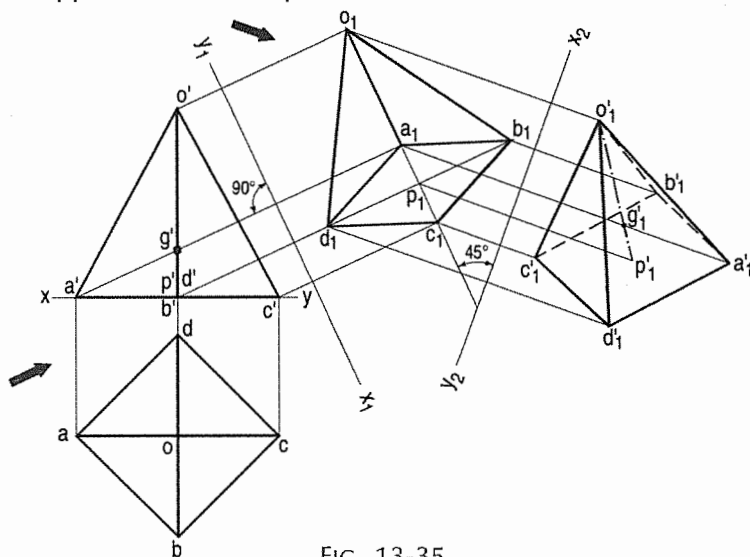


FIG. 13-35

In the initial position, the pyramid should be kept with its base on the ground and the line joining A with the centre of gravity G, parallel to the V.P. In the top view, g will coincide with o the top view of the axis.

- Draw a square $abcd$ (in the top view) with ag , i.e. ao parallel to xy . Project the front view. Making g' at a distance equal to $\frac{1}{4}$ of the axis from xy . Join a' with g' .
- Tilt the front view so that $a'g'$ is perpendicular to xy and project the top view. The axis will still remain parallel to the V.P.
- Reproduce this top view so that o_1p_1 (the top view of the axis) is inclined at 45° to xy . The axis as a vertical plane will thus be making 45° angle with the V.P. Project the final front view.

Fig. 13-35 shows the projections obtained by the change-of-reference-line method.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 30 for the following problem.

Problem 13-22. (fig. 13-36): A hexagonal pyramid, base 25 mm side and axis 55 mm long, has one of its slant edges on the ground. A plane containing that edge and the axis is perpendicular to the H.P. and inclined at 45° to the V.P. Draw its projections when the apex is nearer the V.P. than the base.

Assume the pyramid to be resting on the ground on its base with a slant edge parallel to the V.P.

- Draw the top view of the pyramid with a side of the hexagon parallel to xy . The lines ao and do for the slant edges will also be parallel to xy . Project the front view.
- Tilt this front view so that $a'o'$ or $d'o'$ is in xy . Project the second top view.

- (iii) Draw a new reference line x_1y_1 making 45° angle with o_1d_1 (the top view of the axis) and project the final front view.

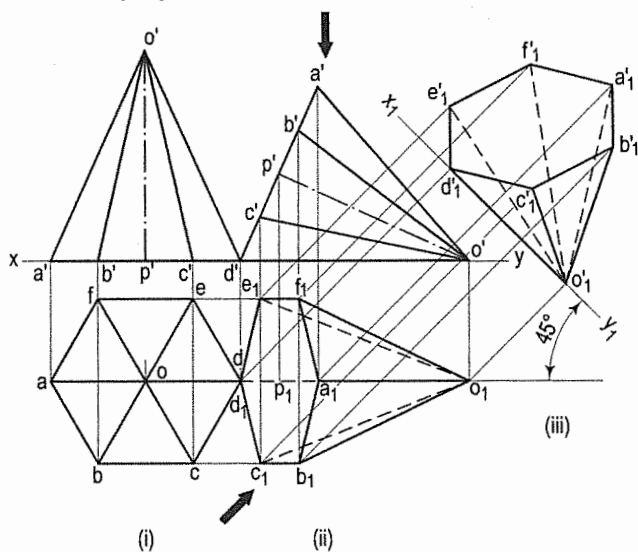


FIG. 13-36

The problem is thus solved by combination of the change-of-position and change-of-reference-line methods.

Problem 13-23. (fig. 13-37): Draw the projections of a cube of 25 mm long edges resting on the H.P. on one of its corners with a solid diagonal perpendicular to the V.P.

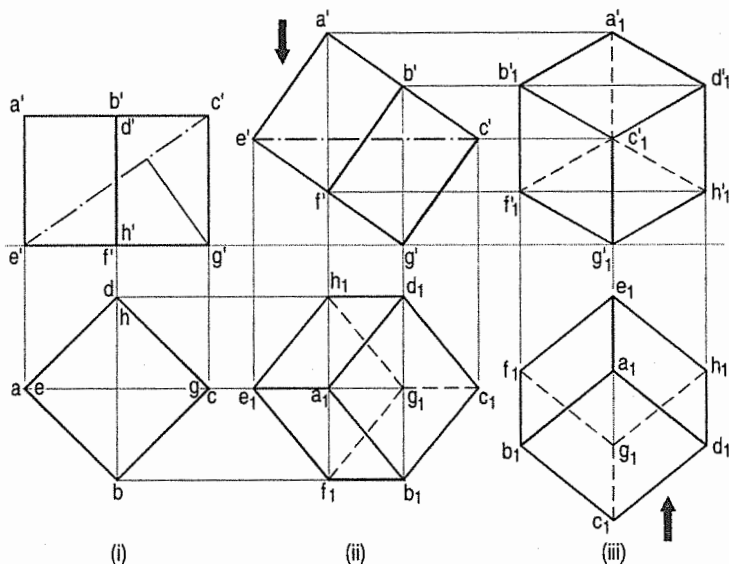


FIG. 13-37

Assume the cube to be resting on one of its faces on the H.P. with a solid diagonal parallel to the V.P.

- (i) Draw a square $abcd$ in the top view with its sides inclined at 45° to xy . The line ac representing the solid diagonals AG and CE is parallel to xy . Project the front view.

- (ii) Tilt the front view about the corner g' so that the line $e'c'$ becomes parallel to xy . Project the second top view. The solid diagonal CE is now parallel to both the H.P. and the V.P.
- (iii) Reproduce the second top view so that the top view of the solid diagonal, viz. e_1c_1 is perpendicular to xy . Project the required front view.

Problem 13-24. (fig. 13-38):

A triangular prism, base 40 mm side and axis 50 mm long, is lying on the H.P. on one of its rectangular faces with the axis perpendicular to the V.P. A cone, base 40 mm diameter and axis 50 mm long, is resting on the H.P. and is leaning centrally on a face of the prism, with its axis parallel to the V.P. Draw the projections of the solids and project another front view on a reference line making 60° angle with xy .

It will first be necessary to draw the cone with its base on the H.P. to determine the length of its generator and to project the top view.

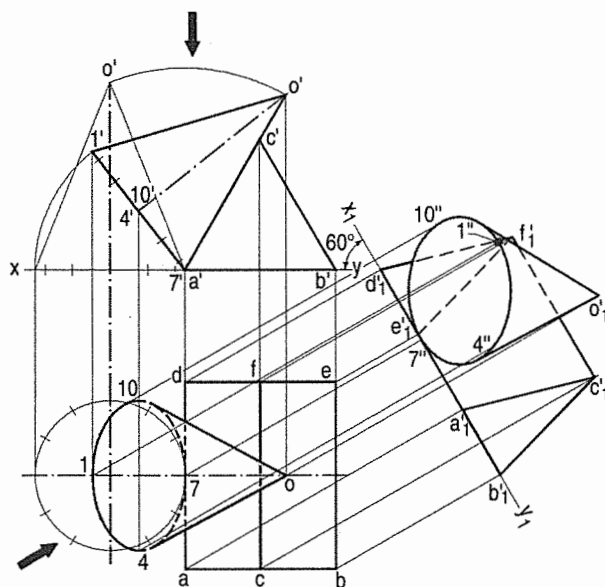


FIG. 13-38

Next, draw a triangle $a'b'c'$ for the prism and a triangle $o'1'7'$ for the cone as shown by the construction lines. Project the top view. Draw a reference line x_1y_1 and project the required front view as shown.

Problem 13-25. A pentagonal prism is resting on one of the corners of its base on the H.P. The longer edge containing that corner is inclined at 45° to the H.P. The axis of the prism makes an angle of 30° to the V.P. Draw the projections of the solid.

Also, draw the projections of the solid when the top view of axis is inclined at 30° to xy . Take the side of base 45 mm and height 70 mm.

- (i) Assuming the prism to be resting on its base on the horizontal plane, draw its projections keeping one of the sides of its base perpendicular to xy .
- (ii) Redraw the front view so that the edge $c'3'$ is inclined at 45° to xy . Project the required top view as shown in fig. 13-39(i).
- (iii) Determine the apparent angle of inclination which the top view of the axis should make with xy when the axis makes an angle of 30° with the V.P.
- (iv) Mark any point p_1 below xy . Draw a line p_1o_2 equal to the true length of the axis (70 mm) and inclined at 30° to xy . With p_1 as centre and radius equal to p_1o_1 (the length of the top view of the axis) draw an arc cutting the locus of o_2 at o_1 . Then β is the required apparent angle of inclination. Considering p_1o_1 as axis, reproduce the second top view and project the final front view as shown in fig. 13-39(i).

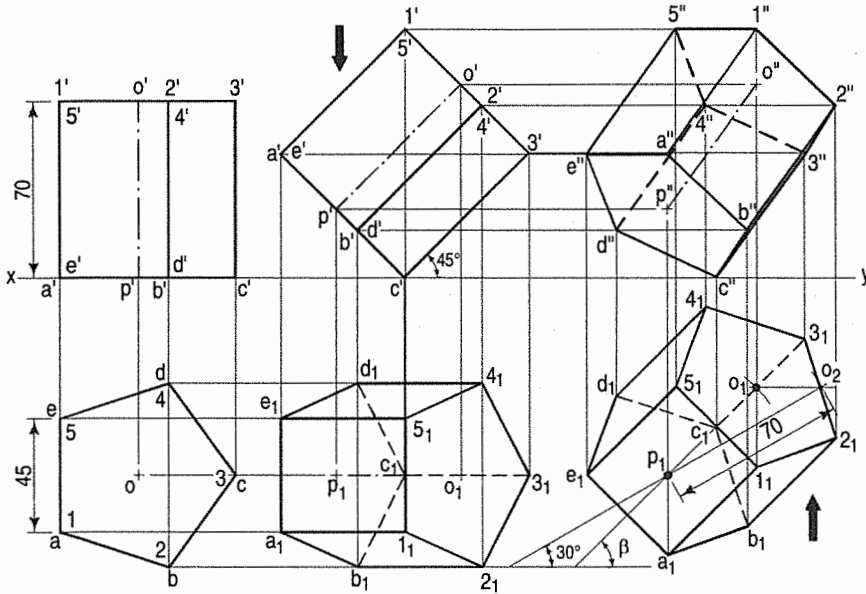


FIG. 13-39(i)

- (v) When the top view of axis makes an angle of 30° with the V.P., it is evident that p_1o_1 is inclined at 30° to xy . Hence, reproduce the top view and the front view as shown in fig. 13-39(ii).

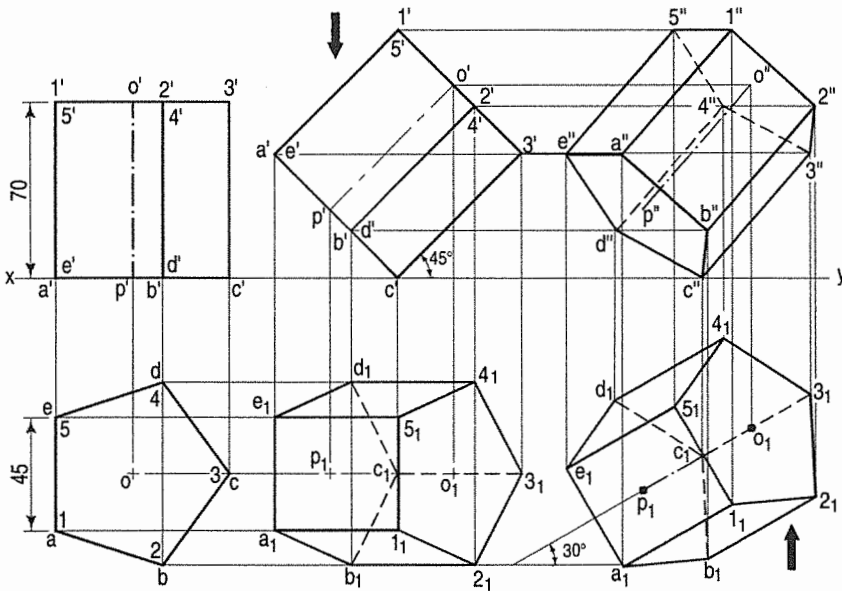


FIG. 13-39(ii)

Problem 13-26 (fig 13-40): A square prism, with the side of its base 40 mm and axis 70 mm long is lying on one of its base edges on the H.P. in such a way that this base edge makes an angle of 45° with the V.P. and the axis is inclined at 30° to the H.P. Draw the projections of the solid using the 'auxiliary plane method'.

- (i) In the initial position assume the axis of the prism to be perpendicular to the H.P. Draw the projections as shown.
- (ii) Draw a new reference line x_1y_1 making an angle of 30° with the front view of the axis, to represent an auxiliary horizontal plane. Draw projectors from $a', b' c', d'$ and $1', 2', 3', 4'$ perpendicular to x_1y_1 and on them, mark these points keeping the distance of each point from x_1y_1 equal to its distance from xy in the top view. Join the points as shown.
- (iii) Draw another reference line x_2y_2 inclined at 45° to the line $a'1'$ (or $b'2'$). From the auxiliary top view, project the required new front view, keeping the distance of each point from x_2y_2 equal to its distance from x_1y_1 , i.e. $q'1'' = q'1'$ etc. Join the points as shown. Note that the view is obtained by observing the auxiliary top view from the top, along the projectors.

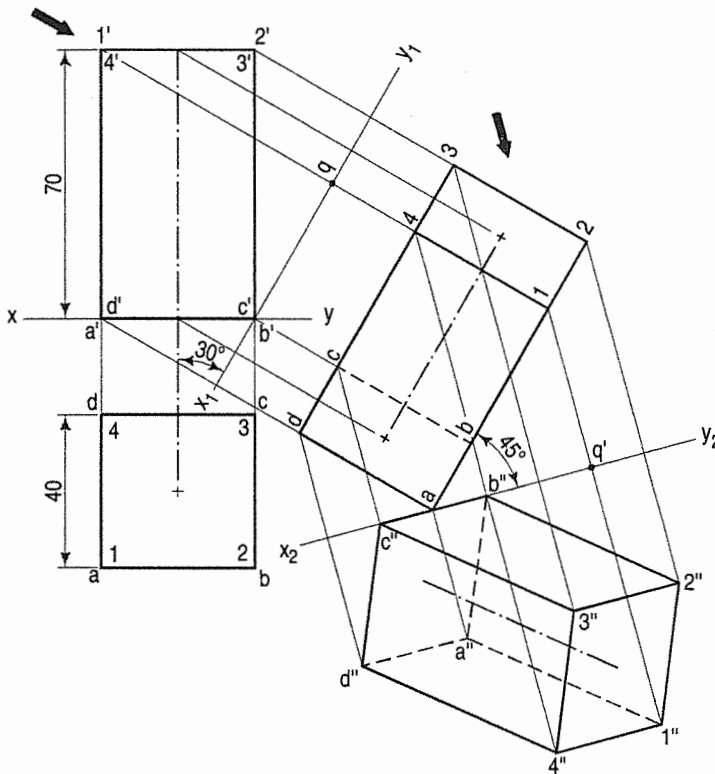


FIG. 13-40

Problem 13-27. A hexagonal prism, with the side of the hexagon 30 mm and height of 70 mm is resting on the H.P. on one of the edges of its hexagonal base in such a way that, the edge is at 60° to the V.P. and the base is at 30° to the H.P. Draw to scale 1:1, the view from the front and the view from the top.

Refer to fig. 13-41.

- (i) Draw the top view and the front view in simple position keeping the axis perpendicular to the H.P.
- (ii) Draw a new reference line x_1y_1 making 60° angle with the axis. On it, project the auxiliary top view.

- (iii) Draw another reference line x_2y_2 inclined at 60° to the edge of base c_1d_1 . From the auxiliary top view, project the required new front view, keeping the distance of each point from x_2y_2 , equal to its distance from x_1y_1 i.e. $q_3' = q_3''$ etc. Join the points as shown. It should be noted that the edge of base away from x_2y_2 will be observed as full lines and nearest lines from x_2y_2 will be dotted lines. i.e. $c''d''$, $d''e''$ and $e''f''$ are full lines while $f''a''$, $a''b''$ and $b''c''$ are dotted lines. Note that the view is drawn by observing the auxiliary top view from the top along the projectors.

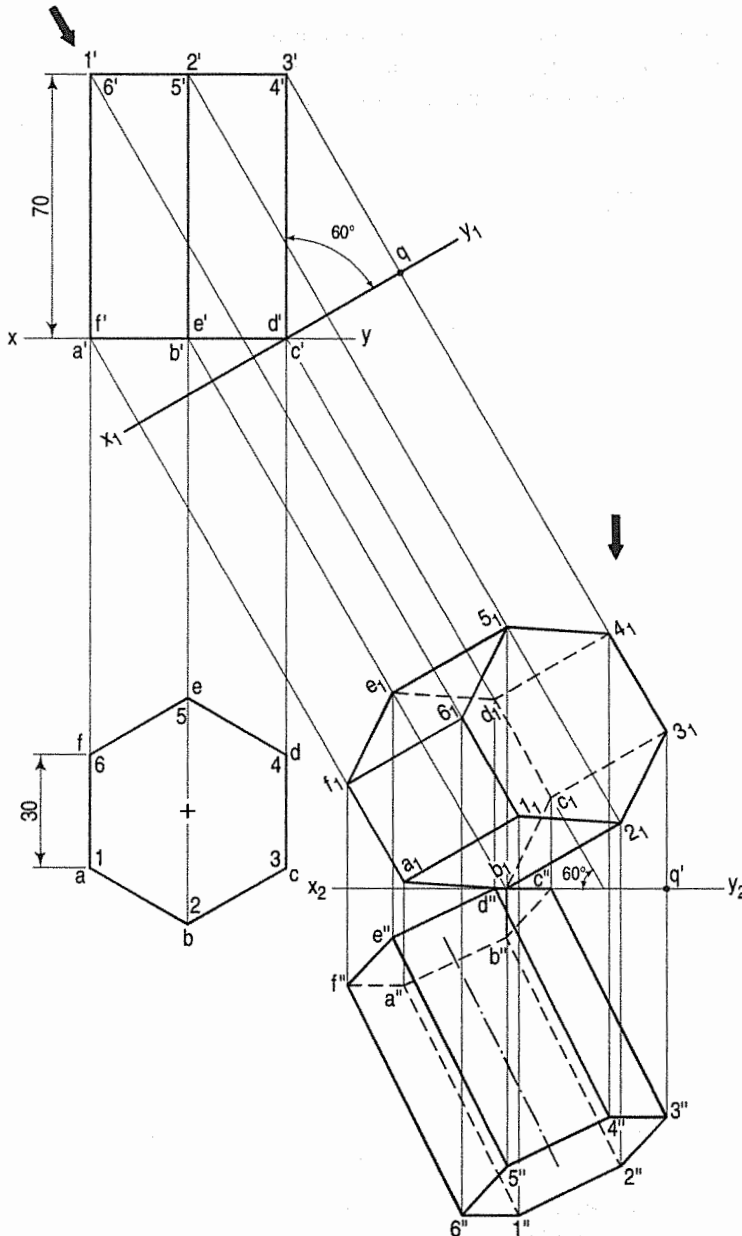


FIG. 13-41

Problem 13-28. A regular pentagonal prism lies with its axis inclined at 60° to the H.P. and 30° to the V.P. The prism is 60 mm long and has a face width of 25 mm. The nearest corner is 10 mm away from the V.P. and the farthest shorter edge is 100 mm from the H.P. Draw the projections of the solid.

- (i) Draw initial position of the prism as shown in fig. 13-42.
- (ii) With $4'$ as centre and radius equal to 100 mm, draw an arc. Mark tangent to the arc making 60° with the axis as shown. This is a new reference line x_1y_1 . Project the required new top view.
- (iii) Draw another reference line x_2y_2 inclined at 30° angle to the axis of new top view. Project the various points to obtain new front view as shown in fig. 13-42. Observe the auxiliary top view from the top along the projectors.

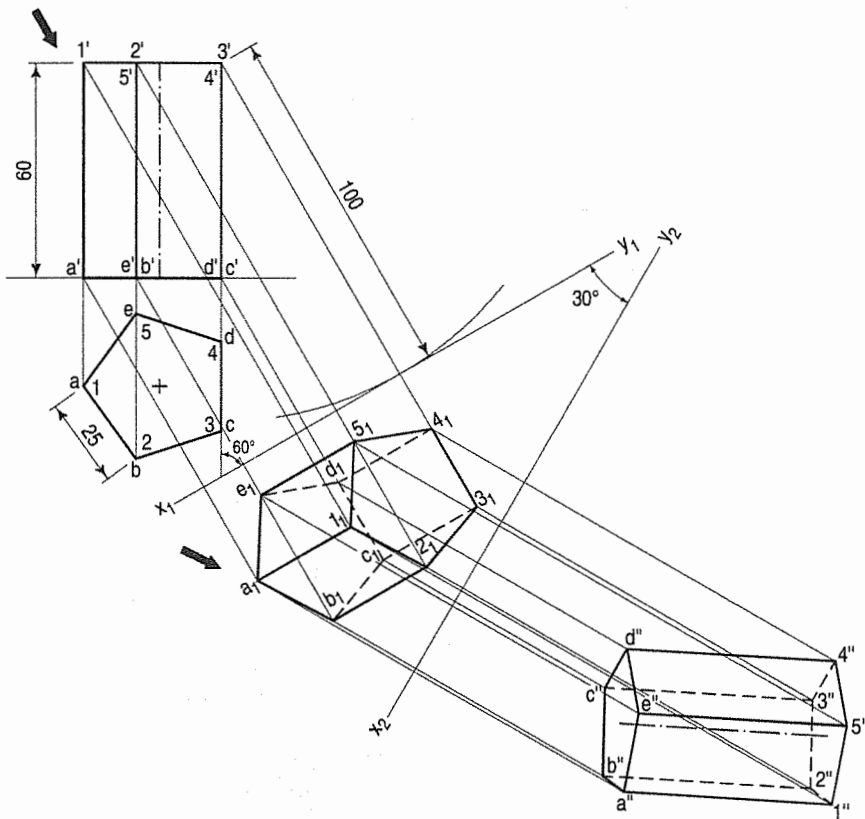


FIG. 13-42

Problem 13-29. A square pyramid of 50 mm side of base and 50 mm length of axis is resting on one of its triangular faces on the H.P. having a slant edge containing that face parallel to the V.P. Draw the projections of the pyramid.

- (i) Assuming the axis of pyramid perpendicular to the H.P., draw the front view and the top view as shown in fig. 13-43.
- (ii) Draw new reference line x_1y_1 coinciding with $o'c'$ in the front view. Project new top view, keeping the distance of $a_1, b_1...o_1$ from x_1y_1 equal to the distance of $a, b,...o$ from xy . Join these points.

- (iii) Draw another reference line x_2y_2 parallel to the slant edge o_1c_1 or o_1b_1 . Project new front view as shown. Observe auxiliary top view from the base a, b, c, d, o along projectors.

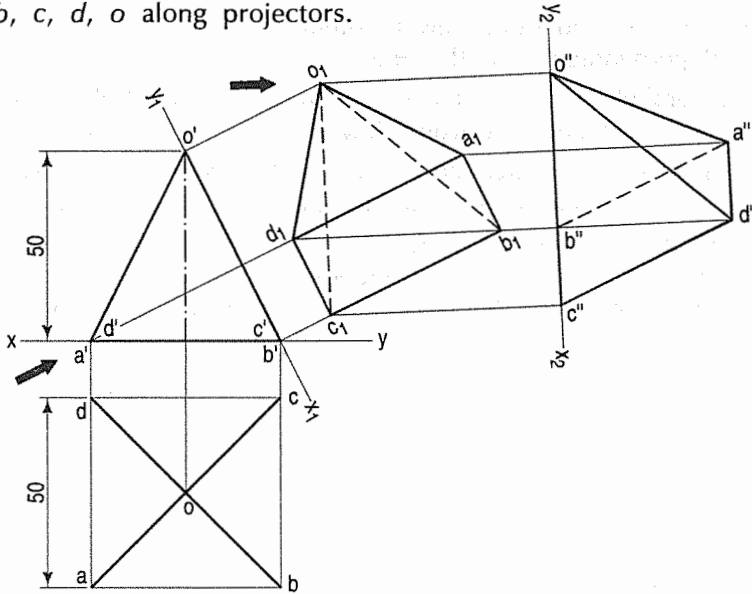


FIG. 13-43

Problem 13-30. A regular pentagonal pyramid, base 30 mm side and height 80 mm rests on one edge of its base on the ground so that the highest point in the base is 30 mm above the ground. Draw its projection when the axis is parallel to the V.P.

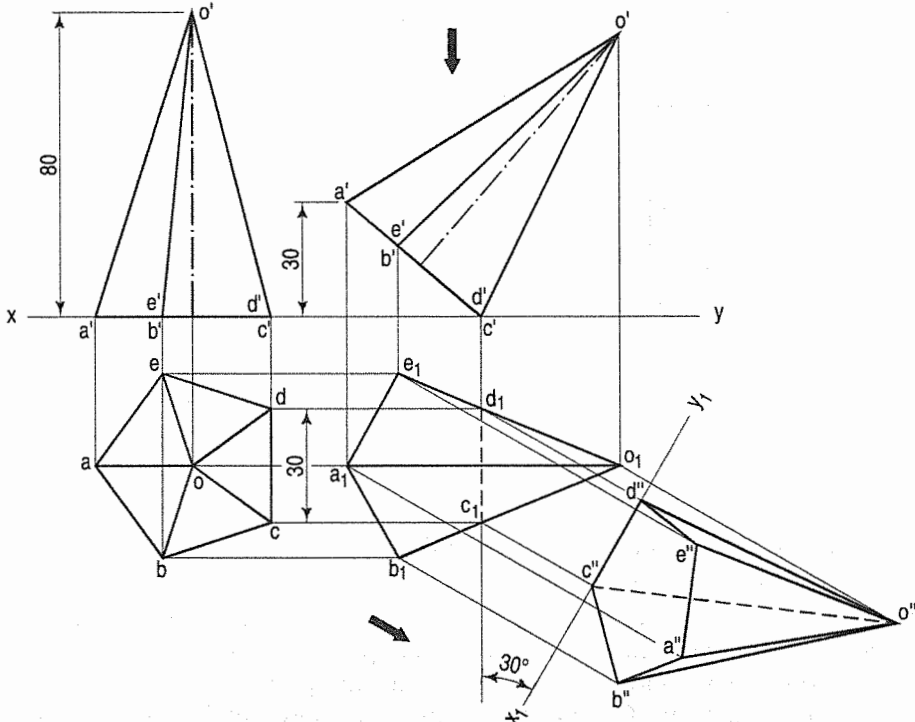


FIG. 13-44

Draw another front view on a reference line inclined at 30° to the edge on which it is resting so that the base is visible.

- Draw top view and front view in simple position assuming the axis of the pyramid perpendicular to the H.P.
- Draw a parallel line at a distance of 30 mm from xy . Mark the point c' on the line xy and reproduce the front view as shown fig. 13-44.
- Project points a' , b' , c' etc. and obtain new top view keeping distance of points a_1 , b_1 , c_1 etc. from xy equal to distance of a , b , c , etc. from the line xy .
- Draw another reference line x_1y_1 making an angle of 30° with the side of base c_1d_1 and obtain a new front view as shown. Note that the base is visible. Observe from the base a , b , c , d , e along the projectors.

Problem 13-31. A regular pentagonal pyramid with the sides of its base 30 mm and height 80 mm rests on an edge of the base. The base is tilted until its apex is 50 mm above the level of the edge of the base on which it rests. Draw the projection of the pyramid when the edge on which it rests, is parallel to the V.P. and the apex of the pyramid points towards V.P.

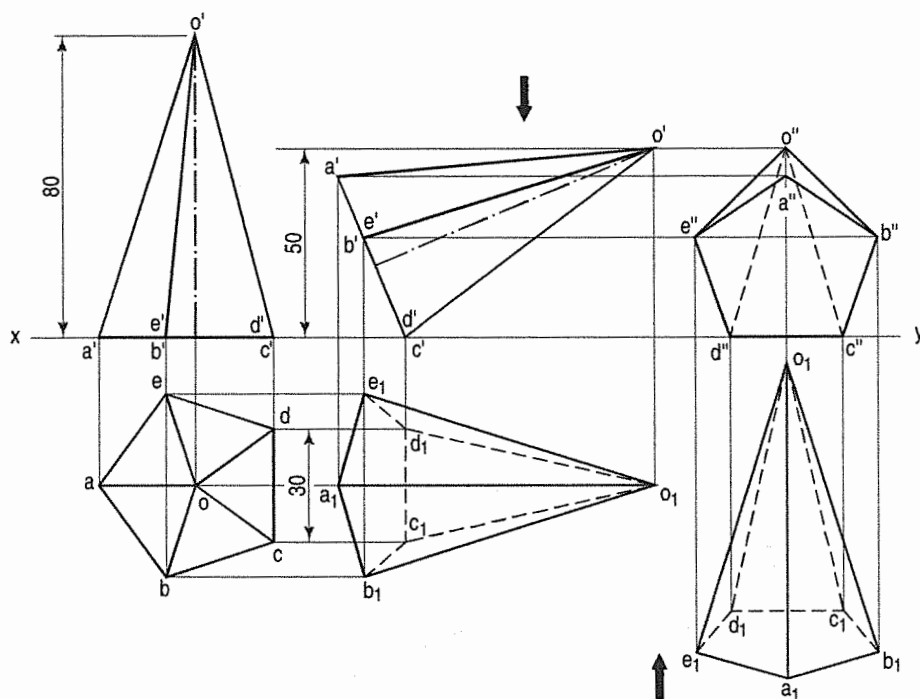


FIG. 13-45

- Draw top view and front view assuming the axis of the pyramid perpendicular to the H.P. as shown in fig. 13-45.
- Draw a parallel line at a distance of 50 mm from xy . Reproduce the front view as shown. Draw projectors from points a' , b' , c' etc. vertically from the front view and horizontally from the points a , b , c etc. from the previous top view. Complete the new top view, joining the intersection of the projectors in the correct sequence as shown.

- (iii) Redraw the top view keeping c_1d_1 parallel to xy . Project the points a_1, b_1, c_1 etc. vertically from the new top view and horizontal projectors from the points a', b', c' etc. of the front view. Join the intersection points of both the projectors in the correct sequence as shown.

Problem 13-32. A right regular pentagonal pyramid, with the sides of the base 30 mm and height 65 mm rests on the edge of its base on the horizontal plane, the base being tilted until the vertex is at 60 mm above the H.P. Draw the projections of the pyramid tilted when the edge on which it rests, is made parallel to FRP. Assuming the pyramid to be resting on its base on the horizontal plane, draw its projections keeping one of the sides of the base perpendicular to xy .

Method I: Changing position of reference line [fig. 13-46(i)]:

- With o' as centre and radius equal to 60 mm, draw an arc. Draw the tangent to the arc passing through c' or d' . This is a new reference line x_1y_1 . Project the required top view.
- Draw another reference line x_2y_2 parallel to c_1d_1 . Project new front view as shown. Observe auxiliary top view from the base a, b, c, d, e along the projectors.

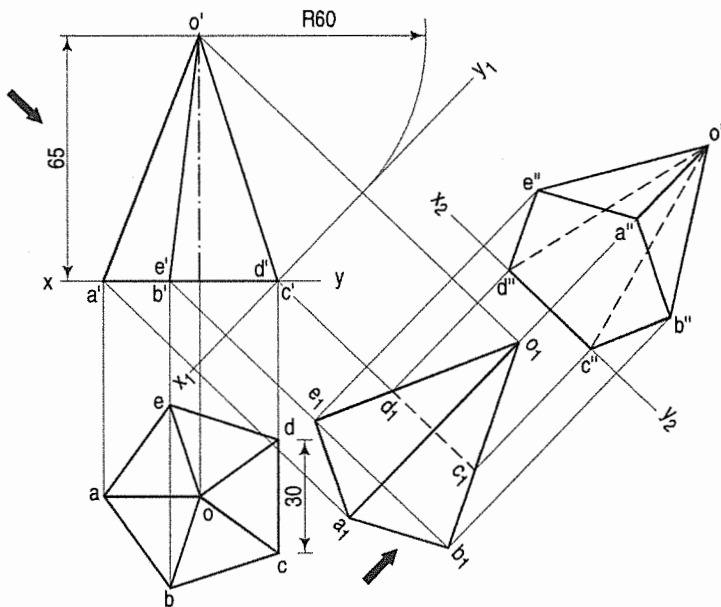


FIG. 13-46(i)

Method II: Changing positions of solid [(fig. 13-46(ii))]:

- Draw a line 60 mm parallel to xy . Mark point c' or d' on xy . With c' as centre and the radius equal to $o'c'$, draw an arc cutting the above line at o' . With o' and c' as centre and radius equal to $o'c'$ and $a'c'$ draw an arc cutting each other at the point a' . Join a', o' and c' as shown. Project the required top view as shown.
- Redraw the top view keeping side of base $c'd'$ parallel to xy . Project new front view as shown.

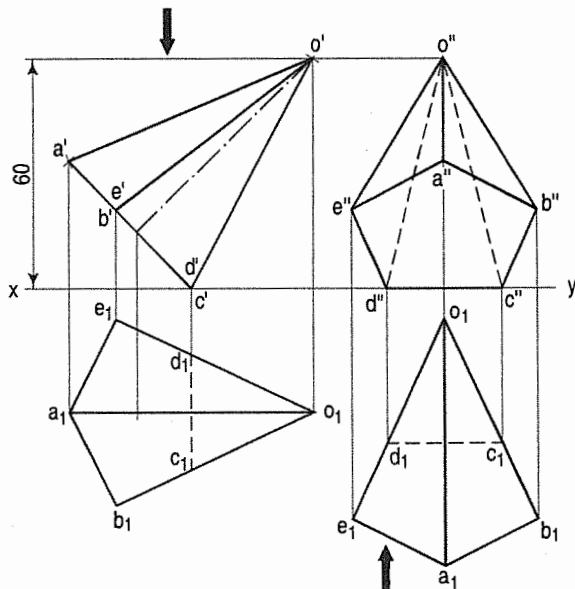


FIG. 13-46(ii)

Problem 13-33. The front view, part top view and part auxiliary view of a casting are given in fig. 13-47(i). Project its side view.

See fig. 13-47(ii).

The construction for the ellipse for 38 mm diameter circle has been shown in detail. Horizontal distances are taken from the auxiliary view. Other ellipses are drawn in the same manner.

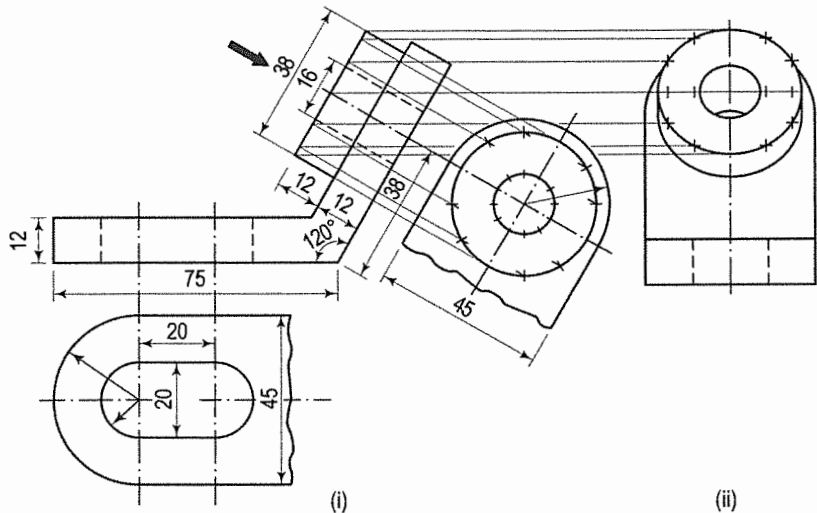


FIG. 13-47

13-5. PROJECTIONS OF SPHERES



The projection of a sphere in any position on any plane is always a circle whose diameter is equal to the diameter of the sphere (fig. 13-48). This circle represents the contour of the sphere.

A flat circular surface is formed when a sphere is cut by a plane. A hemisphere (i.e. a sphere cut by a plane passing through its centre) has a flat circular face of diameter equal to that of the sphere.

When it is placed on the ground on its flat face, its front view is a semi-circle, while its top view is a circle [fig. 13-49(i)].

When the flat face is inclined to the H.P. or the ground and is perpendicular to the V.P. it is seen as an ellipse (partly hidden) in the top view [fig. 13-49(ii)], while the contour of the hemisphere is shown by the arc of the circle drawn with radius equal to that of the sphere.

Fig. 13-50 shows the projections of a sphere, a small portion of which is cut off by a plane. Its flat face is perpendicular to the H.P. and inclined to the V.P. An ellipse is seen in the front view within the circle for the sphere.

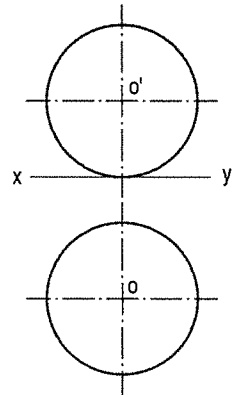


FIG. 13-48

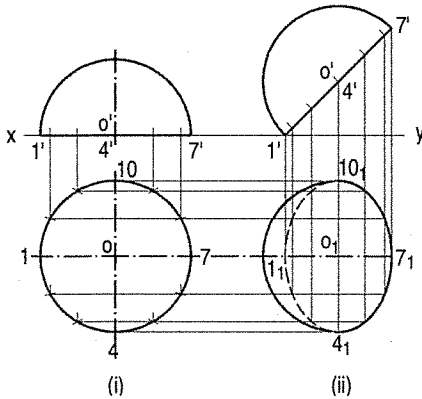


FIG. 13-49

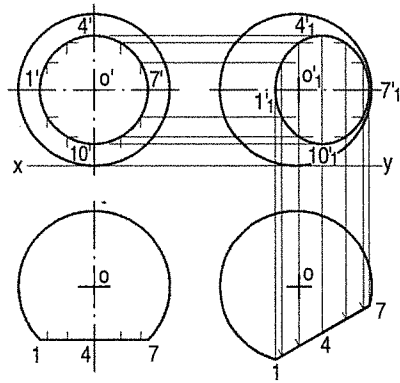


FIG. 13-50

When the flat face of a cut sphere is perpendicular to the V.P. and inclined to the H.P., its projections can be drawn as described in problem 13-34.

Problem 13-34. (fig. 13-51): A brass flower-vase is spherical in shape with flat, circular top 35 cm diameter and bottom 25 cm diameter and parallel to each other. The greatest diameter is 40 cm. Draw the projections of the vase when its axis is parallel to the V.P. and makes an angle of 60° with the ground.

- Draw the front view of the vase resting on its bottom with its axis vertical. Project the top view.
- Tilt the front view so that the axis makes 60° angle with xy and project the top view. Note that a part of the ellipse for the bottom is also visible.

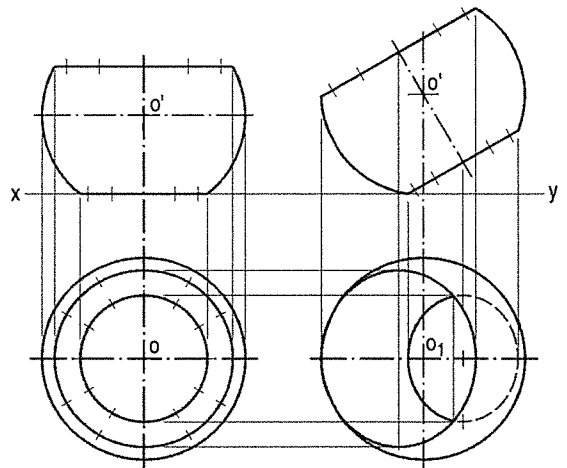


FIG. 13-51

(1) **Spheres in contact with each other:** Projections of two equal spheres resting on the ground and in contact with each other, with the line joining their centres parallel to the V.P., are shown in fig. 13-52.

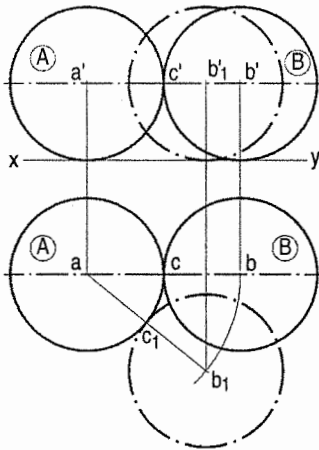


FIG. 13-52

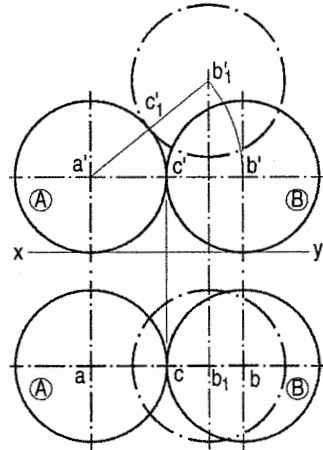


FIG. 13-53

As the spheres are equal in size, the line joining their centres is parallel to the ground also. Hence, both ab and $a'b'$ show the true length of that line (i.e. equal to the sum of the two radii or the diameter of the spheres). The point of contact between the two spheres is also visible in each view.

If the position of one of the spheres, say sphere B , is changed so that the line joining their centres is inclined to the V.P., in the front view, the centre b' will move along the line $a'b'$ to b'_1 . The true length of the line joining the centres and the point of contact are now seen in the top view only.

When the sphere B is so moved that it remains in contact with the sphere A and the line joining their centres is parallel to the V.P., but inclined to the ground (fig. 13-53), the true length of that line and the point of contact are visible in the front view only.

Problem 13-35. (fig. 13-54): *Three equal spheres of 38 mm diameter are resting on the ground so that each touches the other two and the line joining the centres of two of them is parallel to the V.P.*

A fourth sphere of 50 mm diameter is placed on top of the three spheres so as to form a pile. Draw three views of the arrangement and find the distance of the centre of the fourth sphere above the ground.

As the spheres are resting on the ground and are equal in size, the lines joining their centres will be parallel to the ground. In the top view, the centres will lie at the corners of an equilateral triangle of sides equal to the sum of the two radii, i.e. 40 mm.

Draw (in the top view) an equilateral triangle abc of 40 mm long sides with one side, say ab , parallel to xy . At its corners, draw three circles of 40 mm diameter. Project the front view. The centres will lie on a line parallel to and 20 mm above xy .

When the fourth sphere is placed on top, its centre d in the top view will be in the centre of the triangle. In the front view, it will lie on a projector through d .

The true distance between the centre of the top sphere and that of any one of the bottom spheres will be equal to the sum of the two radii, viz. 20 mm + 25 mm, i.e. 45 mm.

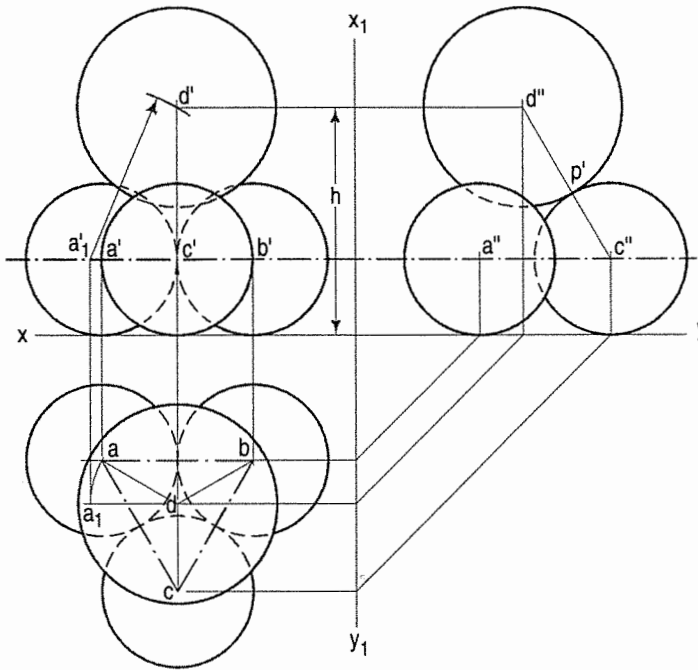


FIG. 13-54

But as none of the lines da , db or dc is parallel to xy , their front views will not show their true lengths. Therefore, to locate the position of the centre of the top sphere in the front view,

- (i) make one of the lines, say da , parallel to xy ;
- (ii) project a_1 to a'_1 on the path of a' and
- (iii) with a'_1 as centre and radius equal to 45 mm, draw an arc cutting the projector through d at the required point d' . With d' as centre and radius equal to 25 mm, draw the required circle which will be partly hidden as shown. h is the distance of the centre of the sphere from the ground.
- (iv) Project the side view. As $c'd'$ is parallel to the new reference line, $c''d''$ will be equal to 45 mm and the point of contact p' between the spheres having centres c and d will be visible.

(2) Unequal spheres: When two unequal spheres are on the ground and are in contact with each other, their point of contact and the true length of the line joining their centres will be seen in the front view if that line is parallel to the V.P. In the top view, the length of the line will be shorter but will remain constant even when it is inclined to the V.P.

Problem 13-36. (fig. 13-55): Three spheres A, B and C of 75 mm, 50 mm and 30 mm diameters respectively, rest on the ground each touching the other two. Draw their projections and show the three points of contact when the line joining the centres of the spheres A and B is parallel to the V.P.

- (i) With centre a' and radius equal to 37.5 mm, draw a circle of sphere A, mark a' at 37.5 mm above xy in front view. With $a'b'$ equal to 62.5 mm, mark point b' 25 mm above xy . With b' as centre and radius equal to 25 mm, draw a circle of sphere B in front view.

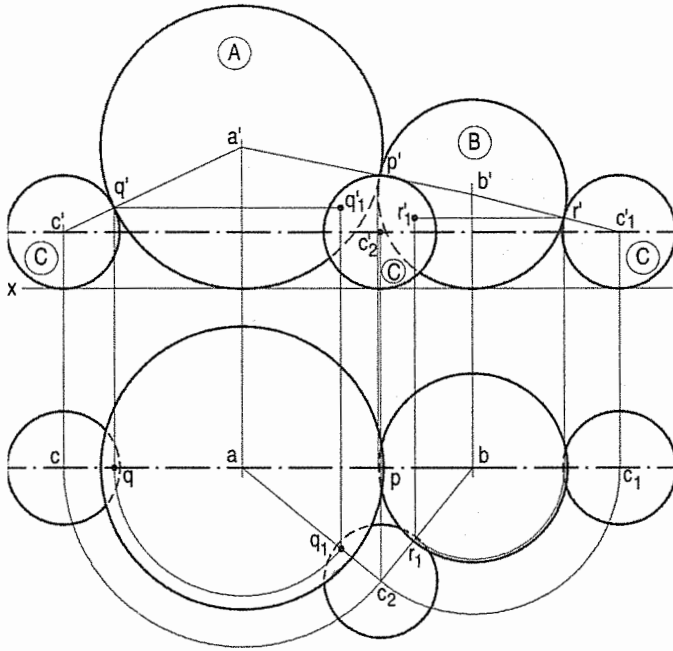


FIG. 13-55

- (ii) Project the centres and obtain points a and b on a line parallel to xy in top view. With a as centre and radius equal to 37.5 mm of sphere A , and with b as centre and radius equal to 25 mm of sphere B , draw circles in the top view.
- (iii) Similarly, draw the views of sphere C in contact with spheres A and B .
- (iv) With a as centre and radius equal to ac , and with b as centre and radius equal to bc_1 , draw arcs intersecting each other at c_2 . With c_2 centre draw top view of the sphere C .
- (v) Draw the projector through c_2 to cut the path of c' at c'_2 . Then c'_2 is the required centre of the sphere C in the front view. p , q_1 and r_1 , and p' , q'_1 and r'_1 are the points of contact in the top view and the front view respectively.

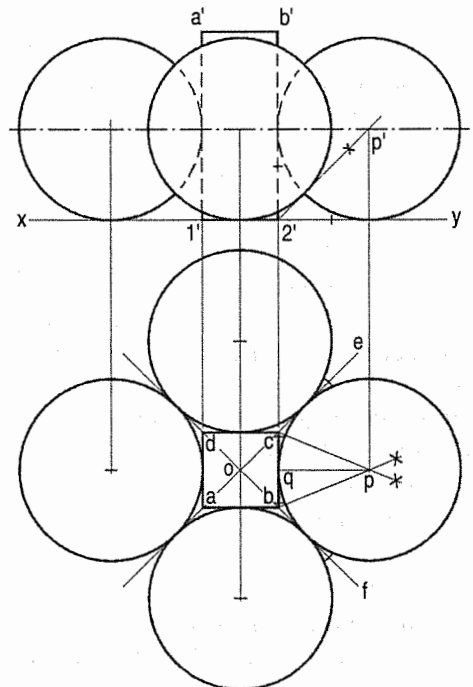


FIG. 13-56

Problem 13-37. (fig. 13-56): A square prism, base 20 mm side and axis 50 mm long, is resting on its base on the ground with two faces perpendicular to the V.P. Determine the radius of four equal spheres resting on the ground, each touching a face of the prism and other two spheres. Draw the projections of the arrangement.

- (i) Draw the front and top views of the prism. In the top view, draw diagonals of the square (intersecting each other at o) and produce them on both sides.
- (ii) Draw the bisectors of angles bce and cbf intersecting each other at p . From p , draw a perpendicular pq to bc . Then pq is the required radius of the sphere and p is the centre of the circle for the sphere.
- (iii) Obtain the other three centres in the same manner. Or, with o as centre and radius equal to op , draw a circle to cut the centre lines through o at the required centres. Draw the four circles.
- (iv) Draw a bisector of angle $b'2'y$ intersecting the projector through p at p' . Then p' is the centre of the sphere in the front view. The centres for the other circles will lie on the horizontal line through p' . Project their exact positions from the top view and draw the circles.

Problem 13-38. (fig. 13-57): Six equal spheres are resting on the ground, each touching other two spheres and a triangular face of a hexagonal pyramid resting on its base on the ground.

Draw the projections of the solids when a side of the base of the pyramid is perpendicular to the V.P.

Determine the diameter of each sphere. Base of the pyramid 20 mm side; axis 50 mm long.

- (i) Draw the projections of the pyramid in the required position. Assuming the solid to be a prism, locate the positions of the centre of one sphere (viz. p and p') in the two views.
- (ii) Draw a line joining p' with a' (the centre of the base) which coincides with $2'$. The centre of the required sphere will lie on this line. Draw a bisector of angle $o'3'y$ cutting $a'p'$ at c' . Draw a line $c'q'$ perpendicular to xy .
- (iii) With c' as centre and radius $c'q'$, draw one of the required circles. Project c' to c on op in the top view. Then c is the centre of the circle in the top view. Other centres may be located in the top view as shown and projected down in the front view.
- (iv) Draw the six circles in the top view and four in the front view as shown in the fig. 13-57.

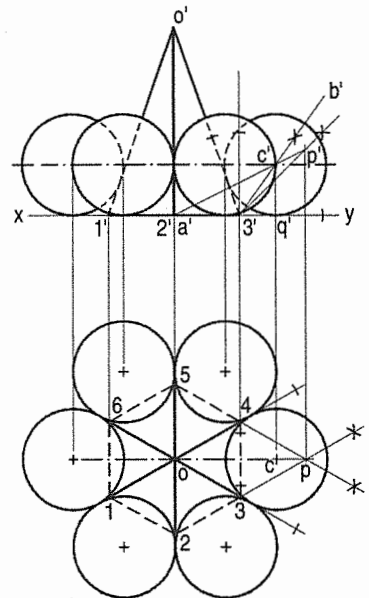


FIG. 13-57

Problem 13-39. The projections of a paper-weight with a spherical knob are given in fig. 13-58(i). Draw the two views and project another top view when its flat base makes an angle of 60° with the H.P.

See fig. 13-58(ii) which is self-explanatory.

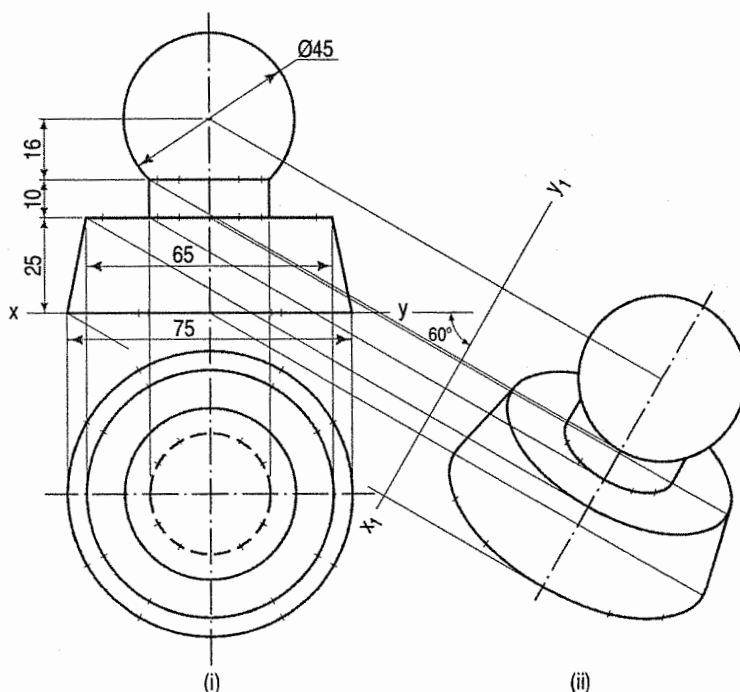


FIG. 13-58

Problem 13-40. (fig. 13-59): A vertical hexagonal prism of base side 20 mm and thickness 15 mm has one side of hexagon perpendicular to the V.P. A right cone of 34 mm diameter and height 40 mm is placed on the top face of prism such that the base of cone touches top surface of prism while the axes of both coincide. Draw the front view and top view of the combined object. Draw also projections when axes of combined solid is inclined at 35° with auxiliary plane.

- (i) Draw the top view of hexagonal prism keeping one of sides perpendicular to xy . (i.e. ab or ed). Project above xy line and draw the front view of prism of height 15 mm.
- (ii) Inscribe circle in the top view touching sides of the prism. Project it in the front view and mark the height of the cone as shown.
- (iii) Draw auxiliary x_1y_1 inclined at 35° with the axes of the combined solids.
- (iv) Draw the projectors from the various points of combined solids in the front view.
- (v) Taking distance of various points from the top view of combined solids from xy and mark same distances along the respective projectors.
- (vi) Complete auxiliary top view as shown.

Problem 13-41. (fig. 13-60): A vertical cylindrical disc of thickness 10 mm and diameter 50 mm is resting on the ground. A vertical frustum of pentagonal pyramid, having bottom of 20 mm sides, top face of 40 mm sides with 60 mm height is resting on the top surface of the disc so that axes of the both solids coincide. Take one of sides of the base of pentagon is perpendicular to V.P. Draw the projections of combined solid when the axis of combined solids is inclined to 30° with the H.P.

- (i) Draw the top view of frustum of pyramid (pentagon) keeping one of the sides perpendicular to xy as shown.

- (ii) Project the front view marking height of cylindrical disc and frustum of the pyramid 10 mm and 60 mm respectively.
- (iii) Draw a line at angle of 30° with xy . (As axis is inclined with H.P., its inclination observed in the front view).
- (iv) Reproduce the front view considering inclined line as axes of the combined solids.
- (v) Draw the vertical projectors from various points of the front view.
- (vi) Draw horizontal projectors to intersect respective vertical projectors. Obtain the auxiliary top view as shown.

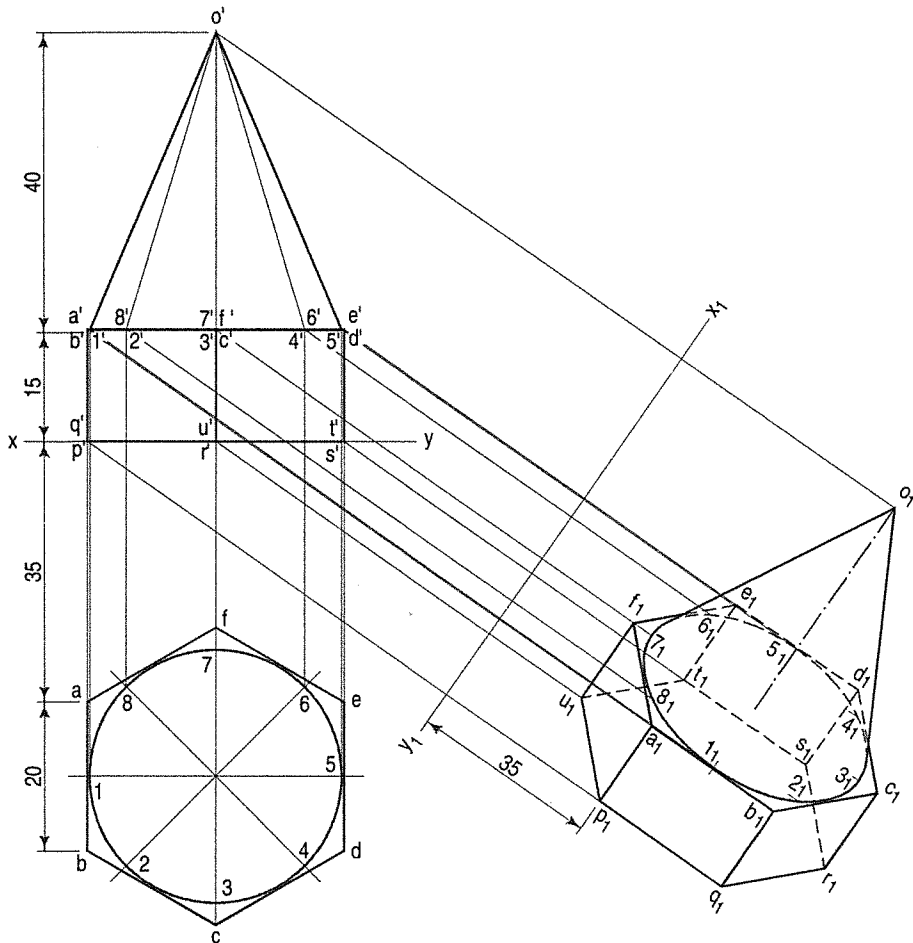


FIG. 13-59

Problem 13-42. (fig. 13-61): A right hexagonal prism of side 25 mm and 20 mm thick with one side of the base is perpendicular to the V.P. resting on the ground. A vertical frustum of square pyramid of base 20 mm sides and top face side 30 mm and height 50 mm is resting on the prism such that one side of square makes 45° with the V.P. Assume that axes of both solids are coinciding. Draw the projections of the combined solids when top corner of the square pyramid is 70 mm above the ground (H.P.). Determine angle of combined solids with the H.P.

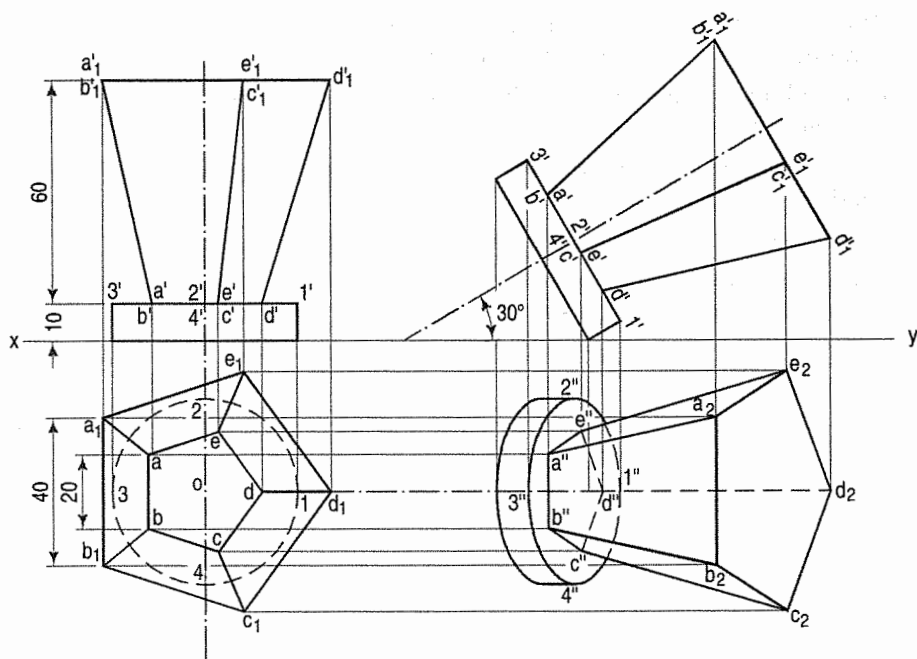


FIG. 13-60

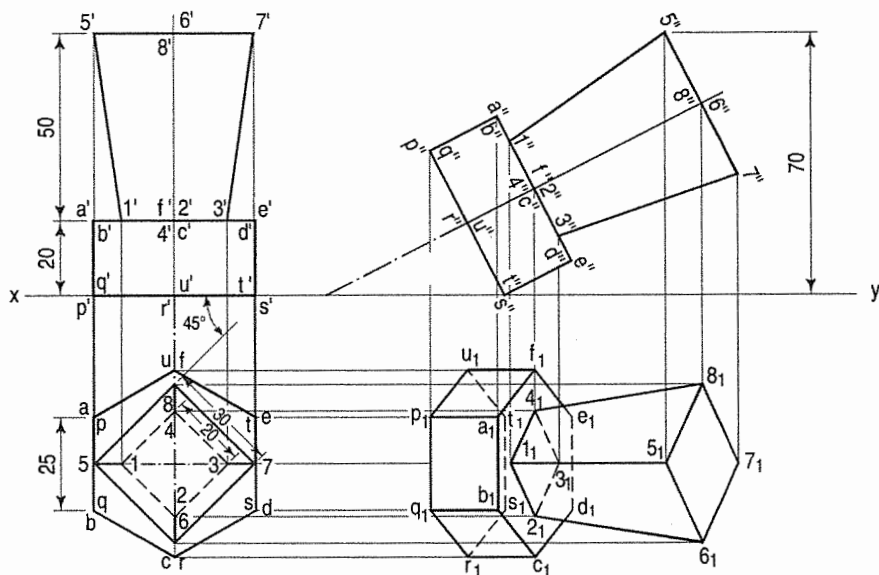


FIG. 13-61

- (i) Draw the top view and front view as shown in figure. Keep one side of the hexagonal perpendicular to the xy .
- (ii) Project the front view as shown.
- (iii) Draw a parallel line at distance 70 mm away from xy . Reproduce the front view of the combined solids as shown.
- (iv) Draw the projectors from the new front view.

- (v) Draw from the top view horizontal projectors to intersect respective projectors drawn from the new front view.
- (vi) Complete the top view as shown.

EXERCISES 13(b)



1. A rectangular block 75 mm $\times$ 50 mm $\times$ 25 mm thick has a hole of 30 mm diameter drilled centrally through its largest faces. Draw the projections when the block has its 50 mm long edge parallel to the H.P. and perpendicular to the V.P. and has the axis of the hole inclined at 60° to the H.P.
2. Draw the projections of a square pyramid having one of its triangular faces in the V.P. and the axis parallel to and 40 mm above the H.P. Base 30 mm side; axis 75 mm long.
3. A cylindrical block, 75 mm diameter and 25 mm thick, has a hexagonal hole of 25 mm side, cut centrally through its flat faces. Draw three views of the block when it has its flat faces vertical and inclined at 30° to the V.P. and two faces of the hole parallel to the H.P.
4. Draw three views of an earthen flower pot, 25 cm diameter at the top, 15 cm diameter at the bottom, 30 cm high and 2.5 cm thick, when its axis makes an angle of 30° with the vertical.
5. A tetrahedron of 75 mm long edges has one edge parallel to the H.P. and inclined at 45° to the V.P. while a face containing that edge is vertical. Draw its projections.
6. A hexagonal prism, base 30 mm side and axis 75 mm long, has an edge of the base parallel to the H.P. and inclined at 45° to the V.P. Its axis makes an angle of 60° with the H.P. Draw its projections.
7. A pentagonal prism is resting on a corner of its base on the ground with a longer edge containing that corner inclined at 45° to the H.P. and the vertical plane containing that edge and the axis inclined at 30° to the V.P. Draw its projections. Base 40 mm side; height 65 mm.
8. Draw three views of a cone, base 50 mm diameter and axis 75 mm long, having one of its generators in the V.P. and inclined at 30° to the H.P., the apex being in the H.P.
9. A square pyramid, base 40 mm side and axis 90 mm long, has a triangular face on the ground and the vertical plane containing the axis makes an angle of 45° with the V.P. Draw its projections.
10. A frustum of a pentagonal pyramid, base 50 mm side, top 25 mm side and axis 75 mm long, is placed on its base on the ground with an edge of the base perpendicular to the V.P. Draw its projections. Project another top view on a reference line parallel to the line which shows the true length of the slant edge. From this top view, project a front view on an auxiliary vertical plane inclined at 45° to the top view of the axis.
11. Draw the projections of a cone, base 50 mm diameter and axis 75 mm long, lying on a generator on the ground with the top view of the axis making an angle of 45° with the V.P.
12. The front view, incomplete top view and incomplete auxiliary top view of a casting are given in fig. 13-47. Draw all the three views completely in the third-angle projection.

13. A line sketch (in two views) of a shed with a curved roof is given in fig. 13-62. Draw its front view on an auxiliary vertical plane inclined at 60° to the V.P. All dimensions are in metres. Scale, 10 mm = 0.5 m.

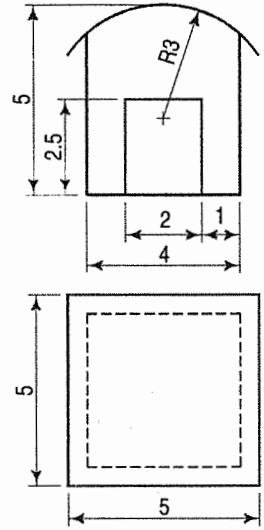


FIG. 13-62

14. The front view of a hexagonal pyramid [base 25 mm side] having one of its triangular faces resting centrally on a triangular face of a square pyramid [base 50 mm side and axis 50 mm long] is given in fig. 13-63. The plane containing the two axes is parallel to the V.P. Draw the top view of the solids. From this top view, project a front view on a reference line x_1y_1 inclined at 30° to xy ; (ii) from the given front view, project another top view on a reference line x_2y_2 inclined at 45° to xy .

15. A cube of 50 mm long edges is resting on the ground with its vertical faces equally inclined to the V.P. A hexagonal pyramid, base 25 mm side and axis 50 mm long, is placed centrally on top of the cube so that their axes are in a straight line and two edges of its base parallel to the V.P. Draw the front and top views of the solids. Project another top view on an A.I.P. making an angle of 45° with the H.P. From this top view project another front view on an auxiliary vertical plane inclined at 30° to the top view of the combined axis.

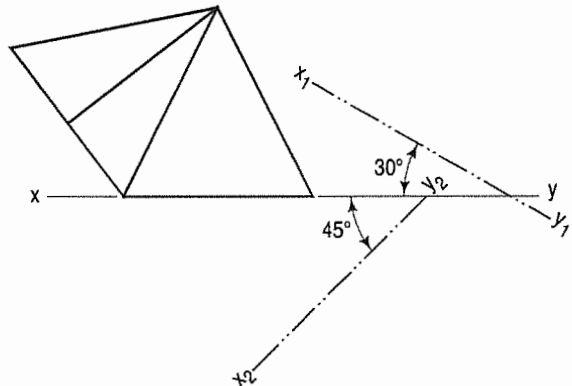


FIG. 13-63

16. Four equal spheres of 25 mm diameter are resting on the ground, each touching the other two spheres, so that a line joining the centres of two touching spheres is inclined at 30° to the V.P. A fifth sphere of 30 mm diameter is placed centrally on top of the four spheres, thus forming a pile. Draw the projections of the spheres and measure the height of the centre of the top sphere above the ground.
17. Three spheres of 25 mm, 50 mm and 75 mm diameter respectively are resting on the ground so that each touches the other two. Draw their projections when the top view of the line joining centres of any two of them is perpendicular to the V.P.
18. Three equal cones, base 50 mm diameter and axis 75 mm long, are placed on the ground on their bases, each touching the other two. A sphere of 40 mm diameter is placed centrally between them. Draw three views of the arrangement and determine the height of the centre of the sphere above the ground.
19. Five equal spheres are resting on the ground each touching the other two spheres and a vertical face of a pentagonal prism of 25 mm side. Determine the diameter of the spheres and draw the projections when a side of the base of the prism is perpendicular to the V.P.

20. Four equal spheres are resting on the ground, each touching the other two spheres and a triangular face of a square pyramid, having base 25 mm side and axis 50 mm long. Draw their projections and find the diameter of the spheres.
21. One of the body diagonals of a cube of 45 mm edge is parallel to the H.P. and inclined at 45° to the V.P. Draw the front view and the top view of the cube.
22. A pentagonal pyramid, base 40 mm side and height 75 mm rests on one edge of its base on the ground so that the highest point in the base is 25 mm above the ground. Draw its projections when the axis is parallel to the V.P. Draw another front view on a reference line inclined at 30° to the edge on which it is resting, and so that the base is visible.
23. A thin lamp shade in the form of a frustum of a cone has its larger end 200 mm diameter, smaller end 75 mm diameter and height 150 mm. Draw its three views when it is lying on its side on the ground and the axis parallel to the V.P.
24. A bucket made of tin sheet has its top 200 mm diameter and bottom 125 mm diameter with a circular ring 40 mm wide attached at the bottom. The total height of the bucket is 250 mm. Draw its projections when its axis makes an angle of 60° with the vertical.
25. A hexagonal pyramid, side of the base 25 mm long and height 70 mm, has one of its triangular faces perpendicular to the H.P. and inclined at 45° to the V.P. The base-side of this triangular face is parallel to the H.P. Draw its projections.
26. A pentagonal pyramid has an edge of the base in the V.P. and inclined at 30° to the H.P., while the triangular face containing that edge makes an angle of 45° with the V.P. Draw three views of the pyramid. Length of the side of the base is 30 mm, while that of the axis is 80 mm.
27. A square pyramid, base 40 mm side and axis 75 mm long is placed on the ground on one of its slant edges, so that the vertical plane passing through that edge and the axis makes an angle of 30° with the V.P. Draw its three views.
28. A hexagonal prism, side of base 40 mm and height 50 mm is lying on the ground on one of its bases with a vertical face perpendicular to the V.P. A tetrahedron is placed on the prism so that the corners of one of its faces coincide with the alternate corners of the top surface of the prism. Draw the projections of the solids. Project another top view on an auxiliary inclined plane making 45° with the H.P.
29. A square duct is in the form of a frustum of a square pyramid. The sides of top and bottom are 150 mm and 100 mm respectively and its length is 150 mm. It is situated in such a way that its axis is parallel to the H.P. and lies in a plane inclined at 60° to the V.P. Draw the projections of the duct, assuming the thickness of the duct-sheet to be negligible.
30. A pentagonal pyramid, base 30 mm edge and axis 75 mm long, stands upon a circular block, 75 mm diameter and 25 mm thick, so that their axes are in a straight line. Draw the projections of the solids when the base of the block is inclined at 30° to the ground, an edge of the base of the pyramid being parallel to the V.P.
31. The body diagonal of a cube is 75 mm long. The cube has a central 25 mm square hole. The faces of the hole make 45° with the side faces of the cube. Draw the projections of the cube when a body diagonal is perpendicular to the H.P.

32. A bucket, 300 mm diameter at the top and 225 mm diameter at the bottom has a circular ring 225 mm diameter and 50 mm wide attached at the bottom. The total height of the bucket is 300 mm. Draw the projections of the bucket when its axis is inclined at 60° to the H.P. and as a vertical plane makes an angle of 45° with the V.P. Assume the thickness of the plate of the bucket to be equal to that of a line.
33. The vertex-angle of the cone just touching the edges of a vertical hexagonal pyramid 125 mm in height is 45° . Draw the projections of the pyramid on a 45° inclined plane when the former is truncated by a plane making 45° with the axis and bisecting the axis.
34. A knob of a machine handle consists of 15 mm diameter $\times$ 150 mm long cylindrical portion and 40 mm diameter spherical portion. The centre of the sphere lies on the axis of the cylindrical portion. Draw the projections if its axis is inclined at 45° to the horizontal plane.
35. Six equal spheres rest on the ground in contact with each other and also with the slanting faces of a regular upright hexagonal pyramid, 25 mm edge of base and 125 mm length of axis. Draw the projections and find the diameter of the sphere.
36. A cylinder, 100 mm diameter and 150 mm long, has a rectangular slot 50 mm $\times$ 30 mm cut through it. The axis of the slot bisects the axis of the cylinder at right angles and the 50 mm side of the slot makes an angle of 60° with the base of the cylinder. Draw three views of the cylinder.
37. A very thin glass shade for a table lamp is the portion of a sphere 125 mm diameter included between two parallel planes at 15 mm and 55 mm from the centre, making the height 70 mm. If the axis of the shade is inclined at 30° to the vertical, obtain the projections of the shade.
38. A cone frustum, base 75 mm diameter, top 35 mm diameter and height 65 mm has a hole of 30 mm diameter drilled through it so that the axis of the hole coincides with that of the cone. It is resting on its base on the ground and is cut by a section plane perpendicular to the V.P., parallel to an end generator and passing through the top end of the axis. Draw sectional top view and sectional side view of the frustum.
39. Three vertical poles AB , CD and EF are respectively 5, 8 and 12 metres long. Their ends B , D and F are on the ground and lie at the corners of an equilateral triangle of 10 metres long sides. Determine graphically the distance between the top ends of the poles, viz. AC , CE and EA .
40. Two cylinders of 80 mm diameter each meet each other at right angles. The axis of one of the cylinders is parallel to both the reference planes and is 40 mm in front of the axis of the other cylinder. Draw three views of the cylinders showing lines of intersection in them. Take any suitable lengths of the cylinders.
41. A tetrahedron of side 40 mm rests on the top face of a hexagonal prism of base and height 25 mm such that their apex coincide. Draw the projections when the combination rests with one of the sides of the prism on the H.P., is perpendicular to the V.P., and the axis is inclined at 30° to the H.P.
42. A pentagonal pyramid, base 30 mm side and axis 70 mm long, has one of its slant edges in the H.P. and inclined at 30° to the V.P. Draw the projections of the solid when the apex is towards the observer.

Chapter

14



SECTIONS OF SOLIDS

14-0. INTRODUCTION

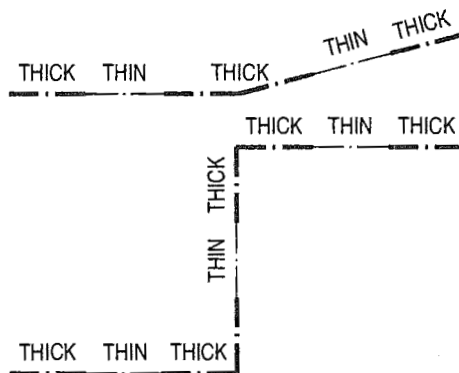
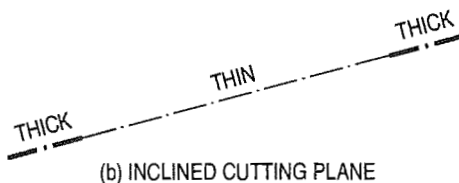
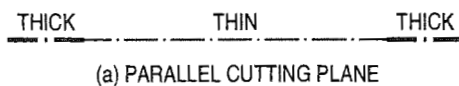
Invisible features of an object are shown by dotted lines in their projected views. But when such features are too many, these lines make the views more complicated and difficult to interpret. In such cases, it is customary to imagine the object as being cut through or *sectioned* by planes. The part of the object between the cutting plane and the observer is assumed to be removed and the view is then shown *in section*.

The imaginary plane is called a *section plane* or a *cutting plane*. The surface produced by cutting the object by the section plane is called the *section*. It is indicated by thin section lines uniformly spaced and inclined at 45° .

The projection of the section along with the remaining portion of the object is called a *sectional view*. Sometimes, only the word *section* is also used to denote a sectional view.

(1) **Section planes:** Section planes are generally perpendicular planes. They may be perpendicular to one of the reference planes and either perpendicular, parallel or inclined to the other plane. They are usually described by their traces. It is important to remember that the projection of a section plane, on the plane to which it is perpendicular, is a straight line. This line will be parallel, perpendicular or inclined to xy , depending upon the section plane being parallel, perpendicular or inclined respectively to the other reference plane.

As per latest B.I.S. convention (SP: 46-2003), the cutting-plane line should be drawn as shown in fig. 3-2 which is reproduced here in fig. 14-1 for ready reference.



(c) CUTTING PLANE AT CHANGING POSITION

FIG. 14-1

(2) **Sections:** The projection of the section on the reference plane to which the section plane is perpendicular, will be a straight line coinciding with the trace of the section plane on it. Its projection on the other plane to which it is inclined is called *apparent section*. This is obtained by

- (i) projecting on the other plane, the points at which the trace of the section plane intersects the edges of the solid and
- (ii) drawing lines joining these points in proper sequence.

(3) **True shape of a section:** The projection of the section on a plane parallel to the section plane will show the true shape of the section. Thus, when the section plane is parallel to the H.P. or the ground, the true shape of the section will be seen in *sectional top view*. When it is parallel to the V.P., the true shape will be visible in the *sectional front view*.

But when the section plane is inclined, the section has to be projected on an auxiliary plane parallel to the section plane, to obtain its true shape. When the section plane is perpendicular to both the reference planes, the sectional side view will show the true shape of the section. In this chapter sections of different solids are explained in stages by means of typical problems as follows:

1. Sections of prisms
2. Sections of pyramids
3. Sections of cylinders
4. Sections of cones
5. Sections of spheres.

14-1. SECTIONS OF PRISMS



These are illustrated according to the position of the section plane with reference to the principal planes as follows:

- (1) Section plane parallel to the V.P.
- (2) Section plane parallel to the H.P.
- (3) Section plane perpendicular to the H.P. and inclined to the V.P.
- (4) Section plane perpendicular to the V.P. and inclined to the H.P.

(1) Section plane parallel to the V.P.

Problem 14-1. (fig. 14-2): A cube of 35 mm long edges is resting on the H.P. on one of its faces with a vertical face inclined at 30° to the V.P. It is cut by a section plane parallel to the V.P. and 9 mm away from the axis and further away from the V.P. Draw its sectional front view and the top view.

In fig. 14-2(i), the section plane is assumed to be transparent and the cube is shown with the cut-portion removed. It can be seen that four edges of the cube are cut and hence, the section is a figure having four sides.

Draw the projections of the whole cube in the required position [fig. 14-2(ii)].

As the section plane is parallel to the V.P., it is perpendicular to the H.P.; hence, the section will be seen as a line in the top view coinciding with the H.T. of the section plane.

- (i) Draw a line H.T. in the top view (to represent the section plane) parallel to xy and 9 mm from o .

- (ii) Name the points at which the edges are cut, viz. ab at 1, bc at 2, gf at 3 and fe at 4.
- (iii) Project these points on the corresponding edges in the front view and join them in proper order.

As the section plane is parallel to the V.P., figure $1' 2' 3' 4'$ in the front view, shows the true shape of the section.

Show the views by dark but thin lines, leaving the lines for the cut-portion fainter.

- (iv) Draw section lines in the rectangle for the section.

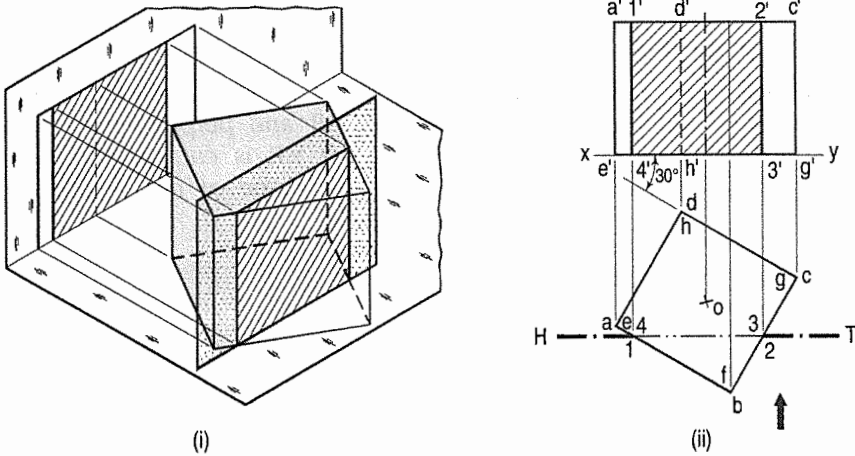


FIG. 14-2

(2) Section plane parallel to the H.P.

Problem 14-2. (fig. 14-3): A triangular prism, base 30 mm side and axis 50 mm long, is lying on the H.P. on one of its rectangular faces with its axis inclined at 30° to the V.P. It is cut by a horizontal section plane, at a distance of 12 mm above the ground. Draw its front view and sectional top view.

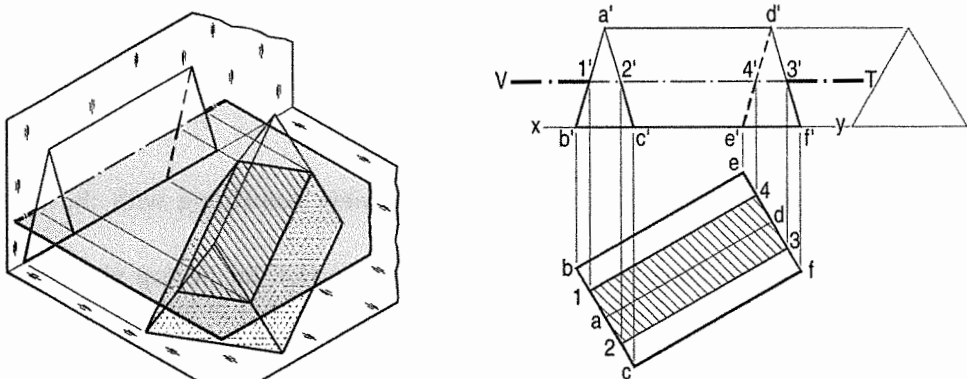


FIG. 14-3

Draw the projections of the prism in the required position.

As the section plane is horizontal, i.e. parallel to the H.P., it is perpendicular to the V.P. Hence, the section will be seen as a line in the front view, coinciding with the V.T. of the section plane.

- (i) Therefore, draw a line V.T. in the front view to represent the section plane, parallel to xy and 12 mm above it.
- (ii) Name in correct sequence, points at which the edges are cut viz. $a'b'$ at $1'$, $a'e'$ at $2'$, $d'f'$ at $3'$ and $d'e'$ at $4'$.
- (iii) Project these points on the corresponding lines in the top view and complete the sectional top view by joining them in proper order.

As the section plane is parallel to the H.P., the figure 1 2 3 4 (in the top view) is the true shape of the section.

(3) Section plane perpendicular to the H.P. and inclined to the V.P.

Problem 14-3. (fig. 14-4): A cube in the same position as in problem 14-1, is cut by a section plane, inclined at 60° to the V.P. and perpendicular to the H.P., so that the face which makes 60° angle with the V.P. is cut in two equal halves. Draw the sectional front view, top view and true shape of the section.

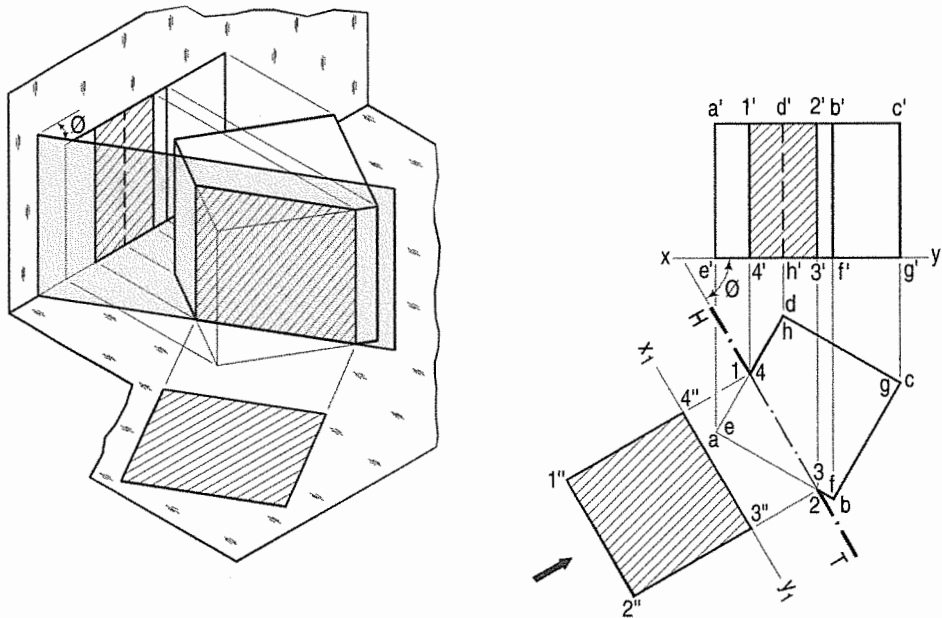


FIG. 14-4

The section will be seen as a line in the top view coinciding with the H.T. of the section plane.

- (i) Draw the projections of the cube. Draw a line H.T. in the top view inclined at 60° to xy and cutting the line ad (or bc) at its mid-point.
- (ii) Name the corners at which the four edges are cut and project them in the front view. As the section plane is inclined to the V.P., the front view of the section viz. $1' 2' 3' 4'$ does not reveal its true shape. Only the vertical lines show true lengths, while the true lengths of the horizontal lines are seen in the top view.

The true shape of the section will be seen when it is projected on an auxiliary vertical plane, parallel to the section plane.

- (iii) Therefore, draw a new reference line x_1y_1 parallel to the H.T. and project the section on it. The distances of the points from x_1y_1 should be taken equal to their corresponding distances from xy in the front view. Thus $4''$ and $3''$ will be on x_1y_1 . $1'' 4''$ and $2'' 3''$ will be equal to $1' 4'$ and $2' 3'$ respectively. Complete the rectangle $1'' 2'' 3'' 4''$ which is the true shape of the section and draw section lines in it.

(4) Section plane perpendicular to the V.P. and inclined to the H.P.

Problem 14-4. (fig. 14-5): A cube in the same position as in problem 14-1 is cut by a section plane, perpendicular to the V.P., inclined at 45° to the H.P. and passing through the top end of the axis. (i) Draw its front view, sectional top view and true shape of the section. (ii) Project another top view on an auxiliary plane, parallel to the section plane.

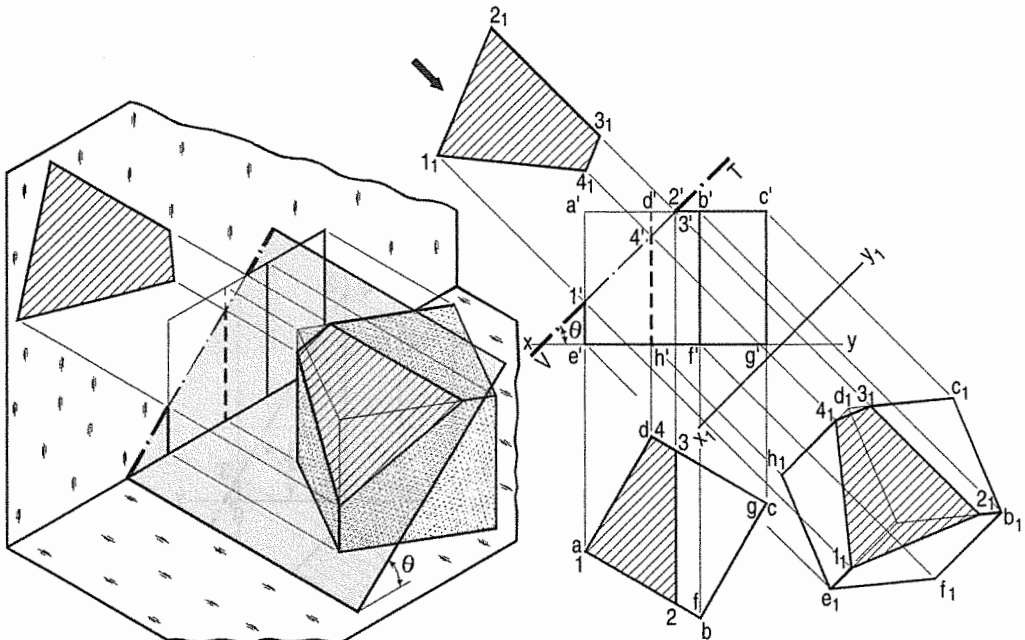


FIG. 14-5

The section will be seen as a line in the front view.

- (i) Draw a line V.T. in the front view, inclined at 45° to xy and passing through the top end of the axis. It cuts four edges, viz. $a' e'$ at $1'$, $a' b'$ at $2'$, $c' d'$ at $3'$ and $d' h'$ at $4'$.
- (ii) Project the top view of the section, viz. the figure $1 2 3 4$. It does not show the true shape of the section, as the section plane is inclined to the H.P. To determine the true shape, an auxiliary top view of the section should be projected on an A.I.P. parallel to the section plane.
- (iii) Assuming the new reference line for the A.I.P. to coincide with the V.T., project the true shape of the section as shown by quadrilateral $1_1 2_1 3_1 4_1$.

The distances of all the points from the V.T. should be taken equal to their corresponding distances from xy in the top view, e.g. $1_1 1' = e' 1$, $4_1 4' = h' 4$ etc.

- (iv) To project an auxiliary sectional top view of the cube, draw a new reference line $x_1 y_1$, parallel to the V.T. The whole cube may first be projected and the points for the section may then be projected on the corresponding lines for the edges. Join these points in correct sequence and obtain the required top view.
- (v) Draw section lines in the cut-surface, in the views where it is seen. Keep the lines for the removed edges thin and fainter.

Additional problems on sections of prisms:

Problem 14-5. (fig. 14-6): A square prism, base 40 mm side, axis 80 mm long, has its base on the H.P. and its faces equally inclined to the V.P. It is cut by a plane, perpendicular to the V.P., inclined at 60° to the H.P. and passing through a point on the axis, 55 mm above the H.P. Draw its front view, sectional top view and another top view on an A.I.P. parallel to the section plane.

The problem is similar to problem 14-4 and needs no further explanation. The true shape of the section is seen in the auxiliary top view.

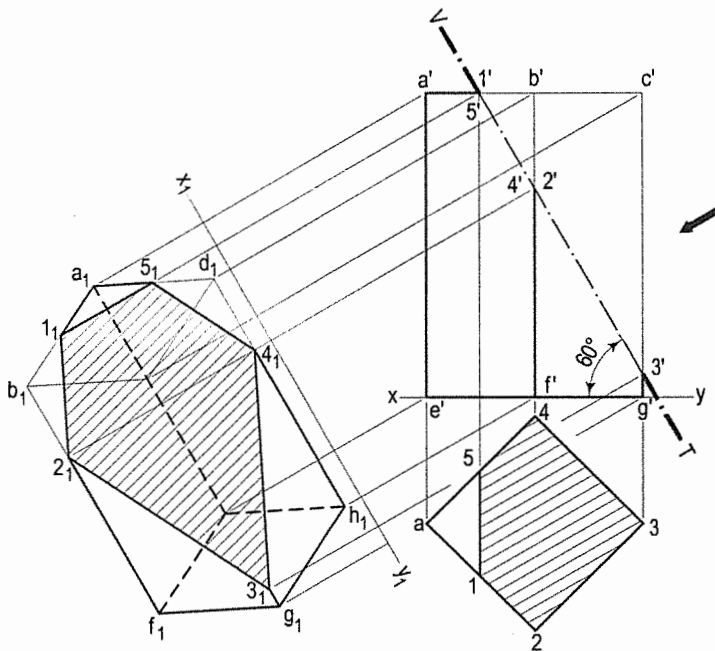


FIG. 14-6

Problem 14-6. (fig. 14-7): A hexagonal prism, has a face on the H.P. and the axis parallel to the V.P. It is cut by a vertical section plane, the H.T. of which makes an angle of 45° with xy and which cuts the axis at a point 20 mm from one of its ends. Draw its sectional front view and the true shape of the section. Side of base 25 mm long; height 65 mm.

- (i) Draw the front view and the top view of the prism and show the H.T. of the section plane in the top view. Name in proper sequence, the points at which the lines are cut.

- (ii) Project them on the corresponding lines in the front view. The positions of points 4 and 5 cannot be located directly. Hence, project them on the first top view to 4_1 on ef and 5_1 on ed . From this top view, obtain their positions $4'_1$ and $5'_1$ on the corresponding lines in the first front view. As the two front views are identical, these points can now be transferred to the second front view by making $e'4'_1$ equal to $e'4'_1$ and $e'5'_1$ equal to $e'5'_1$. $4'$ and $5'$ are the projections of points 4 and 5 respectively. Complete the sectional front view as shown.

- (iii) Obtain the true shape of the section on x_1y_1 as explained in problem 14-3, making $o''1''$ equal to $o'1'$, etc.

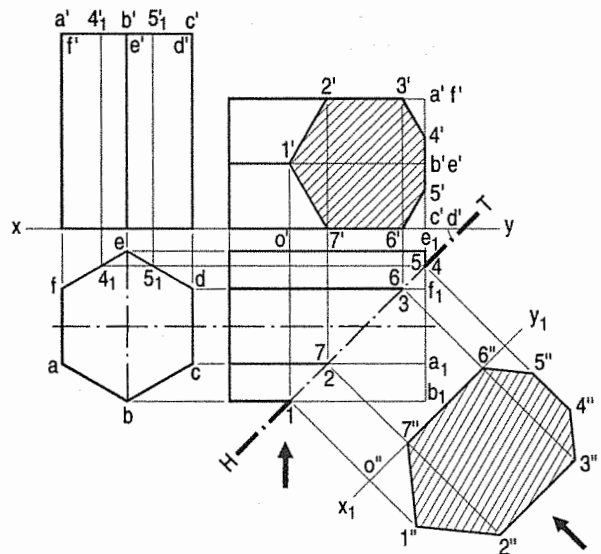


FIG. 14-7

Problem 14-7. (fig. 14-8): A pentagonal prism, base 28 mm side and height 65 mm has an edge of its base on the H.P. and the axis parallel to the V.P. and inclined at 60° to the H.P. A section plane, having its H.T. perpendicular to xy , and the V.T. inclined at 60° to xy and passing through the highest corner, cuts the prism. Draw the sectional top view and true shape of the section.

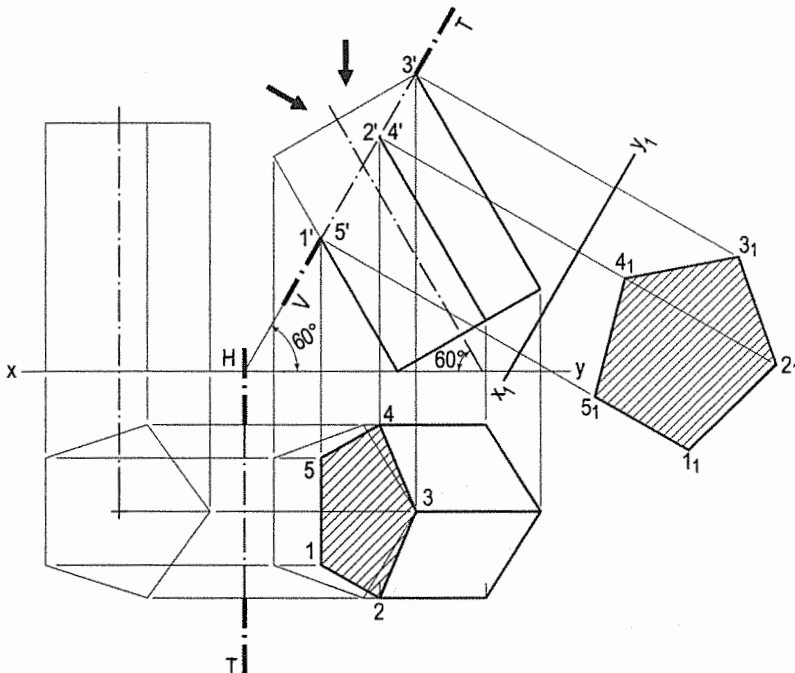


FIG. 14-8

- (i) Draw the projections of the prism in the required position.
- (ii) Draw the line V.T. passing through the highest corner $3'$ and inclined at 60° to xy . A perpendicular to xy through V will be the H.T. of the section plane.
- (iii) Project the sectional top view and the true shape of the section, as shown in the figure.

Problem 14-8. (fig. 14-9): A hollow square prism, base 40 mm side (outside), height 65 mm and thickness 8 mm is resting on its base on the H.P. with a vertical face inclined at 30° to the V.P. A section plane, inclined at 30° to the H.P., perpendicular to the V.P. and passing through the axis at a point 12 mm from its top end, cuts the prism. Draw its sectional top view, sectional side view and true shape of the section.

- (i) Draw the projections of the prism in the given position, showing the hidden edges by dashed lines.
- (ii) Draw a line V.T. for the cutting plane and mark points at which the inside and outside edges are cut.
- (iii) Project the sectional top view, true shape of the section and the sectional side view as shown.

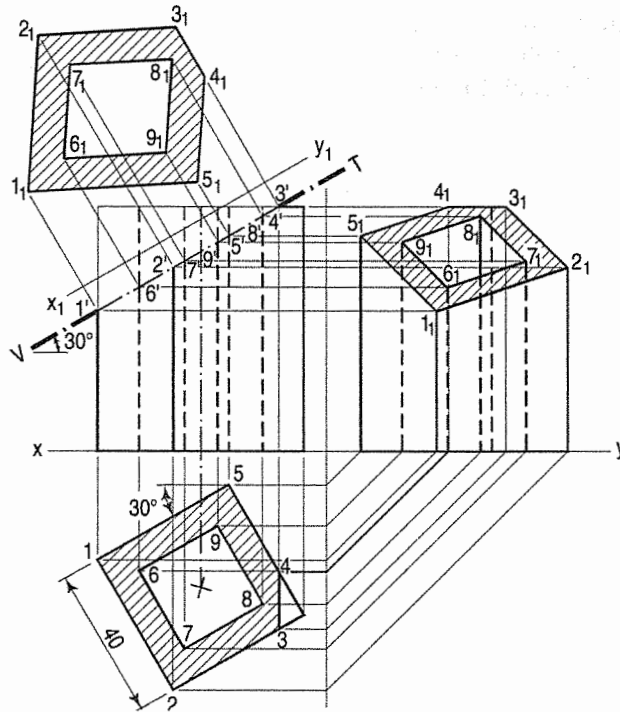


FIG. 14-9

14-2. SECTIONS OF PYRAMIDS

The following cases are discussed in details.

- (1) Section plane parallel to the base of the pyramid.
- (2) Section plane parallel to the V.P.
- (3) Section plane perpendicular to the V.P. and inclined to the H.P.
- (4) Section plane perpendicular to the H.P. and inclined to the V.P.

(1) Section plane parallel to the base of the pyramid.

Problem 14-9. (fig. 14-10): A pentagonal pyramid, base 30 mm side and axis 65 mm long, has its base horizontal and an edge of the base parallel to the V.P. A horizontal section plane cuts it at a distance of 25 mm above the base. Draw its front view and sectional top view.

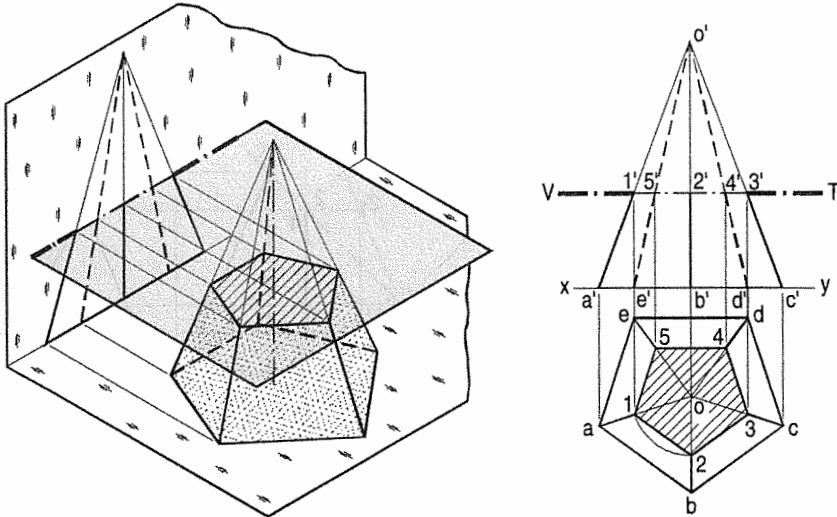


FIG. 14-10

- (i) Draw the projections of the pyramid in the required position and show a line V.T. for the section plane, parallel to and 25 mm above the base. All the five slant edges are cut.
- (ii) Project the points at which they are cut, on the corresponding edges in the top view. The point 2' cannot be projected directly as the line ob is perpendicular to xy . But it is quite evident from the projections of other points that the lines of the section in the top view, viz. 3-4, 4-5 and 5-1 are parallel to the edges of the base in their respective faces and that the points 1, 3, 4 and 5 are equidistant from o .
- (iii) Hence, line 1-2 also will be parallel to ab and $o2$ will be equal to $o1$, $o3$ etc. Therefore, with o as centre and radius $o1$, draw an arc cutting ob at a point 2 which will be the projection of 2'. Complete the sectional top view in which the true shape of the section, viz. the pentagon 1, 2, 3, 4 and 5 is also seen.
- (iv) Hence, when a pyramid is cut by a plane parallel to its base, the true shape of the section will be a figure, similar to the base; the sides of the section will be parallel to the edges of the base in the respective faces and the corners of the section will be equidistant from the axis.

(2) Section plane parallel to the V.P.

Problem 14-10. (fig. 14-11): A triangular pyramid, having base 40 mm side and axis 50 mm long, is lying on the H.P. on one of its faces, with the axis parallel to the V.P. A section plane, parallel to the V.P. cuts the pyramid at a distance of 6 mm from the axis. Draw its sectional front view and the top view.

- (i) Draw the projections of the pyramid in the required position and show a line H.T. (for the cutting plane) in the top view parallel to xy and 6 mm from the axis.
- (ii) Project points 1, 2 and 3 (at which the edges are cut) on corresponding edges in the front view and join them. Figure 1' 2' 3' shows the true shape of the section.

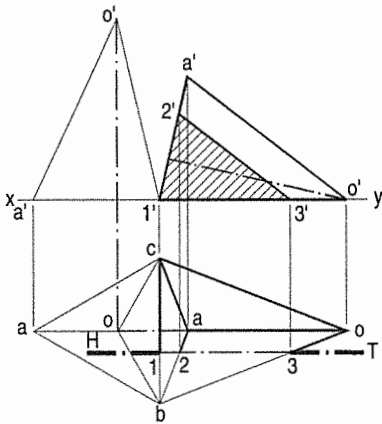


FIG. 14-11

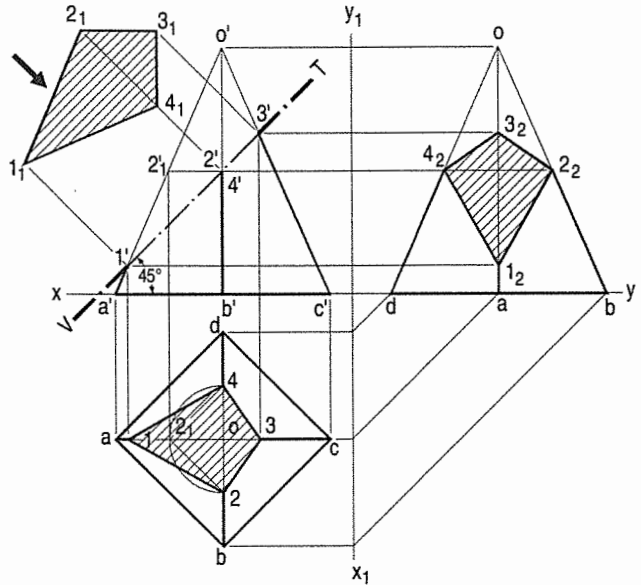


FIG. 14-12

(3) Section plane perpendicular to the V.P. and inclined to the H.P.

Problem 14-11. (fig. 14-12): A square pyramid, base 40 mm side and axis 65 mm long, has its base on the H.P. and all the edges of the base equally inclined to the V.P. It is cut by a section plane, perpendicular to the V.P., inclined at 45° to the H.P. and bisecting the axis. Draw its sectional top view, sectional side view and true shape of the section.

- (i) Draw the projections of the pyramid in the required position. The section plane will be seen as a line in the front view. Hence, draw a line V.T. through the mid-point of the axis and inclined at 45° to xy . Name in correct sequence the points at which the four edges are cut and project them in the top view. Here also, points 2' and 4' cannot be projected directly.

Hence, assume a horizontal section through 2' and draw a line parallel to the base, cutting $o'a'$ at 2'. Project 2' to 2₁ on oa in the top view. From 2₁ draw a line parallel to ab and cutting ob at a point 2. Or, with o as centre and radius $o2_1$, draw an arc cutting ob at 2 and ob at 4. Complete the section 1 2 3 4 by joining the points and draw section lines in it.

- (ii) Assuming the V.T. to be the new reference line, draw the true shape of the section. Project the side view from the two views. The removed portion of the pyramid may be shown by thin and faint lines.

(4) Section plane perpendicular to the H.P. and inclined to the V.P.

Problem 14-12. (fig. 14-13): A pentagonal pyramid has its base on the H.P. and the edge of the base nearer the V.P., parallel to it. A vertical section plane, inclined at 45° to the V.P., cuts the pyramid at a distance of 6 mm from the axis. Draw the top view, sectional front view and the auxiliary front view on an A.V.P. parallel to the section plane. Base of the pyramid 30 mm side; axis 50 mm long.

The section plane will be seen as a line in the top view. It is to be at a distance of 6 mm from the axis.

- (i) Hence, draw a circle with o as centre and radius equal to 6 mm.
- (ii) Draw a line H.T., tangent to this circle and inclined at 45° to xy . It can be drawn in four different positions, of which any one may be selected.
- (iii) Project points 1, 2 etc. from the top view to the corresponding edges in the front view. Here again, point 2 cannot be projected directly. The process shown in problem 14-11 must be reversed. With centre o and radius $o2$ draw an arc cutting any one of the slant edges, say oc at 2_1 . Project 2_1 to $2'_1$ on $o'c'$.
- (iv) Through $2'_1$, draw a line parallel to the base, cutting $o'b'$ at $2'$. Then $2'$ is the required point. Complete the view. It will show the apparent section.
- (v) Draw a reference line x_1y_1 parallel to the H.T. and project an auxiliary sectional front view which will show the true shape of the section also.

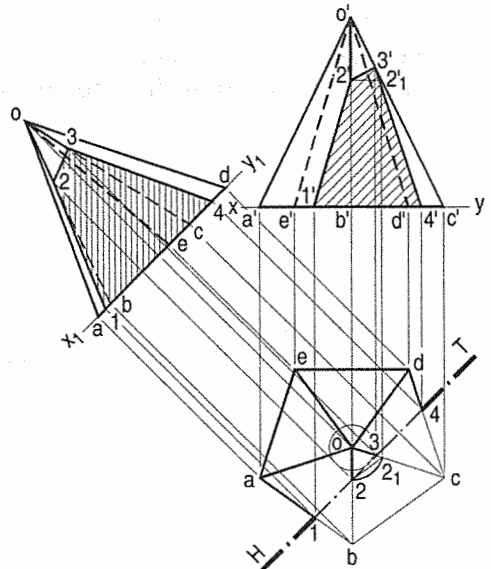


FIG. 14-13

Additional problems on sections of pyramids:

Problem 14-13. (fig. 14-14): A hexagonal pyramid, base 30 mm side and axis 65 mm long, is resting on its base on the H.P. with two edges parallel to the V.P. It is cut by a section plane, perpendicular to the V.P. inclined at 45° to the H.P. and intersecting the axis at a point 25 mm above the base. Draw the front view, sectional top view, sectional side view and true shape of the section.

This problem is similar to problem 14-11. In this case, the base is also cut and hence, the section is a heptagon. Care must be taken to name the points in proper sequence.

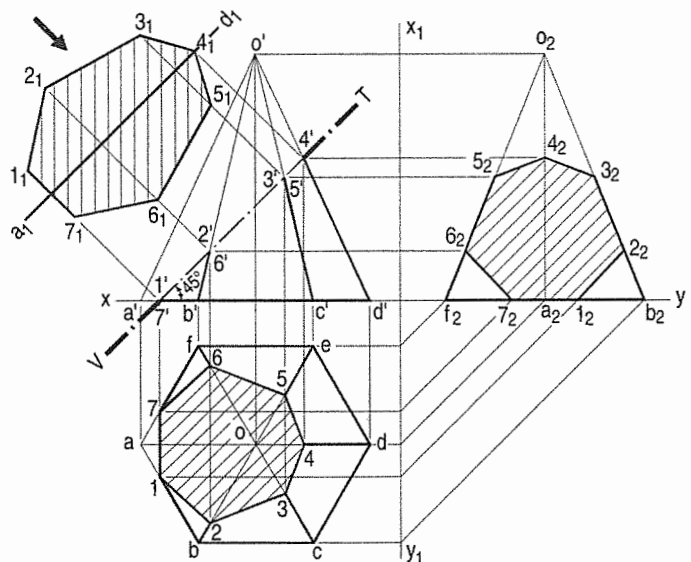


FIG. 14-14

The true shape may be drawn on the V.T. as a new reference line or around the centre line a_1d_1 , drawn parallel to the V.T. as shown.

The distances of the points $1_1, 2_1$ etc. from a_1d_1 are taken equal to the distances of points 1, 2 etc. from the line ad (which is parallel to xy).

Problem 14-14. (fig. 14-15): A pentagonal pyramid, base 30 mm side and axis 60 mm long, is lying on one of its triangular faces on the H.P. with the axis parallel to the V.P. A vertical section plane, whose H.T. bisects the top view of the axis and makes an angle of 30° with the reference line, cuts the pyramid, removing its top part. Draw the top view, sectional front view, true shape of the section and development of the surface of the remaining portion of the pyramid.

(i) Draw the H.T. of the section plane and name the points at which the edges are cut, in correct sequence, i.e. mark the visible edges first and then the hidden edges.

(ii) Project the sectional front view which will show the apparent section.

(iii) Obtain the true shape of the section on x_1y_1 as a new reference line drawn parallel to the H.T.

Development (fig. 14-16): The line $o'a'$ shows the true length of the slant edge.

(i) With any point O as centre and radius $o'a'$, draw an arc and construct the development of the whole pyramid. Mark points on it, taking the positions of 1 and 2 from the first top view and those of other points by projecting them on the true-length-line $o'a'$.

(ii) Draw lines joining these points and complete the development as shown in the figure.

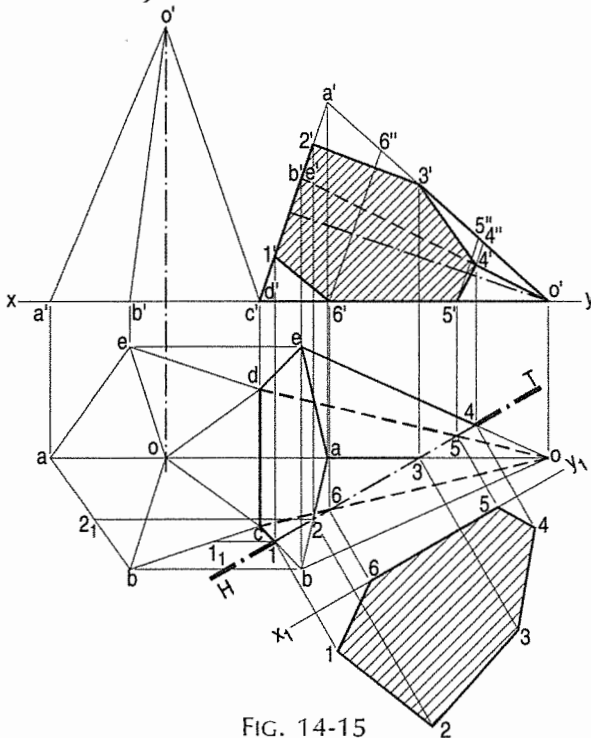


FIG. 14-15

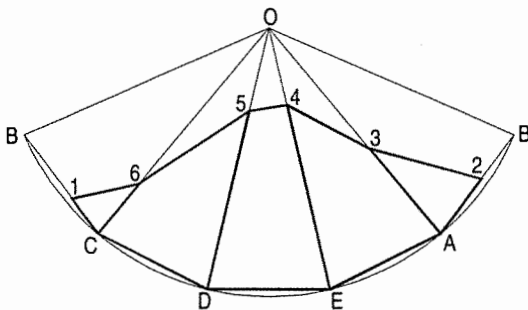


FIG. 14-16

Problem 14-15. (fig. 14-17): A hexagonal pyramid, base 30 mm side and axis 60 mm long, has a triangular face on the H.P. and the axis parallel to the V.P. It is cut by a horizontal section plane which bisects the axis. Draw the front view and sectional top view and develop the surface of the cut-pyramid.

The V.T. cuts six edges. The sectional top view shows the true shape of the section also.

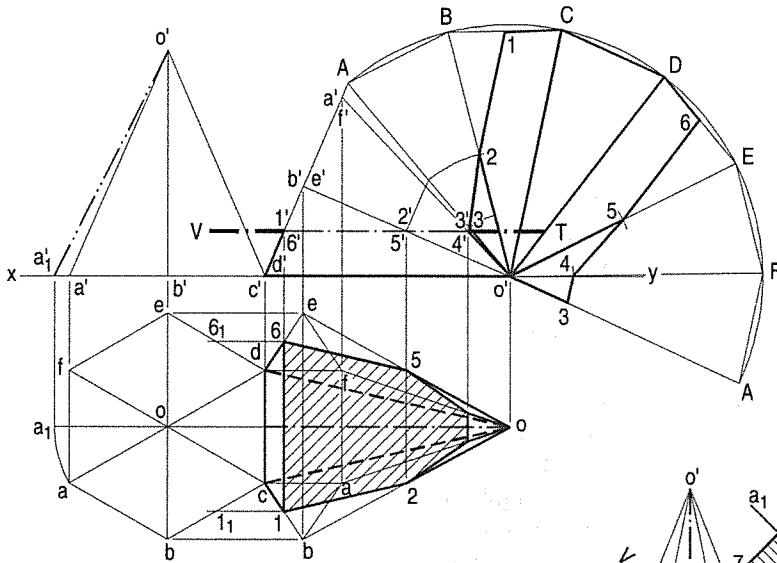


FIG. 14-17

Development: None of the edges shows the true length of the slant edge.

- (i) Hence, determine the true length $o'a_1$ and draw the development of the whole pyramid.
- (ii) Locate positions of the points 1 and 6 by projecting them on the first top view and positions of other points by drawing lines through them, parallel to the base and upto the true length line $o'A$.
- (iii) Mark these points on the development and complete it as shown.

Problem 14-16. (fig. 14-18): A hexagonal pyramid, base 30 mm side and axis 75 mm long, resting on its base on the H.P. with two of its edges parallel to the V.P. is cut by two section planes, both perpendicular to the V.P. The horizontal section plane cuts the axis at a point 35 mm from the apex. The other plane which makes an angle of 45° with the H.P., also intersects the axis at the same point. Draw the front view, sectional top view, true shape of the section and development of the surface of the remaining part of the pyramid.

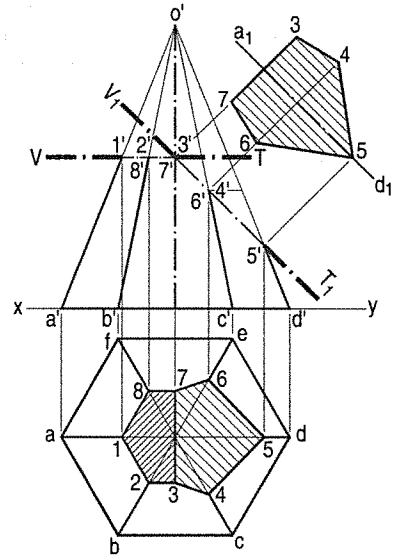


FIG. 14-18

- (i) Draw lines $V.T.$ and V_1T_1 for the two section planes. The top view will show the true shape of the horizontal section, the sides of which are parallel to the respective sides of the base. The true shape of the other section may be obtained on V_1T_1 as the reference line or around a_1d_1 .
- (ii) Draw the development with $o'a'$ or $o'd'$ as radius and locate the points on it, as shown in the figure.

14-3. SECTIONS OF CYLINDERS

We shall now learn the following three cases. They are

- (1) Section plane parallel to the base
- (2) Section plane parallel to the axis
- (3) Section plane inclined to the base.

(1) Section plane parallel to the base:

When a cylinder is cut by a section plane parallel to the base, the true shape of the section is a circle of the same diameter.

(2) Section plane parallel to the axis:

When a cylinder is cut by a section plane parallel to the axis, the true shape of the section is a rectangle, the sides of which are respectively equal to the length of the axis and the length of the section plane within the cylinder (fig. 14-19). When the section plane contains the axis, the rectangle will be of the maximum size.

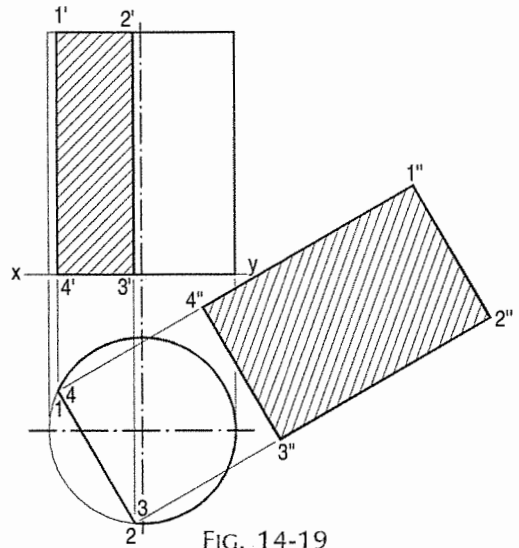


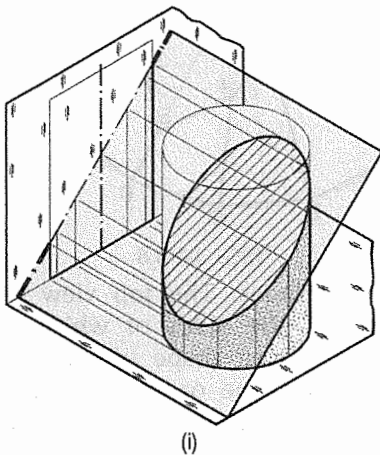
FIG. 14-19

(3) Section plane inclined to the base:

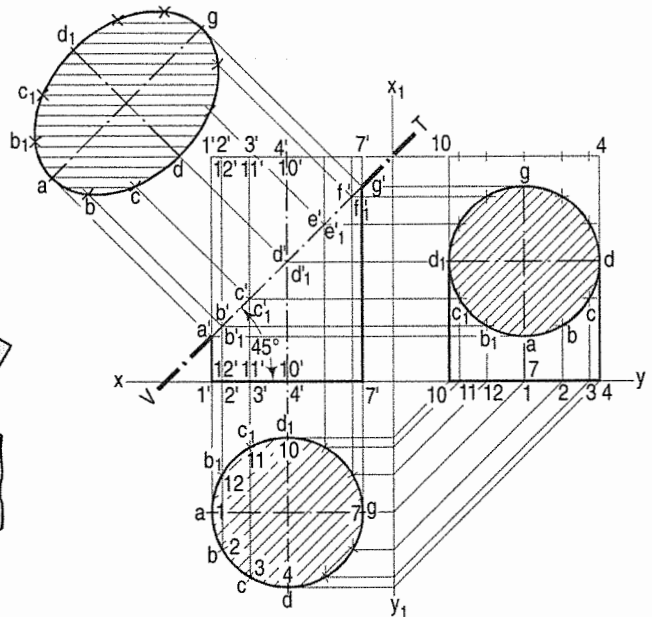


This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 31 for the following problem.

Problem 14-17. (fig. 14-20): A cylinder of 40 mm diameter, 60 mm height and having its axis vertical, is cut by a section plane, perpendicular to the V.P., inclined at 45° to the H.P. and intersecting the axis 32 mm above the base. Draw its front view, sectional top view, sectional side view and true shape of the section.



(i)



(ii)

FIG. 14-20

As the cylinder has no edges, a number of lines representing the generators may be assumed on its curved surface by dividing the base-circle into, say 12 equal parts.

- (i) Name the points at which these lines are cut by the V.T. In the top view, these points lie on the circle and hence, the same circle is the top view of the section. The width of the section at any point, say c' , will be equal to the length of the chord cc_1 in the top view.
- (ii) The true shape of the section may be drawn around the centre line ag drawn parallel to V.T. as shown. It is an ellipse the major axis of which is equal to the length of the section plane viz. $a'g'$, and the minor axis equal to the diameter of the cylinder viz. dd_1 .
- (iii) Project the sectional side view as shown. The section will be seen as a circle because the section plane makes 45° angle with xy .

Additional problems on sections of cylinders:

Problem 14-18. (fig. 14-21): A cylinder 50 mm diameter and 60 mm long, is resting on its base on the ground. It is cut by a section plane perpendicular to the V.P., the V.T. of which cuts the axis at a point 40 mm from the base and makes an angle of 45° with the H.P. Draw its front view, sectional top view and another sectional top view on an A.I.P. parallel to the section plane.

In this case, the top end of the cylinder is also cut. Hence, the true shape of the section is a part of an ellipse as shown in the auxiliary top view.

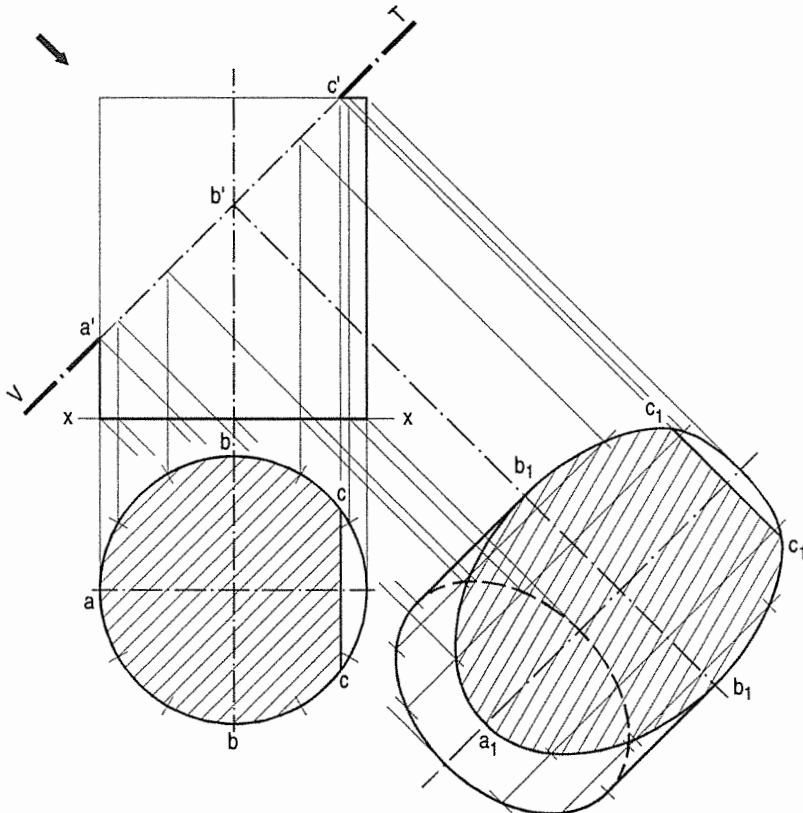


FIG. 14-21

Problem 14-19. (fig. 14-22): A cylinder, 55 mm diameter and 65 mm long, has its axis parallel to both the H.P. and the V.P. It is cut by a vertical section plane inclined at 30° to the V.P., so that the axis is cut at a point 30 mm from one of its ends and both the bases of the cylinder are partly cut. Draw its sectional front view and true shape of the section.

Draw the projections of the cylinder and a line H.T. for the section plane. Project the points at which the bases and the lines are cut. The points on the bases cannot be projected directly. Therefore, project them

- (i) To the first top view i.e. a to a_1 and e to e_1 .
- (ii) Then to the first front view, i.e. a_1 to a'_1 and e_1 to e'_1 .
- (iii) Finally, transfer them to the second front view to a' and e' each, at two places as shown.

- (iv) Draw the true shape of the section either on a new reference line or symmetrically around the centre line and making aa' equal to $a'a'$, cc' equal to $c'c'$ etc.

Problem 14-20. (fig. 14-23): A hollow cylinder, 50 mm outside diameter, axis 70 mm long and thickness 8 mm has its axis parallel to the V.P. and inclined at 30° to the vertical. It is cut in two equal halves by a horizontal section plane. Draw its sectional top view.

The figure is self-explanatory. Note that a part of the ellipse for the inside bottom will also be visible.

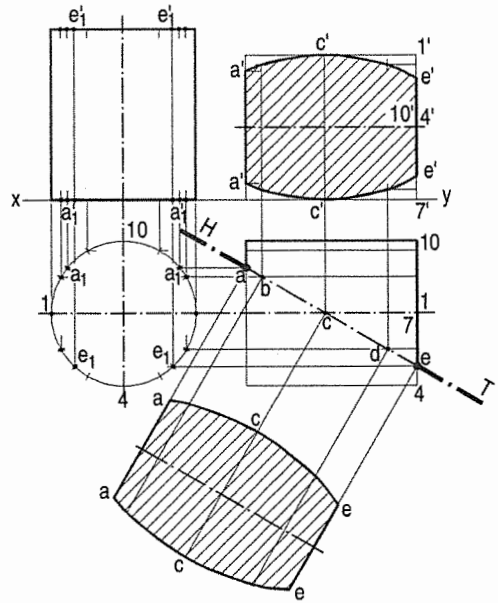


FIG. 14-22

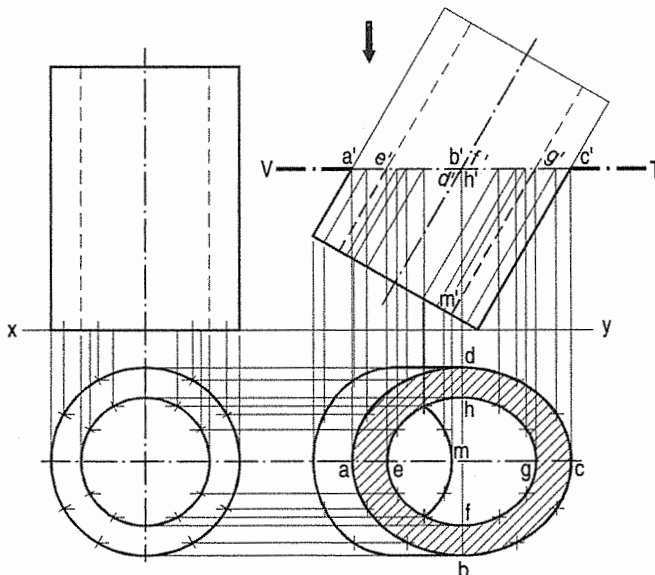


FIG. 14-23

14-4. SECTIONS OF CONES



This is discussed in details as follows:

- (1) Section plane parallel to the base of the cone.
- (2) Section plane passing through the apex of the cone.
- (3) Section plane inclined to the base of the cone at an angle smaller than the angle of inclination of the generators with the base.
- (4) Section plane parallel to a generator of the cone.
- (5) Section plane inclined to the base of the cone at an angle greater than the angle of inclination of the generators with the base.

(1) Section plane parallel to the base of the cone:

The cone resting on the H.P. on its base [fig. 14-24(i)] is cut by a section plane parallel to the base. The true shape of the section is shown by the circle in the top view, whose diameter is equal to the length of the section viz. $a'a'$. The width of the section at any point, say b' , is equal to the length of the chord bb_1 .

Problem 14-21. [fig. 14-24(ii)]: To locate the position in the top view of any given point p' in the front view of the above cone.

Method I:

- (i) Through p' , draw a line $r'r'$ parallel to the base.
- (ii) With o as centre and diameter equal to $r'r'$, draw a circle in the top view.
- (iii) Project p' to points p and p_1 on this circle. p is the top view of p' . p_1 is the top view of another point p_1 on the back side of the cone and coinciding with p' . The chord pp_1 shows the width of the horizontal section of the cone at the point p' . This method may be called the *circle method*.

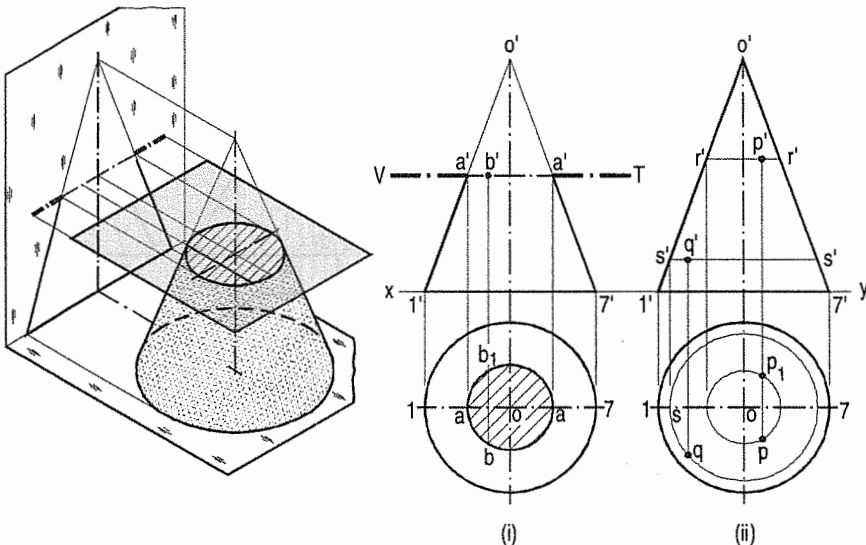


FIG. 14-24

Problem 14-23. [fig. 14-25(ii)]: To determine by generator method, the position in the top view of a given point p' in the front view of the above cone.

Draw the line $o'p'$ and produce it to cut the base at r' . Project r' to points r and r_1 on the base-circle in the top view. Draw lines or and or_1 . Thus, or is the top view of the generator $o'r'$, and or_1 that of the generator (at the back) which coincides with $o'r'$. Project p' to p and p_1 on or and or_1 respectively. Thus, p is the top view of p' , and p_1 is the top view of another point on the other side of the cone and coinciding with p' . The line pp_1 is the width of the horizon-section of the cone at p' .

The position in the front view of any point in the top view, say q , may be determined by reversing the process. Draw the line oq and produce it to cut the base-circle at s . Project s to s' on the base in the front view. Join o' with s' . Through, q , draw a projector to cut $o's'$ at the required point q' .

Sectional views of cones may be obtained by applying any one of the above two methods for locating the positions of points. The generator method is more suitable particularly when the cone is in inclined positions.

(3) Section plane inclined to the base of the cone at an angle smaller than the angle of inclination of the generators with the base:

Problem 14-24. A cone, base 75 mm diameter and axis 80 mm long is resting on its base on the H.P. It is cut by a section plane perpendicular to the V.P., inclined at 45° to the H.P. and cutting the axis at a point 35 mm from the apex. Draw its front view, sectional top view, sectional side view and true shape of the section.

Draw a line V.T. in the required position in the front view of the cone. The positions of points on this line and the width of section at each point can be determined by one of the methods explained in problem 14-21 and problem 14-23 and as described below.

(i) Generator method [fig. 14-26(i) and fig. 14-26(ii)]:

- (a) Divide the base-circle into a number of equal parts, say 12. Draw lines (i.e. generators) joining these points with o . Project these points on the line representing the base in the front view.
- (b) Draw lines $o'2'$, $o'3'$ etc. cutting the line for the section at points b' , c' etc. Project these points on the corresponding lines in the top view. For example, point b' on $o'2'$, also represents point b'_1 on $o'-12'$ which coincides with $o'-12'$. Therefore, project b' to b on $o2$ and to b_1 on $o'-12'$. b and b_1 are the points on the section (in the top view).

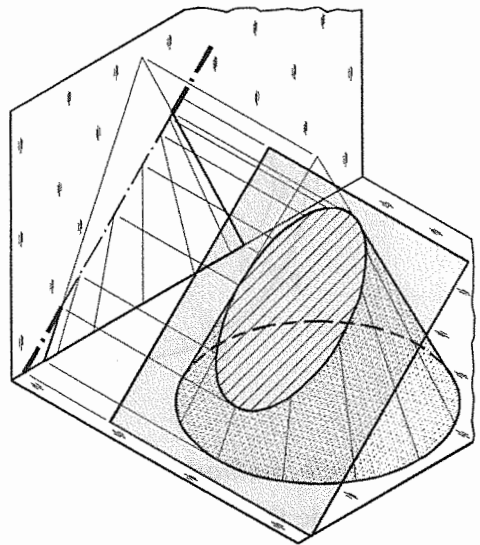


FIG. 14-26(i)

- (c) Similarly, obtain other points. Point d' cannot be projected directly. Hence, the same method as in case of pyramids should be employed to determine the positions d and d_1 , as shown. In addition to these, two more points

for the maximum width of the section at its centre should also be obtained. Mark m' , the mid-point of the section and obtain the points m and m_1 . Draw a smooth curve through these points.

- (d) The true shape of the section may be obtained on the V.T. as a new reference line or symmetrically around the centre line ag , drawn parallel to the V.T. as shown. It is an ellipse whose major axis is equal to the length of the section and minor axis equal to the width of the section at its centre.

Draw the sectional side view by projecting the points on corresponding generators, as shown.

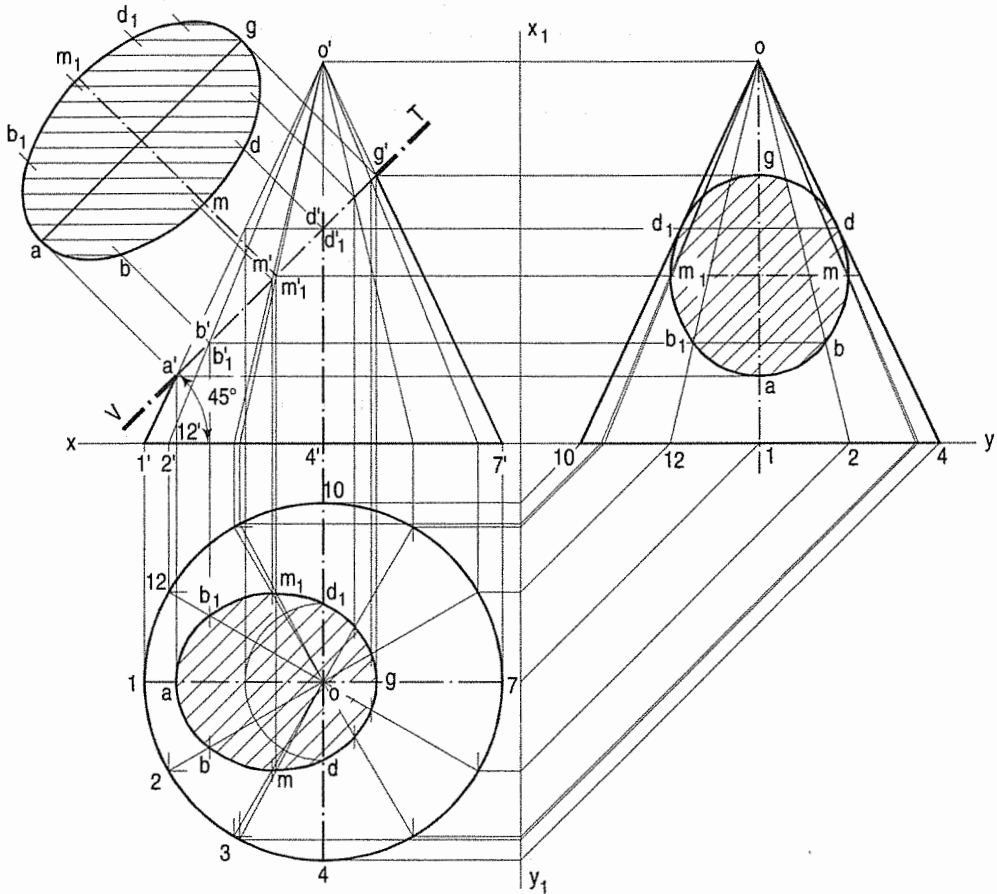


FIG. 14-26(ii)

(ii) **Circle method** (fig. 14-27):

- Divide the line of section into a number of equal parts. Determine the width of section at, and the position of each division-point in the top view by the circle method. For example, through c' , draw a line $c''c'''$ parallel to the base.
- With o as centre and radius equal to half of $c''c'''$, draw an arc. Project c' to c and c_1 on this arc. Then c and c_1 are the required points. The straight line joining c and c_1 will be the width of the section at c' .

- (c) Similarly, obtain all other points and draw a smooth curve through them. This curve will show the apparent section. The maximum width of the section will be at the mid-point e' . It is shown in the top view by the length of the chord joining e and e_1 .
- (d) Draw a reference line $x_1 y_1$ parallel to the V.T. and project the true shape of the section. In the figure, the auxiliary sectional top view of the truncated cone is shown. It shows the true shape of the section.

The sectional side view (not shown in the figure) may be obtained by projecting all the division-points horizontally and then marking the width of the section at each point, symmetrically around the axis of the cone.

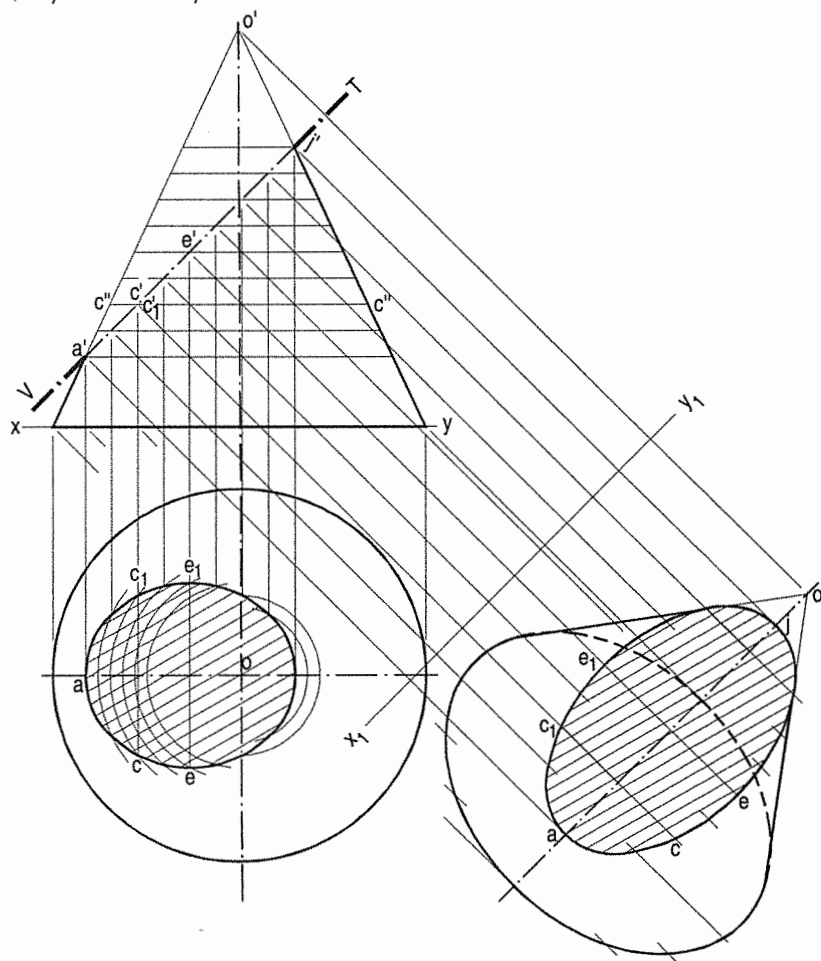


FIG. 14-27

(4) Section plane parallel to a generator of the cone:

Problem 14-25. (fig. 14-28): The cone in same position as in problem 14-24, is cut by a section plane perpendicular to the V.P. and parallel to and 12 mm away from one of its end generators. Draw its front view, sectional top view and true shape of the section.

- (i) Draw a line V.T. (for the section plane) parallel to and 12 mm away from the generator $o'1'$.

- (ii) Draw the twelve generators in the top view and project them to the front view. All the generators except $o'1'$, $o'2'$ and $o'12'$ are cut by the section plane. Project the points at which they are cut, to the corresponding generators in the top view. The width of the section at the point where the base is cut will be the chord aa_1 . Draw a curve through $a...f...a_1$. The figure enclosed between aa_1 and the curve is the apparent section.
- (iii) Obtain the true shape of the section as explained in the previous problem. It will be a parabola.

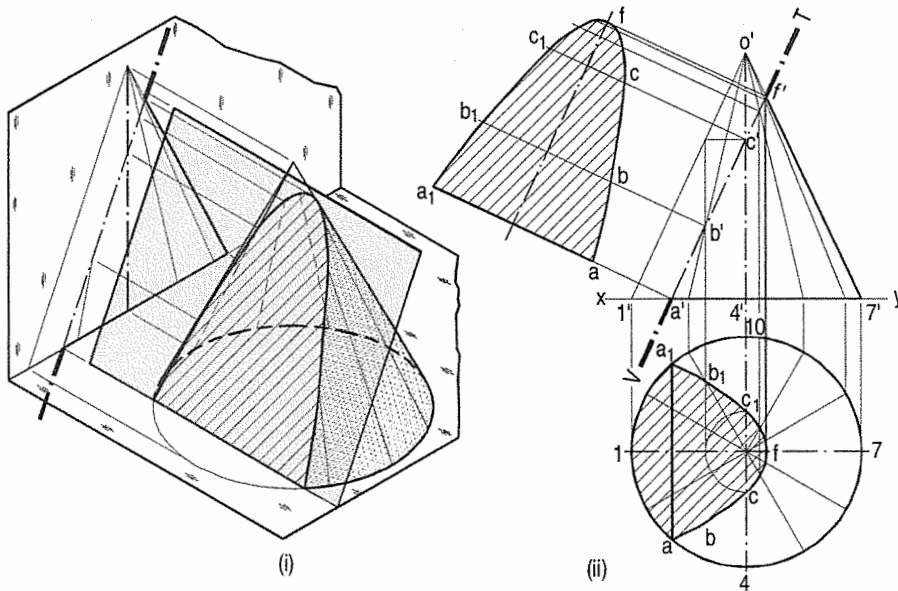


FIG. 14-28

- (5) **Section plane inclined to the base of the cone at an angle greater than the angle of inclination of the generators with the base:**



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 32 for the following problem.

Problem 14-26. [fig. 14-29(i) and fig. 14-29(ii)]: A cone, base 45 mm diameter and axis 55 mm long is resting on the H.P. on its base. It is cut by a section plane, perpendicular to both the H.P. and the V.P. and 6 mm away from the axis. Draw its front view, top view and sectional side view.

The section will be seen as a line, perpendicular to xy , in both the front view and the top view. The side view will show the true shape of the section. The width of the section at any point, say c' , will be equal to cc_1 obtained by the circle method [fig. 14-29(i)].

- (i) Draw the side view of the cone.
- (ii) Project the points (on the section) in the side view taking the widths from the top view. For example, through c' draw a horizontal line. Mark on it points c'' and c''_1 equidistant from and on both sides of the axis so that $c''c''_1 = cc_1$.
- (iii) Draw a curve through the points thus obtained. It will be a hyperbola.

Fig. 14-29(ii) shows the views obtained by the generator method.

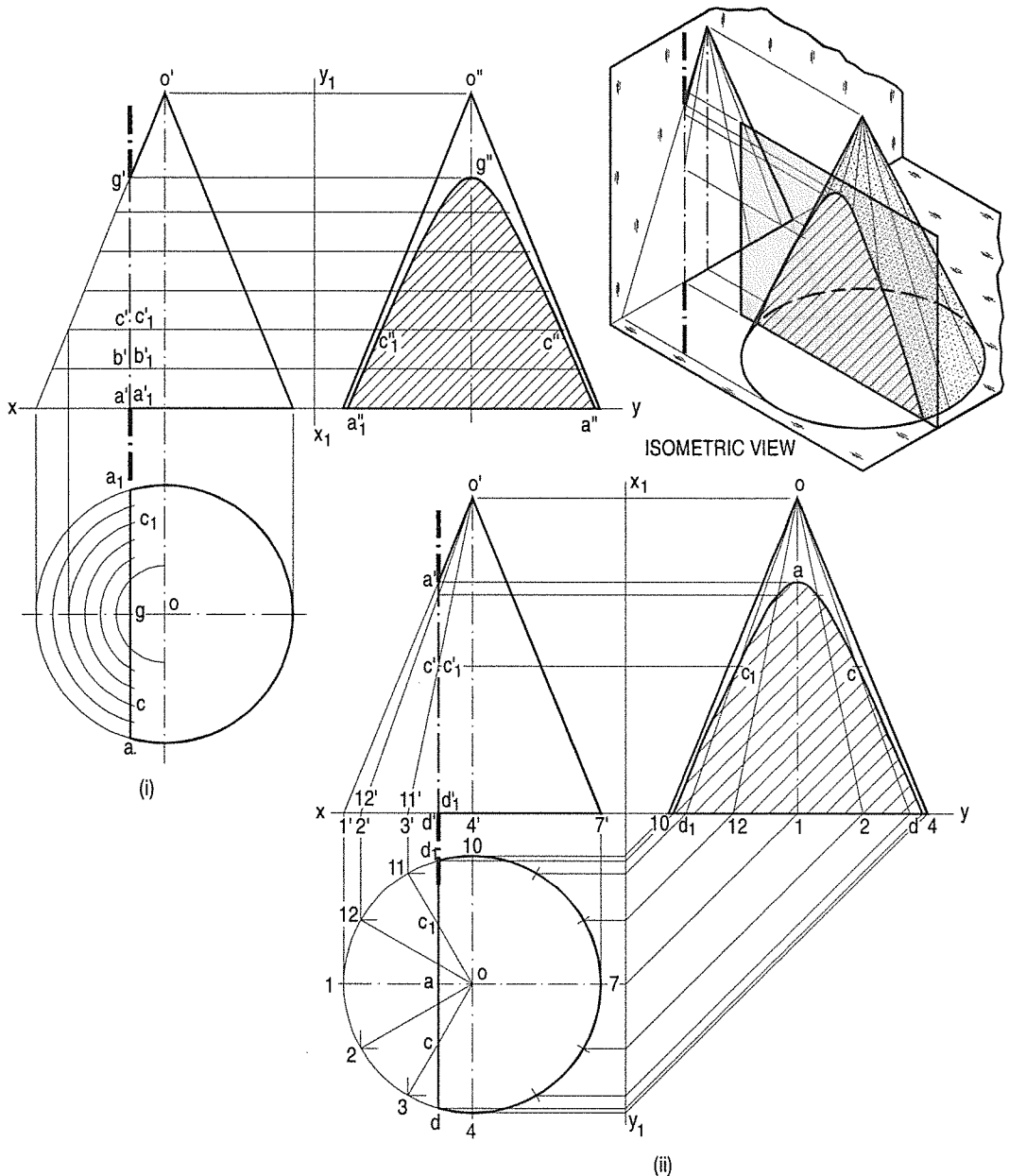


FIG. 14-29

Additional problems on sections of cones:

Problem 14-27. (fig. 14-30): A cone, diameter of base 50 mm and axis 65 mm long, is lying on the H.P. on one of its generators with the axis parallel to the V.P. It is cut by a horizontal section plane 12 mm above the ground. Draw its front view, sectional top view, and development of its surface.

Use the generator method and project the points in the top view. The curve will show the true shape of the section viz. a parabola.

For development, the true lengths of the cut-generators are obtained by drawing lines parallel to the base. Positions of points a and a_1 are determined by projecting them on the base-circle in the first top view.

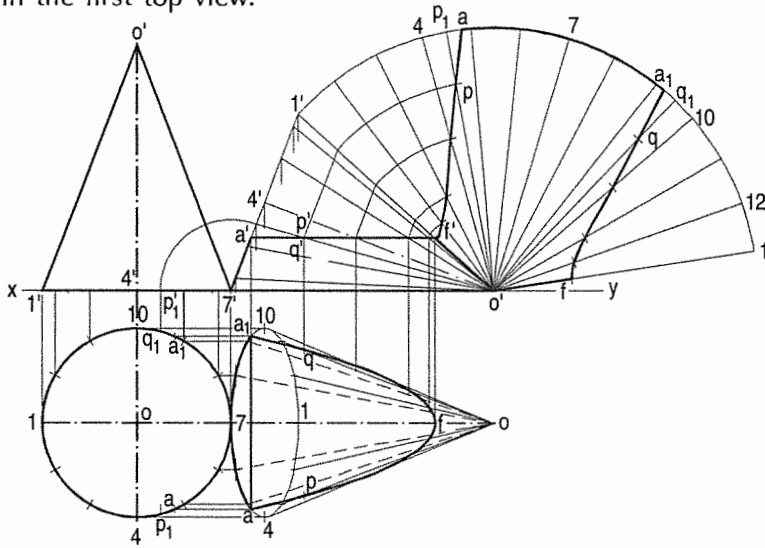


FIG. 14-30

Problem 14-28. (fig. 14-31): A cone, base 70 mm diameter, axis 75 mm long and resting on its base on the H.P., is cut by a vertical section plane, the H.T. of which makes an angle of 60° with the reference line and is 12 mm away from the top view of the axis. (i) Draw the sectional front view and the true shape of the section. (ii) Also draw the sectional front view and the top view when the same section plane is parallel to the V.P.

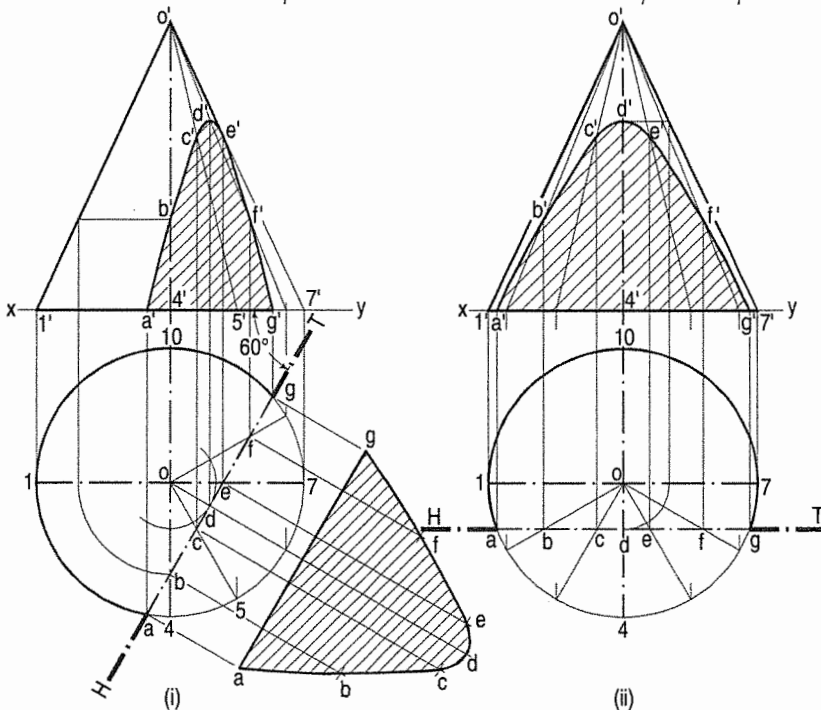


FIG. 14-31

- (i) Draw a circle with centre o and radius equal to 12 mm.
- (ii) Draw a line for the section plane, tangent to this circle and inclined at 60° to xy [fig. 14-31(i)].
- (iii) Project the front view by the generator method as shown. Note how the point f' is obtained.

Fig. 14-31(ii) shows the two views when the section plane is parallel to the V.P.

Problem 14-29. (fig. 14-32): A cone, base 60 mm diameter and axis 60 mm long is lying on the H.P. on one of its generators with the axis parallel to the V.P. A vertical section plane parallel to the generator which is tangent to the ellipse (for the base) in the top view, cuts the cone bisecting the axis and removing a portion containing the apex. Draw its sectional front view and true shape of the section.

- (i) Name in correct sequence the points at which the base and the generators are cut and project them in the front view.
- (ii) Project the true shape of the section on the new reference line $x_1 y_1$ drawn parallel to the H.T.

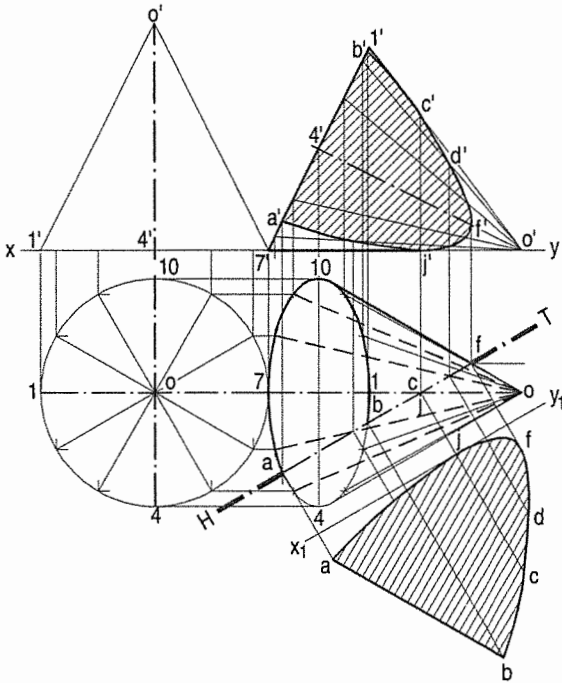


FIG. 14-32

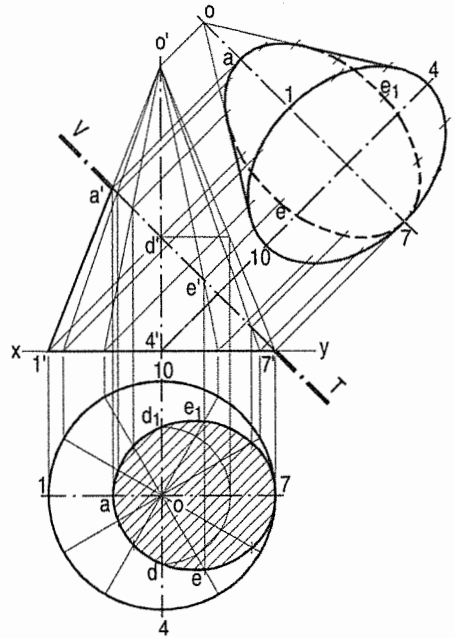


FIG. 14-33

Problem 14-30. (fig. 14-33): A cone, base 60 mm diameter and axis 75 mm long, is resting on the H.P. on its base. It is cut by a section plane, perpendicular to the V.P., inclined at 45° to the H.P. and intersecting the axis 30 mm above the base. Draw its front view and sectional top view. Also draw its top view when it is lying on the ground on its cut-surface with the axis parallel to the V.P.

See the figure which is self-explanatory. Note that, when the cone is tilted so as to lie on the cut-surface, its base is fully visible, while the section is hidden in the top view.

14-5. SECTIONS OF SPHERES



These are discussed in details as under.

- (1) Section plane parallel to the H.P.
- (2) Section plane parallel to the V.P.
- (3) Section plane perpendicular to the V.P. and inclined to the H.P.
- (4) Section plane perpendicular to the H.P. and inclined to the V.P.

(1) Section plane parallel to the H.P.: When a sphere is cut by a plane, the true shape of the section is always a circle.

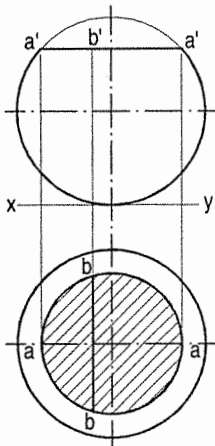


FIG. 14-34

The sphere in fig. 14-34 is cut by a horizontal section plane. The true shape of the section (seen in the top view) is a circle of diameter $a'a'$. The width of the section at any point say b' , is equal to the length of the chord bb .

(2) Section plane parallel to the V.P.: When the sphere is cut by a section plane parallel to the V.P. (fig. 14-35), the true shape of the section, seen in the front view, is a circle of diameter cc . The width of section at any point d is equal to the length of the chord $d'd'$.

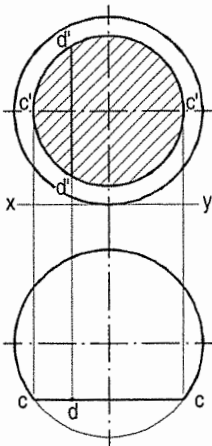


FIG. 14-35

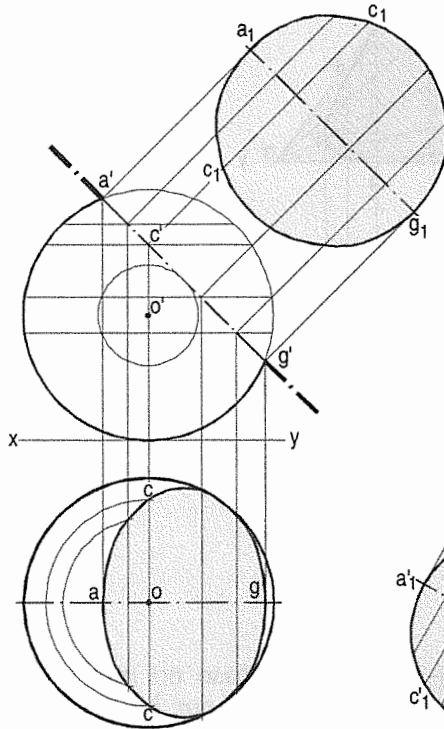


FIG. 14-36

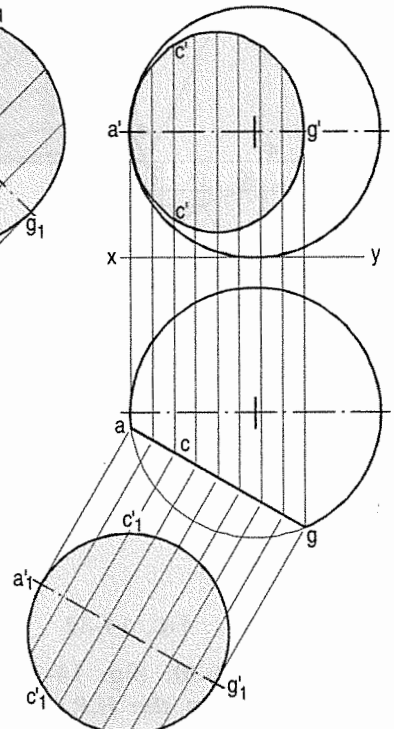


FIG. 14-37

(3) Section plane perpendicular to the V.P. and inclined to the H.P.:

Problem 14-31. (fig. 14-36): A sphere of 50 mm diameter is cut by a section plane perpendicular to the V.P., inclined at 45° to the H.P. and at a distance of 10 mm from its centre. Draw the sectional top view and true shape of the section.

Draw a line (for the section plane) inclined at 45° to xy and tangent to the circle of 10 mm radius drawn with o' as centre. Mark a number of points on this line.

Method I:

- (i) Find the width of section at each point in the top view as shown in fig. 14-34. For example, the chord cc is the width of section at the point c' .
- (ii) Draw a curve through the points thus obtained. It will be an ellipse. The true shape of the section will be a circle of diameter $a'g'$.

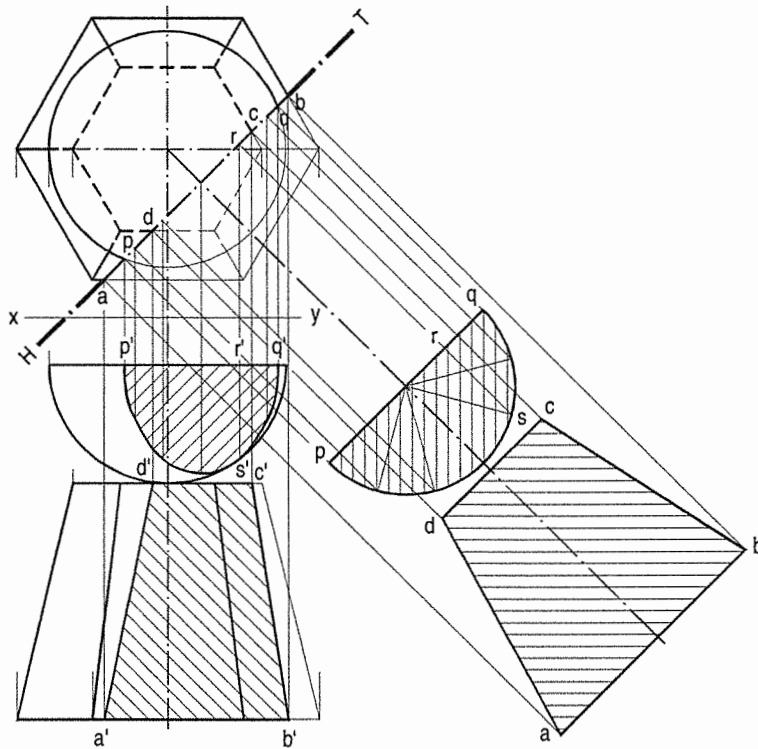
Method II:

It is known that the true shape of the section is a circle of diameter equal to $a'g'$. The width of section at any point say c' is equal to the chord c_1c_1 on this circle. Therefore, project c' to points c in the top view so that $cc = c_1c_1$. Similarly, obtain other points and draw the ellipse through them.

Fig. 14-37 shows the sectional front view and true shape of the section when the section plane is vertical and inclined to the V.P.

(4) Section plane perpendicular to the H.P. and inclined to the V.P.:

Problem 14-32. (fig. 14-38): The projections of a hemisphere 50 mm diameter, placed centrally on the top of a frustum of a hexagonal pyramid, base 32 mm side, top 20 mm side and axis 50 mm long are given. Draw the sectional front view when the vertical section plane H.T. inclined at 45° to the V.P. and 10 mm from the axis, cuts them. Also draw the true shapes of the sections of both the solids.



(Third-angle projection)

FIG. 14-38

The widths of the section of the sphere at various points are obtained from the semi-circle drawn in the top view.

14-6. TYPICAL PROBLEMS OF SECTIONS OF SOLIDS

Problem 14-33. A solid composed of a half-cone and a half-hexagonal pyramid is shown in fig. 14-39. It is cut by a section plane, which makes an angle of 30° with the base, is perpendicular to the V.P. and contains an edge of the base of the pyramid. Draw its sectional top view, true shape of the section and development of the surface of the remaining portion. Base of cone 60 mm diameter; axis 70 mm long.

- Draw a line V.T. inclined at 30° to the base and passing through a' .
- Project the sectional top view. Note how points b and b_1 are obtained. The true shape of the section will be partly elliptical.
- Draw the development of the half-cone and half-pyramid and show the lines for the section in it.

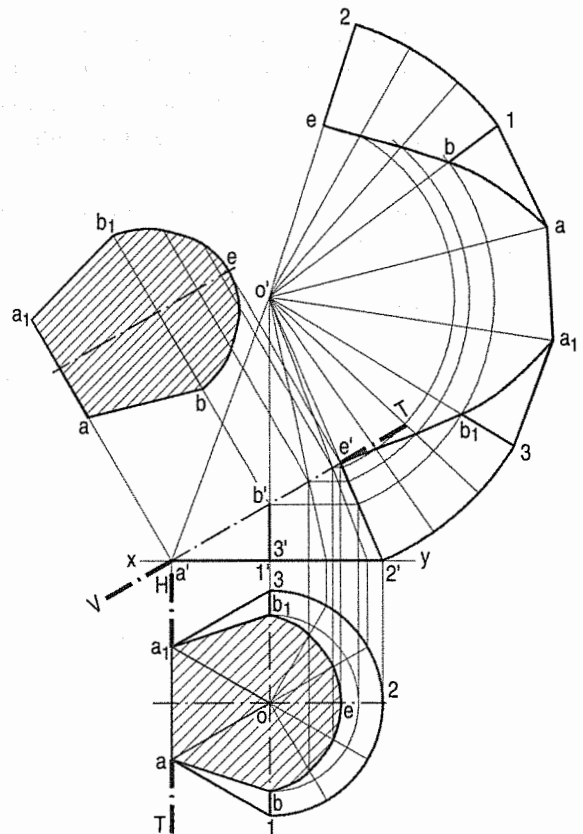


FIG. 14-39

Problem 14-34. (fig. 14-40): A cylinder, base 50 mm diameter and axis 75 mm long, has a square hole of 25 mm side cut through it so that the axis of the hole coincides with that of the cylinder. The cylinder is lying on the H.P. with the axis perpendicular to the V.P. and the faces of the hole equally inclined to the H.P.

A vertical section plane, inclined at 60° to the V.P. cuts the cylinder in two equal halves. Project the front view of the cylinder on an A.V.P. parallel to the section plane.

- Assuming the cylinder to be whole, draw its auxiliary front view.
- Project the points at which the generators of the cylinder and the edges of the hole are cut. The section of the cylinder will be a part of an ellipse. Join the points at which the edges of the hole are cut. The back edges

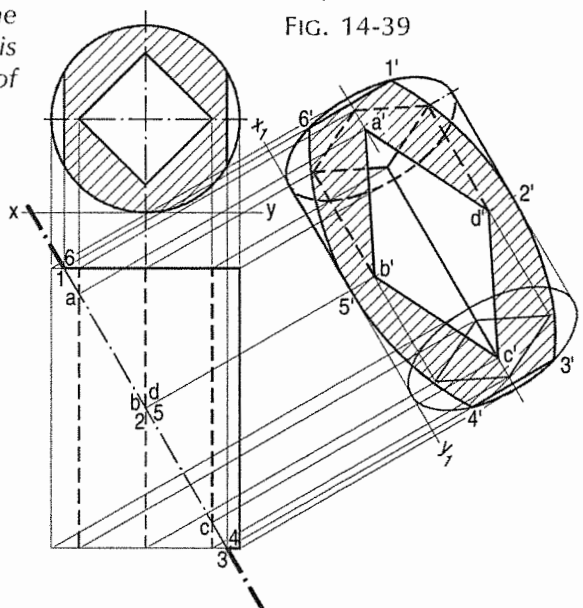


FIG. 14-40

of the hole will be visible within the section and hence, must be shown as full lines.

- (iii) Complete the view by showing the section and the remaining portion of the cylinder with dark lines.

Problem 14-35. (fig. 14-41):

A square prism axis 110 mm long is resting on its base in the H.P. The edges of the base are equally inclined to V.P.

The prism is cut by an A.I.P. passing through the mid-point of the axis in such a way that the true shape of the section is rhombus having diagonals of 100 mm and 50 mm. Draw the projections and determine the inclination of A.I.P. with the H.P.

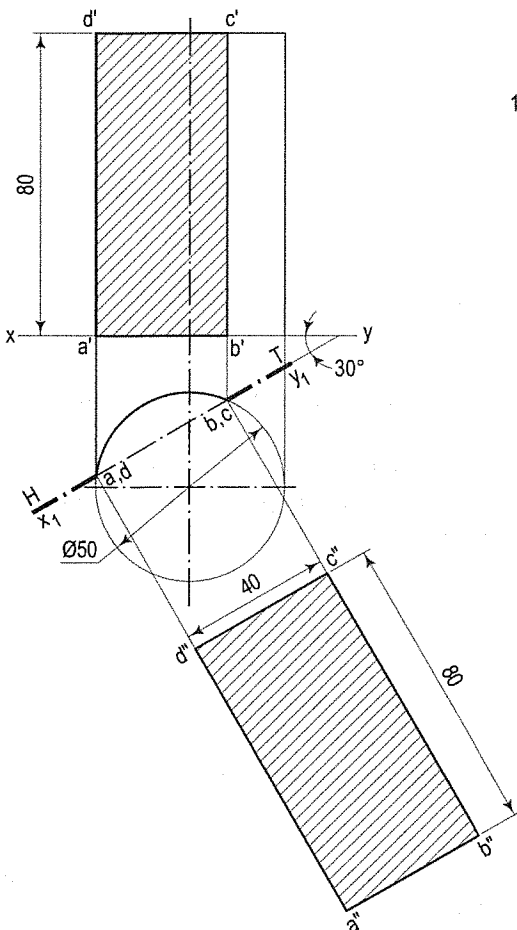


FIG. 14-41

- (i) Draw the top view and the front view as shown.
- (ii) Mark the mid-point of the axis in the front view.
- (iii) With the mid-point of the axis as centre and radius equal to 50 mm (half of the longer diagonal i.e. 100 mm), draw an arc cutting the two opposite vertical sides of the prism. Project the points and complete the true shape as shown.

Problem 14-36. (fig. 14-42): A vertical cylinder 50 mm diameter is cut by an A.V.P. making 30° to the V.P. in such a way that the true shape of the section is a rectangle of 40 mm × 80 mm sides. Draw the projections and the true shape of the section.

FIG. 14-42

- (i) Draw a top view and an front view of the cylinder.
- (ii) Draw x_1y_1 at 30° to xy in such a way that the chord length in the top view is 40 mm. Project points 1, 2, 3 and 4 and draw the rectangle of 40 mm $\times$ 80 mm as shown.

Problem 14-37. (fig. 14-43): A hexagonal pyramid, base 30 mm side and axis 70 mm long is resting on its slant edge of the face on the horizontal plane.

A section plane, perpendicular to the V.P., inclined to the H.P. passes through the highest corner of the base and intersecting the axis at 25 mm from the base. Draw the projections of the solid and determine the inclination of the section plane with the H.P.

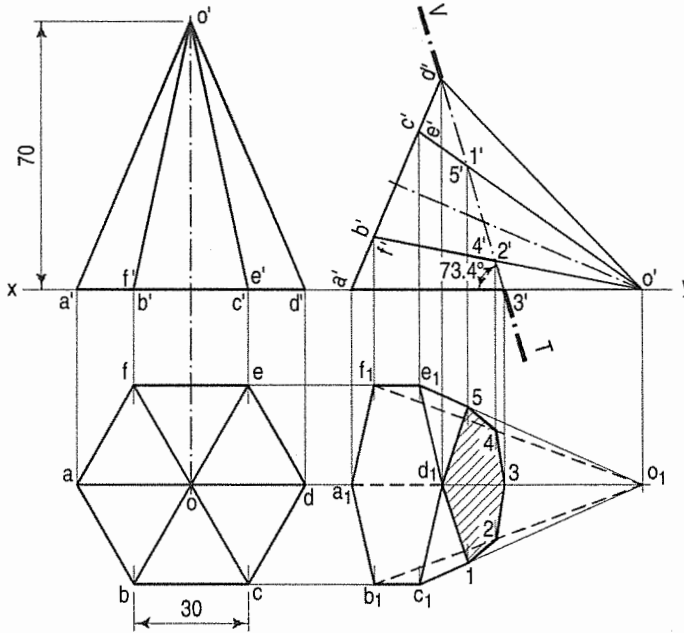


FIG. 14-43

- (i) Draw the top view and the front view keeping one of the sides of the base parallel to xy .
- (ii) With a' and o' , as centres and radii equal to $a'd'$ and $o'd'$ draw arcs intersecting each other at point d' . Draw a section plane passing through d' and point 25 mm away from the base along the axis as shown.
- (iii) Measure the angle by V.T. with xy .

Problem 14-38. (fig. 14-44): A pentagonal pyramid, base side 30 mm, length of axis 80 mm is resting on a base edge on the H.P. with a triangular face containing that edge being perpendicular to the V.P. and inclined to the H.P. at 60° . It is cut by a horizontal section plane whose V.T. passes through the mid-point of the axis. Draw the front view, sectional top view and add a profile view.

- (i) Draw the top view and the front view keeping one of the sides of the base perpendicular to xy .
- (ii) Tilt the front view on the points c' , d' as shown.
- (iii) Draw a line parallel to xy and passing through the mid-point of the axis representing V.T. of the section plane.
- (iv) Complete the projection as shown.

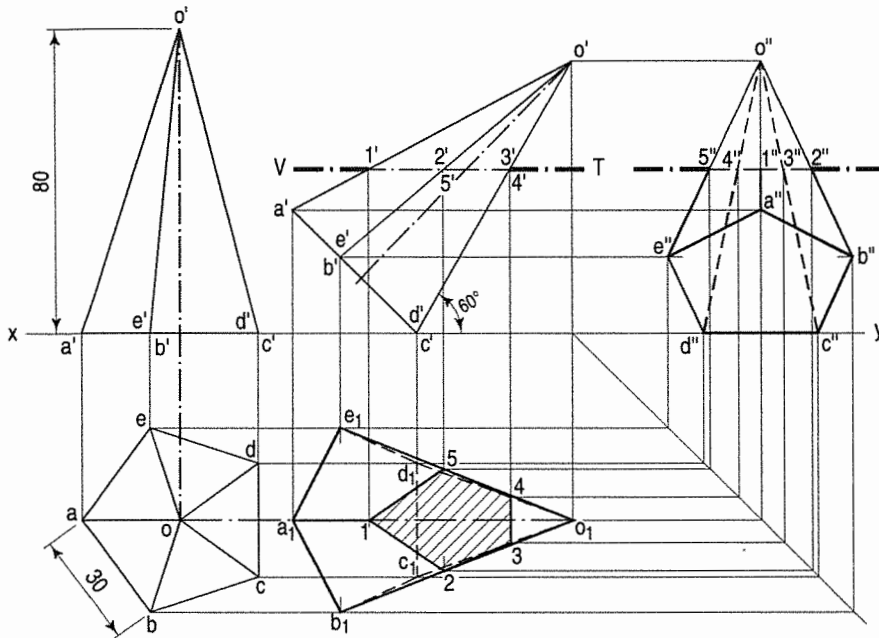


FIG. 14-44

Problem 14-39. (fig. 14-45): A cone, diameter of the base 60 mm and axis 70 mm long is resting on its base on the H.P. It is cut by an A.I.P. so that the true shape of the section is an isosceles triangle having 50 mm base. Draw the top view, the front view and the true shape of the section.

When a section plane passes through an apex of a cone and cuts the base of the cone, the true-shape of section is a triangle.

- (i) Draw a top view and an front view as shown.
- (ii) Mark chord ab of 50 mm (the base of triangle) in the top view. Project points a and b in the front view intersecting base at a' or b' . Join points a' and o' .

This represents V.T. of the section plane.

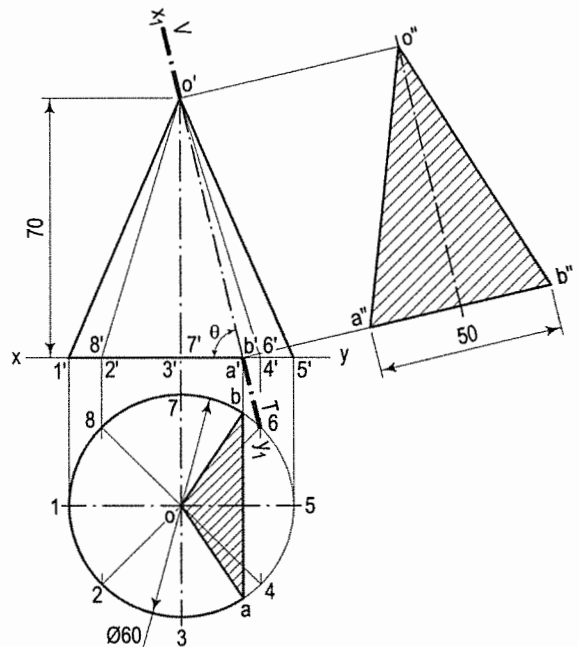


FIG. 14-45

- (iii) Considering line V.T. as new line $x_1 y_1$, draw the projectors from o' , a' and b' .
- (iv) Construct the true shape of triangle as shown.

Problem 14-40. (fig. 14-46): A square pyramid of 60 mm side of base and 70 mm length of axis is resting on its base on the H.P., having a side of base perpendicular to the V.P. It is cut by two cutting planes. One is parallel to its extreme right face and 10 mm away from it. While the other is parallel to the extreme left face.

Both the cutting planes intersect each other on the axis of the pyramid. Draw the sectional top view, front view and the left hand side view.

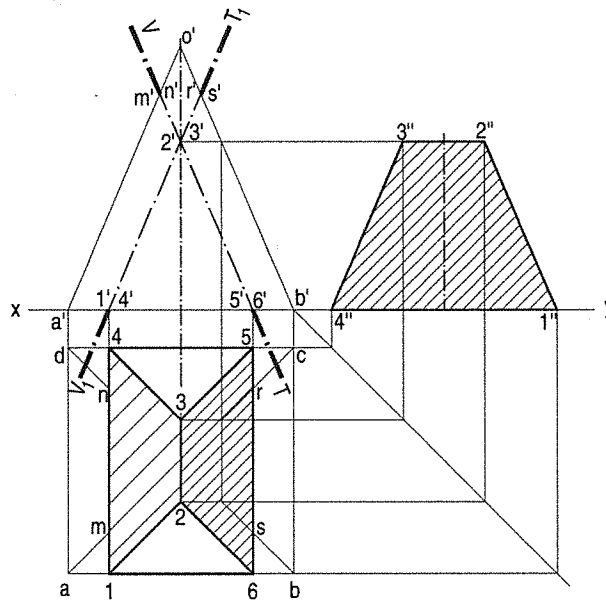


FIG. 14-46

- (i) Draw the top view and the front view as shown.
- (ii) Draw lines V.T. and V_1T_1 representing section planes in the front view intersecting at $2'$ and $3'$.
- (iii) Complete the projections. Note that the intersection points $2'$ and $3'$ are transferred on the slant edge $o'b'$ and then projected in the top view.

Problem 14-41. (fig. 14-47): A cylinder of diameter 50 mm and axial height 90 mm having 34 mm square hole centrally along the axis, rests on a point on the circular edge of the base remaining on the H.P. The axis of the cylinder is parallel to the V.P. and inclined at 30° to the H.P. and the rectangular faces of the square hole remain equally inclined to the V.P. A section plane perpendicular to the V.P. and inclined to the horizontal plane passing through the mid-point of the axis, such that the apparent section in the top view is a circle of 50 mm diameter cuts the cylinder. Draw the front view, section plane, true shape of the section and find the inclination of the section plane with H.P.

Fig. 14-47 shows the projections of the solid. Note that the generators 4 and 6 cut the apparent section at the points s , q , s_1 and q_1 . These points are transferred in the front view as shown. The required section plane must pass through the midpoint of the axis, and s , q (s_1 and q_1).

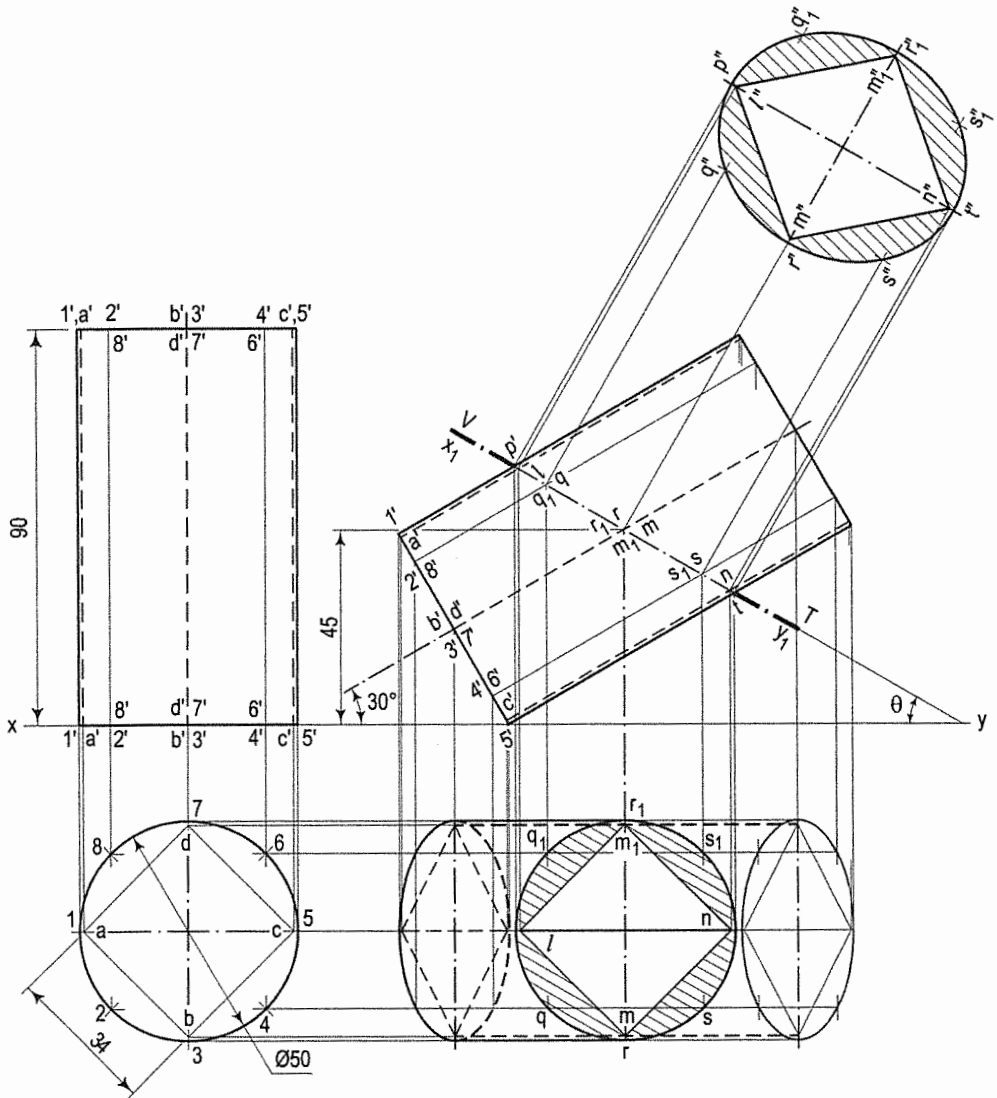


FIG. 14-47

Problem 14-42. (fig. 14-48): A cylindrical disc of 46 mm diameter is resting on the H.P. and has height of 20 mm. A hexagonal prism of side 23 mm and height of 30 mm is resting on the disc such that their axes are in one line and its faces are equally inclined with the V.P. It is cut by the auxiliary plane, offset 10 mm in front of the centre of the disc and is inclined at 40° with xy . Draw the projection of combined solids and obtain true shape of the section.

- Draw the projections of the combined solids with given positions.
- Draw an offset circle of 10 mm radius and mark a cutting plane inclined at 40° , touching this circle.
- Project various points in the front view as shown.
- Draw x_1y_1 parallel to the cutting plane.
- Obtain true shape as shown.

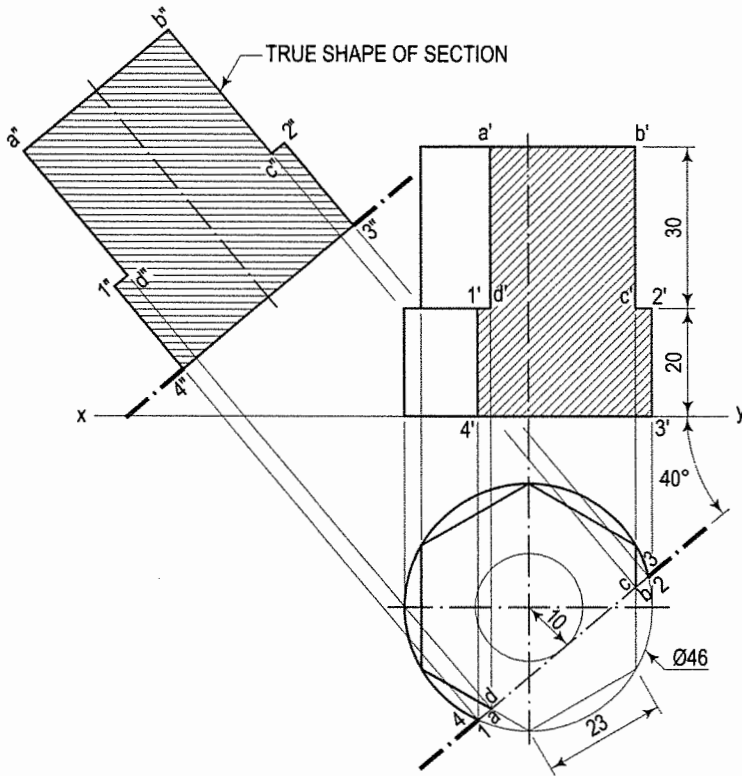


FIG. 14-48

Problem 14-43. (fig. 14-49): A square prism of 40 mm side is resting on H.P. and has height of 30 mm. Its faces are equally inclined with the V.P. A frustum of cone having base diameter 40 mm and top 30 mm diameter with height 30 mm is kept on the prism such that axes of the both solids are coinciding. A sectional plane cuts the combined axes and inclined at 55° with H.P. passes through left corners of the prism. Draw the front view and section top view. Draw also true-shape of the section.

- Draw the top view and front view of combined solid as shown.
- In front view, draw a section plane at angle of 55° with xy passing through the corner of square prism. Mark points $1'$, $2'$, $3'$, $4'$, $5'$ and $6'$ as shown.
- Project these points in the top view and draw section line as shown in fig. 14-49.

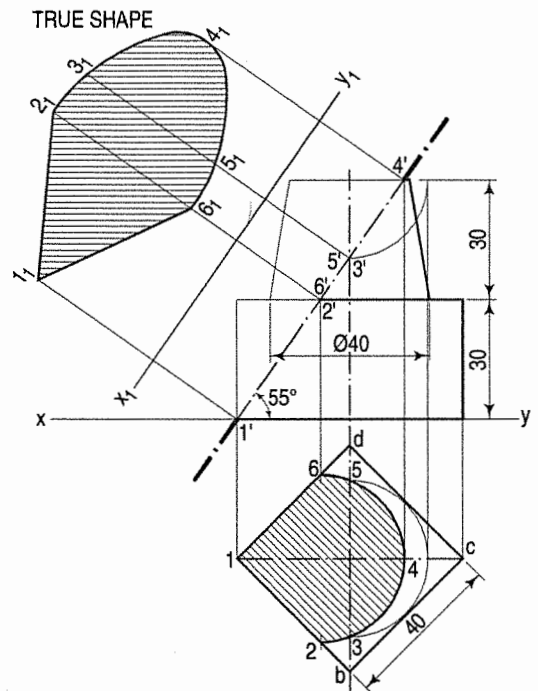


FIG. 14-49

- (iv) Draw a new reference line x_1y_1 parallel to the section plane and project the section on it.
- (v) The distances of the points from x_1y_1 should be taken equal to their corresponding distances from xy in the top view of combined solids.

Problem 14-44. (fig. 14-50): A horizontal frustum of square pyramid having front square of 20 mm side and back square of 30 mm at length of 60 mm has axis perpendicular to the V.P. and the side of square is inclined to 45° with H.P. It is cut by section plane making 40° with V.P. and passing through a point on axis 30 mm from the surface of large square side. Draw projections of the frustum.

- (i) Mark point b' and construct the square $a' b' c' d'$ with sides equally inclined to the xy . This is a front view of the frustum of square pyramid.
- (ii) Project the top view such that the axis remains perpendicular to xy as shown in fig. 14-50.
- (iii) Draw a section plane at distance of 20 mm along axis from large square at angle of 40° with xy .
- (iv) Mark the points p, q, r, s and project them in the front view. Draw section lines in the front view for the section and complete the view.

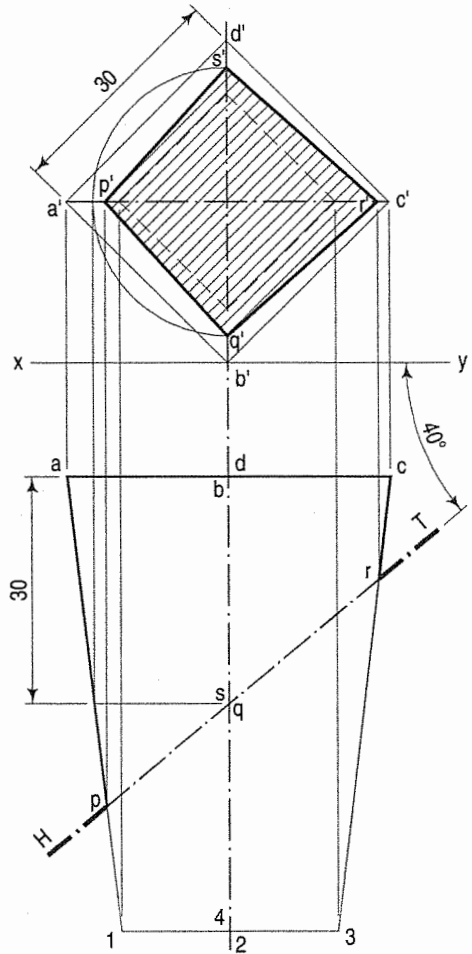


FIG. 14-50

EXERCISES 14

1. A cube of 50 mm long edges is resting on the H.P. with a vertical face inclined at 30° to the V.P. It is cut by a section plane, perpendicular to the V.P., inclined at 30° to the H.P. and passing through a point on the axis, 38 mm above the H.P. Draw the sectional top view, true shape of the section and development of the surface of the remaining portion of the cube.
2. A hexagonal prism, side of base 35 mm and height 75 mm is resting on one of its corners on the H.P. with a longer edge containing that corner inclined at 60° to the H.P. and a rectangular face parallel to the V.P. A horizontal section plane cuts the prism in two equal halves.
 - (i) Draw the front view and sectional top view of the cut prism.
 - (ii) Draw another top view on an auxiliary inclined plane which makes an angle of 45° with the H.P.
3. A pentagonal prism, side of base 50 mm and length 100 mm has a rectangular face on the H.P. and the axis parallel to the V.P. It is cut by a vertical section

plane, the H.T. of which makes an angle of 30° with xy and bisects the axis. Draw the sectional front view, top view and true shape of the section. Develop the surface of the remaining half of the prism.

4. A hollow square prism, base 50 mm side (outside), length 75 mm and thickness 9 mm is lying on the H.P. on one of its rectangular faces, with the axis inclined at 30° to the V.P. A section plane, parallel to the V.P. cuts the prism, intersecting the axis at a point 25 mm from one of its ends. Draw the top view and sectional front view of the prism.
5. A cylinder, 65 mm diameter and 90 mm long, has its axis parallel to the H.P. and inclined at 30° to the V.P. It is cut by a vertical section plane in such a way that the true shape of the section is an ellipse having the major axis 75 mm long. Draw its sectional front view and true shape of the section.
6. A cube of 65 mm long edges has its vertical faces equally inclined to the V.P. It is cut by a section plane, perpendicular to the V.P., so that the true shape of the section is a regular hexagon. Determine the inclination of the cutting plane with the H.P. and draw the sectional top view and true shape of the section.
7. A vertical hollow cylinder, outside diameter 60 mm, length 85 mm and thickness 9 mm is cut by two section planes which are normal to the V.P. and which intersect each other at the top end of the axis. The planes cut the cylinder on opposite sides of the axis and are inclined at 30° and 45° respectively to it. Draw the front view, sectional top view and auxiliary sectional top views on planes parallel to the respective section planes.
8. A square pyramid, base 50 mm side and axis 75 mm long, is resting on the H.P. on one of its triangular faces, the top view of the axis making an angle of 30° with the V.P. It is cut by a horizontal section plane, the V.T. of which intersects the axis at a point 6 mm from the base. Draw the front view, sectional top view and the development of the sectioned pyramid.
9. A pentagonal pyramid, base 30 mm side and axis 75 mm long, has its base horizontal and an edge of the base parallel to the V.P. It is cut by a section plane, perpendicular to the V.P., inclined at 60° to the H.P. and bisecting the axis. Draw the front view and the top view when the pyramid is tilted so that it lies on its cut-face on the ground with the axis parallel to the V.P. Show the shape of the section by dotted lines. Develop the surface of the truncated pyramid.
10. A tetrahedron of 65 mm long edges is lying on the H.P. on one of its faces, with an edge perpendicular to the V.P. It is cut by a section plane which is perpendicular to the V.P. so that the true shape of the section is an isosceles triangle of base 50 mm long and altitude 40 mm. Find the inclination of the section plane with the H.P. and draw the front view, sectional top view and the true shape of the section.
11. A hexagonal pyramid, base 50 mm side and axis 100 mm long, is lying on the H.P. on one of its triangular faces with the axis parallel to the V.P. A vertical section plane the H.T. of which makes an angle of 30° with the reference line, passes through the centre of the base and cuts the pyramid, the apex being retained. Draw the top view, sectional front view, true shape of the section and the development of the surface of the cut-pyramid.
12. A cone, base 75 mm diameter and axis 75 mm long, has its axis parallel to the V.P. and inclined at 45° to the H.P. A horizontal section plane cuts the

- cone through the mid-point of the axis. Draw the front view, sectional top view and an auxiliary top view on a plane parallel to the axis.
13. A cone, base 65 mm diameter and axis 75 mm long, is lying on the H.P. on one of its generators with the axis parallel to the V.P. A section plane which is parallel to the V.P. cuts the cone 6 mm away from the axis. Draw the sectional front view and development of the surface of the remaining portion of the cone.
 14. The cone in above problem 13 is cut by a horizontal section plane passing through the centre of the base. Draw the sectional top view and another top view on an auxiliary plane parallel to the axis of the cone.
 15. A hemisphere of 65 mm diameter, lying on the H.P. on its flat face, is cut by a vertical section plane inclined to the V.P. so that the semi-ellipse seen in the front view has its minor axis 45 mm long and half major axis 25 mm long. Draw the top view, sectional front view and true shape of the section.
 16. The top view of a cylinder 75 mm diameter, 125 mm long, placed on top of the frustum of a cone, base 100 mm diameter, top 50 mm diameter and axis 125 mm long is shown in fig. 14-51. Both the solids are cut by a vertical section plane, the H.T. of which is 12 mm from the axis of the frustum and makes 30° angle with xy . Draw the sectional front view and true shape of the sections.
 17. A sphere of 75 mm diameter is cut by a section plane, perpendicular to the V.P. and inclined at 30° to the H.P. in such a way that the true shape of the section is a circle of 50 mm diameter. Draw its front view, sectional top view and sectional side view.
 18. A frustum of a cone, base 75 mm diameter, top 50 mm diameter and axis 75 mm long, has a hole of 30 mm diameter drilled centrally through its flat faces. It is resting on its base on the H.P. and is cut by a section plane, the V.T. of which makes an angle of 60° with xy and bisects the axis. Draw its sectional top view and an auxiliary top view on a reference line parallel to the V.T., showing clearly the shape of the section.
 19. A hexagonal prism, side of the base 25 mm long and axis 65 mm long is resting on an edge of the base on the H.P., its axis being inclined at 60° to the H.P. and parallel to the V.P. A section plane, inclined at 45° to the V.P. and normal to the H.P., cuts the prism and passes through a point on the axis at a distance of 20 mm from the top end of the axis. Draw its sectional front view and true shape of the section.
 20. A pentagonal pyramid, edge of base 25 mm long and height 50 mm is resting on the H.P. on a corner of its base in such a way that the slant edge containing that corner makes an angle of 60° with the H.P. and is parallel to the V.P. It is cut by a section plane making an angle of 30° with the V.P., perpendicular to the H.P. and passing through a point on the axis at a distance of 6 mm from its base. Draw its sectional front view and true shape of the section.
 21. The distance between the opposite parallel faces of a 50 mm thick hexagonal block is 75 mm. The block has one of its rectangular faces parallel to the H.P. and its axis makes an angle of 30° with the V.P. It is cut by a section plane making

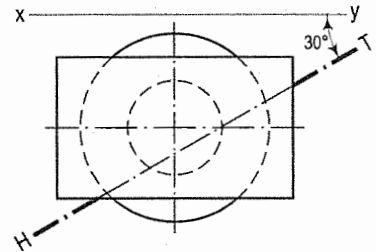


FIG. 14-51

an angle of 30° with the H.P., normal to the V.P. and bisecting the axis. Draw its sectional top view and another top view on a plane parallel to the section.

22. PQR is an isosceles triangle having base PR horizontal and 50 mm long, and altitude 50 mm. A point A is taken on PR at a distance of 15 mm from P and a straight line AB is drawn parallel to PQ cutting QR at B . If AB is regarded as the V.T. of an inclined plane perpendicular to the V.P., cutting a cone of which PQR is the front view, draw the sectional top view, sectional side view and true shape of the section.
23. A cone of 55 mm diameter and 75 mm height is resting on the H.P. on one of its generators in such a way that, the generator is parallel to the V.P. It is cut by a plane parallel to the V.P. and inclined at 90° to the H.P. and passing through a point 15 mm in front of its axis. Draw the sectional front view and the top view of the cone.
24. The true section of a vertical square prism cut by an inclined plane is a rectangle of 75 mm $\times$ 40 mm. The plane cuts one of the side faces at a height of 40 mm from the base. Draw three views of the cut prism when it rests on the cut face on the H.P. with its axis remaining parallel to the V.P.
25. An equilateral triangular prism, base 50 mm side and height 100 mm is standing on the H.P. on its triangular face with one of the sides of that face inclined at 90° to the V.P. It is cut by an inclined plane in such a way that the true shape of the section is a trapezium of 50 mm and 12 mm parallel sides. Draw the projections and true shape of the section and find the angle which the cutting plane makes with the H.P.
26. A horizontal cylinder, 30 mm diameter and length 60 mm, is placed centrally on the top of a frustum of a cone, diameter of the base 45 mm, diameter of the top 25 mm and height 45 mm. Draw a sectional front view of the two solids on a vertical plane, distance 12 mm from the axis of the cone and making an angle of 60° with the axis of the cylinder.
27. A cone, base 75 mm diameter and axis 100 mm long, has its base on the H.P. A section plane, parallel to one of the end generators and perpendicular to the V.P., cuts the cone intersecting the axis at a point 75 mm from the base. Draw the sectional top view and project another top view on a plane parallel to the section plane, showing the shape of the section clearly.
28. A solid is made up of a cylinder, 30 mm diameter and 75 mm long, which joins another cylinder, 75 mm diameter and 25 mm long, by a fillet of 20 mm radius, the axes of the two cylinders being in a straight line. Draw the top view of a horizontal section of the solid made by a plane parallel to and 15 mm above the axis.
29. A cone frustum, base 75 mm diameter, top 35 mm diameter and height 65 mm has a hole of 30 mm diameter drilled through it so that the axis of the hole coincides with that of the cone. It is resting on its base on the H.P. and is cut by a section plane perpendicular to the V.P., parallel to an end generator and passing through the top end of the axis. Draw sectional top view and sectional side view of the frustum.
30. A cube of 25 mm edge rests on one of its corners on the H.P. so that a solid diagonal is vertical and two of its faces are perpendicular to the V.P. A vertical section plane parallel to the V.P. cuts the cube at a distance of 8 mm from the solid diagonal and nearer to the V.P. Draw its sectional front view.

Chapter 15



DEVELOPMENT OF SURFACES

15-0. INTRODUCTION



Imagine that a solid is enclosed in a wrapper of thin material, such as paper. If this covering is opened out and laid on a flat plane, the flattened-out paper is the development of the solid. Thus, when surfaces of a solid are laid out on a plane, the figure obtained is called its development.

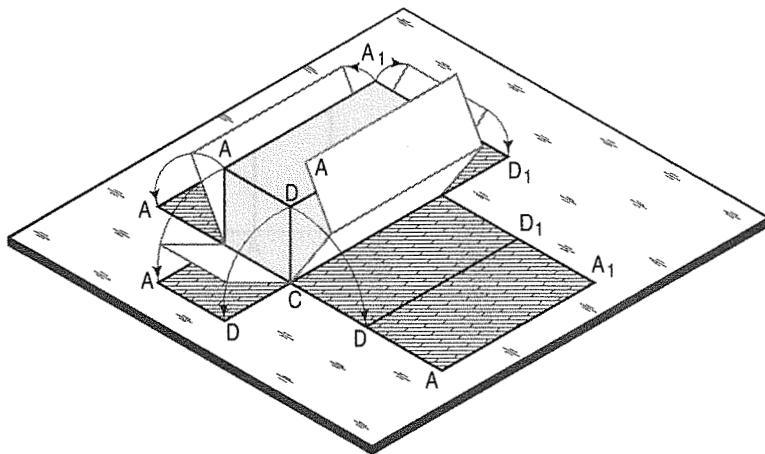


FIG. 15-1

Fig. 15-1 shows a square prism covered with paper in process of being opened out. Its development (fig. 15-2) consists of four equal rectangles for the faces and two similar squares for its ends. Each figure shows the true size and shape of the corresponding surface of the prism. The development of a solid, thus represents the actual shape of all its surfaces which, when bent or folded at the edges, would form the solid.

Hence, it is very important to note that every line on the development must be the true length of the corresponding edge on the surface.

The knowledge of development of surfaces is essential in many industries such as automobile, aircraft, ship building, packaging and sheet-metal work. In construction of boilers, bins, process-vessels, hoppers, funnels, chimneys etc., the plates are marked and cut according to the developments which, when folded, form the desired objects. The form of the sheet obtained by laying all the outer surfaces of the solid with suitable allowances for the joints is known as pattern.

Only the surfaces of polyhedra (such as prisms and pyramids) and single-curved surfaces (as of cones and cylinders) can be accurately developed. Warped and double-curved surfaces are undevelopable. These can however be approximately developed by dividing them up into a number of parts.

This chapter deals with the following topics:

1. Methods of development.
2. Developments of lateral surfaces of right solids.
3. Development of transition pieces.
4. Spheres (approximate method).

15-1. METHODS OF DEVELOPMENT



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 33 for the methods of development of surfaces.

The following are the principal methods of development:

(1) Parallel-line development:

It is employed in case of prisms and cylinders in which stretch-out-line principle is used. Lines A-A and A₁-A₁ in fig. 15-2 are called the stretch-out lines.

(2) Radial-line development: It

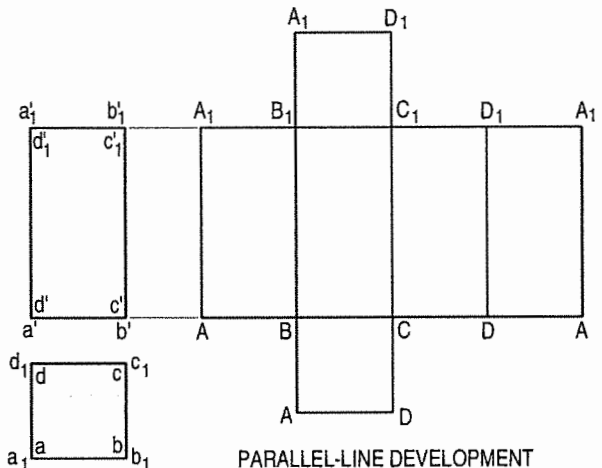
is used for pyramids and cones in which the true length of the slant edge or the generator is used as radius.

(3) Triangulation development:

This is used to develop transition pieces. This is simply a method of dividing a surface into a number of triangles and transferring them into the development.

(4) Approximate method: It

is used to develop objects of double curved or warped surfaces as sphere, paraboloid, ellipsoid, hyperboloid and helicoid.



PARALLEL-LINE DEVELOPMENT

FIG. 15-2

15-2. DEVELOPMENTS OF LATERAL SURFACES OF RIGHT SOLIDS



The methods of drawing developments of surfaces of various solids are explained by means of the following typical problems. Only the lateral surfaces of the solids (except the cube) have been developed. The ends or bases have been omitted. They can be easily incorporated if required.

15-2-1. CUBE

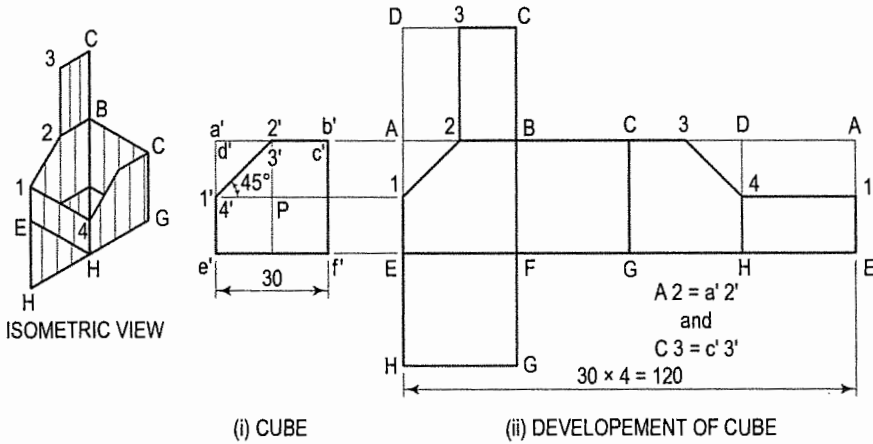


The development of the surface of a cube consists of six equal squares, the length of the side of the squares being equal to the length of the edge of the cube.

Problem 15-1. Draw the development of the surface of the part *P* of the cube, the front view of which is shown in fig. 15-3(i).

Name all the corners of the cube and also the points at which the edges are cut.

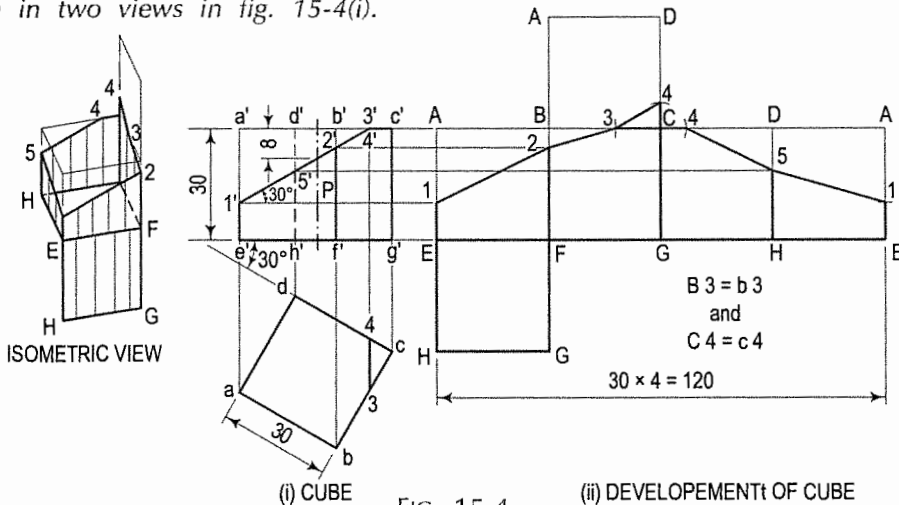
- (i) Draw the stretch-out lines *A-A* and *E-E* directly in line with the front view, and assuming the cube to be whole, draw four squares for the vertical faces, one square for the top and another for the bottom as shown in fig. 15-3(ii).



- (ii) Name all the corners. Draw a horizontal line through $1'$ to cut *AE* at 1 and *DH* at 4. $a' b'$ is the true length of the edge. Hence, mark a point 2 on *AB* and 3 on *CD* such that $A 2 = a' 2'$ and $C 3 = c' 3'$. Mark the point 3 on *CD* in the top square also.

- (iii) Draw lines 1-2, 2-3, 3-4 and 4-1, and complete the development as shown. Keep lines for the removed portion, viz. *A1*, *A2*, *3D*, *D4* and *DA* thin and fainter.

Problem 15-2. Draw the development of the surface of the part *P* of the cube shown in two views in fig. 15-4(i).



Name all the corners of the cube and also the points at which the edges are cut. Draw the development assuming the cube to be whole [fig. 15-4(ii)] as explained in problem 15-1.

- (i) Draw horizontal lines through points 1', 2' and 5' to cut AE in 1, BF in 2 and DH in 5 respectively. Lines b'c' and c'd' do not show the true lengths of the edges. The sides of the square in the top view show the true length. Therefore, mark points 3 in BC and 4 in CD such that $B3 = b3$ and $C4 = c4$.
- (ii) Draw lines joining 1, 2, 3 etc. in correct sequence and complete the required development. Keep the lines for the removed part fainter.

15-2-2. PRISMS

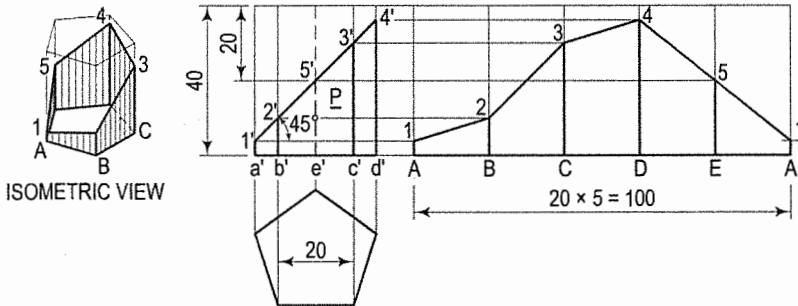


Development of the lateral surface of a prism consists of the same number of rectangles in contact as the number of the sides of the base of the prism. One side of the rectangle is equal to the length of the axis and the other side equal to the length of the side of the base.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 34 for the following problem.

Problem 15-3. Draw the development of the lateral surface of the part P of the pentagonal prism shown in fig. 15-5(i).



(i) PENTAGONAL PRISM (ii) DEVELOPEMENT OF PRISM

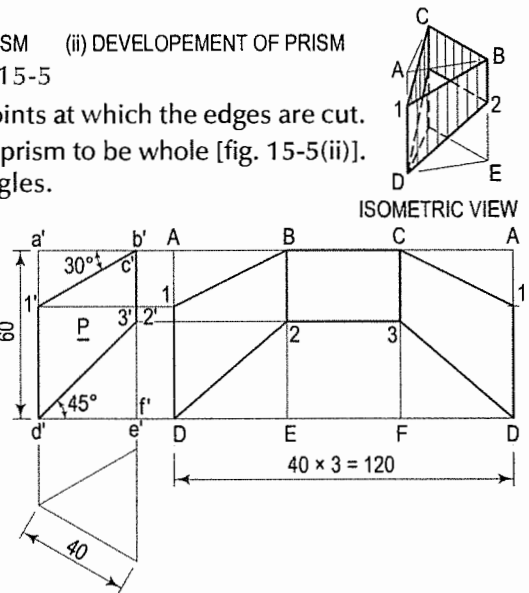
FIG. 15-5

Name the corners of the prism and the points at which the edges are cut.

- (i) Draw the development assuming the prism to be whole [fig. 15-5(ii)]. It is made up of five equal rectangles.
- (ii) Draw horizontal lines through points 1', 2' etc. to cut the lines for the corresponding edges in the development at points 1, 2 etc.
- (iii) Draw lines joining these points and complete the development as shown.

Problem 15-4. Draw the development of the lateral surface of the part P of the triangular prism shown in fig. 15-6(i).

Draw the development of the lateral surface of the whole prism [fig. 15-6(ii)] and obtain points 1, 2 and 3 on it. Draw lines 1B, C1, D2, 2-3 and 3D, and complete the development as shown.



(i) TRIANGULAR PRISM (ii) DEVELOPEMENT OF PRISM

FIG. 15-6

Problem 15-5. Draw the development of the lateral surface of the part *P* of the hexagonal prism shown in fig. 15-7(i).

Name the points at which the edges are cut and draw the development assuming the prism to be whole [fig. 15-7(ii)].

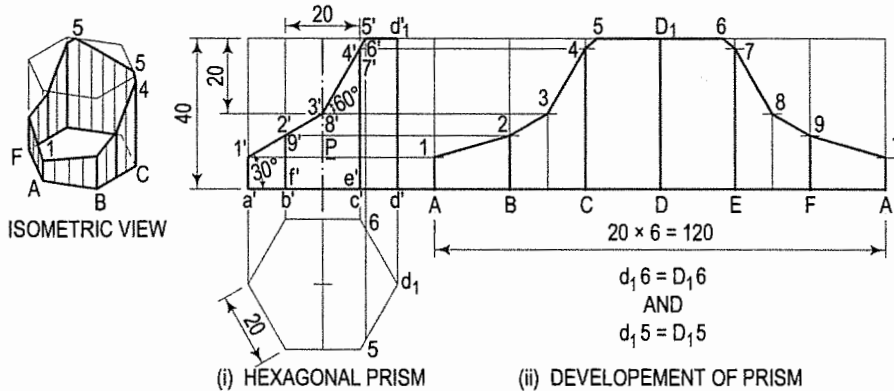


FIG. 15-7

- (i) Obtain all the points except 5 and 6 by drawing horizontal lines. Note that points 3 and 8 lie on vertical lines drawn through the mid-points of *BC* and *EF*.
- (ii) Mark points 5 and 6 such that $5D_1 = 5d_1$ and $D_1 6 = d_1 6$.
- (iii) Draw lines joining points 1, 2, 3 etc. in correct sequence and complete the required development as shown.

Problem 15-6. Draw the development of the lateral surface of the truncated prism shown in fig. 15-8(i). Also, draw the front view of the line joining the points *P* and *Q* (whose projections *p*, *p'* and *q*, *q'* are given) along the surface of the prism by the shortest distance.

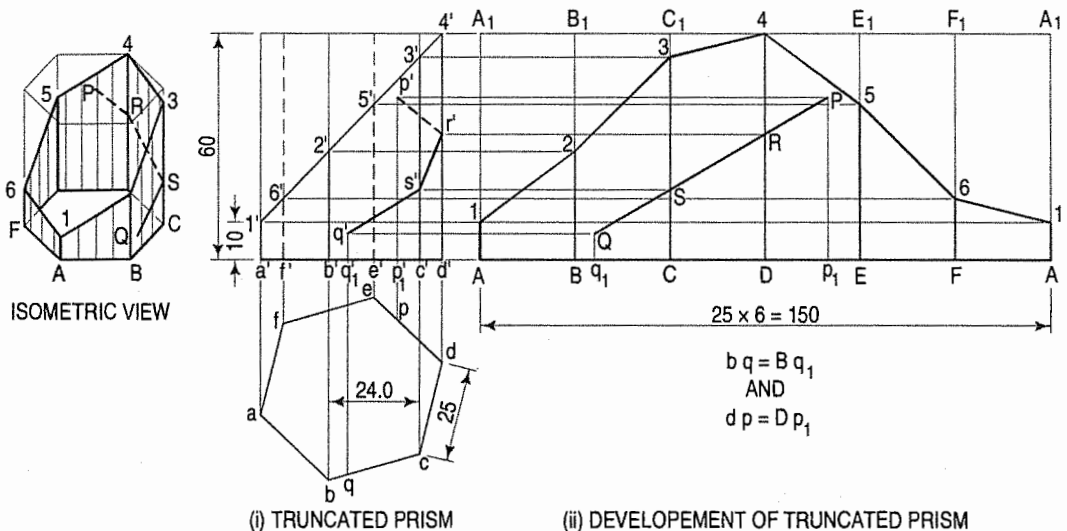


FIG. 15-8

- (i) Divide the circle in the top view into twelve equal parts. Project the division points to the front view and draw the generators. Mark points a' , b' and b'_1 , c' and c'_1 etc. in which the generators are cut.
- (ii) Draw the development of the lateral surface of the whole cylinder along with the generators [fig. 15-10(ii)]. The length of the line 1-1 is equal to $\pi \times D$ (circumference of the circle). This length can also be marked approximately by stepping off with a bow divider, twelve divisions, each equal to the chord-length ab . (The length thus obtained is about 1% shorter than the exact length; but this is permitted in drawing work.)

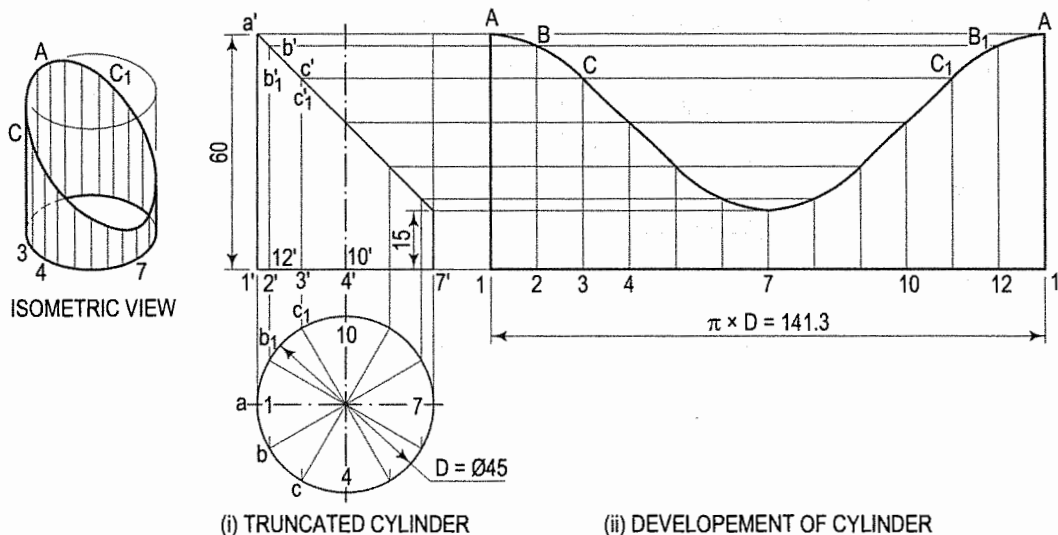


FIG. 15-10

- (iii) Draw horizontal lines through points a' , b' and b'_1 etc. to cut the corresponding generators in points A, B and B_1 etc. Draw a smooth curve through the points thus obtained. The figure 1-A-A-1 is the required development.

Problem 15-9. Draw the development of the lateral surface of the part P of the cylinder shown in fig. 15-11(i).

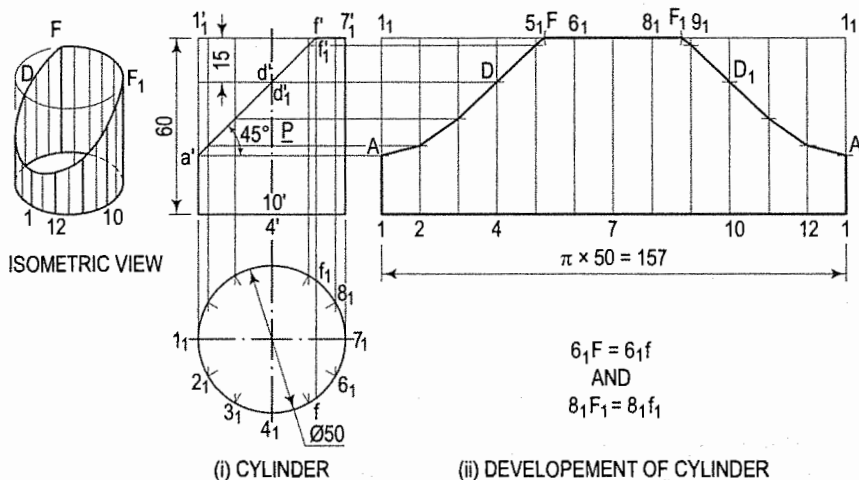


FIG. 15-11

Draw the development as explained in problem 15-8. Positions of the points at which the upper end of the cylinder is cut should be obtained from the top view. Mark these points, viz. F and F_1 on the line 1_1-1_1 between points 5_1 and 6_1 and between 8_1 and 9_1 in such a way that $6_1F = 6_1f$ and $8_1F_1 = 8_1f_1$. Draw curves FA and F_1A passing through these points and complete the required development as shown.

Problem 15-10. Draw the development of the lateral surface of the cylinder cut by three planes as shown in fig. 15-12(i).

See fig. 15-12(ii).

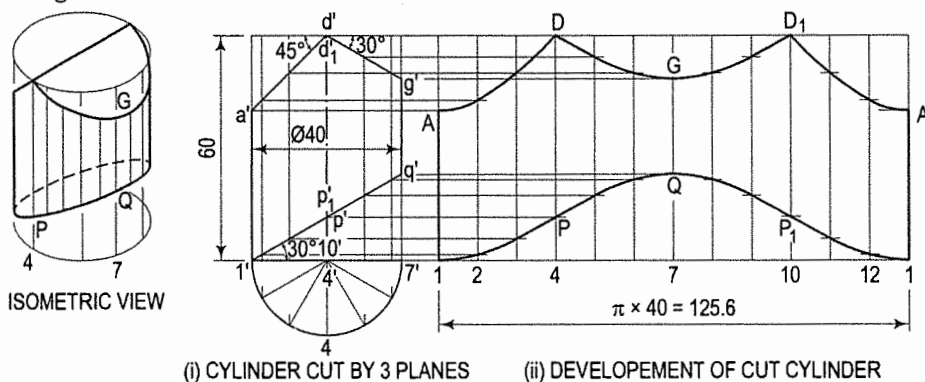


FIG. 15-12

Problem 15-11. (fig. 15-13(ii): Draw the development of the lateral surface of the cylinder having a square hole in it as shown in fig. 15-13(i).

See fig. 15-13(ii).

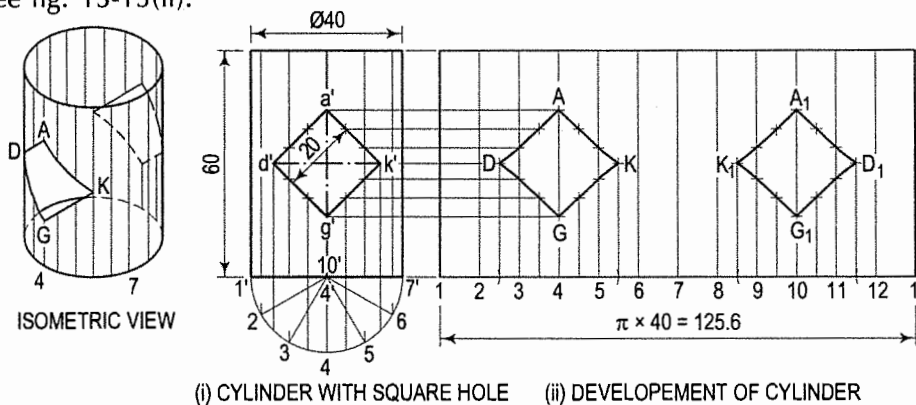


FIG. 15-13

Problem 15-12. Draw the development of the lateral surface of the cylinder cut as shown in fig. 15-14(i).

See fig. 15-14(ii).

Problem 15-13. Develop the surface of the three-piece cylindrical pipe elbow shown in fig. 15-15(i).

The front view is drawn as shown below:

Draw a square $efgh$ of sides equal to the diameter of the pipes (i.e. 50 mm). Bisect angles ehf and ghf by lines Ph and Qh to intersect ef and fg at x and y respectively. Then xy is the axis of the middle piece B .

Parts A and C are similar and are in the form of cylinders truncated at one end only. The part A and care as is shown in fig 15-15(ii) and fig. 15-15(iv) are developed on a stretch-out line drawn directly in line with its base. The twelve divisions are obtained from the semi-circle drawn on the base as diameter. The part B is truncated at both the ends. It is developed on a stretch-out line drawn through *h* and at right angles to its axis *xy* as shown in fig 15-15(iii).

As the curves of the part B are exactly similar to those of A and C, the three developments may also be drawn combined to minimize wastage on plate as shown in fig. 15-15(v).

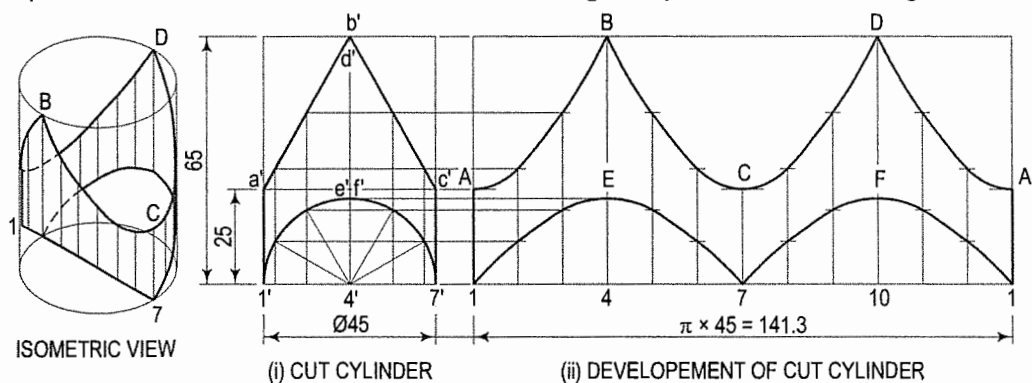


FIG. 15-14

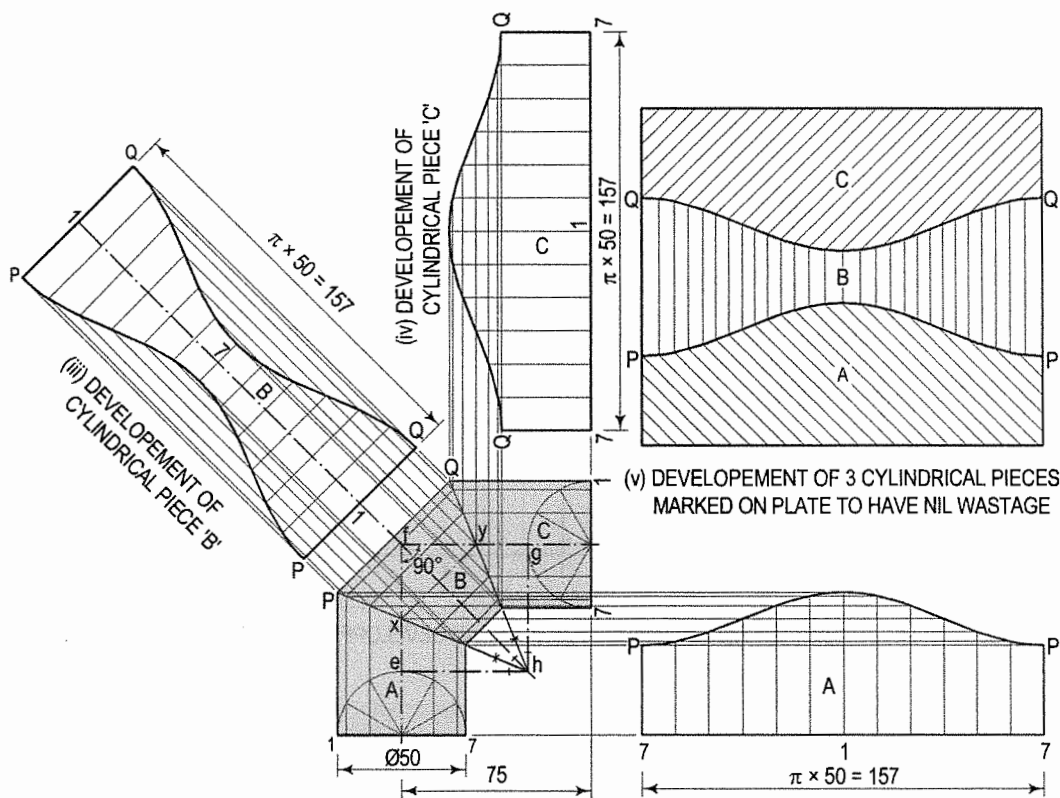


FIG. 15-15

Problem 15-14. Three cylindrical pipes of 50 mm diameter form a Y-piece as shown in the front view in fig. 15-16(i). Draw the development of the surface of each pipe.

- (i) Draw a semi-circle on the base of pipe A as diameter and obtain twelve divisions. Draw the development [fig. 15-16(ii)] as in the previous problem.
- (ii) Draw any convenient line at right angles to the axis of the pipe B [fig. 15-16(iii)]. On this line as a stretch-out line, draw the development as shown. Pipes B and C are similar and hence, their developments will also be similar.

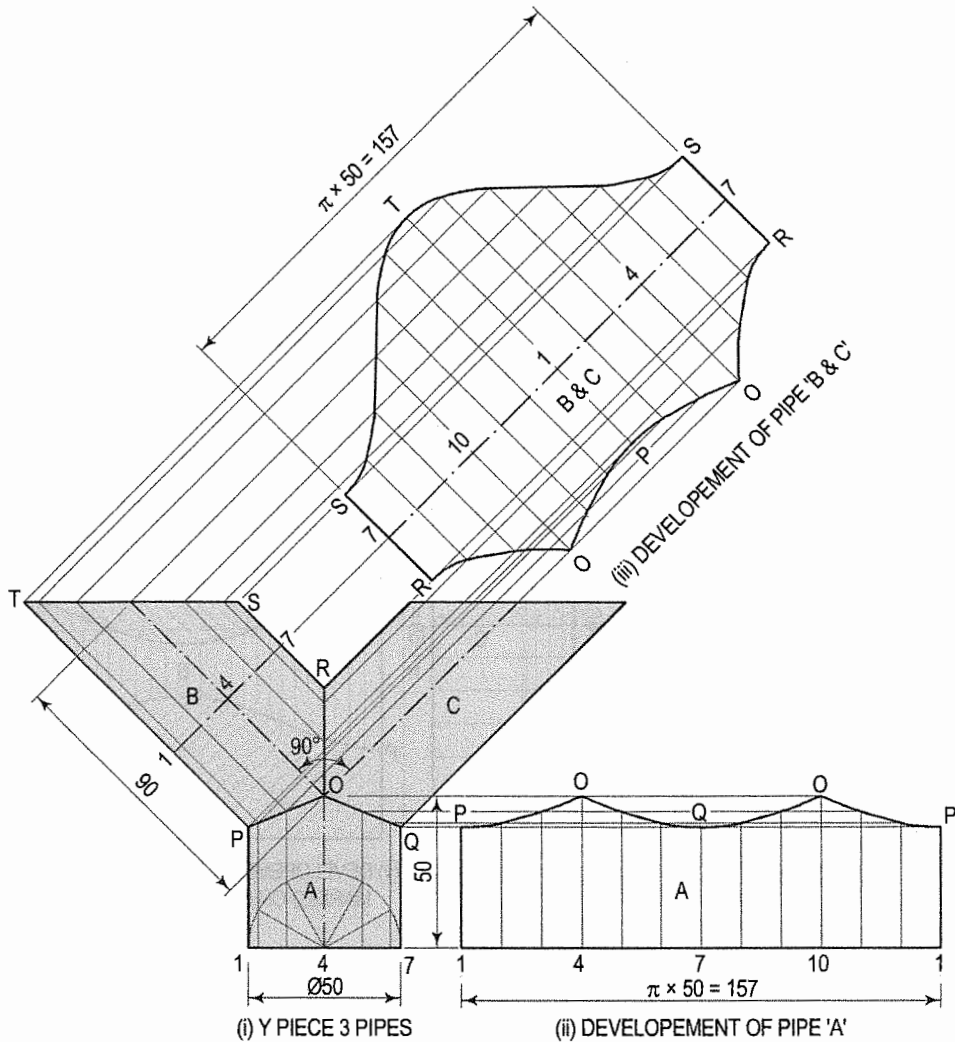


FIG. 15-16

15-2-4. PYRAMIDS

The development of the lateral surface of a pyramid consists of a number of equal isosceles triangles in contact. The base and the sides of each triangle are respectively equal to the edge of the base and the slant edge of the pyramid

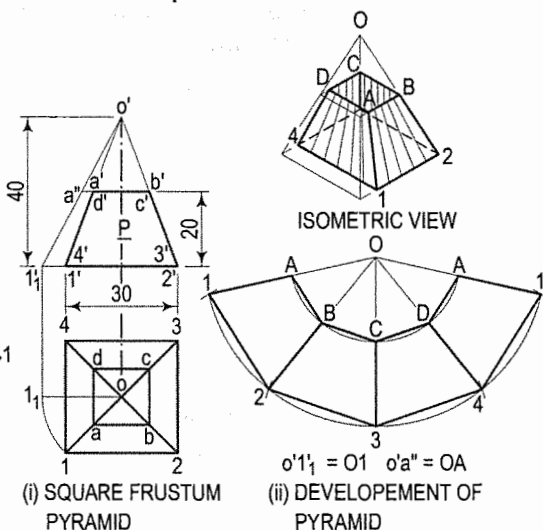
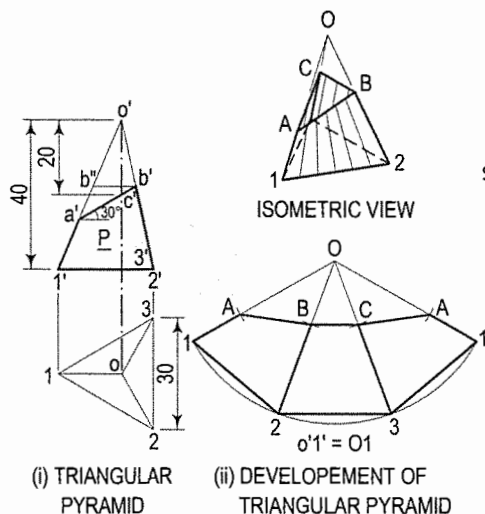
Note: The true length of a slant edge of pyramid can be measured from the front view, if the top view of that edge is parallel to xy ; and it can be measured from the top view, if the slant edge is parallel to xy in the front view.

Method of drawing the development of the lateral surface of a pyramid:

- (i) With any point O as centre and radius equal to the true length of the slant edge of the pyramid, draw an arc of the circle. With radius equal to the true length of the side of the base, step-off (on this arc) the same number of divisions as the number of sides of the base.
- (ii) Draw lines joining the division-points with each other in correct sequence and also with the centre for the arc. The figure thus formed (excluding the arc) is the development of the lateral surface of the pyramid.

Problem 15-15. Draw the development of the lateral surface of the part P of the triangular pyramid shown in fig. 15-17(i). The line $o'1'$ in the front view is the true length of the slant edge because it is parallel to xy in the top view. The true length of the side of the base is seen in the top view.

- (i) Draw the development of the lateral surface of the whole pyramid [fig. 15-17(ii)] as explained above. On $O1$ mark a point A such that $OA = o'a'$. $o'2'$ (with which $o'3'$ coincides) is not the true length of the slant edge.
- (ii) Hence, through b' , draw a line parallel to the base and cutting $o'a'$ at b'' . $o'b''$ is the true length of $o'b'$ as well as $o'c'$. Mark a point B in $O2$ and C in $O3$ such that $OB = OC = o'b''$.
- (iii) Draw lines AB , BC and CA and complete the required development as shown. Keep the arc and the lines for the removed part fainter.



Problem 15-16. Draw the development of the lateral surface of the frustum of the square pyramid shown in fig. 15-18(i).

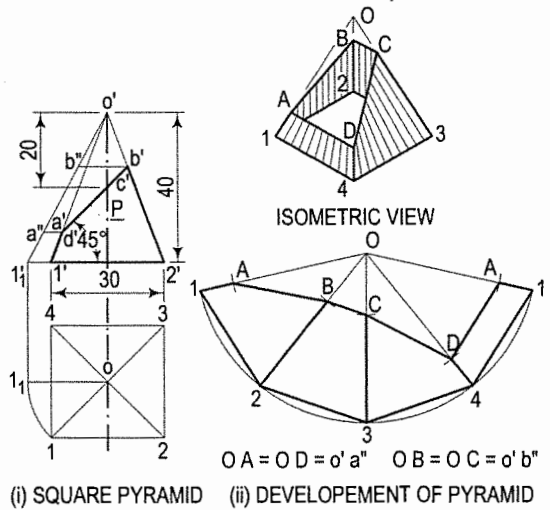
- (i) Determine the position of the apex. None of the lines in the front view shows the true length of the slant edge. Therefore, draw the top view and make any one line (for the slant edge) horizontal, i.e. parallel to xy and determine the true length $o'1'_1$. Through a' , draw a line parallel to the base and obtain the true length $o'a''$.

- (ii) With O as centre and radius $o'1'_1$, draw an arc and obtain the development of the lateral surface of the whole pyramid [fig. 15-18(ii)].
- (iii) With centre O and radius $o'a''$, draw an arc cutting $O1, O2$ etc. at points A, B etc. respectively.
- (iv) Draw lines AB, BC, CD and DA and complete the required development. Note that these lines are respectively parallel to lines 1-2, 2-3 etc.

Problem 15-17. Draw the development of the lateral surface of the part P of the square pyramid shown in fig. 15-19(i).

Draw the top view of the pyramid and determine the true length $o'1'_1$ of the slant edges as explained in problem 15-16. On this line, obtain the true lengths $o'a''$ and $o'b''$.

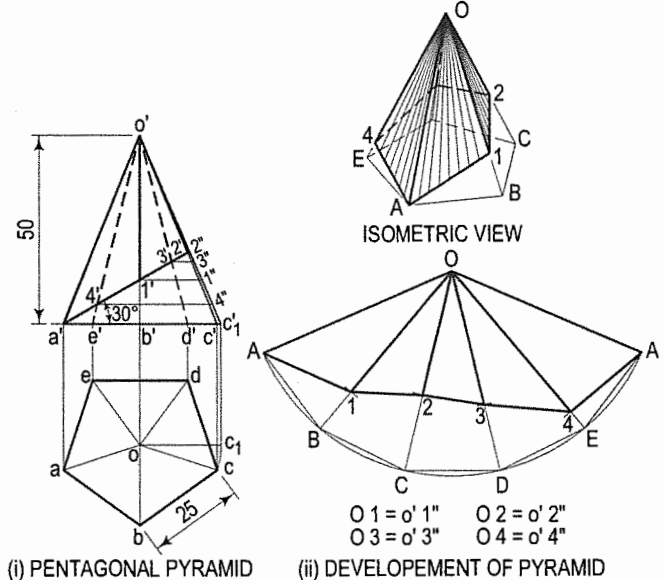
- (i) Draw the development of the lateral surface of the whole pyramid [fig. 15-19(ii)].
- (ii) Mark a point A in $O1$ and D in $O4$ such that $OA = OD = o'a''$. Similarly, mark B in $O2$ and C in $O3$ such that $OB = OC = o'b''$. Draw lines AB, BC etc. and complete the required development as shown.



Problem 15-18. Draw the development of the lateral surface of the pentagonal pyramid, the lower part of which is removed as shown in fig. 15-20(i).

None of the lines in the front view shows the true length of the slant edges. Hence, draw the top view and determine the true length $o'c'_1$. Through points $1', 2'$ etc., draw lines parallel to the base and obtain the true lengths $o'1''$, $o'2''$ etc.

- (i) Draw the development of the lateral surface of the whole pyramid [fig. 15-20(ii)]. Mark points 1, 2, 3 etc. on lines OB, OC, OD etc. such that $O1 = o'1''$, $O2 = o'2''$, $O3 = o'3''$ etc.



- (ii) Draw lines joining points $A, 1, 2$ etc. in correct sequence and complete the required development. Note that the lines for the lower part should be fainter.

Problem 15-19. Draw the development of the lateral surface of the part *P* of the pentagonal pyramid shown in fig. 15-21(i).

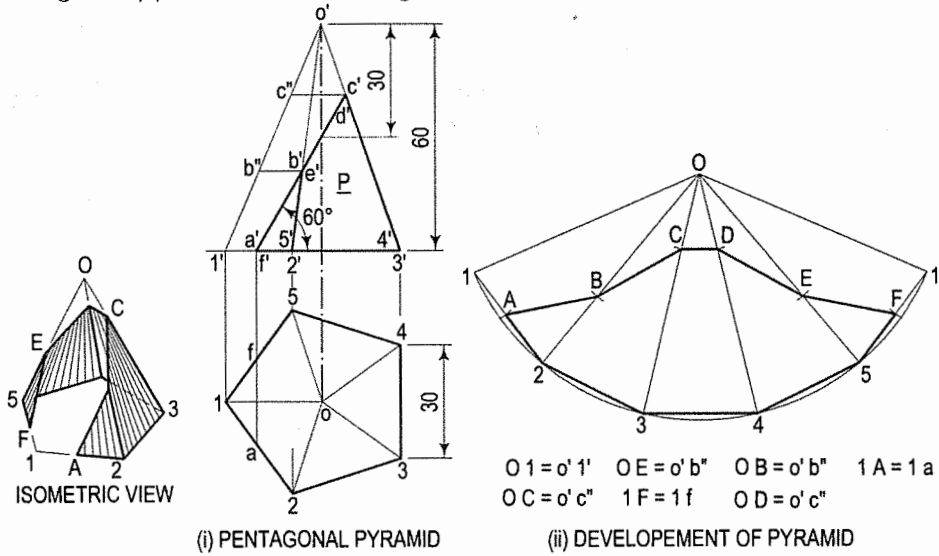


FIG. 15-21

The line $o'1'$ shows the true length of the slant edges. On it, obtain true lengths $o'b''$ and $o'c''$. Two edges of the base are also cut.

- Draw the top view and obtain the true lengths $1a$ and $1f$ as shown. Draw the development of the lateral surface of the whole pyramid.
- Obtain points B, C, D and E as explained in problem 15-18. Mark a point A in $1-2$ and F in $5-1$ such that $1A = 1a$ and $1F = 1f$.
- Draw lines joining points A, B, C etc. and complete the required development.

Problem 15-20. Draw the development of the lateral surface of the part *P* of the hexagonal pyramid shown in fig. 15-22(i).

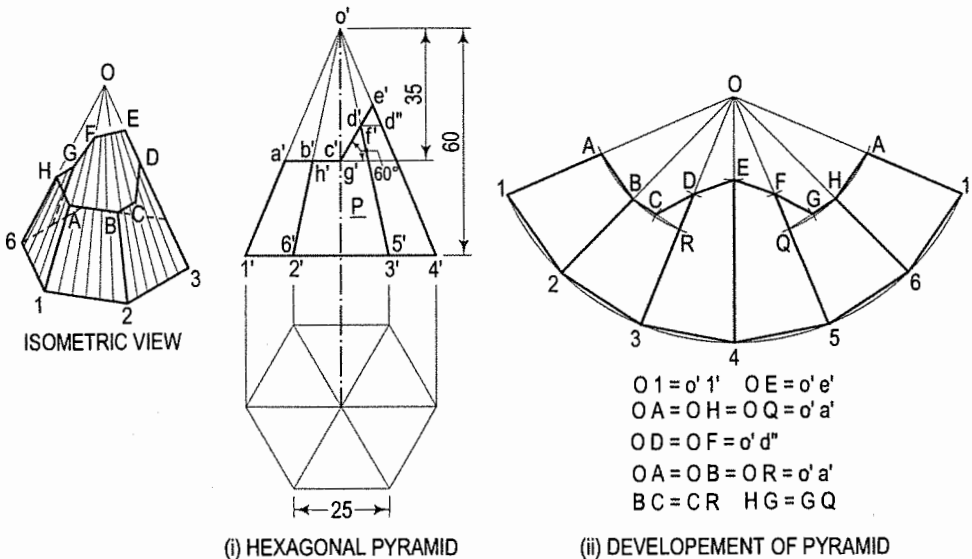


FIG. 15-22

Lines $o'1'$ and $o'4'$ show the true length of the slant edges. Draw the development of the lateral surface of the whole pyramid [fig. 15-22(ii)]. Obtain the development of the left-half of the pyramid as explained in problem 15-16 and that of the right-half as explained in problem 15-15.

Note that the points C and G are the mid-points of the lines BR and HQ respectively.

Problem 15-21. (fig. 15-23): A frustum of a square pyramid has its base 50 mm side and top 25 mm side and height 75 mm. Draw the development of its lateral surface.

Also, draw the projections of the frustum (when its axis is vertical and a side of its base is parallel to the V.P.), showing the line joining the mid-point of a top edge of one face with the mid-point of the bottom edge of the opposite face, by the shortest distance.

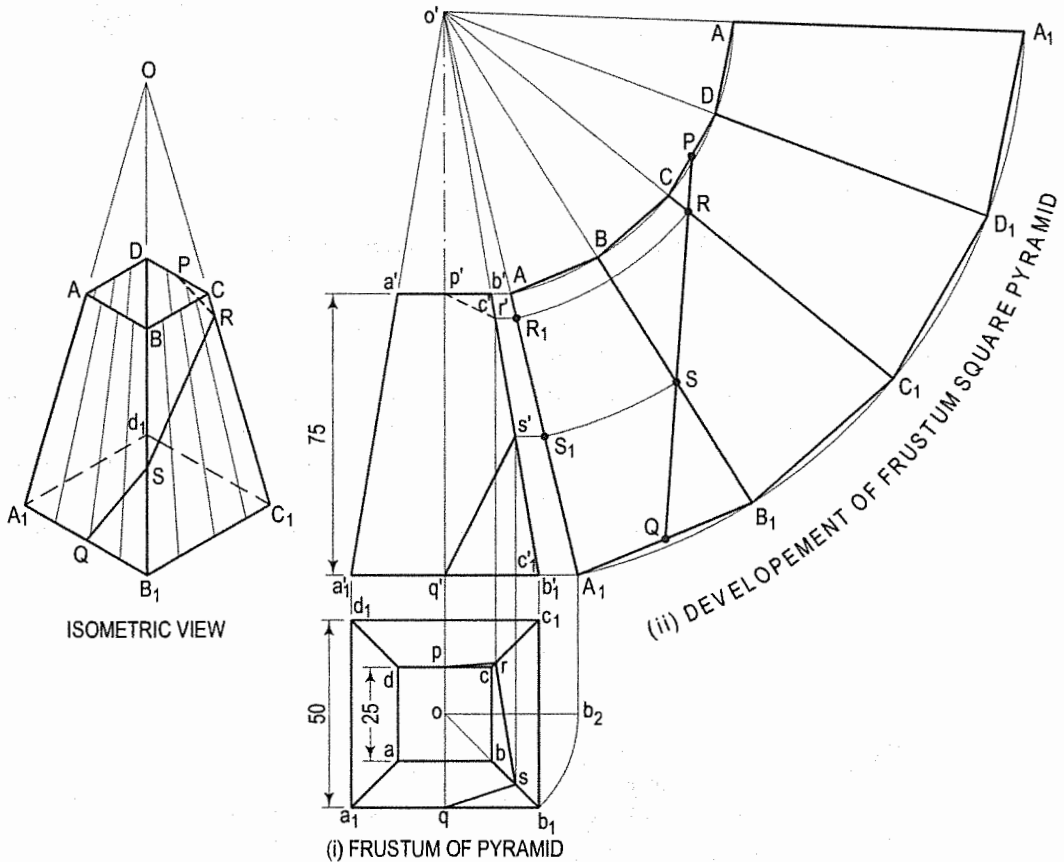


FIG. 15-23

Draw the development as explained in problem 15-16.

- (i) Mark the mid-point P of CD and Q of A_1B_1 . Draw a line joining P and Q and cutting CC_1 at R and BB_1 at S . Transfer these points to the front view and the top view. For example, with o' as centre and radius $o'R$, draw an arc cutting $o'A_1$ at R_1 . Through R_1 , draw a line parallel to the base and cutting $c'c_1$ at r' . Project r' to r on cc_1 in the top view. r' and r are the projections of R .
- (ii) Similarly, obtain s' and s on $b'b_1$ and bb_1 respectively. Draw lines pr , rs and sq which will show the top view of the line PQ . $p'r's'q'$ will be the path of the line PQ in the front view.

15-2-5. CONE

The development of the curved surface of a cone is a sector of a circle, the radius and the length of the arc of which are respectively equal to the slant height and the circumference of the base-circle of the cone.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 35 for the following problem.

Problem 15-22. Draw the development of the lateral surface of the truncated cone shown in fig. 15-24(i).

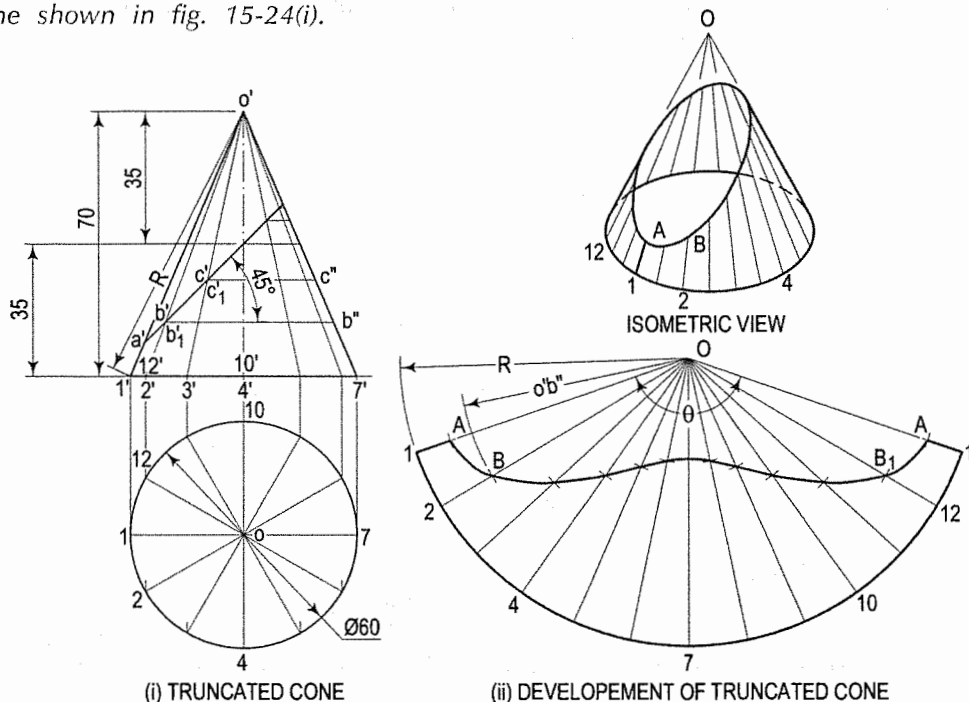


FIG. 15-24

Assuming the cone to be whole, let us draw its development.

- (i) Draw the base-circle in the top view and divide it into twelve equal parts.
- (ii) With any point O as centre and radius equal to $o'1'$ or $o'7'$, draw an arc of the circle [fig. 15-24(ii)]. The length of this arc should be equal to the circumference of the base circle. This can be determined in two ways.
- (iii) Calculate the subtended angle θ by the formula,

$$\theta = 360^\circ \times \frac{\text{radius of the base circle}}{\text{slant height}}$$

Cut-off the arc so that it subtends the angle θ at the centre and divide it into twelve equal parts.

- (iv) Step-off with a bow-divider, twelve equal divisions on the arc, each equal to one of the divisions of the base-circle.

(This will give an approximate length of the circumference. Note that the base-circle should not be divided into less than twelve equal parts.)

- (v) Join the division-points with O , thus completing the development of the whole cone with twelve generators shown in it [fig. 15-24(ii)].
- (vi) The truncated portion of the cone may be deduced from this development by marking the positions of points at which generators are cut and then drawing a curve through them. For example, generators $o'2'$ and $o'12'$ in the front view are cut at points b' and b'_1 which coincide with each other. The true length of $o'b'$ may be obtained by drawing a line through b' , parallel to the base and cutting $o'7'$ at b'' . Then $o'b''$ is the true length of $o'b'$.
- (vii) Mark points B and B_1 on generators $O2$ and $O-12$ respectively, such that $OB = OB_1 = o'b''$. Locate all points in the same way and draw a smooth curve through them. The figure enclosed this curve and the arc is the development of the truncated cone.

Problem 15-23. Draw the development of the lateral surface of the part P of the cone shown in fig. 15-25(i).

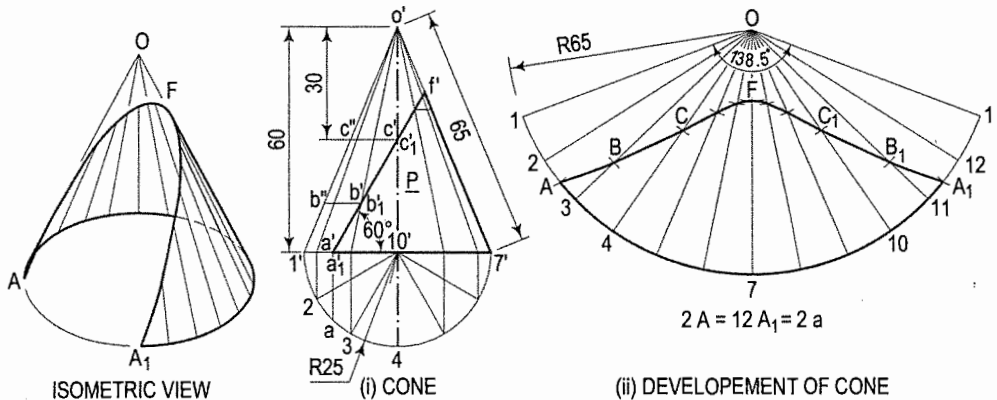


FIG. 15-25

Draw the development as explained in problem 15-22 [fig. 15-25(ii)]. For the points at which the base of the cone is cut, mark points A and A_1 on the arcs 2-3 and 11-12 respectively, such that $A2 = A_1-12 = a_2$. Draw the curve passing through the points A, B, C etc. The figure enclosed between this curve and the arc $A-A_1$ is the required development.

Problem 15-24. Draw the development of the lateral surface of the part P of the cone shown in fig. 15-26(i).

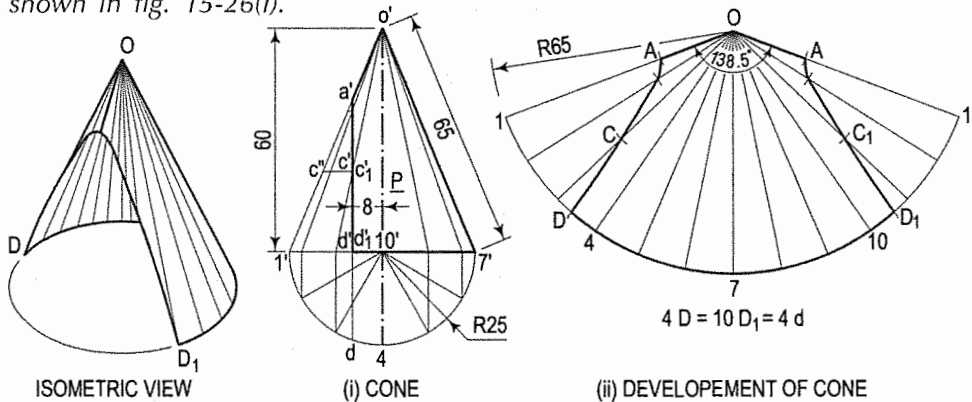


FIG. 15-26

Draw the development as explained in problem 15-23 and as shown in fig. 15-26(ii).

Problem 15-25. Draw the development of the lateral surface of the part P of the cone shown in fig. 15-27(i).

Draw the development of the lateral surface of the whole cone [fig. 15-27(ii)]. With O as centre and radius $o'q'$, draw the arc QQ cutting O_1 at Q .

Obtain the curve for the lower part as explained in problem 15-22. The figure enclosed between this curve and the arc QQ is the required development.

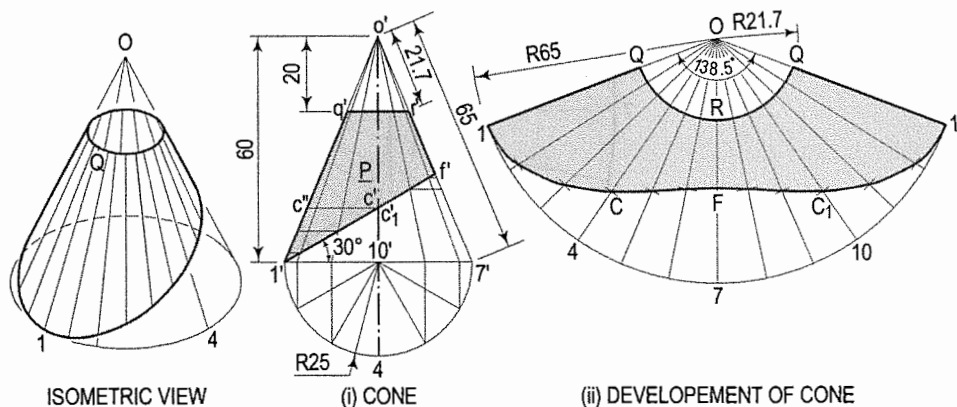


FIG. 15-27

Problem 15-26. Draw the projections of a cone resting on the ground on its base and show on them, the shortest path by which a point P , starting from a point on the circumference of the base and moving around the cone will return to the same point. Base of cone 61 mm diameter; axis 75 mm long.

- (i) Draw the projections and the development of the surface of the cone showing all twelve generators (fig. 15-28). The development may be drawn attached to $o'1'$.
- (ii) Assume that P starts from the point 1 (i.e. point $1'$ in the front view). Draw a straight line $1'1'$ on the development. This line shows the required shortest path.

To transfer this line to the front view the process adopted in problem 15-22 must be reversed. Let us take a point P_4 at which the path cuts the generator $o'4$. Mark a point P''_4 on $o'1'$ such that $o'P''_4 = o'P_4$. This can be done by drawing an arc with o' as centre and radius equal to $o'P_4$ cutting $o'1'$ at P''_4 . Through P''_4 , draw a line parallel to the base cutting $o'4$ at P'_4 . Then p'_4 is the position of the point p_4 in the front view.

Similarly, transfer all the points to the front view and draw the required curve through them. The curve at the back will coincide with the front curve.

- (iii) Project these points to the top view on the respective generators. p'_4 and p'_{10} cannot be projected directly. Hence, project p''_4 to a point q on o_1 . With o as centre and radius equal to oq , draw an arc cutting o_4 at p_4 and $o-10$ at p_{10} . Thus $op_4 = op_{10} = oq$. A curve drawn through the points thus obtained will show the path in the top view.

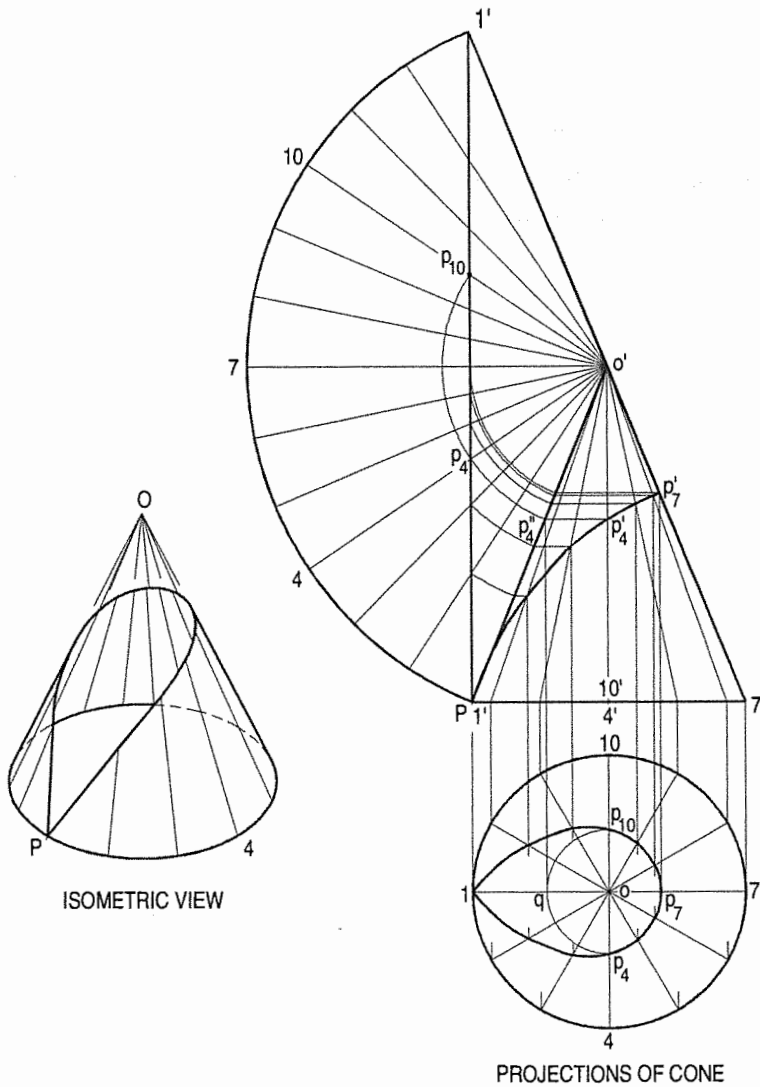


FIG. 15-28

Problem 15-27. (fig. 15-29): Draw the development of the lateral surface of the part of the cone, the front view of which is shown in fig. 15-29.

- Draw a semi-circle on the base as a diameter and divide it into six equal parts for positions of generators.
- Draw the development assuming the cone to be whole. Obtain points on the generators in the development as explained in problem 15-22. Additional points such as a' may also be marked to determine the correct shape of the curve.
- Draw curves through these points and complete the development as shown.

Problem 15-28. (fig. 15-30): The development of the conical surface is a sector of the semi-circular plane. The radius of the sector is 75 mm and the angle subtended by the arc at the centre is 120° . Construct the cone and draw its projections when the apex is 35 mm above the H.P. and its axis is parallel to the V.P.

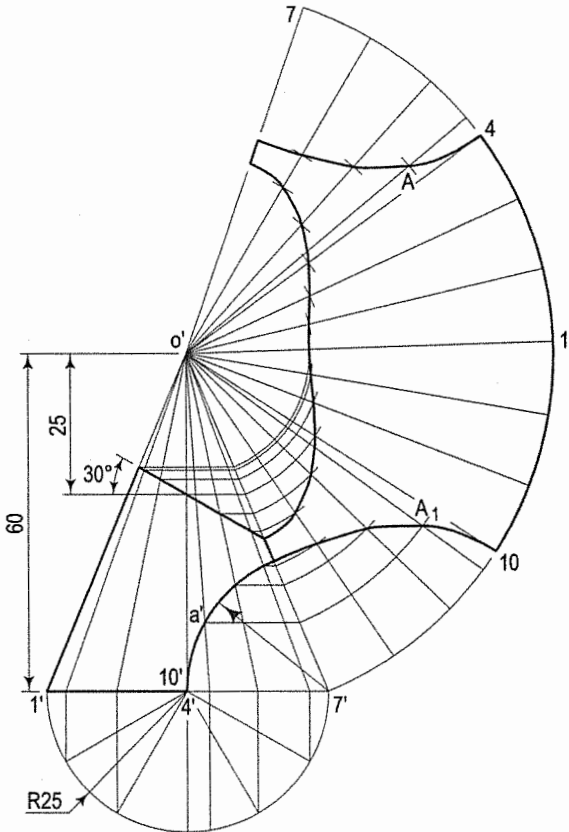
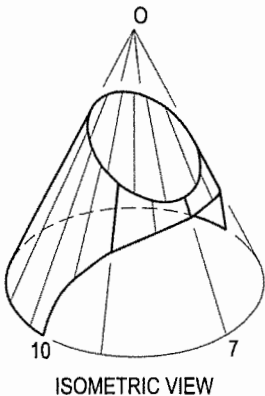


FIG. 15-29

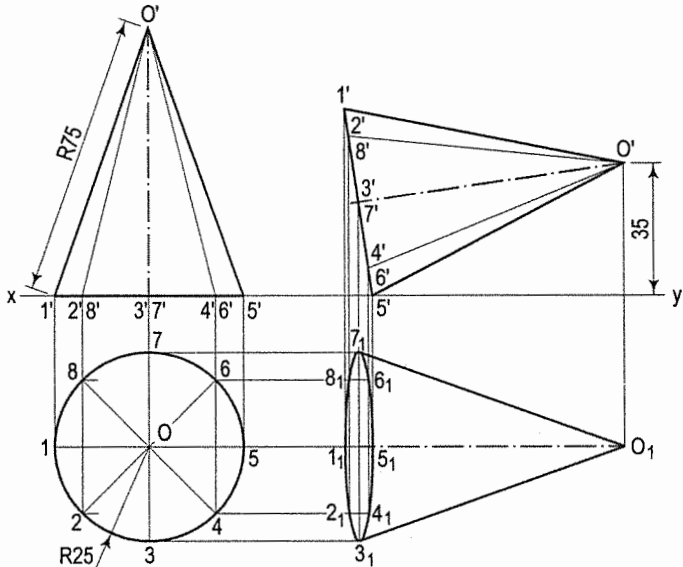
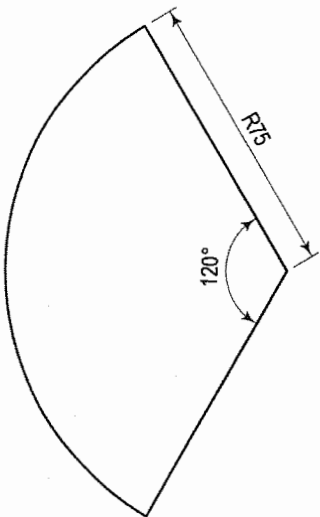


FIG. 15-30

- (i) Determine the radius of the base circle of the cone using following formula:

$$S = R \cdot \theta = 2 \pi r \dots\dots\dots (i)$$

where

S – The arc of circle

R – Radius of the arc

θ – Angle subtended by the arc at the centre

r – Radius of base circle of the cone.

Substitute $R = 75 \text{ mm}$, $\theta = \frac{\pi \times 120}{180}$ radian, we obtain
 $r = 25 \text{ mm}$.

- (ii) Draw the circle of radius of 25 mm representing the top view of the cone.
 (iii) Project the front view with the height equal to 75 mm.
 (iv) Tilt the front view such that the apex is 35 mm above xy line.
 (v) Complete the projection as shown in fig. 15-30.

Problem 15-29. Draw the shape of the sheet metal required for the funnel shown in fig. 15-31.

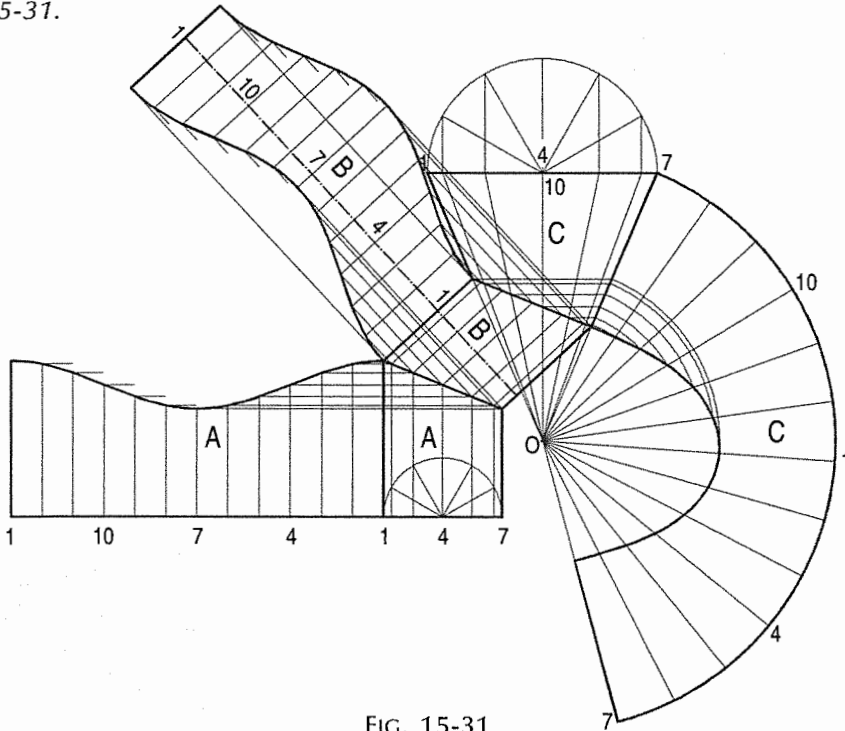


FIG. 15-31

The funnel consists of a conical part C and two cylindrical pipes A and B, all truncated. Draw the developments of the pipes as shown in problem 15-8, and that of the conical piece, as shown in problem 15-22. These are the required shapes.

Problem 15-30. The projections of a solid composed of a truncated half-cylinder and a cut half-prism are given in fig. 15-32(i). Draw the development of its lateral surface.

Assuming the solid to be whole; draw the development of its surface [fig. 15-32(ii)]. Draw a stretch-out line and on it, step-off

- (i) 1A equal to the arc 1a,
 (ii) AB, BC and CD, each equal to the edge ab of the base, and
 (iii) D1 equal to the arc d1. Complete the rectangle.

Draw perpendiculars at A, B etc. and at other intermediate points. Locate on them, positions of points at which they are cut and draw the curves and straight lines as shown.

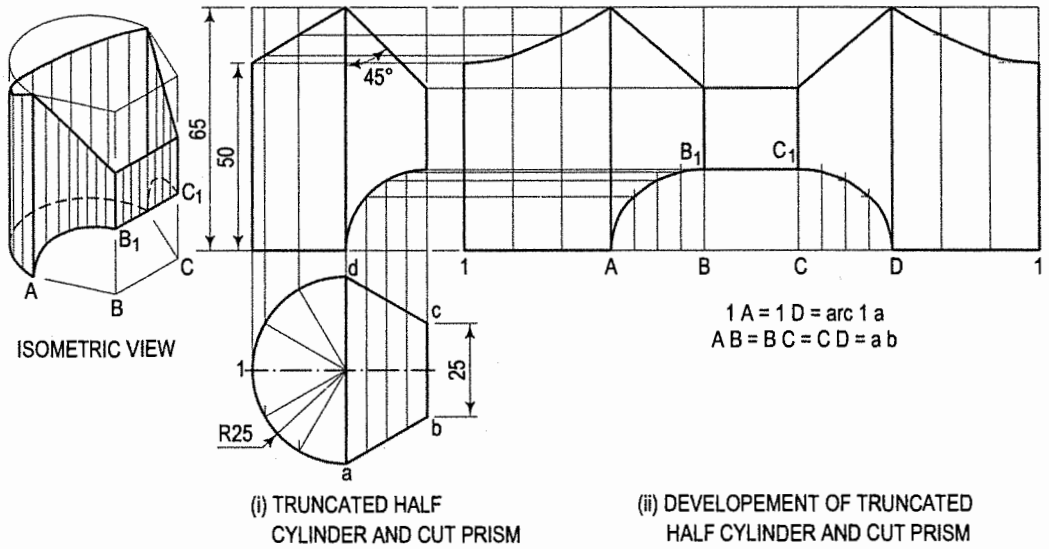


FIG. 15-32

Problem 15-31. Draw the development of the lateral surface of the solid shown in two views (drawn in first-angle projection) in fig. 15-33(i).

The solid is made-up of portions of frusta of a cone and a hexagonal pyramid.

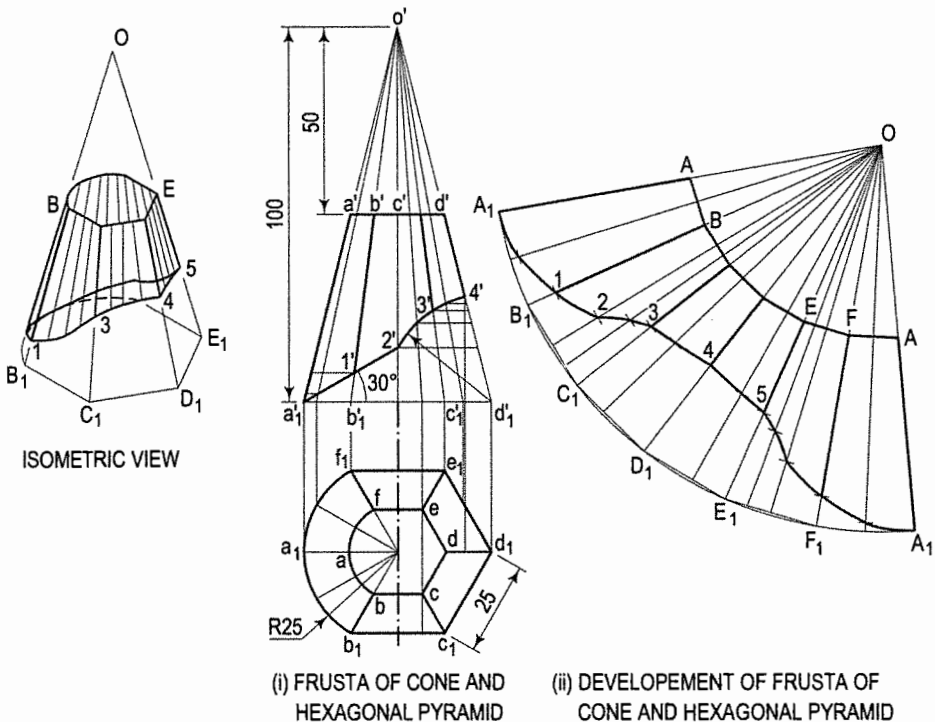


FIG. 15-33

The slant height of the cone is equal to the slant edge of the pyramid.

- (i) Therefore, with any point O as centre and radii $o'a'$ and $o'a'_1$, draw arcs [fig. 15-33(ii)].
- (ii) On the outer arc, step-off distances (i) A_1B_1 equal to the arc a_1b_1 , (ii) B_1C_1 , C_1D_1 , D_1E_1 and E_1F_1 , each equal to the side of the base, viz. b_1c_1 and (iii) F_1A_1 equal to the arc f_1a_1 .
- (iii) Join these points with O , cutting the inner arc at points A , B etc. Locate positions of various points as explained in problems 15-18 and 15-22 and complete the development as shown.

15-3. DEVELOPMENT OF TRANSITION PIECES

Pipes are used in many industries to convey hot or cold fluids. When two different sizes and shapes of pipes are joined using special pipe joint which is known as transition piece. In most cases, transition pieces are composed of plane surfaces and conical surfaces, the latter being developed by triangulation. The procedure of development of few transition pieces is illustrated in the following problems.

Problem 15-32. In air-conditioning system a rectangular duct of $100 \text{ mm} \times 50 \text{ mm}$ connects another rectangular duct of $50 \text{ mm} \times 25 \text{ mm}$ through the transition piece as shown in fig. 15-34(i). Neglecting thickness of a metal sheet, develop the lateral surface of the transition piece as shown in fig. 15-34(ii).

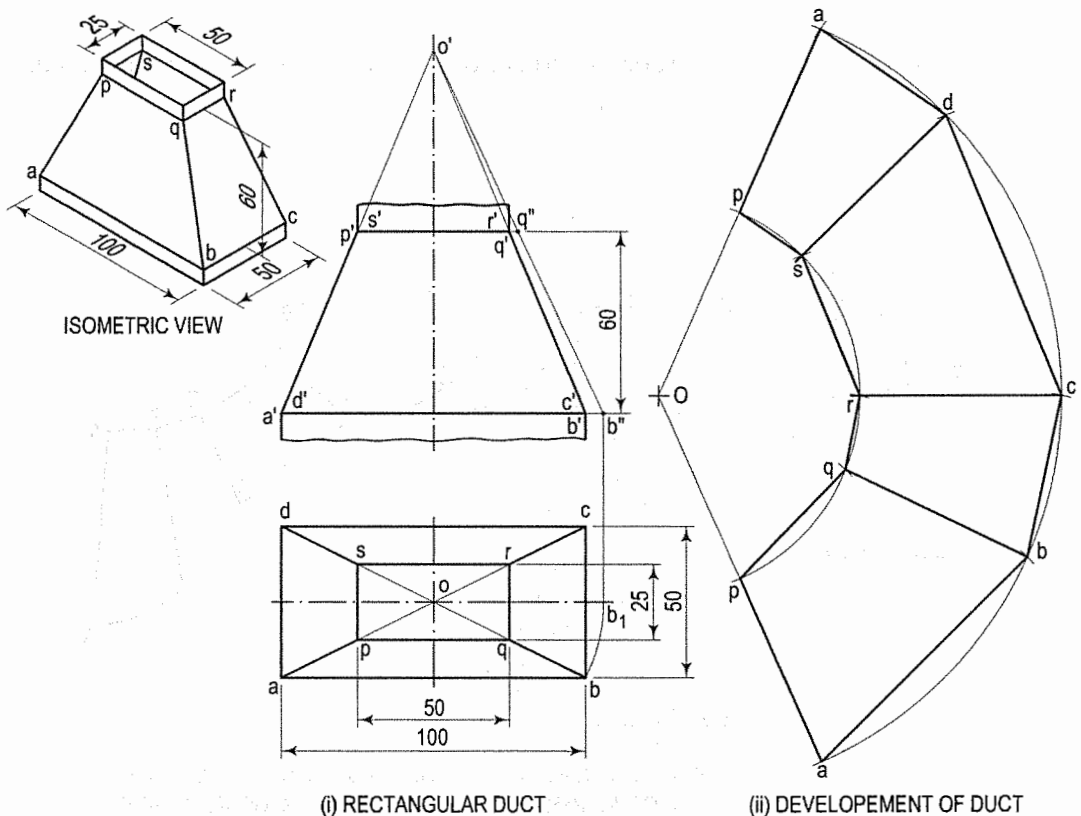


FIG. 15-34

The transition piece is a frustum of a rectangular pyramid.

- (i) Determine the position of the apex of the pyramid by extending $a'p'$ and $b'q'$ as shown. None of the lines in the front view shows the true length of the slant edge. Therefore, draw the top view and make any slant line parallel to xy and determine its true length $o'b''$.
- (ii) With O as centre and radius $o'b''$, draw an arc and obtain the development of a whole pyramid as shown.
- (iii) With O as centre and radius $o'q''$, draw an arc cutting oa , ob , oc at points p , q , etc. respectively. Join them in sequence and complete the development as shown.

Problem 15-33. (fig. 15-35): An air-conditioning duct of a square cross-section $70 \text{ mm} \times 70 \text{ mm}$ connects a circular pipe of 40 mm diameter through the transition piece. Draw the projections and develop the lateral surface of the transition piece.

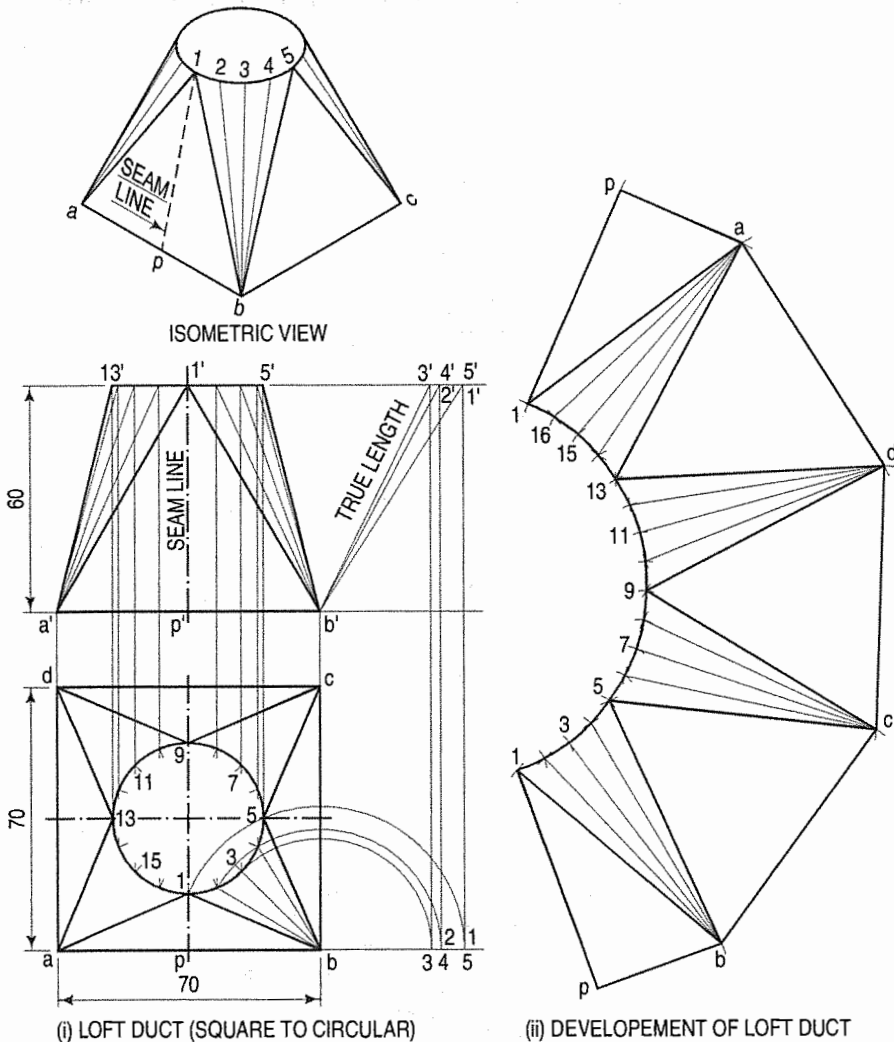


FIG. 15-35

- (i) Draw the front view and the top view as shown in fig. 15-35(i).
- (ii) Divide the top view of circle into some convenient divisions, say 16 parts as shown.

- (iii) Note that the transition piece is composed of four isosceles triangles and four conical surfaces. The seam is along line 1-P.
- (iv) Begin the development from the seam line 1-P ($1'-P'$). As shown in fig. 15-35(ii) draw the right angle triangle 1-P-b, whose base pb is equal to half the side ab and whose hypotenuse 1-b is equal to the true length $1'-b'$ of side 1-b.
- (v) The conical surfaces are developed by the triangulation method as follows.
- (vi) In the top view, join division of the circle 1, 2, 3 etc. with the corner a, b, c and d. Project them in the front view as shown. Obtain the true length of sides of each triangle as shown.
- (vii) With b as centre and $2'b'$ (true length) radius draw an arc, cutting the arc drawn with 1' as centre and $1'2'$ as radius. Similarly, obtain the points 3', 4', 5' etc. Join them in the proper order as shown.

Problem 15-34. A steam pipe bends at certain angle is connected by a transition piece as shown in fig. 15-36(i). Draw the development of lateral surface of the transition piece A.

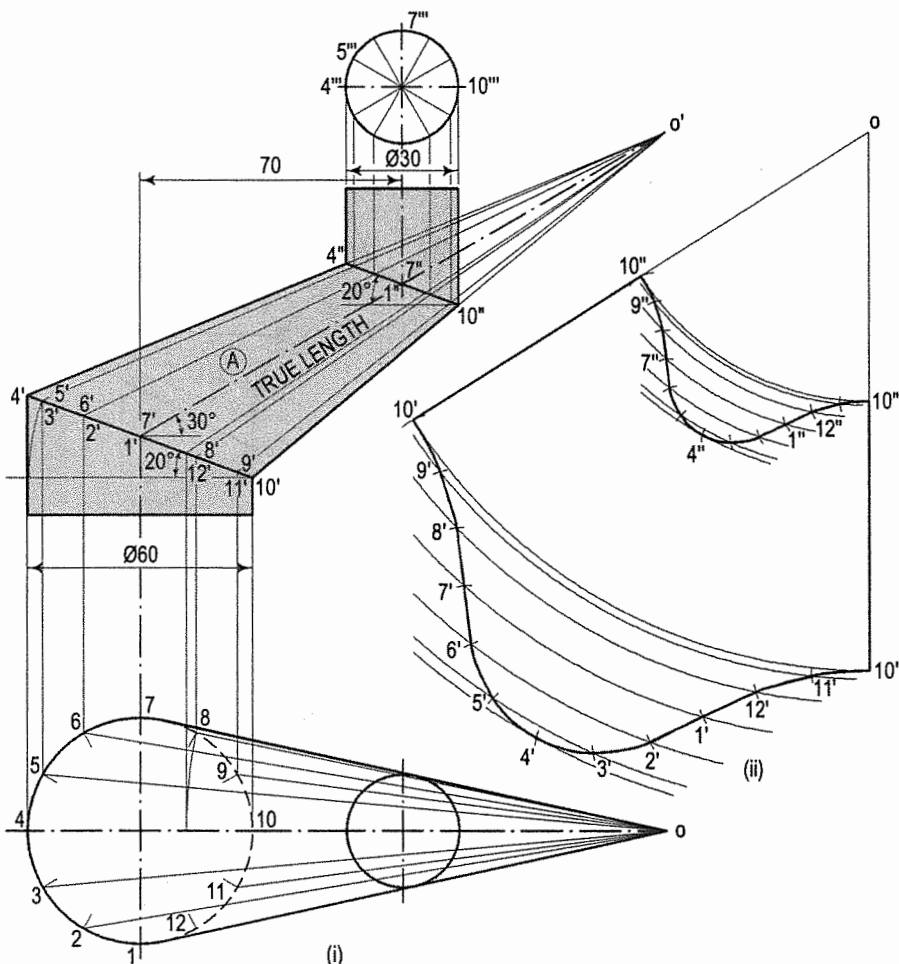


FIG. 15-36

- (i) Draw the top view and the front view. The transition piece is a truncated cone. The apex of the cone is determined by extending extreme generators $4' 4''$ and $10' 10''$. The projections of the base and the top of truncated cone are inclined to xy , therefore in the top view their projections do not show the true shapes.

(The true-shape is an ellipse.) For approximate method of development, these projections in the top view can be taken as the true shape.

- (ii) Draw the projections and the development of the cone showing all generators. Obtain the true length of generators.
- (iii) With O as centre and radius $O10'$, draw an arc of circle. Similarly draw the arcs of circle taking radii $O9'$, $O8'$ etc.
- (iv) With $10'$ as centre and radius equal to $4-3$ (a division of base circle in the top view), draw an arc cutting the previously drawn arc. Similarly obtain points $8'$, $7'$, $6'$ etc. Join them by smooth curve. Obtain the development of the top circle similar way by taking $4''5''$ radius (a division of top circle).

Problem 15-35. The orthographic projections of an exhaust pipe required for an engine is shown in fig. 15-37. Draw development of the transition connector by triangulation method.

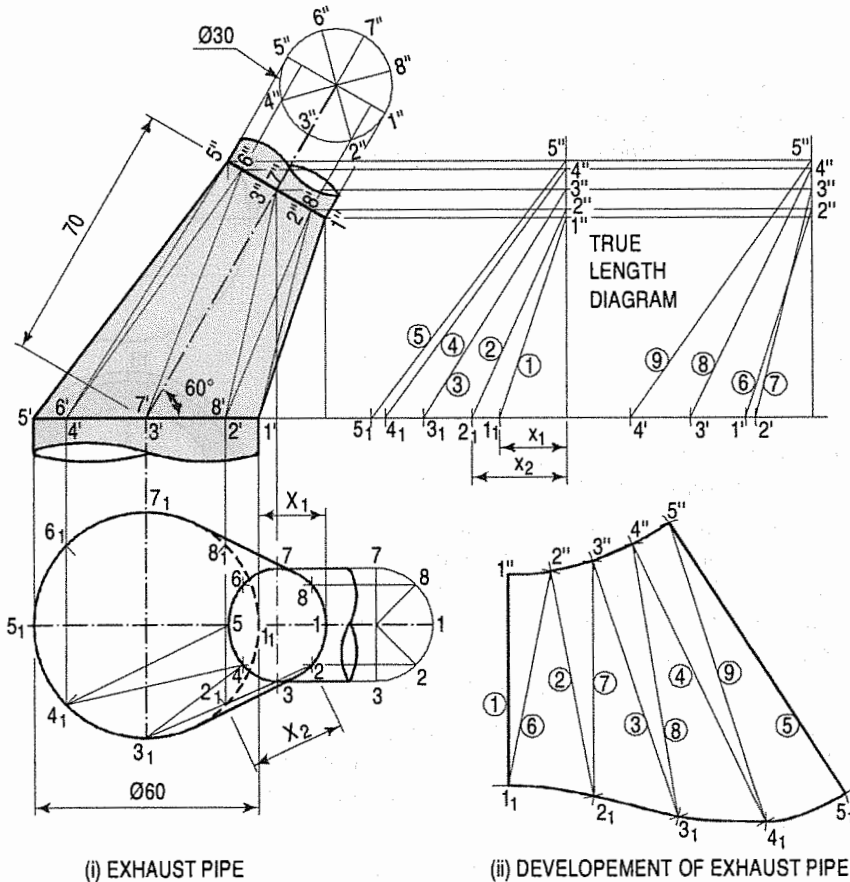


FIG. 15-37

- (i) Draw the given projections. Divide the circle of base in the top view into eight equal parts, say $1_1, 2_1, \dots$ etc. Project them in the front view $1', 2', \dots$ etc. as shown.
- (ii) Draw an auxiliary view of the top pipe as shown and divide it into eight equal parts. Number them $1'', 2'', \dots$ etc.
- (iii) Form the triangles on the lateral surface as shown.
- (iv) Determine true length of each side of a triangle as shown.
- (v) Mark the length of $1''1_1$. With 1 as centre and radius equal to $1'2''$ [line-(6)] draw an arc cutting the arc drawn with radius $1'2''$ (division of auxiliary circle) and the centre as $1''$. A half development is shown.

15-4. SPHERES (APPROXIMATE METHOD)

The surface of a sphere can be approximately developed by dividing it into a number of parts. The divisions may be made in two different ways:

- (i) in zones (ii) in lunes.

A *zone* is a portion of the sphere enclosed between two planes perpendicular to the axis. A *lune* is the portion between two planes which contain the axis of the sphere.

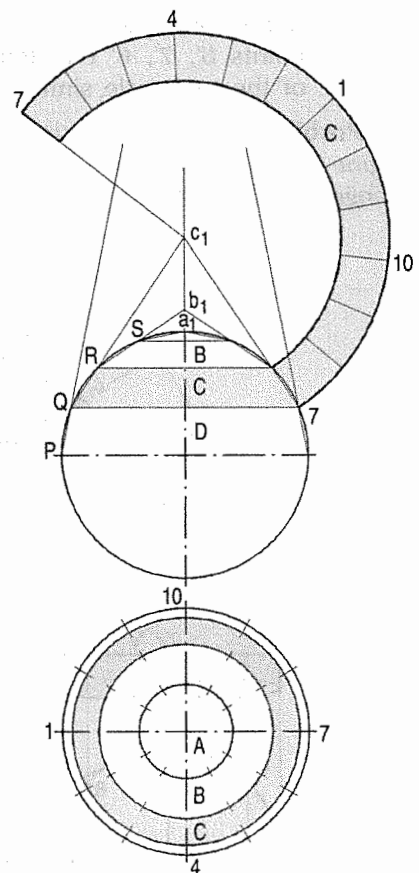
(1) Zone method: Fig. 15-38 shows the top half of a sphere divided into four zones of equal width. By joining the points P, Q, R etc. by straight lines, each zone becomes a cone frustum, except the upper-most zone which becomes a cone of small altitude.

Developments of these cone frusta and the upper cone will give the development of the half sphere. For example, take the zone C. It is a frustum of a cone whose vertex is at c_1 . The surface of this frustum is shown developed in the front view. The length of the divisions on the arc is obtained from the top view. All the zones can be developed in the same manner.

(2) Lune method: A sphere may be divided into twelve lunes, one of which is shown in the front view in fig. 15-39. The semi-circle qr is the top view of the centre line of that lune. It is evident that the length of the lune is equal to the length of the arc qr and its maximum width is equal to gh .

Divide the semi-circle into a number of equal parts say 8 and project the division-points on the front view to points $1', 2'$ etc. With q' as centre and radii equal to $q'1', q'2'$ and $q'3'$, draw arcs ab, cd and ef which will show the widths of the lune at points 1 and 7, 2 and 6, and 3 and 5 respectively.

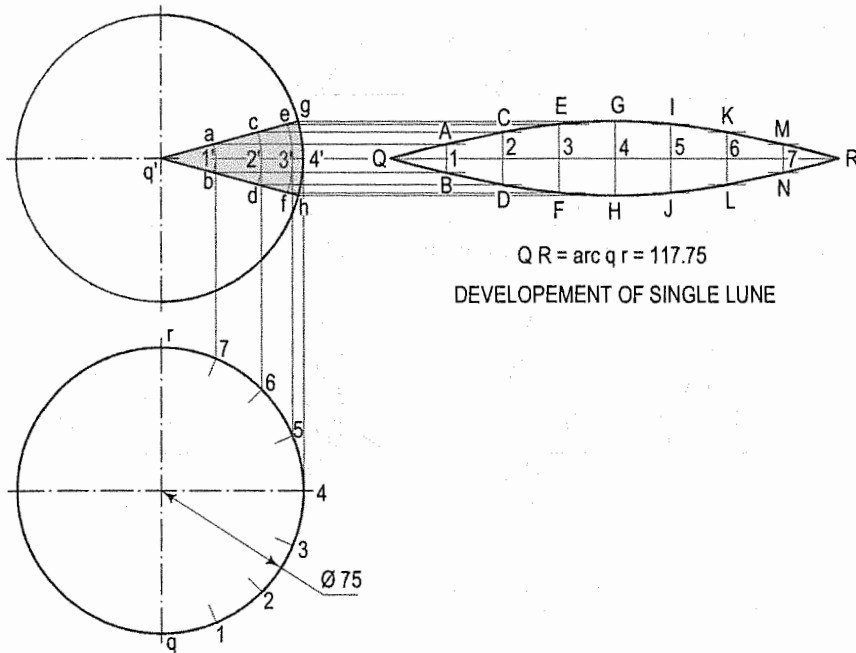
Draw a line QR equal to the length of the arc qr . This may be obtained by stepping-off eight divisions, each equal to the chord-length $q1$.



Zone method

FIG. 15-38

Draw perpendiculars at each division-point and make AB and MN equal to ab at points 1 and 7, CD and KL equal to cd at points 2 and 6 etc. Draw smooth curves through points Q, A, C etc. The figure thus obtained will be the approximate development of one-twelfth of the surface of the sphere. Development of surfaces of some more solids cut by different planes, and solids with holes cut or drilled through them are treated in chapters 14 and 16.



Lune method
FIG. 15-39

EXERCISES 15

1. Draw the development of the lateral surface of the part P of each of the solids, the front views of which are shown in fig. 15-40 and described below.

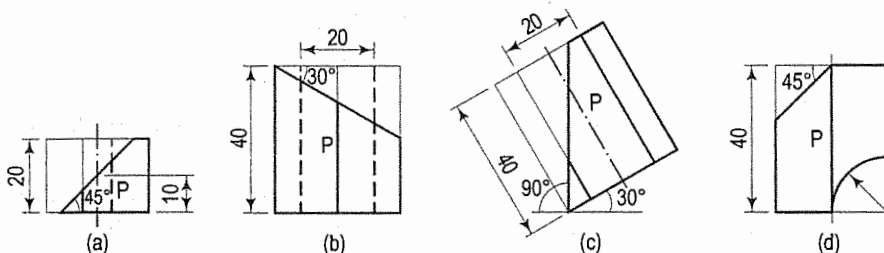


FIG. 15-40

- (a) A cube, one vertical face inclined at 30° to the V.P.
- (b) A pentagonal prism, a side of the base parallel to the V.P.
- (c) A hexagonal prism, two faces parallel to the V.P.
- (d) A square prism, length of the side of the base 20 mm and all faces equally inclined to the V.P.

2. Draw the development of the lateral surface of the part *P* of each of the cylinders, the front views of which are shown in fig. 15-41.

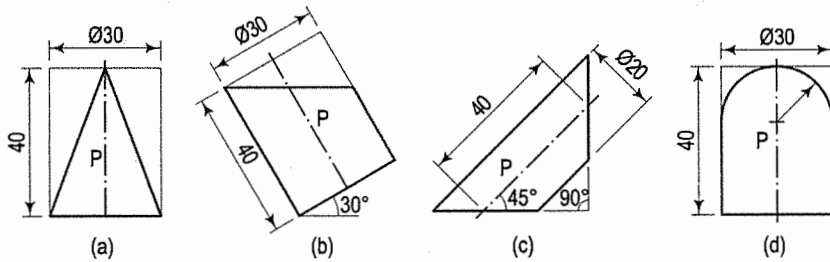


FIG. 15-41

3. Draw the development of the lateral surface of the part *P* of each of the pyramids, the front views of which are shown in fig. 15-42, and described below.

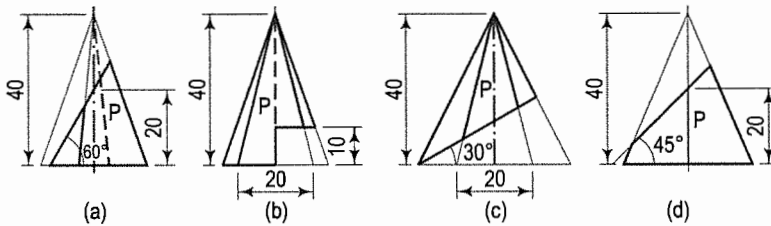


FIG. 15-42

- (a) A square pyramid, side of the base 20 mm long and one side of the base inclined at 30° to the V.P.
 (b) A pentagonal pyramid, one side of the base parallel to the V.P.
 (c) A hexagonal pyramid, two sides of the base parallel to the V.P.
 (d) A square pyramid, side of the base 20 mm long and all the sides of the base equally inclined to the V.P.
4. Draw the development of the lateral surface of the part *P* of each of the cones, front views of which are shown in fig. 15-43.

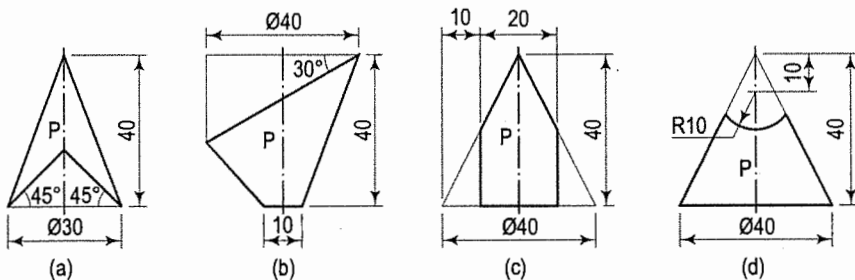


FIG. 15-43

Refer to fig. 15-44 for the following exercises and for dimensions, assume each square to be of 10 mm side.

5. Draw the development of the surfaces of the portions of the following prisms, front views of which are shown in the top row:

- (a) A hexagonal prism having a face parallel to the V.P.
- (b) A square prism, all faces equally inclined to the V.P.
- (c) A pentagonal prism having a vertical face parallel to the V.P.
- (d) A triangular prism having a vertical face parallel to the V.P.
- (e) A hexagonal prism having two faces perpendicular to the V.P.

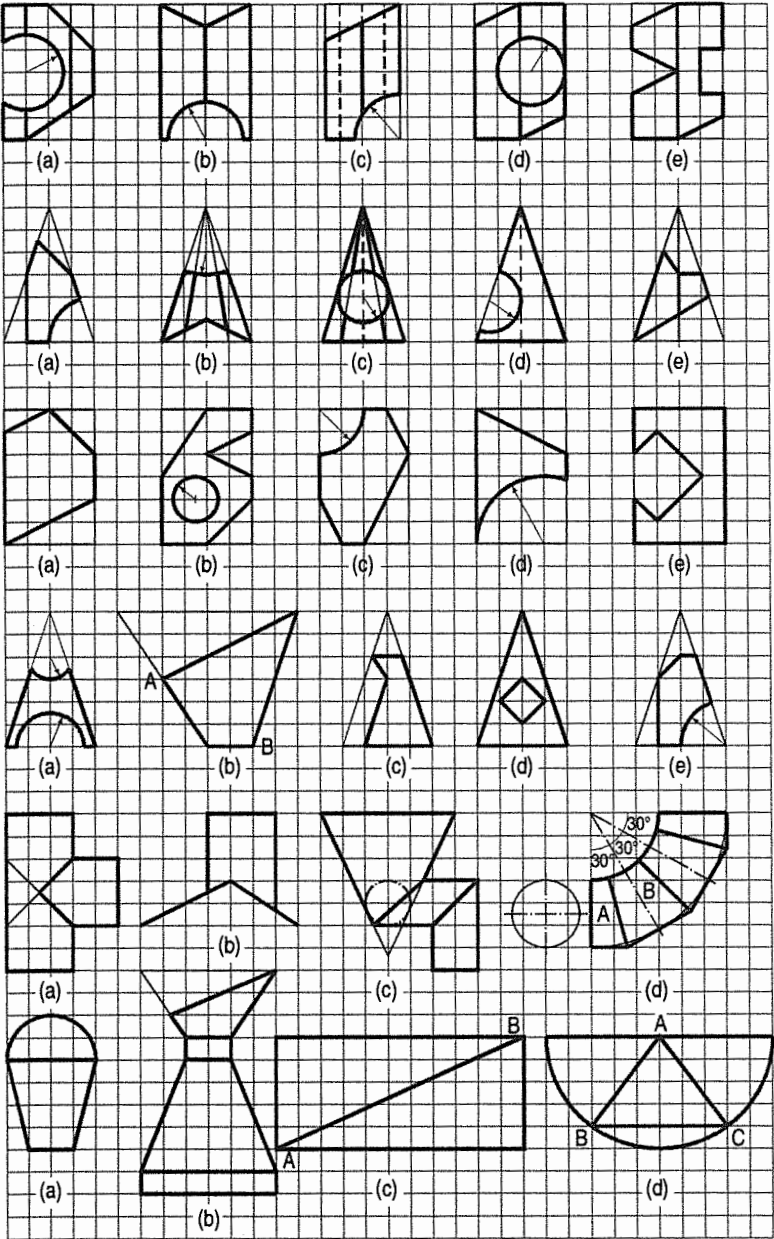


FIG. 15-44

6. Draw the development of the surfaces of the portions of the following pyramids, front views of which are shown in the second row:

- (a) A square pyramid having a side of base perpendicular to the V.P.
 - (b) A hexagonal pyramid having a side of base parallel to the V.P.
 - (c) A pentagonal pyramid having a side of base parallel to the V.P.
 - (d) A triangular pyramid having a side of base parallel to the V.P.
 - (e) A square pyramid having all sides of the base equally inclined to the V.P.
7. Draw the development of the surfaces of the portions of the cylinders shown in the third row.
 8. Draw the development of the surfaces of the portions of the cones shown in the fourth row.
 9. Refer to the fifth row, and
 - (i) Draw the development of the pipes forming a Tee shown at (a).
 - (ii) Draw the development of the cylindrical steel chimney erected on a roof [shown at (b)], assuming the squares to be of 30 cm side.
 - (iii) Draw the development of the three parts of the funnel shown at (c).
 - (iv) Develop parts A and B of the transition piece shown at (d).
 10. Refer to the last row, and
 - (i) Develop the surface of the conical buoy with a hemispherical top shown at (a).
 - (ii) Determine the shape of the tin sheet required to prepare the can shown at (b).
 - (iii) The development of the surface of a cylinder is given at (c). Draw the front view of the cylinder showing the line AB in it.
 - (iv) The development of the surface of a cone is shown at (d). Draw the projections of the cone showing lines AB, BC and CA in each view.
 11. A pipe 40 mm diameter and 120 mm long (along the axis) is welded to the vertical side of a tank. Show the development of the pipe, if it makes an angle of 60° with the side to which it is welded, the other end of the pipe making an angle of 30° with its own axis. Neglect thickness of the pipe.
 12. The inside of a hopper of a floor mill is to be lined with tin sheet. The top and bottom of the hopper are regular pentagons with each side equal to 450 mm and 300 mm respectively (internally). The height of the hopper is 450 mm. Draw the shape to which the tin sheet is to be cut so as to fit in the hopper. Scale, 1:10.
 13. A 50 mm cylindrical pipe branches off at 90° from a 75 mm cylindrical main pipe as shown in fig. 15-45. Draw the developments of both the pipes at the joint. Assume suitable lengths for the main pipe as well as for the branch pipe.
 14. A cone of 90 mm diameter of base and 90 mm height stands on its base on the ground. A semi-circular hole of 50 mm diameter is cut through the cone. The axis of the hole is horizontal and intersects the axis of the cone. It is 30 mm above the base of the cone. The flat surface of the hole contains the axis of the cone and is perpendicular to the V.P. Draw three views of the cone and also develop the surface of the cone.

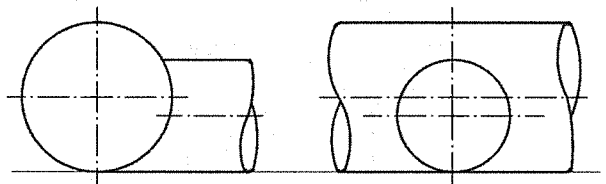


FIG. 15-45

Chapter 16



INTERSECTION OF SURFACES

16-0. INTRODUCTION

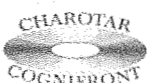


The intersecting surfaces may be two plane surfaces or two curved surfaces of solids. The lateral surface of every solid taken as a whole is a curved surface. This surface may be made of only curved surface as in case of cylinders, cones etc. or of plane surfaces as in case of prisms, pyramids etc. In the former case, the problem is said to be on the intersection of surfaces and in the latter case, it is commonly known as the problem on interpenetration of solids. It may, however, be noted that when two solids meet or join or interpenetrate, it is the curved surfaces of the two that intersect each other. The latter problem also is, therefore, on the intersection of surfaces.

In this chapter, we shall learn about the intersection of surfaces as shown below:

1. Line of intersection
2. Methods of determining the line of intersection between surfaces of two interpenetrating solids
3. Intersection of two prisms
4. Intersection of cylinder and cylinder
5. Intersection of cylinder and prism
6. Intersection of cone and cylinder
7. Intersection of cone and prism
8. Intersection of cone and cone
9. Intersection of sphere and cylinder or prism.

16-1. LINE OF INTERSECTION



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 36 for the line of intersection.

In engineering practice, objects constructed may have constituent parts, the surfaces of which intersect one another in lines which are called *lines of intersection*. A dome fitted on a boiler is one such example. The surface of the dome extends upto the line of intersection only. For accurate development of the surface of the dome, this line of intersection must be accurately located and shown in two orthographic views. The shape of the hole to be cut in the boiler-shell is also determined from the shape of the same line of intersection.

Thus, the line of intersection of the two surfaces is a line common to both. It is composed of points at which the lines of one surface intersect those on the other surface. The line of intersection may be straight or curved, depending upon the nature of intersecting surfaces.

Two plane surfaces (e.g. faces of prisms and pyramids) intersect in a straight line. The line of intersection between two curved surfaces (e.g. of cylinders and cones) or between a plane surface and a curved surface is a curve.

When a solid completely penetrates another solid, there will be two lines of intersection. These lines are, sometimes, called *the lines or curves of interpenetration*. The portion of the penetrating solid which lies hidden within the other solid is shown by dotted lines.

16-2. METHODS OF DETERMINING THE LINE OF INTERSECTION BETWEEN SURFACES OF TWO INTERPENETRATING SOLIDS



(1) **Line method:** A number of lines are drawn on the lateral surface of one of the solids and in the region of the line of intersection. Points of intersection of these lines with the surface of the other solid are then located. These points will obviously lie on the required line of intersection. They are more easily located from the view in which the lateral surface of the second solid appears edgewise (i.e. as a line). The curve drawn through these points will be the line of intersection.

(2) **Cutting-plane method:** The two solids are assumed to be cut by a series of cutting planes. The cutting planes may be vertical (i.e. perpendicular to the H.P.), edgewise (i.e. perpendicular to the V.P.) or oblique. The cutting planes are so selected as to cut the surface of one of the solids in straight lines and that of the other in straight lines or circles.

Each method is explained in detail while solving illustrative problems. Sound knowledge of projections of solids in various positions is quite essential while dealing with these problems.

16-3. INTERSECTION OF TWO PRISMS



Prisms have plane surfaces as their faces. The line of intersection between two plane surfaces is obtained by locating the positions of points at which the edges of one surface intersect the other surface and then joining the points by a straight line. These points are called *vertices* (plural of vertex). The line of intersection between two prisms is therefore a closed figure composed of a number of such lines meeting at the vertices.

It is determined by locating the points at which edges of one prism intersect edges or faces of the other prism and then joining them in correct sequence.

Problem 16-1. (fig. 16-1): A vertical square prism, base 50 mm side, is completely penetrated by a horizontal square prism, base 35 mm side, so that their axes intersect. The axis of the horizontal prism is parallel to the V.P., while the faces of the two prisms are equally inclined to the V.P. Draw the projections of the solids, showing lines of intersection. (Assume suitable lengths for the prisms.)

- (i) Draw the projections of the prisms in the required position. The faces of the vertical prism are seen as lines in the top view. Hence, let us first locate the points of intersection in that view.
- (ii) Lines 1-1 and 3-3 intersect the edge of the vertical prism in points p_1 and p_3 (coinciding with a). Lines 2-2 and 4-4 intersect the faces at p_2 and p_4 respectively. The exact positions of these points along the length of the prism may now be determined by projecting them on corresponding lines in the front view. For example, p_2 is projected to p'_2 on the line $2'2'$. Note that p'_4 coincides with p'_2 .
- (iii) Draw lines $p'_1p'_2$ and $p'_2p'_3$. Lines $p'_1p'_4$ and $p'_3p'_4$ coincide with the front lines. These lines show the line of intersection. Lines $q'_1q'_2$ and $q'_2q'_3$ on the other side are obtained in the same manner. Note that the lines for the hidden portion of the edges are shown as dashed lines. The portions $p'_1p'_3$ and $q'_1q'_3$ of vertical edges $a'a'$ and $c'c'$ do not exist and hence, must be removed or kept fainter.

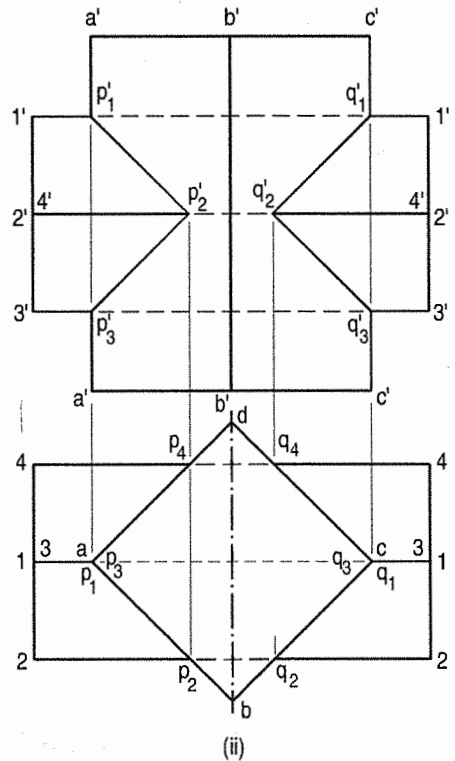
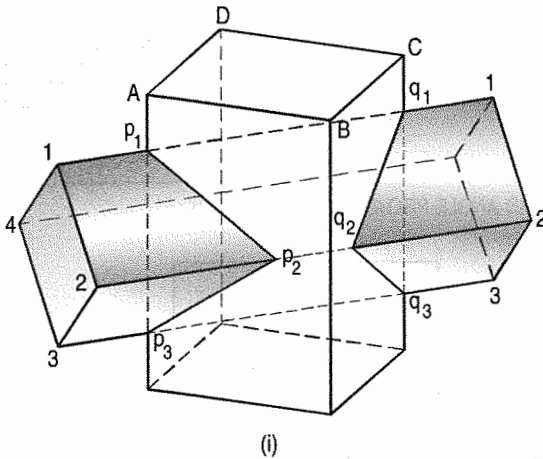
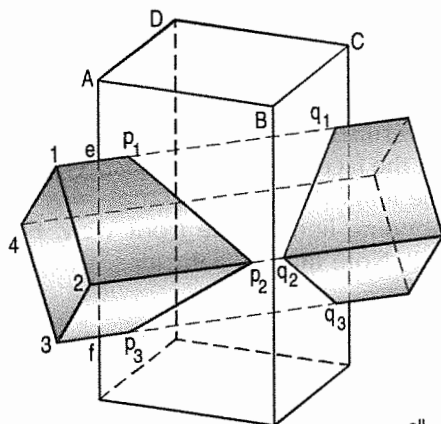


FIG. 16-1

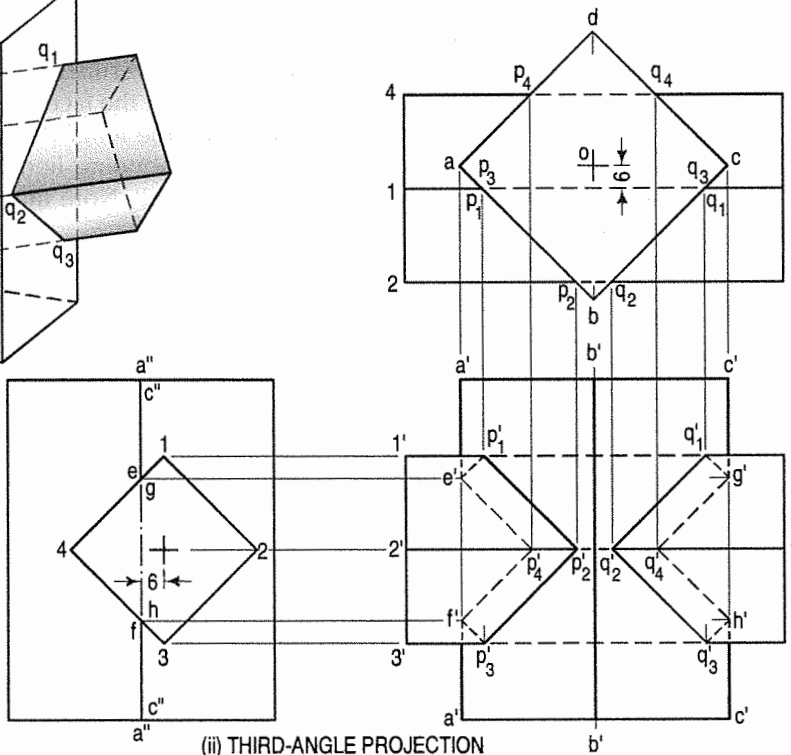
Problem 16-2. (fig. 16-2): A vertical square prism, base 50 mm side is completely penetrated by a horizontal square prism, base 35 mm side so that their axes are 6 mm apart. The axis of the horizontal prism is parallel to the V.P., while the faces of both prisms are equally inclined to the V.P. Draw the projections of the prisms showing lines of intersection.

Points $p'_1...p'_4$ at which edges of the horizontal prism intersect faces of the vertical prism may be located from the top view. In addition to these points, it will be necessary to find points at which edges of the vertical prism are cut. They will be the points at which these edges intersect the faces of the horizontal prism.

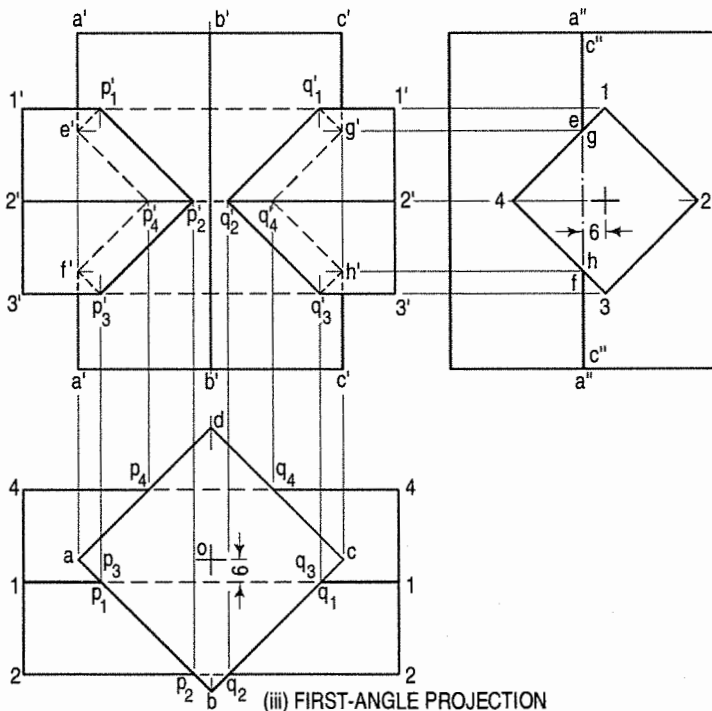
- (i) For this purpose, draw the side view. In this view, all faces of the horizontal prism are seen as lines. Mark points e and f at which the line $a''a''$ intersects the faces. Project these two points to e' and f' on the line $a'a'$ in the front view. Join all the points of intersection in correct sequence. Care must be taken to determine visible and hidden lines. Only two lines viz. $p'_1p'_2$ and $p'_2p'_3$ are visible.



(i) ISOMETRIC VIEW



(ii) THIRD-ANGLE PROJECTION



(iii) FIRST-ANGLE PROJECTION

FIG. 16-2

- (ii) Locate points (on the other side) at which the edges come out and also the two points g' and h' at which the edge $c'c'$ is cut.
- (iii) Draw lines joining these points. They will be exactly similar to lines $p'_1 p'_2$ etc. on the left-hand side.

Problem 16-3. (fig. 16-3): A vertical square prism, base 50 mm side and height 90 mm has a face inclined at 30° to the V.P. It is completely penetrated by another square prism, base 38 mm side and axis 100 mm long, faces of which are equally inclined to the V.P. The axes of the two prisms are parallel to the V.P. and bisect each other at right angles. Draw the projections showing lines of intersection.

Adopt the same method as explained in problem 16-2.

- (i) The edges 1-1 and 3-3 enter one face of the vertical prism and come out of its opposite face.
- (ii) Obtain the points (from the top view) at which all edges intersect the faces and also the four key points (from the side view).

Note carefully the lines for visible and hidden edges, shown as full lines and dotted lines respectively.

Although the two axes are intersecting, the visible portions of the lines of intersection, when the penetrating prism enters and comes out differ because the penetrated prism has its faces inclined to the V.P.

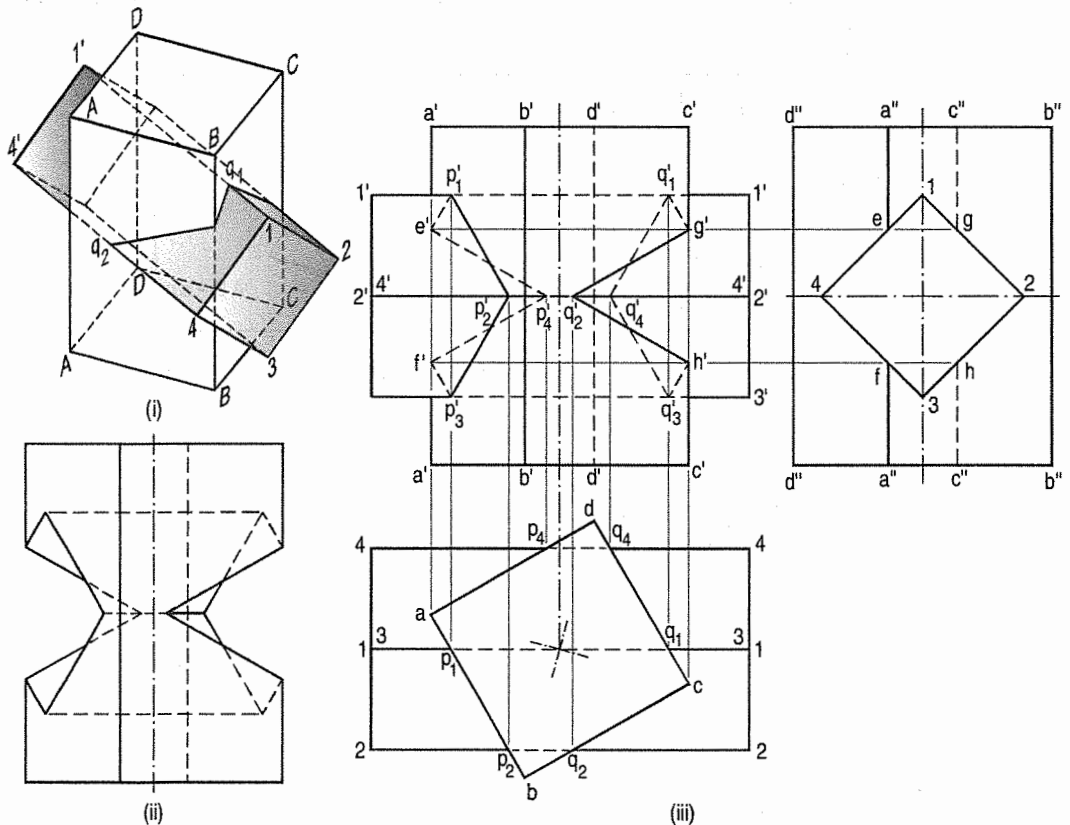


FIG. 16-3

Lines p_1p_2 and p_2p_3 are visible on the left side while q_2g' and q_2h' are visible on the right side. Edges $a'e'$ and $a'f'$ are partly hidden, while $c'g'$ and $h'c'$ are fully visible.

Fig. 16-3(b) shows the front view of the vertical prism, when the penetrating prism has been removed. Note that the edges of the back portions of the hole are partly visible.

Problem 16-4. (fig. 16-4): A square pipe of 50 mm side has a similar branch of 40 mm side. The axis of the main pipe is vertical and is intersected by the axis of the branch at an angle of 45° . All the faces of both the pipes are equally inclined to the V.P. Draw the projections of the pipes, showing lines of intersection. Also develop the surfaces of both the pipes.

The line of intersection between the two pipes is obtained [fig. 16-4(i)] in the same manner as shown in problem 16-1. As the axes are intersecting, the edge $a'a'$ is cut by the two edges of the branch at points p_1 and p_3 . The other two edges of the branch enter the faces of the main pipe at points p_2 and p_4 .

Developments of the surfaces of the two pipes are shown in fig. 16-4(ii).

- (i) Heights of all the points for fig. 16-4(ii) are obtained from the front view, e.g. $P_1A = p_1a'$, $P_11 = p_11'$ etc.
- (ii) The exact position of the point P_2 is located from the top view by making $AE = ap_2$ and then erecting a perpendicular at E. The point P_4 is similarly located.

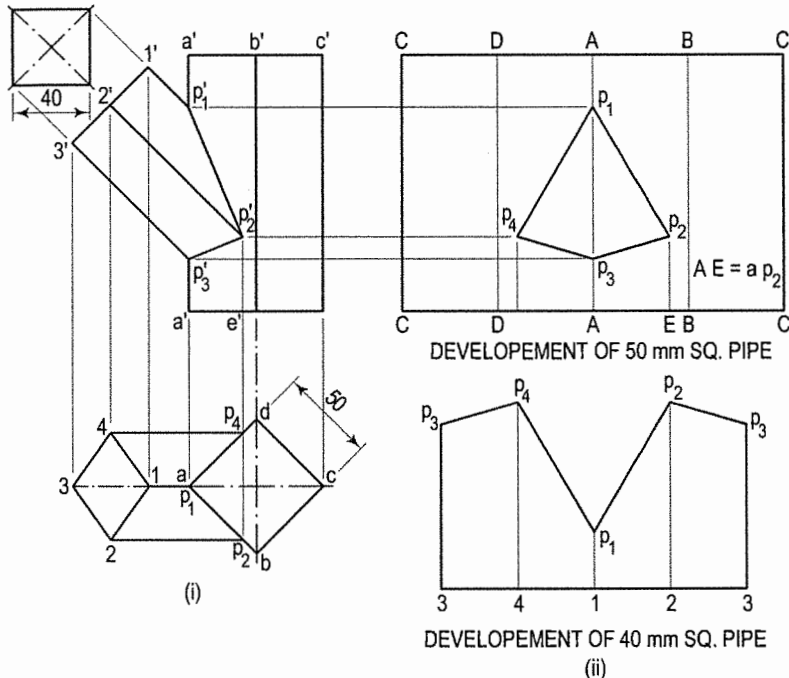


FIG. 16-4

Problem 16-5. (fig. 16-5): A vertical square prism, base 50 mm side, is intersected by another square prism, base 35 mm side, the axis of which is parallel to the V.P. and inclined at 30° to the H.P. The axes of the two prisms are 6 mm apart and their faces are equally inclined to the V.P. Draw the projections showing the line of intersection.

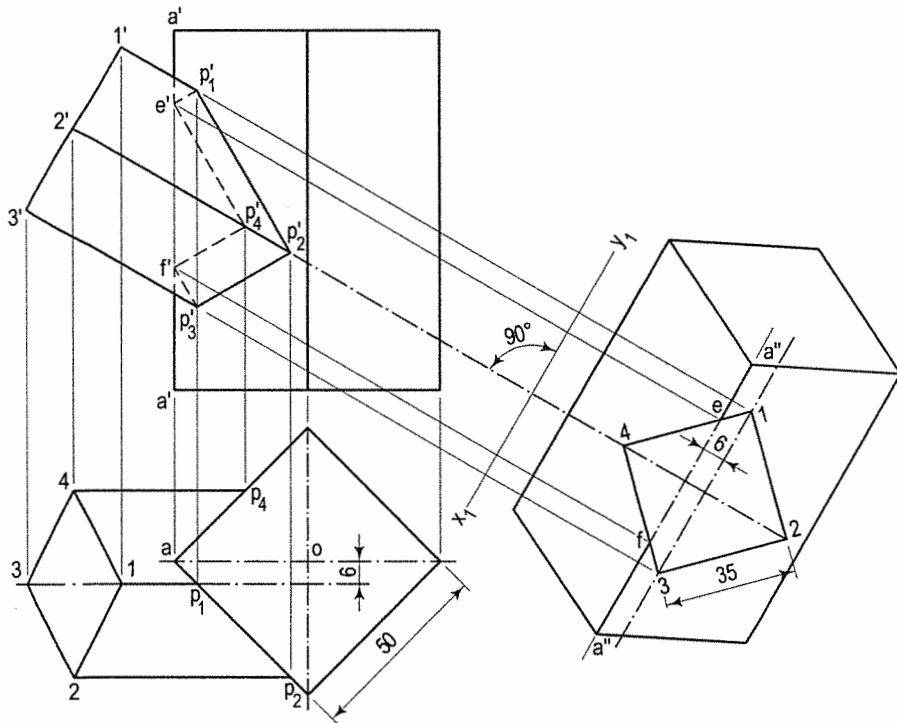


FIG. 16-5

- (i) Obtain points of intersection of the edges of the inclined prism from the top view. For the points at which the edge $a'a'$ of the vertical prism is cut, it will be necessary to project a view in which faces of the inclined prism will be seen as lines.
- (ii) Therefore, project an auxiliary top view on a reference line x_1y_1 drawn perpendicular to the axis of the inclined prism.
- (iii) Mark points e and f at which $a''a'''$ is pierced by the faces and project them to points e' and f' on the corresponding line $a'a'$ in the front view. Draw straight lines joining the six points in correct sequence.

Problem 16-6. (fig. 16-6): A square pyramid is of base sides 40 mm and height 55 mm. The sides of the base are equally inclined with the V.P. It is penetrated by a horizontal triangular prism of sides 40 mm and 70 mm axis long. The axis is perpendicular to the V.P. 25 mm above the base of the pyramid and 8 mm away from the axis of the pyramid. Assume that one of the faces of the prism is vertical and passes through the pyramid. Draw the front view, top view and side-view showing the intersection curve.

- (i) Draw the projections of the square pyramid and triangular prism in the required position.
- (ii) In the front view, the edges $1'-1'$, $2'-2'$ and $3'-3'$ of horizontal prism, intersects the faces of the vertical pyramid at P'_1 , P'_2 , Q'_2 , P'_3 , P'_4 , Q'_4 , P'_5 , Q'_5 , and P'_6 , respectively.
- (iii) Project points P'_1 , to P'_6 , in the top view and obtain intersection point P_1 , P_2 , Q_2 , P_3 , P_4 , Q_4 , P_5 , Q_5 , and P_6 .
- (iv) Similarly take projections from top view and front view and complete the side view.

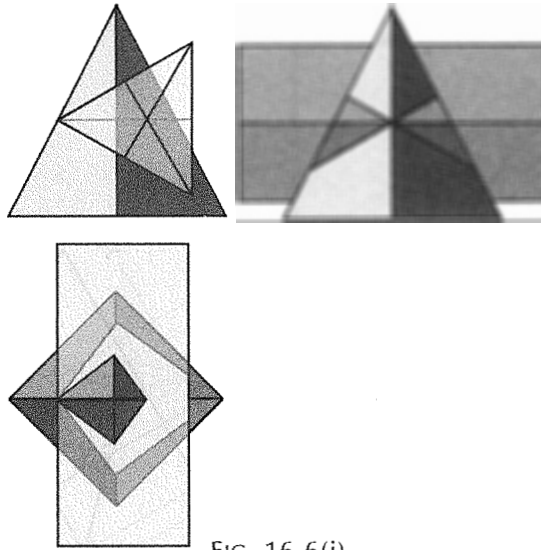


FIG. 16-6(i)

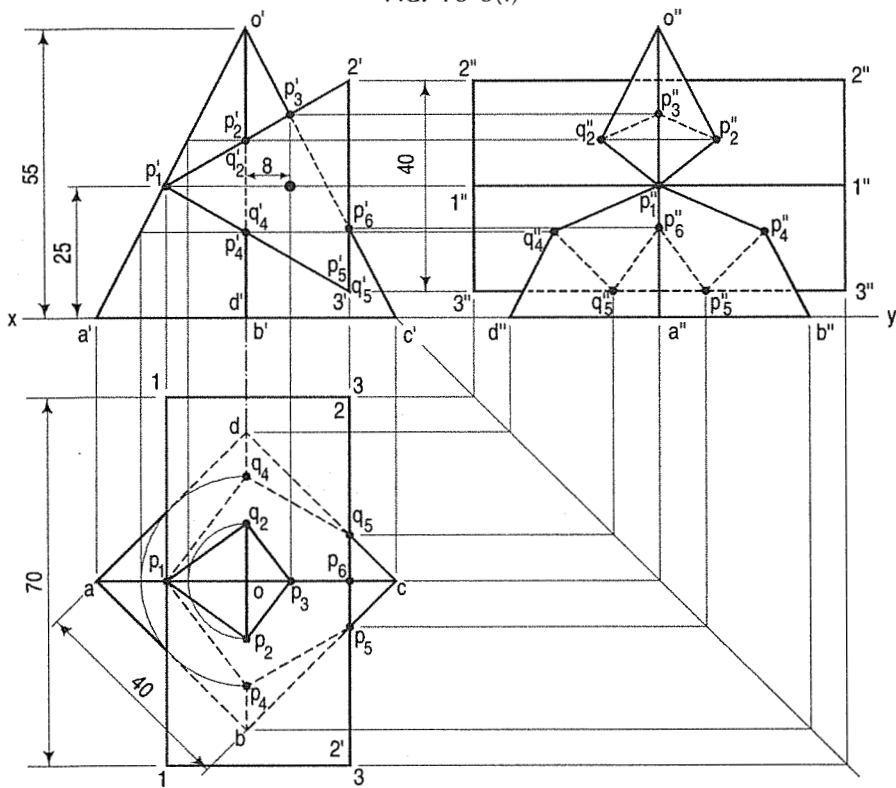


FIG. 16-6(ii)

Problem 16-7. (fig. 16-7): A square pyramid of base sides 50 mm and height 60 mm. The sides of base are equally inclined with the V.P. It is penetrated by a horizontal triangular prism of sides 30 mm and 80 mm axis long. The axes of both solids are intersecting each other. The axis of triangular prism is 22 mm above H.P. and perpendicular to the V.P. One of faces of triangular prism is perpendicular to the V.P. Draw the top view, front view and side view showing the curve of the penetration.

- (i) Draw the projections of the square pyramid and triangular prism in the required position.
- (ii) In the front view, draw lines $o'e'$, $o'b'$, $o'g'$ and $o'h'$ passing through $1'$, $2'$, $3'$ and intersecting points respectively.
- (iii) As the axes are intersecting, the edges $o'b'$ and $o'd'$ are cut by the two edges of the horizontal prism at P'_2 , q'_2 , P'_4 and q'_4 .
- (iv) The third edge of the horizontal prism enters the faces of vertical prism at P'_3 , q'_3 , P'_5 and q'_5 .
- (v) Project these points in the top view and obtain the intersecting points P_1 , P_2 , P_3 , P_4 , P_5 , q_1 , q_2 , q_3 , q_4 and q_5 .
- (vi) Take projections from front view and top view and complete side view.

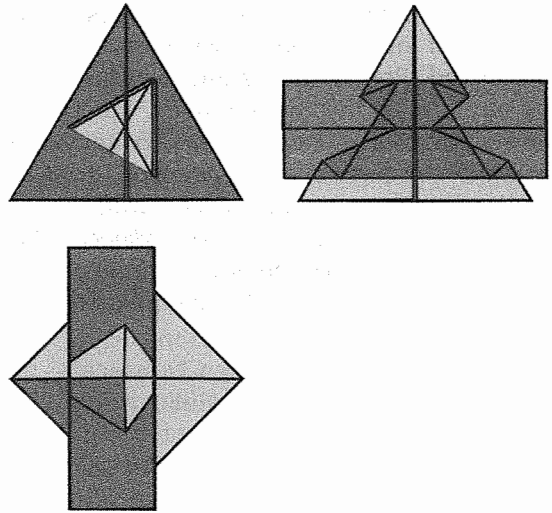


FIG. 16-7(i)

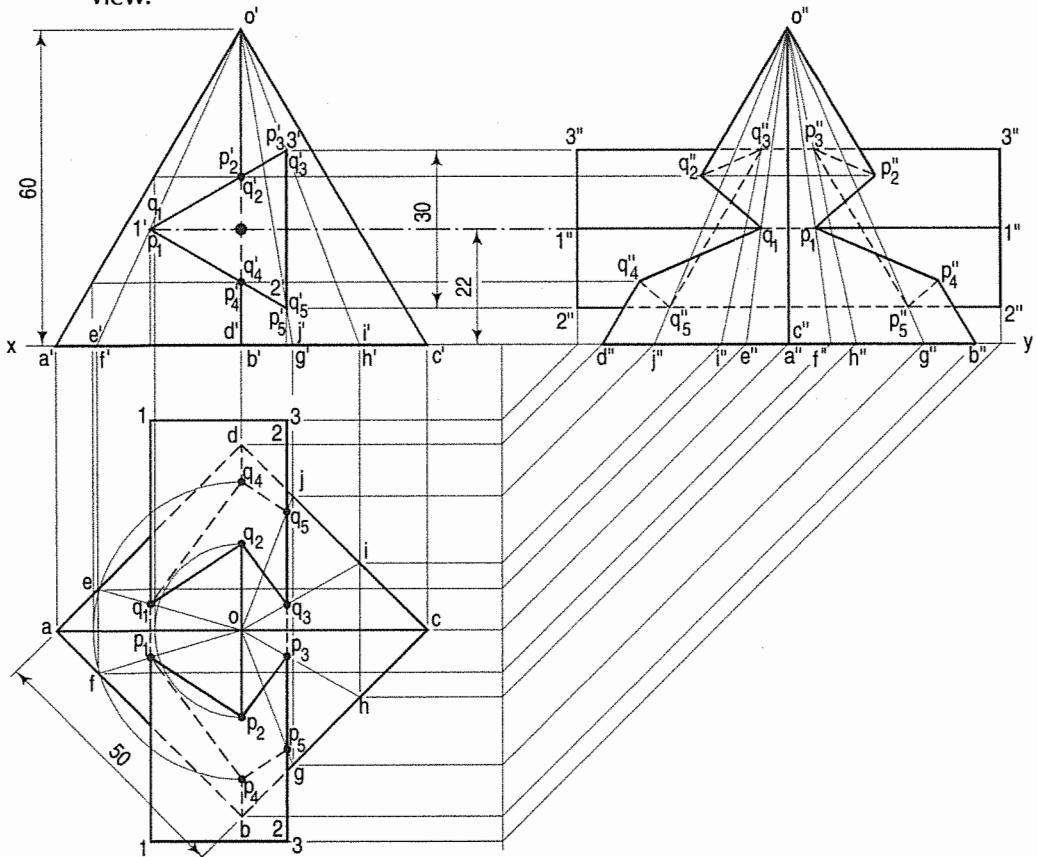


FIG. 16-7(ii)

16-4. INTERSECTION OF CYLINDER AND CYLINDER

As cylinders have their lateral surfaces curved, the line of intersection between them will also be curved. Points on this line may be located by any one of the two methods. For plotting an accurate curve, certain *critical or key points*, at which the curve changes direction, must also be located. *These are the points at which outermost or extreme lines of each cylinder pierce the surface of the other cylinder.* In prisms, vertices are the key points.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 37 for the line of intersection.

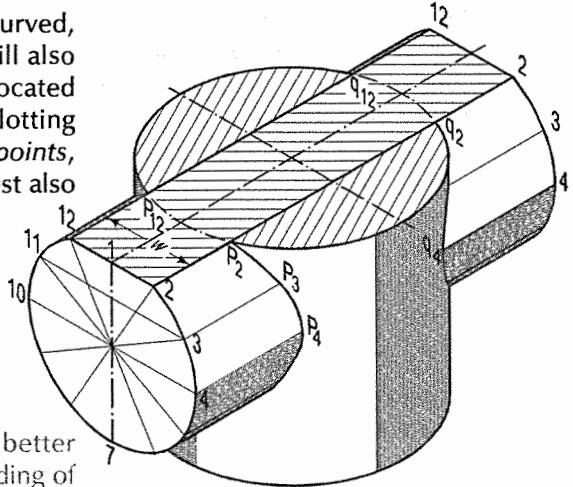


FIG. 16-8(i)

Problem 16-8. (fig. 16-8): A vertical cylinder of 80 mm diameter is completely penetrated by another cylinder of 60 mm diameter, their axes bisecting each other at right angles. Draw their projections showing curves of penetration, assuming the axis of the penetrating cylinder to be parallel to the V.P.

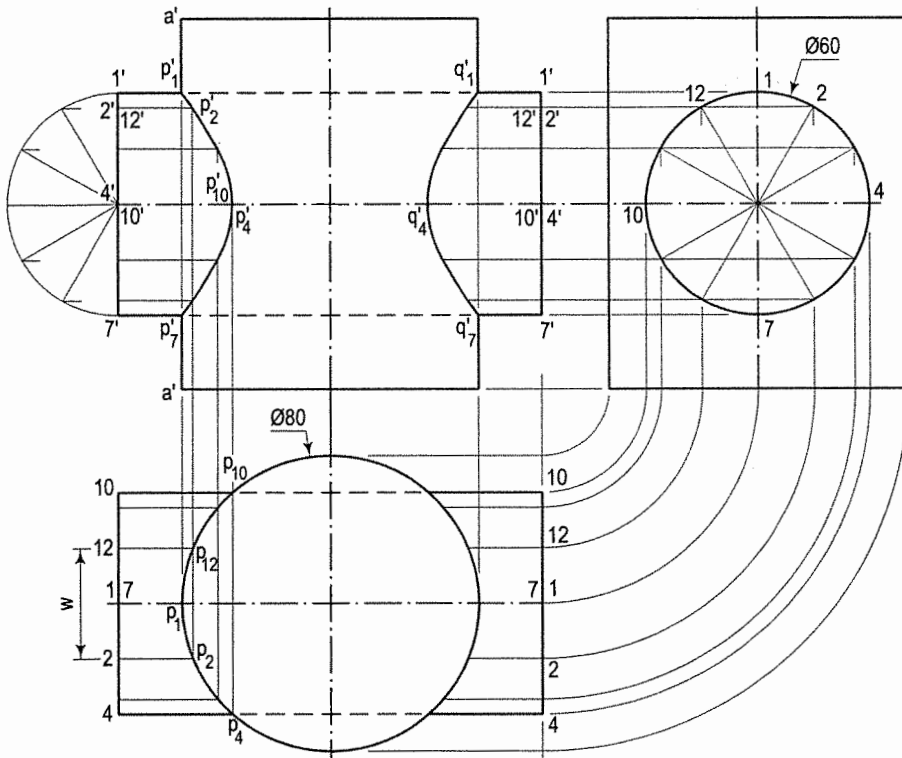


FIG. 16-8(ii)

Draw the front view and the top view and show lines for twelve generators in the horizontal cylinder in both views.

(a) **Line method:**

- (i) Mark points p_1, p_2 etc. at which lines 1-1, 2-2 etc. intersect the circle (showing the surface of the vertical cylinder) in the top view and project them to p'_1, p'_2 etc. on corresponding lines $1'1', 2'2'$ etc. in the front view.
- (ii) Draw the required curves on both sides of the axis through points thus located. Hidden portions of the curves coincide with the visible portions. Points p'_1, p'_4, p'_7 and p'_{10} are the key points where the curve changes direction.

(b) **Cutting-plane method:** It will be seen that in this problem, there is practically no difference between the line method and the cutting-plane method. But the latter method proves more useful in solving problems in which none of the projections shows a line-view of the surface of a solid. Assume a series of horizontal cutting planes passing through the lines on the horizontal cylinder and cutting both cylinders.

Sections of the horizontal cylinder will be rectangles, while those of the vertical cylinder will always be circles of the same diameter as its own. Points at which sides of the rectangles intersect the circle will lie on the curve of intersection. For example, let a horizontal section plane through points 2 and 12 [fig. 16-8(i)].

In the front view, it will be seen as a line coinciding with the line $2'2'$. The section of the horizontal cylinder will be a rectangle of width w (i.e. the line 2-12). The section of the vertical cylinder will be a circle. Points p_2 and p_{12} at which the sides (2-2 and 12-12) of the rectangle cut the circle, lie on the curve. These points are first marked in the top view [fig. 16-8(ii)] and then projected to points p'_2 and p'_{12} on lines $2'2'$ and $12'12'$ in the front view. Points on the other side of the vertical axis are located in the same manner.

The problem may also be solved by assuming cutting planes to be vertical and parallel to both axes. They will be seen as lines in the top view and the side view. Sections of both cylinders will be rectangles and will be seen in their true sizes in the front view. Points at which sides of sections of one cylinder intersect sides of corresponding sections of the other, will lie on the curve of intersection.

Problem 16-9. (fig. 16-9): A cylindrical pipe of 30 mm diameter has a similar branch of the same size. The axis of the main pipe is vertical and is intersected by that of the branch at right-angles. Draw the projections of the pipes, assuming suitable lengths, when the two axes lie in a plane parallel to the V.P. Develop the surfaces of the two pipes.

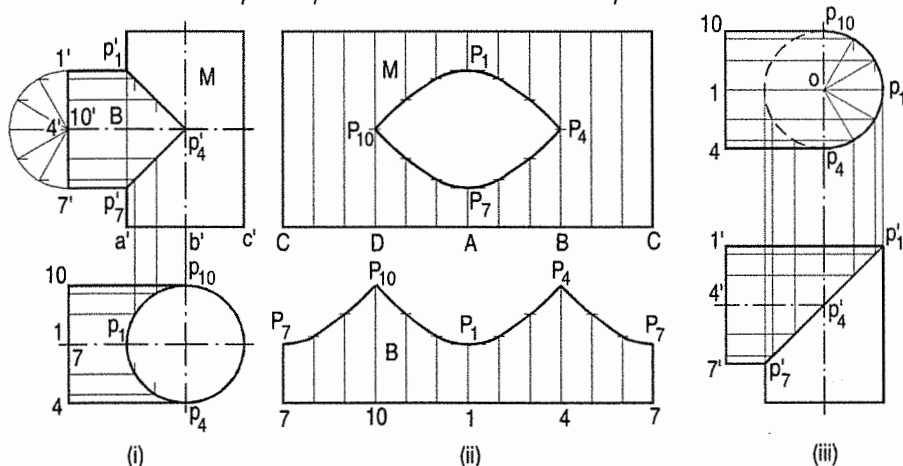


FIG. 16-9

- (iii) Project these points to e' and f' on the line $a'a'$.
- (iv) Draw the curve passing through all the points in correct sequence, showing the hidden portion by dotted lines. Plot a similar curve on the other side of the axis.

Problem 16-11. (fig. 16-11): A vertical cylinder of 75 mm diameter is penetrated by another cylinder of the same size. The axis of the penetrating cylinder is parallel to both the H.P. and the V.P. and is 9 mm away from the axis of the vertical cylinder. Draw the projections showing curves of intersection.

As the cylinders are of the same size and their axes are apart, a portion of the surface of the penetrating cylinder will be outside the vertical cylinder.

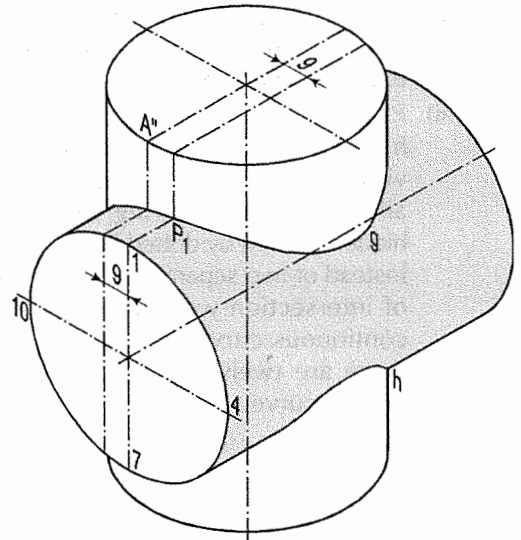


FIG. 16-11(i)

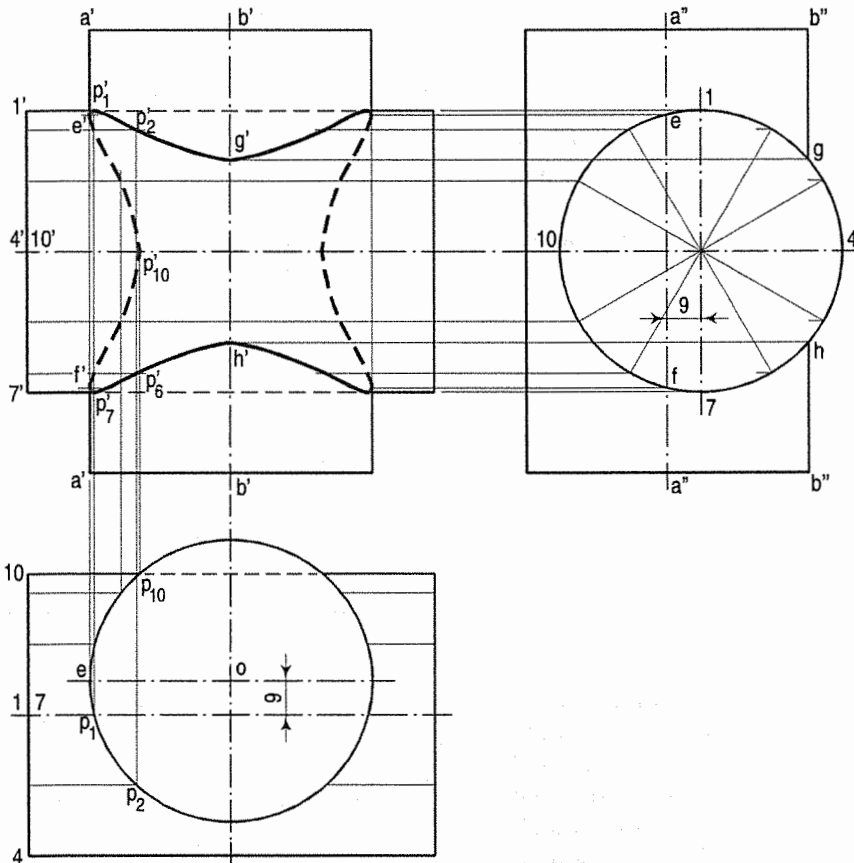


FIG. 16-11(ii)

- (i) Draw the three views and project points of intersection of lines which lie within the vertical cylinder.

- (ii) Locate key points e' and f' as shown in problem 16-10. In addition to these, mark two more key points g and h (in the side view) where the circle cuts the extreme line $b''b''$ of the vertical cylinder.
- (iii) Project these points to g' and h' on the line $b'b'$ in the front view. Draw a curve through all the points, showing the hidden portion by dotted lines. Instead of two separate curves of intersection we have one continuous curve. Note that there are twelve key points in this curve.

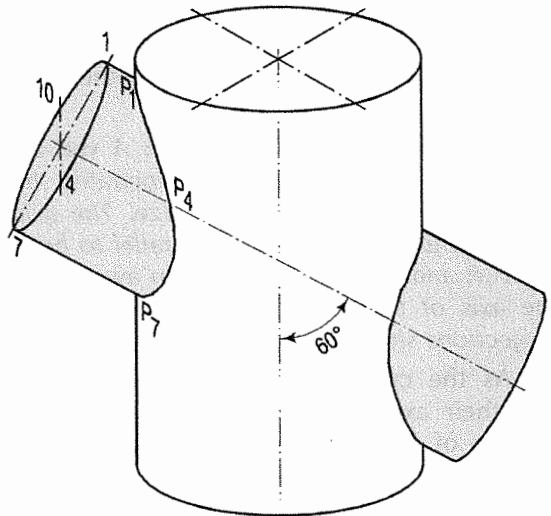


FIG. 16-12(i)

Draw the projections of the cylinders in the required position and proceed to locate the points on curves of intersection as shown in problem 16-8. The back and front curves will coincide with each other.

Problem 16-13. (fig. 16-13): A vertical pipe of 60 mm diameter has a branch of 30 mm diameter. The axis of the branch is inclined at 45° to the ground, parallel to the V.P. and is 15 mm away from the axis of the main pipe. Draw the projections of the pipes showing the curve of intersection. Also, develop the surfaces of the two pipes, assuming any lengths.

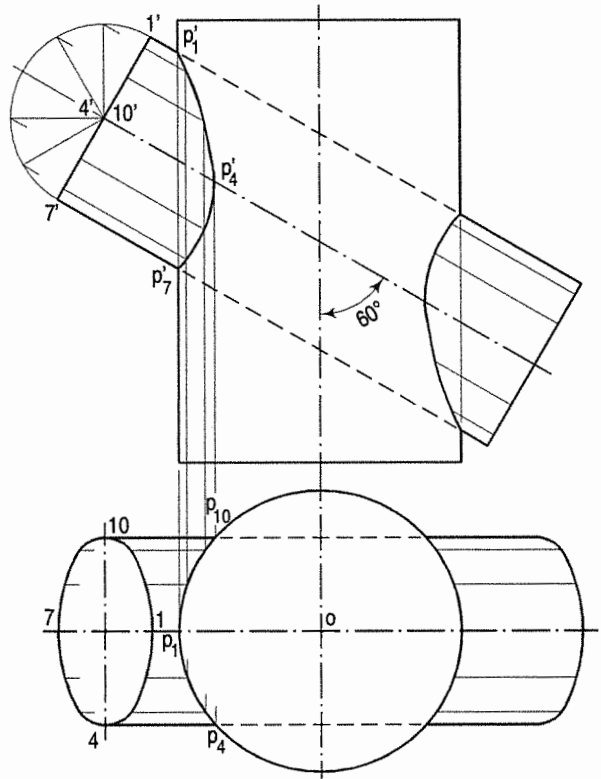


FIG. 16-12(ii)

Apply the same method as in problem 16-9. As the diameter of the branch and the distance between the two axes are respectively one-half and one-fourth of the diameter of the main pipe, one extreme line of the branch (in the top view) will be tangent to the circle. In the front view, the visible part of the curve will extend up to the centre line of the main pipe, while the hidden part will just touch the line $a'a'$.

the centre line of the main pipe, while the hidden

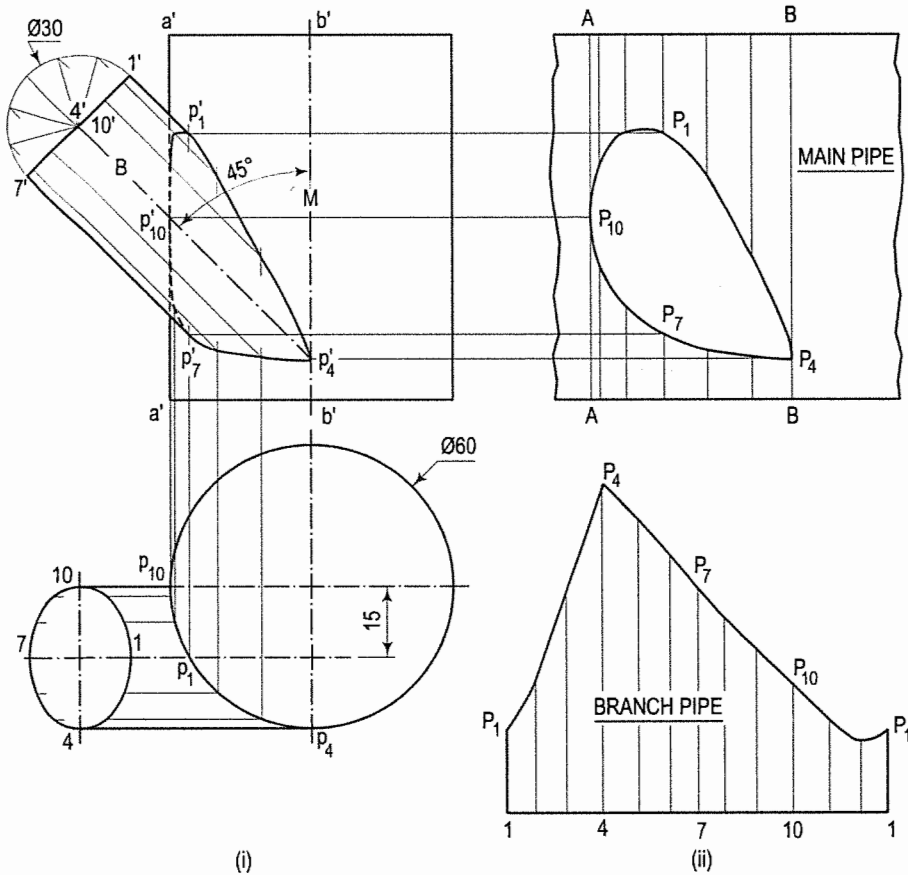


FIG. 16-13

Only a part of the development of the main pipe, just sufficient to show the shape of the hole in it, is shown in fig. 16-13(ii). Distances along the length of the stretch-out line are taken from the top view and positions of points are projected from the front view. In the development of the branch, the stretch-out line is divided into twelve equal parts and heights of points are taken from the front view, e.g. $4p_4 = 4'p'_4$ etc.

Problem 16-14. (fig. 16-14): A cylindrical pipe of 30 mm diameter has a similar branch of the same size. The axis of the branch intersects that of the main pipe at an angle of 45° . Draw the projections, when the two axes lie in a plane parallel to the V.P., and the axis of the main pipe is vertical. Also, develop the surfaces of the two pipes, assuming suitable lengths.

The curve of intersection in the front view is made up of two straight lines meeting at right-angles [fig. 16-14(i)].

Developments of surfaces of the pipes are shown in fig. 16-14(ii).

If instead of the branch, two similar pipes interpenetrate, the curves of intersection in the front view would be seen as two straight lines joining the opposite corners and bisecting each other at right-angles.

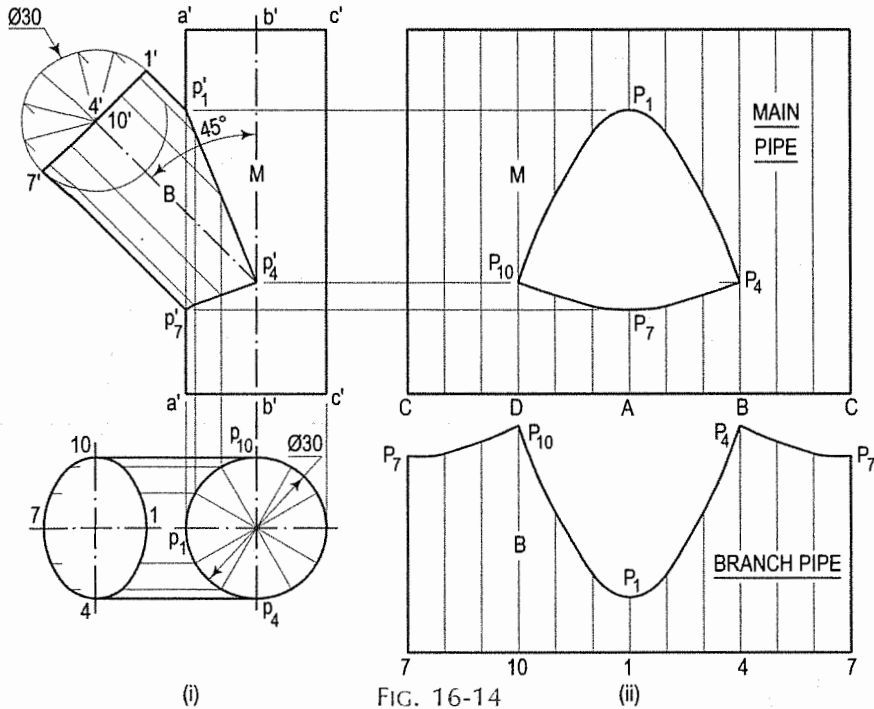
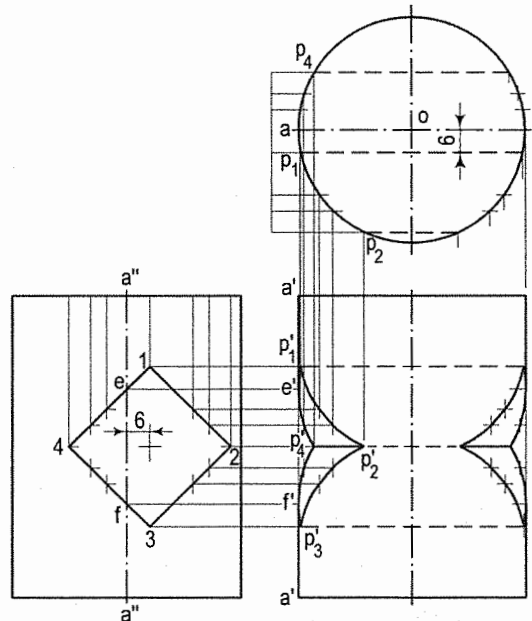


FIG. 16-14

16-5. INTERSECTION OF CYLINDER AND PRISM

Problem 16-15. (fig. 16-15): A vertical cylinder of 60 mm diameter has a square hole of 30 mm sides cut through it. The axis of the hole is horizontal, parallel to the V.P. and 6 mm away from the axis of the cylinder. The faces of the hole are equally inclined to the H.P. and the V.P. Draw the projections of the cylinder showing the hole in it.

- (i) Draw three views of the cylinder showing the lines for the hole in given position. Project all the key points, i.e. points of intersection of the edges of the hole with the surface of the cylinder, viz. p_1' , p_2' etc. and those of the extreme lines of the cylinder with the surfaces of the hole, viz. e' , f' etc. Project a few intermediate points also.
- (ii) Draw the required curve through these points on both sides of the axis. Note that the back portion of the hole is also visible. The curves from p_1' to e' and p_3' to f' bend in opposite directions.



(Third-angle projection)

FIG. 16-15

Problem 16-16. (fig. 16-16): A vertical cylinder of 60 mm diameter is penetrated by a horizontal square prism, base 40 mm side, the axis of which is parallel to the V.P. and 10 mm away from the axis of the cylinder. A face of the prism makes an angle of 30° with the H.P. Draw their projections, showing curves of intersection.

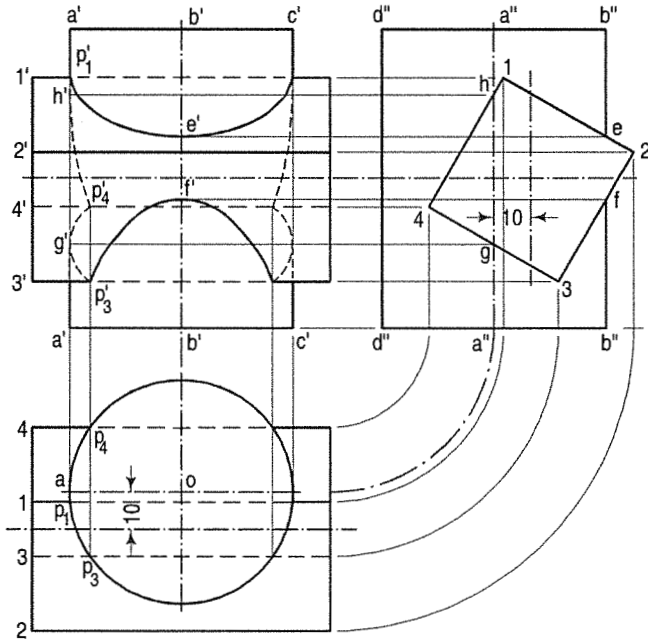


FIG. 16-16

- (i) Draw the views in the required position. One longer edge of the prism will remain outside the cylinder. Project all the key points, viz. points of intersection of the edges of the prism with the cylinder and those of the extreme lines of the cylinder with the surface of the prism, as shown in the figure.
- (ii) To obtain accurate shape of the curves, project few more intermediate points also. These are omitted from the figure. Draw the required curve of intersection through these points, taking precaution to show the hidden part by dashed lines.

Problem 16-17. (fig. 16-17):

A vertical square prism having its faces equally inclined to the V.P. is completely penetrated by a horizontal cylinder, the axis of which is parallel to the V.P. and 6 mm away from that of the prism. Draw the projections of the solids showing curves of intersection. The length of the sides of the base of the prism is 50 mm and the diameter of the cylinder is 40 mm.

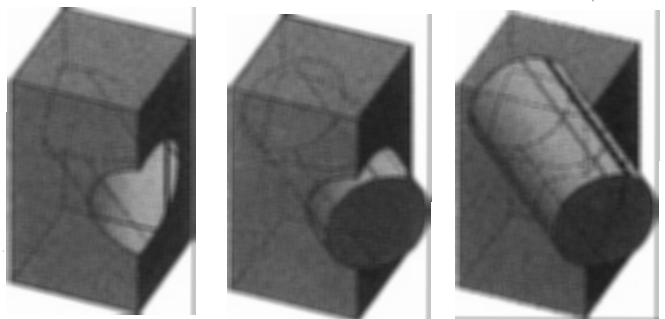


FIG. 16-17(i)

Obtain the twelve points as explained in problem 16-10. In addition, project the key points e' and f' and draw the curves as shown in the figure.

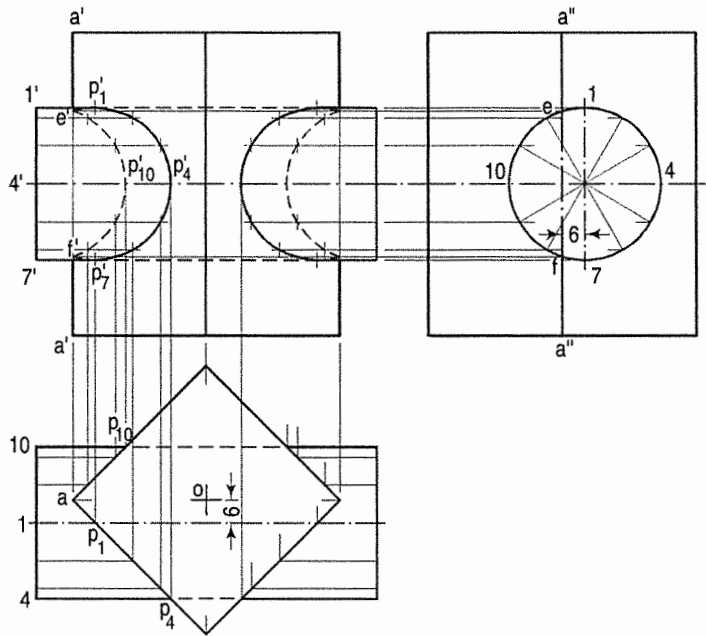
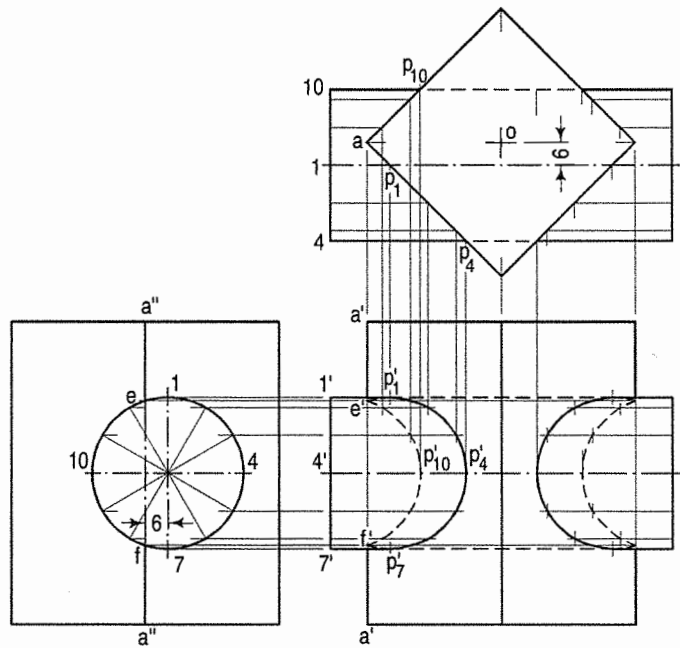


FIG. 16-17(ii)



(Third-angle projection)

FIG. 16-17(iii)

Problem 16-18. (fig. 16-18): A vertical square prism, base 50 mm side, has a face inclined at 30° to the V.P. It has a hole of 65 mm diameter drilled through it. The centre line of the hole is parallel to both the H.P. and the V.P. and is 5 mm away from the axis of the prism. Draw the projections of the prism.

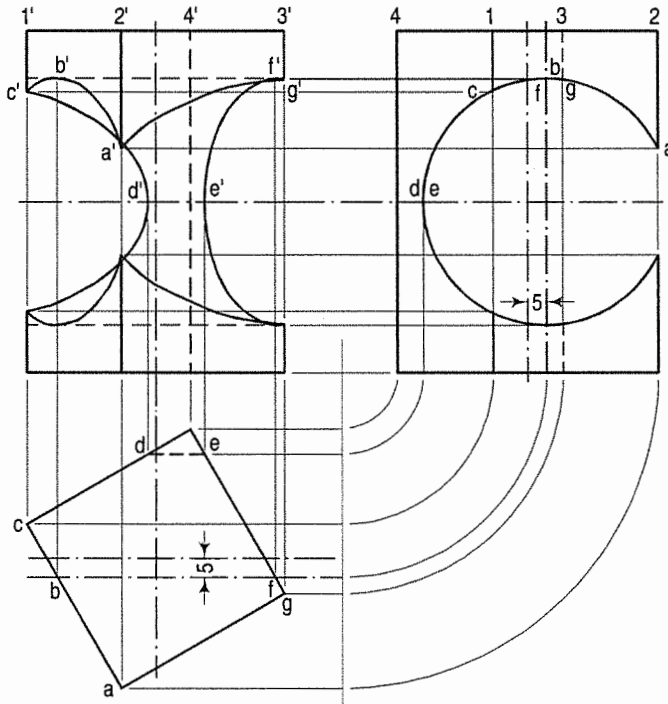


FIG. 16-18

- (i) Draw the views and project all the key points as shown. Plot few intermediate points (between the key points) also.
- (ii) Draw a curve joining the points in correct sequence. As it is a hole cut in the prism, the curve at the back will also be visible. Only a small portion of the curve will not be seen.

Problem 16-19. (fig. 16-19): A connecting rod, 50 mm diameter, has a rectangular block 65 mm wide and 25 mm thick, forged at its end. The rod joins the block with a turned radius of 25 mm. Draw the projections of the rod showing curve of intersection.

The rod increases in diameter as it approaches the block. This forms what is called a fillet of 25 mm radius. As the width of the block is not as large as the biggest diameter of the rod, a curve of intersection is formed. Points on the curve are found as shown below.

- (i) Assume a horizontal cutting plane passing through a line, say $a'a'$. The section of the rod is a circle (see the top view) which cuts the sides of the block at points b .
- (ii) Project these points to points b' on the line $a'a'$. Then the points b' lie on the curve. Obtain more points by assuming additional sections. The highest point on the curve will be on the line where the diameter of the section of the rod is equal to the width of the block.

Fig. 16-20 shows the shape of the curve when the diameter of the rod is equal to the width of the block. In this case, the highest point on the curve is on the horizontal line through the centre of the arc for the fillet.

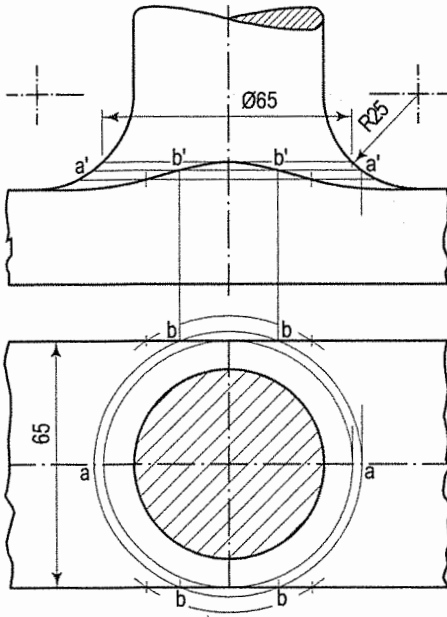


FIG. 16-19

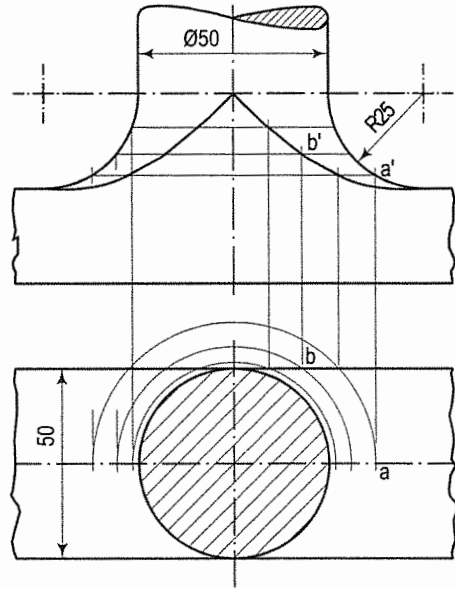


FIG. 16-20

Problem 16-20. Draw the front view and the side view of the forked end of the connecting rod shown in fig. 16-21, showing the curve of intersection.

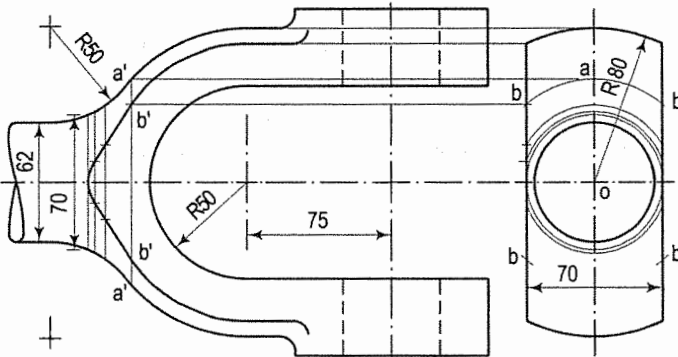
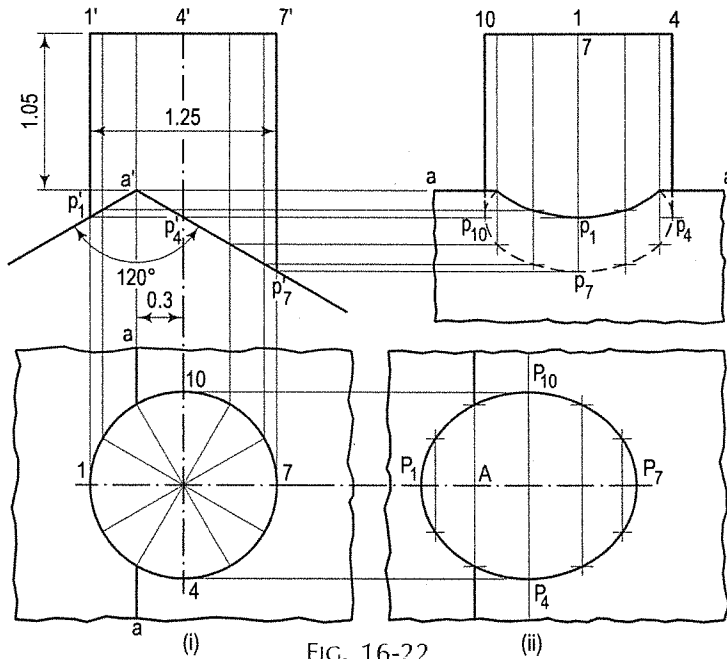


FIG. 16-21

In this case, a cylindrical rod meets a forked piece, the sides of which are flat surfaces, while the top and bottom surfaces are cylindrical.

- (i) Assume a series of section planes perpendicular to the axis of the rod. Points at which circles for the sections intersect the flat faces of the fork, lie on the curve.
- (ii) Take a section plane say at $a'a'$.
- (iii) With centre o and radius equal to one-half of $a'a'$, draw arcs cutting the sides of the fork at points b (in the side view). Project these points to points b' on $a'a'$. Obtain other points in the same manner and draw the required curve through them.

Problem 16-21. Projections of a steel chimney erected on a roof, are given in fig. 16-22(i). Project its side view showing the curve of intersection. Determine the real shape of the hole in the roof. (All dimensions are in metres.)



- (i) Draw the given views and project the side view.
- (ii) To obtain points on the curve, draw lines on the surface of the chimney by dividing the circle (in the top view) into twelve equal parts. Project points at which these lines intersect the two lines for the roof in the front view to corresponding lines in the side view.
- (iii) Draw a curve through the points thus obtained. The upper part of the curve will be hidden.

The real shape of the hole in the roof is shown above the side view. Horizontal distances are taken from the front view and widths are projected from the top view.

16-6. INTERSECTION OF CONE AND CYLINDER

Problem 16-22. A vertical cone, diameter of base 75 mm and axis 100 mm long, is completely penetrated by a cylinder of 45 mm diameter. The axis of the cylinder is parallel to the H.P. and the V.P. and intersects the axis of the cone at a point 28 mm above the base. Draw the projections of the solids showing curves of intersection.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 38 for the cutting-plane method.

- (a) **Cutting-plane method** (fig. 16-23 and fig. 16-24):
- (i) Draw lines dividing the surface of the cylinder into twelve equal parts.
 - (ii) Assume a horizontal cutting plane passing through say, point 2 (fig. 16-23). The section of the cylinder will be a rectangle of width w (i.e. the line 2-12), while that of the cone will be a circle of diameter ee . These two sections intersect at points p_2 and p_{12} . These sections are clearly indicated in the top view by the rectangle 2-2-12-12 and the circle of diameter ee (fig. 16-24).

In the front view, the cutting plane is seen as a line coinciding with $2'2'$. Points p_2 and p_{12} when projected on the line $2'2'$ (with which the line $12'-12'$ coincides) will give a point p'_2 (with which p'_{12} will coincide). Then p'_2 and p'_{12} are the points on the curve of intersection.

(iii) To obtain the points systematically, draw circles with centre o and diameters dd , ee , ff etc. cutting lines through 1, 2 and 12, 3 and 11 etc. at points p_1 , p_2 and p_{12} , p_3 and p_{11} etc.

(iv) Project these points to the corresponding lines in the front view. Two more key points at which the curve changes direction must also be located. Their positions are determined from the side view. They are the points of nearest approach viz. m'' and n'' at which, lines drawn from the centre of the circle (i.e. the axis of the cylinder) and perpendicular to the extreme generators of the cone, cut the circle.

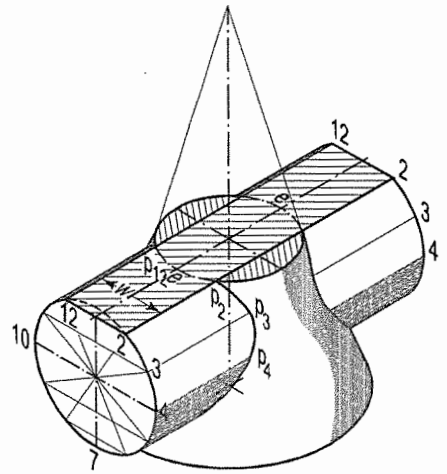


FIG. 16-23

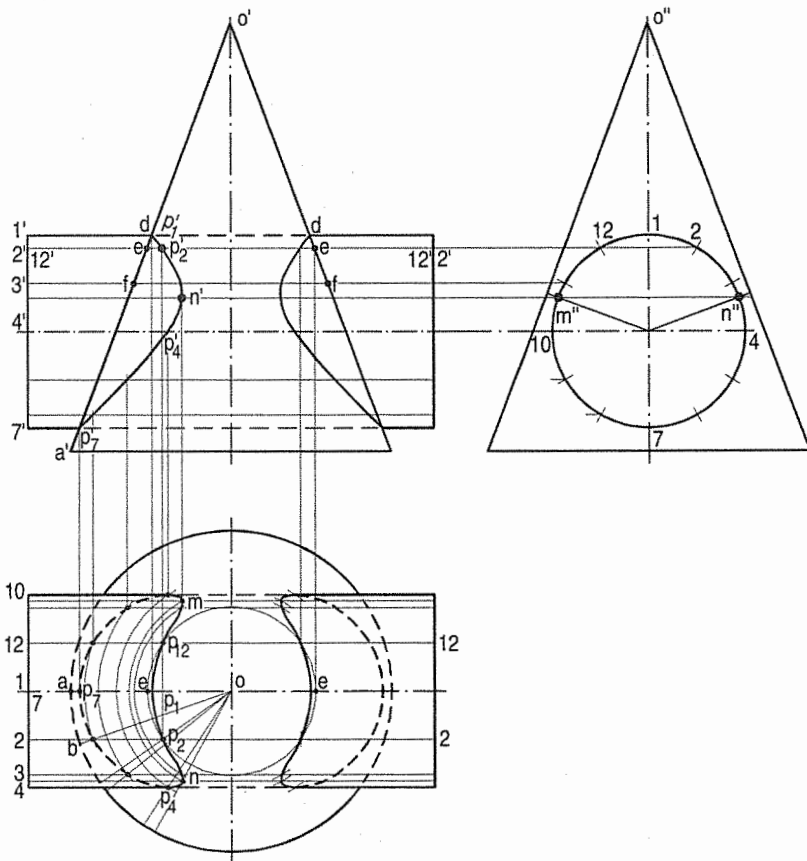


FIG. 16-24

- (v) Project these points to m' and n' in the front view and to m and n in the top view on the corresponding lines. Draw curves through these points in both the views. The back curve in the front view will coincide with the front curve. In the top view a part of the curve will lie hidden and hence, it will be dotted. Draw similar curves on the right-hand side of the axis of the cone.

(b) **Line method** (fig. 16-25): The surface of the cylinder is seen as a circle in the side view.

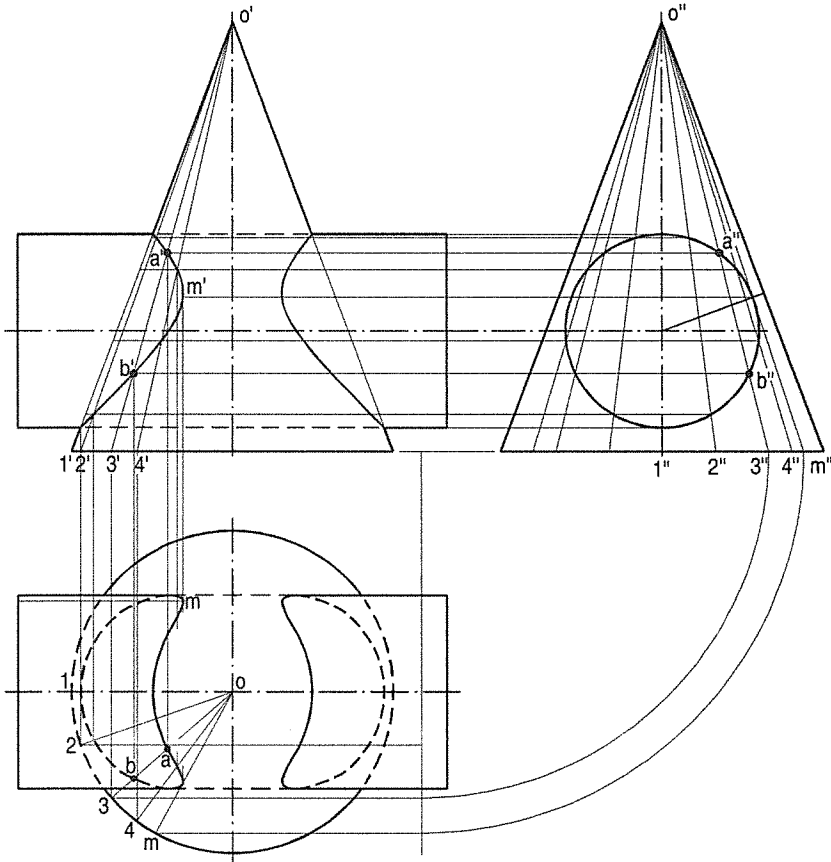


FIG. 16-25

- (i) Hence, draw a number of lines (representing generators of the cone) $o''1''$, $o''2''$ etc. in the region of the circle and symmetrical on both sides of the axis. Points where these lines intersect the circle, lie on the curve of intersection.
- (ii) To project them in the front and top views, first project the lines in both the views and then, locate the positions of these points on them. Let us take the line $o''3''$ in the side view. Locate its position $o3$ in the top view as shown and project its front view $o'3'$. Project points a'' and b'' to points a' and b' on $o'3'$ and from there, to a and b on $o3$.
- (iii) Project all points in the same manner and draw the required curves through them. This method is a type of cutting-plane method in which cutting planes pass through the apex and are parallel to the axis of the cylinder.

Problem 16-23. (fig. 16-26): A vertical cone, base 80 mm diameter, axis 100 mm long, is penetrated by a horizontal cylinder of 40 mm diameter, the axis of which is 25 mm above the base of the cone, parallel to the V.P. and 6 mm away from the axis of the cone. Draw the projections, showing curves of intersection.

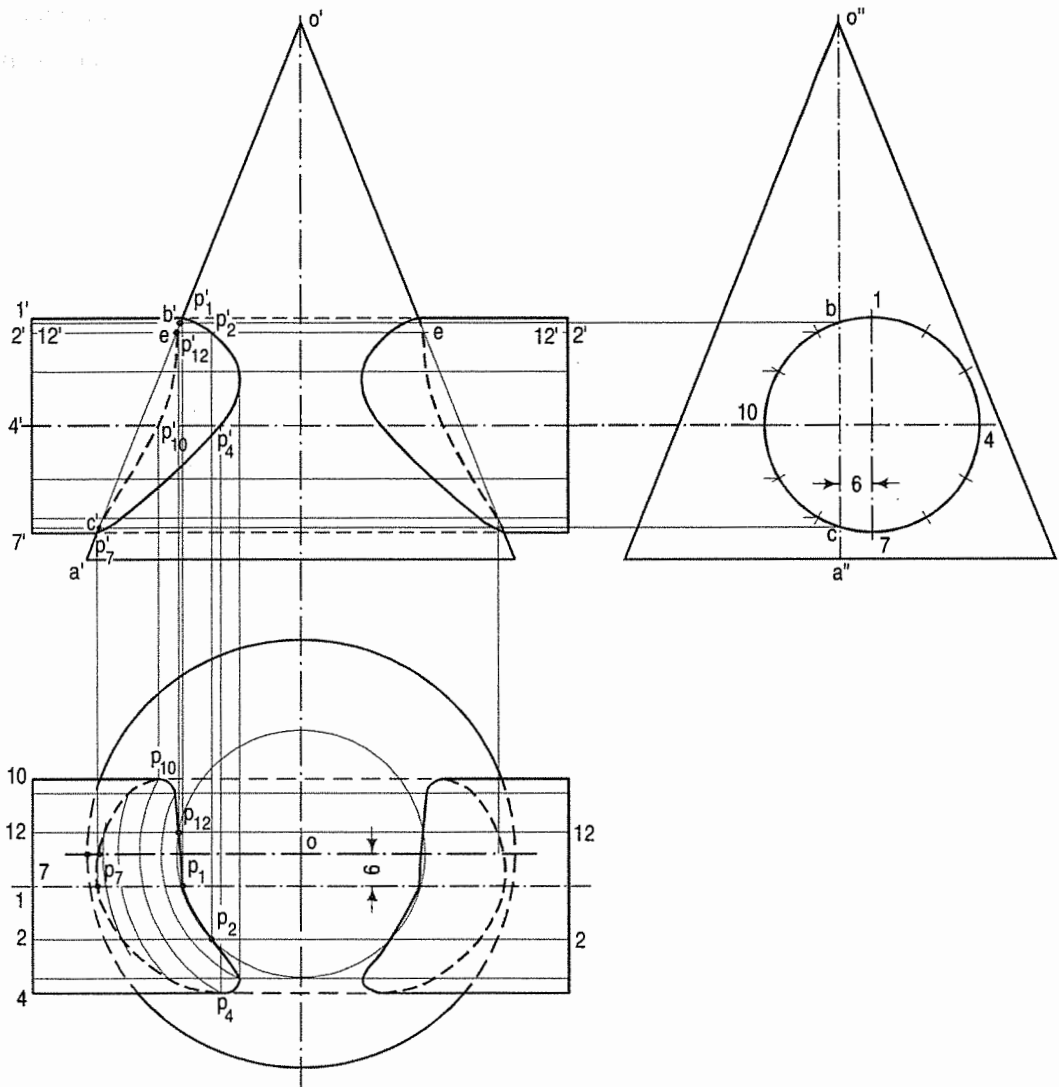


FIG. 16-26

Draw the three views of the solids.

- (i) Assuming a horizontal cutting plane through the line $2'2'$, draw a circle in the top view with centre o and diameter ee , cutting lines $2-2$ and $12-12$ at points p_2 and p_{12} .
- (ii) Project these points to p'_2 and p'_{12} on the line $2'2'$ (with which $12'-12'$ coincides). Obtain other points and the key points b' and c' in the same manner and draw the curves as shown. As the axes do not intersect, the back curves in the front view are different from the front curves.

Problem 16-24. (fig. 16-27): A vertical cone, base 80 mm diameter and axis 110 mm long is penetrated by a horizontal cylinder, 45 mm diameter. The axis of the cylinder is 25 mm above the base of the cone, is parallel to the V.P. and is 10 mm away from the axis of the cone. Draw the projections of the solids showing curves of intersection.

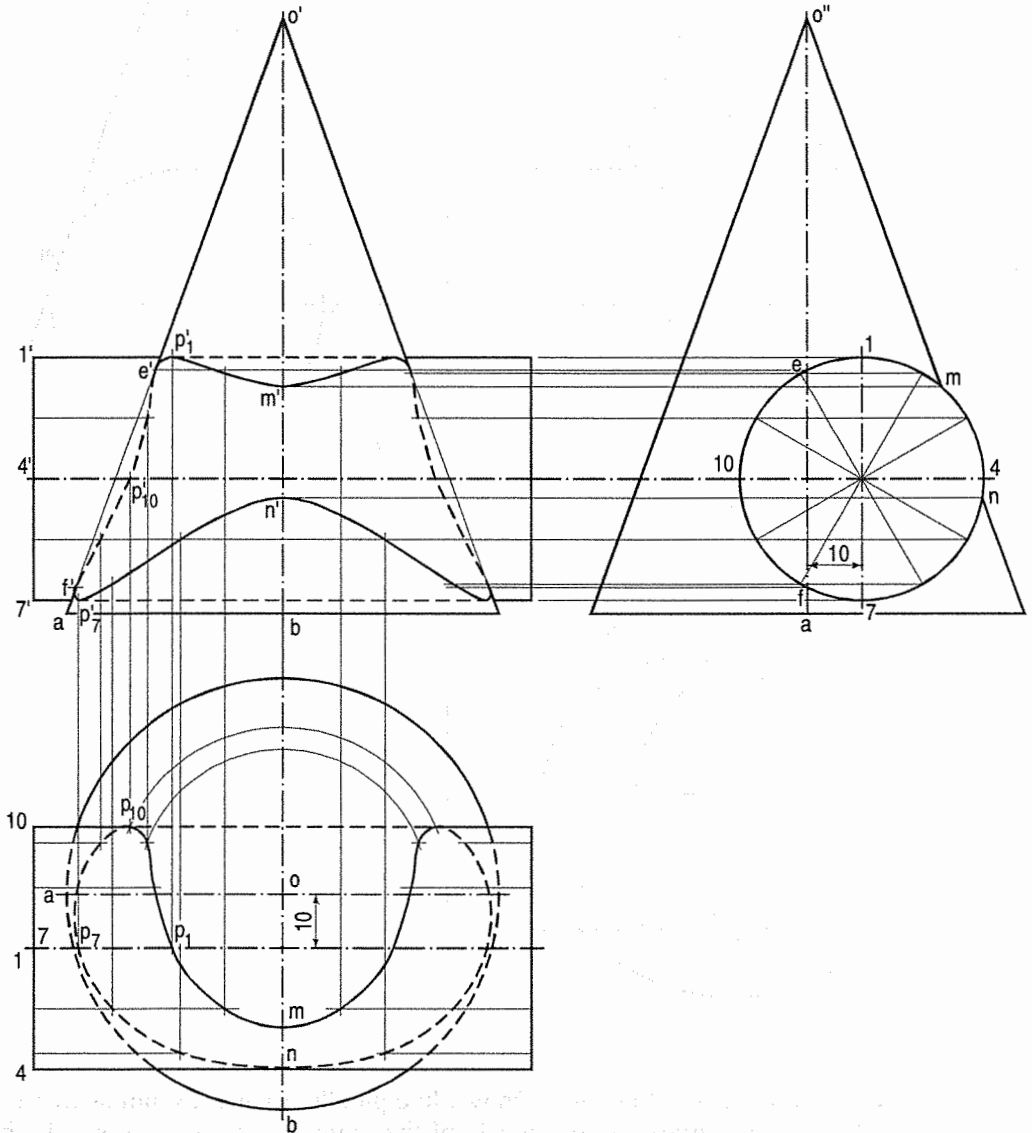


FIG. 16-27

The axis of the penetrating cylinder is so displaced that a portion of the cylinder remains outside the cone. The curve changes direction at key points m' and n' and becomes a single continuous curve.

Problem 16-25. (fig. 16-28): A vertical cone, base 75 mm diameter and axis 110 mm long is penetrated by a horizontal cylinder of 50 mm diameter in such a way that both the solids envelope an imaginary common sphere and their axes intersect each other. Draw the projections of the solids when their axes lie in a plane parallel to the V.P.

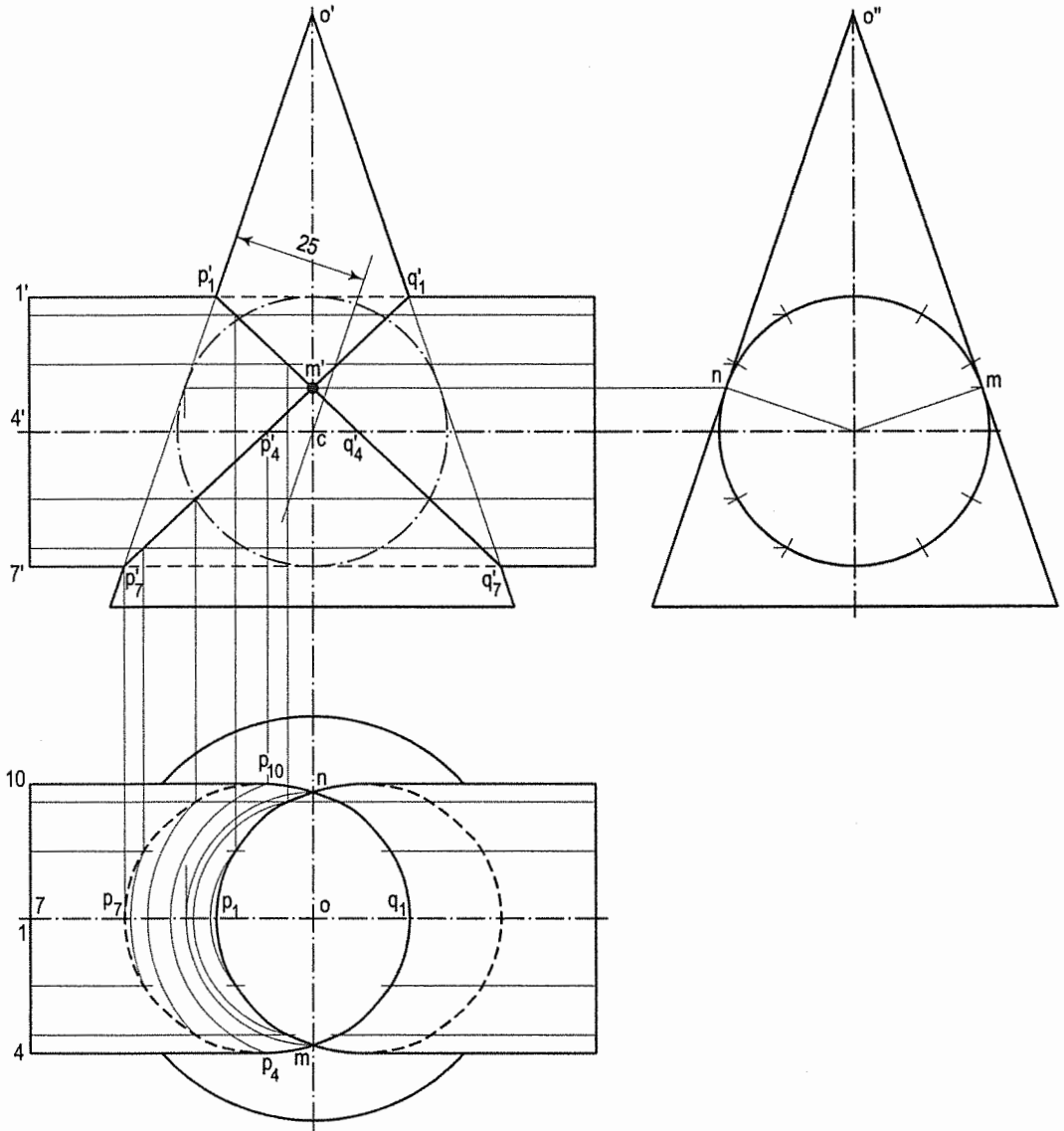


FIG. 16-28

- (i) Draw the front view of the cone. Draw a line parallel to and 25 mm away from an end generator, intersecting the axis of the cone at a point c . Then c is the centre of the circle for the common sphere. Through c , draw the horizontal axis and complete the front view of the cylinder.
- (ii) Draw the top and side views and determine the lines of intersection as explained in problem 16-22. In the front view they appear as straight lines intersecting each other at right-angles.

Problem 16-26. (fig. 16-29): A funnel is made of two constituent parts: (i) a cylindrical pipe and (ii) a conical part, both enveloping a common sphere of 40 mm diameter, with their axes intersecting at right-angles. The diameter at the mouth of the funnel is 80 mm and

the total height of the funnel is 60 mm. Draw projections showing the curve of intersection when it is placed with its mouth on the ground and the two axes parallel to the V.P.

Draw the projection of the two parts. To form the funnel, the pipe must be extended beyond the centre line of the conical part. The points of intersection of the upper part of the pipe with the cone will therefore be on the right-hand side of the centre line.

The curve of intersection will be seen as a straight line in the front view and will be elliptical in the top view.

Problem 16-27. A vertical cone, diameter of base 75 mm and axis 90 mm long is penetrated by a cylinder of 50 mm diameter, the axis of which is parallel to and 10 mm away from that of the cone. Draw the projections showing curves of intersection, when (i) the plane containing the two axes is parallel to the V.P.; (ii) the plane containing the two axes is inclined at 45° to the V.P.

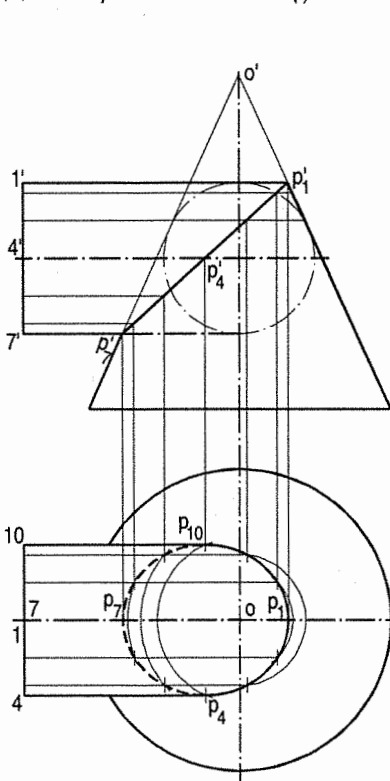


FIG. 16-29

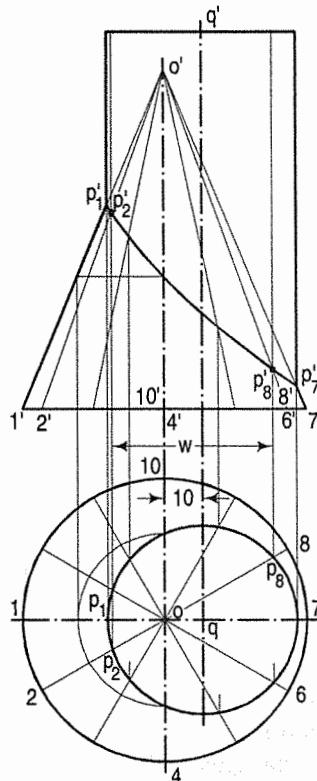


FIG. 16-30

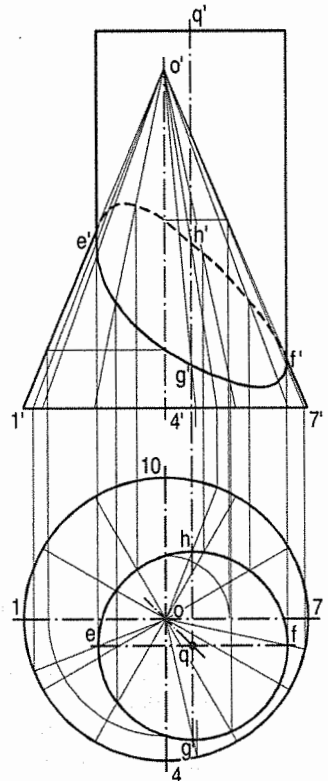


FIG. 16-31

- (i) Draw the projections (fig. 16-30). The centre of the circle for the cylinder (in the top view) will lie on the horizontal centre line and 10 mm away from the centre of the circle for the cone. Divide the base-circle of the cone into twelve equal parts and draw twelve generators in both the views. The surface of the cylinder is seen as a circle in the top view. It cuts the lines on the surface of the cone at points p_1, p_2 etc. Project these points to p'_1, p'_2 etc. on the corresponding lines in the front view. Draw the required curve of intersection through these points. The curve at the back will coincide with the front curve.

The cutting-plane method is exactly similar. Assume a series of vertical cutting planes passing through the apex. The sections of the cone will be triangles and those of the cylinder will be rectangles. Points of intersection between these sections will lie on the curve. For example, take a cutting plane coinciding with the line 2-8. In the front view, the section of the cone will be shown by triangle $o'2'8'$, while that of the cylinder will be a rectangle of width w . Points p'_2 and p'_8 at which lines $o'2'$ and $o'8'$ cut the sides of the rectangle, lie on the curve of intersection.

- (ii) As the plane containing the two axes makes an angle of 45° with the V.P., the centre q of the circle for the cylinder will lie on a line drawn through o , inclined at 45° to xy and 10 mm away from o (fig. 16-31).

Draw the projections and adopt the same method as in the case (i). In addition to the twelve points, four key points e, f, g and h (where the extreme lines of the cylinder intersect the cone) must be located. Plot the curve which will be partly hidden. If a hole is drilled through the cone instead of a cylinder penetrating it, the curve will be fully visible.

Problem 16-28. (fig. 16-32): A vertical cylinder of 75 mm diameter is penetrated by a cone, base 75 mm diameter and axis 110 mm long, the two axes bisecting each other at right-angles. Draw the front view showing lines of intersections.

- (i) Divide the base circle of the cone into twelve equal parts and draw lines for the generators in both views. Assume a series of cutting planes perpendicular to the V.P. and passing through the apex. They will be seen as lines in the front view. The sections of the cone will be triangles and those of the cylinder will be ellipses.

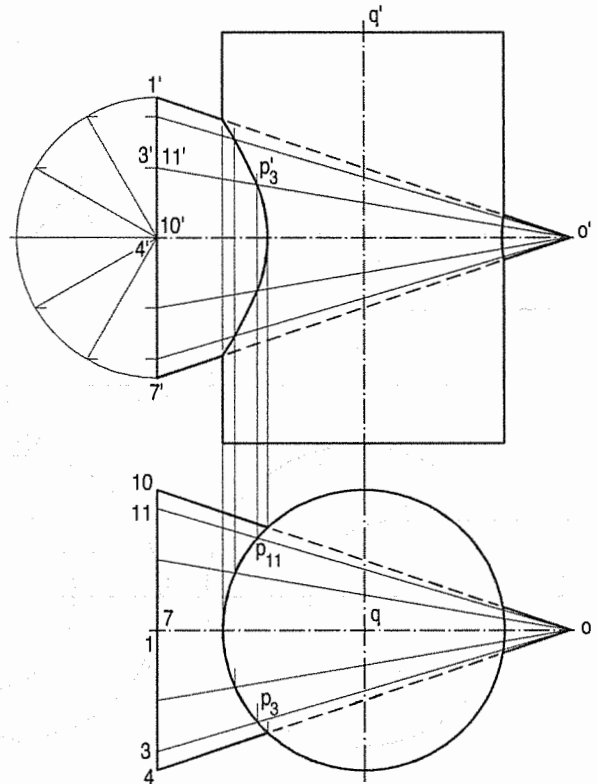


FIG. 16-32

Take a cutting plane coinciding with a line 3'o'. In the top view, the section of the cone is shown by the triangle $o-3-11$ and that of the cylinder by the circle (the top view of the cylinder). The sides of the triangle intersect the circle at points p_3 and p_{11} .

- (ii) Project these points in the front view to p_3 on the line 3'o' with which 11'o' coincides. Obtain other points in the same manner and draw a curve through them. Points on the curve on the right-hand side of the axis of the cylinder are obtained simultaneously and in the same manner.

16-7. INTERSECTION OF CONE AND PRISM



Problem 16-29. (fig. 16-33): Draw an equilateral triangle of 100 mm side with one side horizontal. Draw a square of 35 mm side in its centre with its sides inclined at 45° to the base of the triangle. The figure shows the front view of a cone standing on its base on the ground and having a square hole cut through it. Draw three views of the cone.

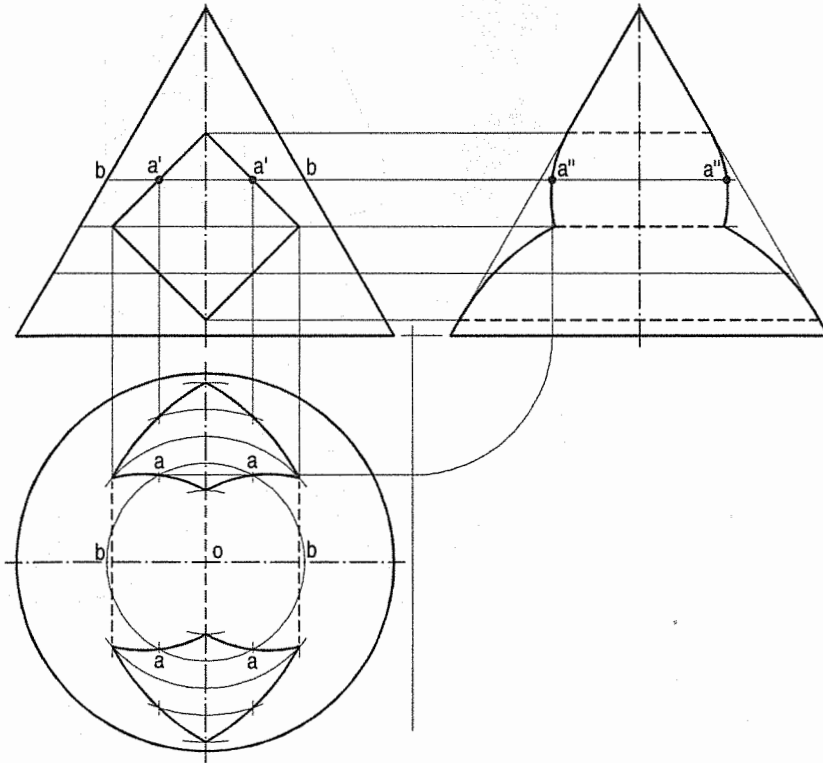


FIG. 16-33

- (i) Project the top and side views from the given view. Assume the cone as cut by a horizontal cutting plane passing through points a' . The section of the cone will be a circle of diameter bb . The hole will be cut in two straight lines through points a' and perpendicular to the V.P.
- (ii) Therefore, with centre o (the apex in the top view) and diameter bb , draw a circle cutting the projectors through a' at points a . Assume additional cutting planes, particularly those which pass through corners of the square and find other points. Draw curves through these points.

The method of locating points a'' in the side view is clearly indicated by construction lines. Obtain all points in the same manner.

Problem 16-30. (fig. 16-34): A vertical cone, base 80 mm diameter and axis 90 mm long, is penetrated by a square prism of base 35 mm side. The axis of the prism is parallel to and 12 mm away from that of cone. Draw the projections when the plane containing the two axes is parallel to the V.P. and the faces of the prism are equally inclined to the V.P.

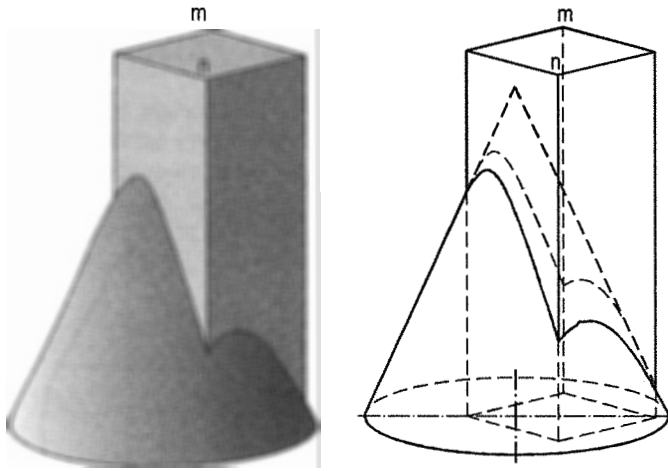


FIG. 16-34(i)

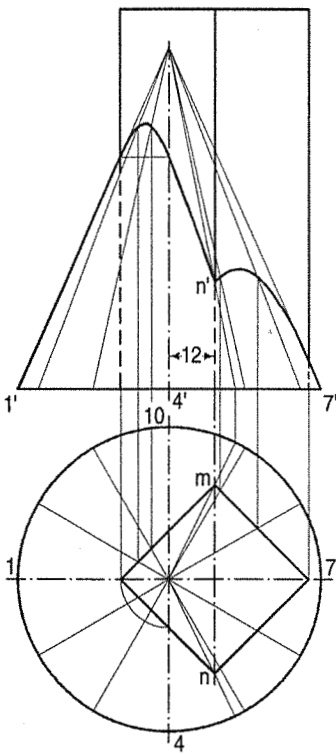
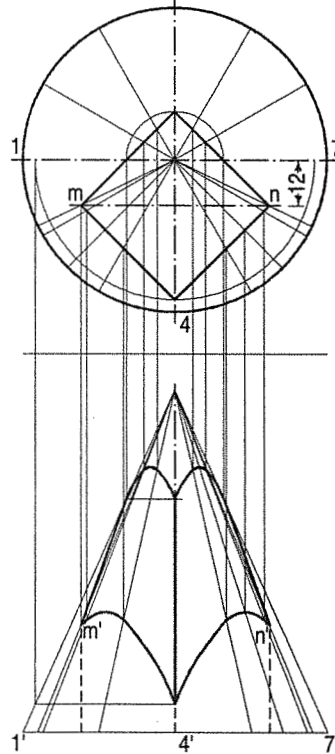
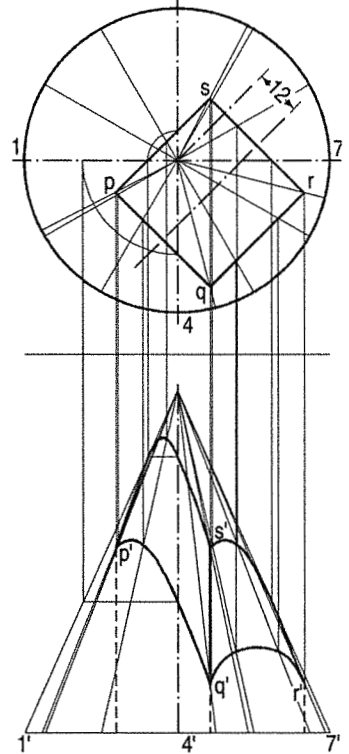


FIG. 16-34(ii)



(Third-angle projection)

FIG. 16-35



(Third-angle projection)

FIG. 16-36

Adopt the same method as shown in problem 16-27.

Fig. 16-35 and fig. 16-36 show two views of a cone having a square hole cut through it. The axes of the cone and the hole are vertical and 12 mm apart. The faces of the hole are equally inclined to the V.P. In fig. 16-35, the vertical plane containing the two axes is perpendicular to the V.P., while in fig. 16-36 it is inclined at 45° to the V.P.

16-8. INTERSECTION OF CONE AND CONE

Problem 16-31. (fig. 16-37): A vertical cone, base 90 mm diameter and axis 90 mm long, is penetrated by another cone of base 75 mm diameter and height 100 mm. The axes of the two cones bisect each other at right-angles. Draw the front view showing curves of intersection, when the axis of the penetrating cone is parallel to the V.P.

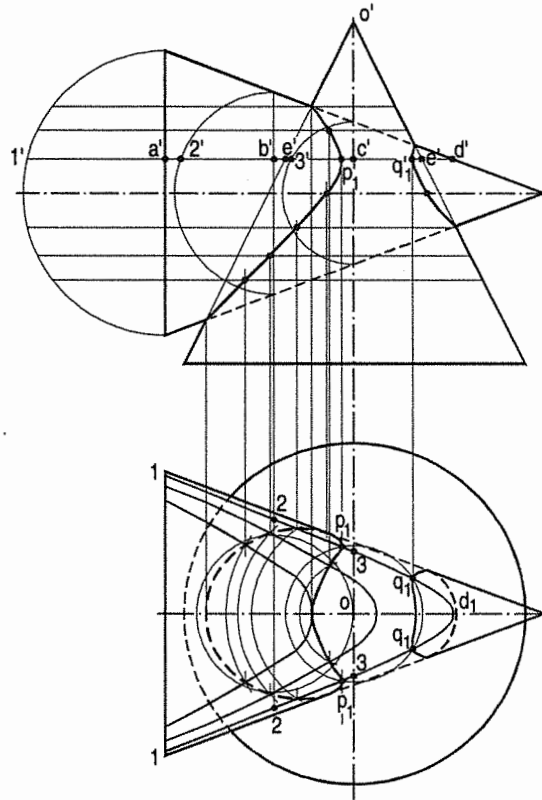


FIG. 16-37

- (i) Assume a horizontal cutting plane coinciding with a line $a'd'$. The section of the vertical cone will be a circle of diameter ee . The section of the horizontal cone will be a hyperbola. The width of the hyperbola at the point a' will be equal to twice the length of the line $a'1'$; that at the point b' will be twice $b'2'$ etc.
- (ii) In the top view, mark points on the projector through a' and symmetrical on both sides of the horizontal axis, so that $1-1 = \text{twice } a'1'$. Similarly, mark points on the projector through b' so that $2-2 = \text{twice } b'2'$ etc. Draw the hyperbola through the points thus obtained.
- (iii) Draw a circle with centre o and radius $c'e'$, cutting the hyperbola at points p_1 and q_1 . Project p_1 to p'_1 and q_1 to q'_1 on the cutting-plane line $a'd'$.

Then p_1 , q_1 , p'_1 and q'_1 are the points on the curves in the two views. Similarly, assume additional sections, preferably at equal distances on both sides of the axis (so that the sections will be the same). Draw the hyperbolas and circles in the top view and determine the points of intersection. Draw curves in both the views and on both sides of the vertical axis. The curves can be determined in the same manner even when the axes do not intersect.

16-9. INTERSECTION OF SPHERE AND CYLINDER OR PRISM



Problem 16-32. (fig. 16-38): A hole of 50 mm diameter is drilled through a sphere of 75 mm diameter. The axis of the hole is 10 mm away from the centre of the sphere. Draw three views of the sphere when a vertical plane containing the centre of the sphere and the axis of the hole is inclined at 60° to the V.P.

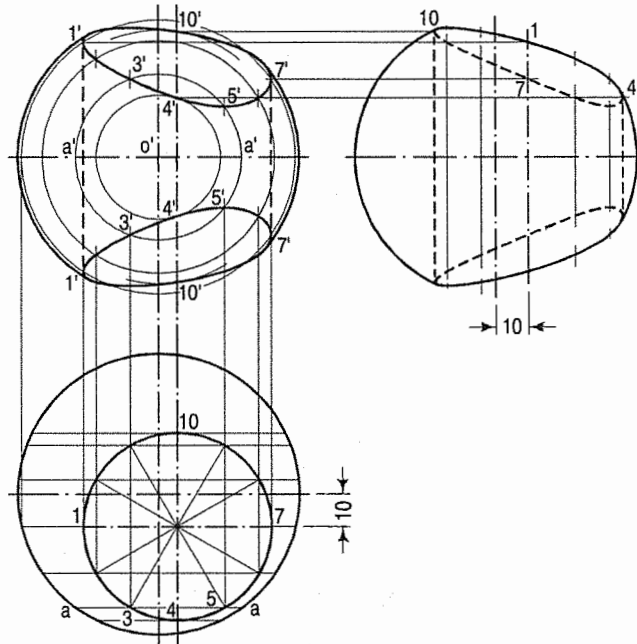


FIG. 16-38

- (i) Draw the top view of the sphere and show the circle for the hole in the required position.
- (ii) Project the front view and the side view. Divide the circle for the hole into twelve equal parts.
- (iii) Assume a number of cutting planes parallel to the V.P. The sections of the sphere will be circles, while the hole will be cut in straight lines. Intersection of the circles with corresponding lines will give points on the curves.
- (iv) Let us take a section plane passing through points 3 and 5. The section of the sphere will be a circle of diameter aa . The lines through 3 and 5 intersect this circle at points $3'$ and $5'$ which lie on the curve. Obtain the twelve points in the same manner. Also plot the key points at which the section through the centre of the sphere is cut by the lines of the hole. Draw curves through all the points. Project the points on the side view horizontally from the front view on corresponding lines of the hole and draw curves through them.

Problem 16-33. (fig. 16-39): A sphere of 80 mm diameter is penetrated by a square prism, base 45 mm side, the axis of which passes through its centre. Draw the front view of the solids showing curves of intersection when the axis of the prism is parallel to both the planes and the faces are equally inclined to the V.P.

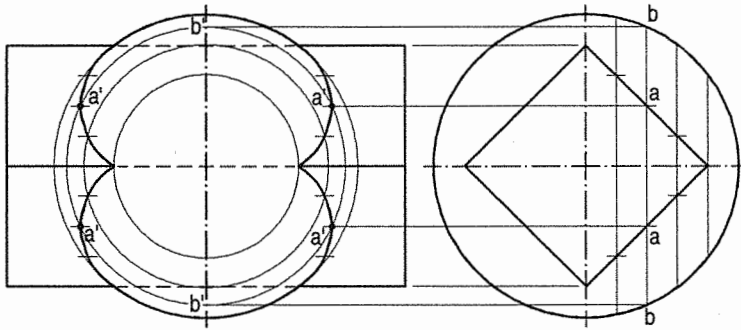


FIG. 16-39

- (i) Draw the front view and project the side view. Assume a vertical cutting plane passing through, say points a . The section of the sphere will be a circle of diameter bb . It is cut by the section of the prism (a rectangle of width aa) at points a' as shown in the front view. Then points a' are on the curve of intersection.
- (ii) Obtain more points together with the key points in the same manner and draw curves through them.

Problem 16-34. (fig. 16-40): Draw the development of a portion of the surface of the cone given in problem 16-22, showing the hole in it.

In fig. 16-40(i), the positions of points are taken from fig. 16-24, while in fig. 16-40(ii) they are obtained from fig. 16-25. In each case, distances along the arc are measured from the top view, while the distances of points (on the generators) from o are taken from the front view, after projecting them on the true length line.

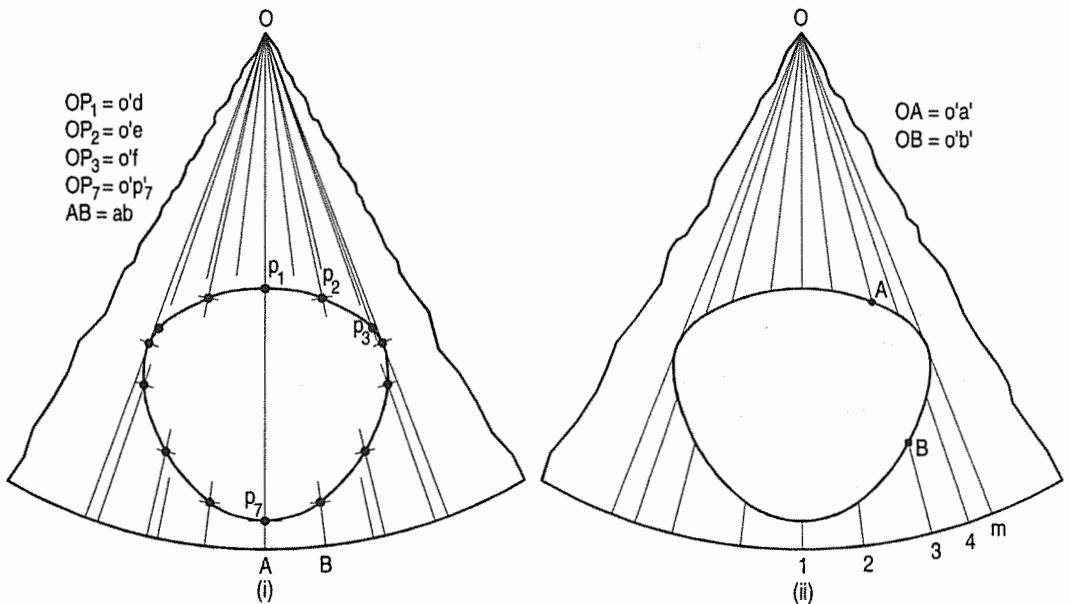


FIG. 16-40

**EXERCISES 16**

1. A vertical square prism, base 50 mm side has its faces equally inclined to the V.P. It is completely penetrated by another square prism of base 30 mm side, the axis of which is parallel to both the planes and is 6 mm away from the axis of the vertical prism. The faces of the horizontal prism also are equally inclined to the V.P. Draw the projections of the solids showing lines of intersection.
2. Two equal prisms whose ends are equilateral triangles of 40 mm side and axes 100 mm long, intersect at right angles. One face of each prism is on the ground. The axis of one of the prisms makes 30° with the V.P. Draw three views of the solids.
3. A square prism of base 50 mm side and height 125 mm stands on the ground with a side of the base inclined at 30° to the V.P. It is penetrated by a cylinder, 50 mm diameter and 125 mm long, whose axis is parallel to both the H.P. and the V.P. and bisects the axis of the prism. Draw the projections showing fully the curves of intersection.
4. A cylindrical boiler is 2 m in diameter and has a cylindrical dome 0.8 m diameter and 0.6 m high. The axis of the dome intersects the axis of the boiler. Draw three views of the arrangement. Develop the surface of the dome. Scale 1 cm = 0.2 m.
5. A vertical pipe, 75 mm diameter and 150 mm long, has two branches, one on each side. The horizontal branch is of 60 mm diameter while the other is of 50 mm diameter and inclined at 45° to the vertical. Assume the axis of 50 mm branch and the main pipe to be in the same plane, and that of 60 mm branch at 6 mm from the axis of the main pipe and parallel to the V.P. Draw the views of the pipe showing curves of intersection. Draw the developments of three pipes, assuming suitable lengths.
6. A cylinder of 75 mm diameter and 125 mm height, stands on its base on the ground. It is penetrated centrally by a cylinder, 50 mm diameter and 125 mm long, whose axis is parallel to the H.P. but inclined at 30° to the V.P. Draw the projections showing curves of intersection. Draw also the development of the penetrated cylinder.
7. A right circular cylinder of 75 mm diameter penetrates another of 100 mm diameter, their axes being at right angles to each other but 10 mm apart. Draw the projections of the curves of intersection on a plane parallel to the axes of the cylinders.
8. Two circular pipes of 75 mm and 50 mm diameters (inside) meet at 30° . The axes of both the pipes are in one plane and the 75 mm pipe is vertical. The thickness of the pipes is 6 mm in both cases. Draw the projections showing curves of intersection.
9. A square hole of 35 mm side is cut in a cylindrical shaft 75 mm diameter and 125 mm long. The axis of the hole intersects that of the shaft at right angles. All faces of the hole are inclined at 45° to the H.P. Draw three views of the shaft when the plane of the two axes is parallel to the V.P.
10. A cylinder of 50 mm diameter, branches off another cylinder of 75 mm diameter. The axis of the smaller cylinder is vertical and that of the other is horizontal. If the distance between the two axes is 10 mm, draw three views of the cylinders.

11. A cone frustum is 125 mm high, 85 mm diameter at the top and 115 mm diameter at the bottom. It is vertically placed and is completely penetrated by a horizontal cylinder 75 mm diameter and 125 mm long, the axis of which bisects the axis of the frustum. Draw the projections of the solids showing curves of intersection.
12. A conical funnel, 150 mm diameter at the mouth and 125 mm high has a pipe of 40 mm diameter attached at its apex. The length of the pipe is 100 mm and its axis is at right-angles to the axis of the funnel and in the same plane, with the outer side of the pipe in line with the apex of the funnel. Draw full size, the arrangement with sufficient number of views to show the joint.
13. A cone, 90 mm diameter of base, axis 110 mm long, stands on the ground and is completely penetrated by a cylinder, 50 mm diameter and 110 mm long. The axis of the cylinder is horizontal, parallel to the V.P. and passes through the axis of the cone, 75 mm from the apex. Draw the projections of both curves of intersection. Develop the surface of the cone.
14. A hole of 25 mm diameter is drilled in a cone having 75 mm diameter of base and 60 mm height. The axis of the hole is parallel to that of the cone and 6 mm away from it. Draw three views of the cone when a vertical plane containing the two axes is perpendicular to the V.P.
15. A conical hopper is fitted in vertical position on a horizontal pipe, 75 mm in diameter and 150 mm in length. The diameter of the hopper at the top is 110 mm and changes by 25 mm in every 25 mm of vertical distance. Draw the projections showing the curves of intersection if the top of the hopper is 70 mm above the axis of the pipe. Develop the surfaces of the hopper and the pipe.
16. A cylinder, 50 mm diameter of base and 100 mm height is centrally penetrated by a cone, 50 mm diameter of base and 75 mm height. The axis of the cylinder which is vertical, cuts the axis of the cone which is horizontal at 30 mm from the base of the cone. Draw the front view and the side view, showing curves of penetration.
17. A cone, base 75 mm diameter and axis 100 mm long, has an equilateral triangular hole of 40 mm side cut through it. The axis of the hole coincides with the axis of the cone. Draw three views of the cone when it is resting on its base on the ground with a face of the hole parallel to the V.P. Develop the surface of the cone.
18. A vertical cone, base 100 mm diameter and height 125 mm is penetrated by another cone, 50 mm diameter and axis 100 mm long. The axis of the penetrating cone is parallel to the H.P. and the V.P., 40 mm above the base and 10 mm from the axis of the vertical cone. It comes out equally on both sides of the cone. Draw the projections showing curves of intersection.
19. A sphere of 100 mm diameter is penetrated vertically by a triangular prism of base 40 mm side. Their axes coincide with each other. Draw the curves of penetration when two faces of the prism are equally seen in the front view.
20. A cylinder, 50 mm diameter and axis vertical, penetrates a pentagonal pyramid having its base on the ground and one side of the base parallel to the V.P. The diameter of the circumscribing circle of the base of the pyramid is 75 mm and its height is 100 mm. The two axes are 10 mm apart. Draw the projections showing curves of intersection, when the plane containing the two axes is parallel to the V.P. Develop the surface of the pyramid.

21. A rectangular slot for a cotter is cut in the tapering portion of a piston rod as shown in fig. 16-41. Draw three views of the rod, showing the curves of penetration in each view.
22. A frustum of a cone appears in the front view as a trapezium of 50 mm and 75 mm parallel sides, 90 mm apart. A cylindrical pipe joins the frustum and in the side view, the circle representing the pipe is found to touch the 50 mm side and the sides of the frustum. Draw three views of the frustum and the portion of the pipe joined as stated.
23. In a circle of 50 mm radius, draw a square of 30 mm side symmetrically around its centre. The figure is the top view of a sphere completely penetrated by a 75 mm long square prism. From this top view, project a front view on a reference line xy parallel to a diagonal of the square. From this front view, project another top view on a reference line making 45° angle with the axis of the prism.
24. A square prism of 50 mm side of a base and 100 mm length of axis is resting on its base on the ground having a side of base at 30° to the V.P. It is completely penetrated by a horizontal square prism of 40 mm side of a base and 100 mm length of axis; the axis of which is parallel to the V.P. and bisects the axis of the vertical prism at right-angle. Draw the projections of the solids showing the lines of intersection when a face of the penetrating solid is at 30° to the H.P.
25. A vertical cylinder of 60 mm diameter is penetrated by a square prism of 35 mm side, the axis of which is inclined at 30° to the ground but parallel to the V.P. The faces of the prism are equally inclined to the V.P. and the axis of the prism is 10 mm in front of the axis of the cylinder. Draw the projections of the solids showing the curves of interpenetration.
26. A cone, diameter of base 70 mm and axis 70 mm long is kept on the ground on its base. A vertical hole of 50 mm is drilled through the cone in such a way that the axis of the hole is 10 mm away from that of the cone. The plane containing both axes is parallel to the V.P. Draw three views of the solids.

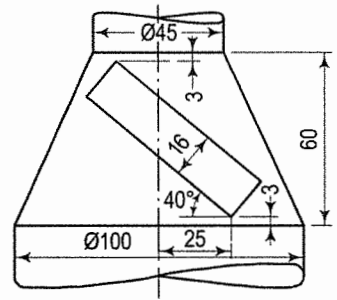


FIG. 16-41



ISOMETRIC PROJECTION

17-1. INTRODUCTION

Isometric projection is a type of pictorial projection in which the three dimensions of a solid are not only shown in one view, but their actual sizes can be measured directly from it.

If a cube is placed on one of its corners on the ground with a solid diagonal perpendicular to the V.P., the front view is the *isometric projection* of the cube. The step-by-step construction is shown in fig. 17-1.

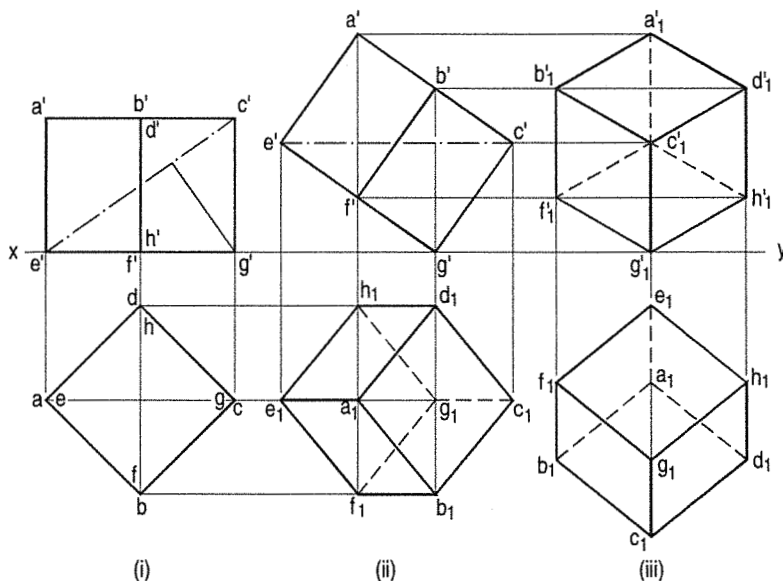


FIG. 17-1

To draw the projections of a cube of 25 mm long edges resting on the ground on one of its corners with a solid diagonal perpendicular to the V.P., assume the cube to be resting on one of its faces on the ground with a solid diagonal parallel to the V.P.

- (i) Draw a square $abcd$ in the top view with its sides inclined at 45° to xy . The line ac representing the solid diagonals AG and CE is parallel to xy . Project the front view.

- (ii) Tilt the front view about the corner g' so that the line $e'c'$ becomes parallel to xy . Project the second top view. The solid diagonal CE is now parallel to both the H.P. and the V.P.
- (iii) Reproduce the second top view so that the top view of the solid diagonal, viz. $e_1 c_1$ is perpendicular to xy . Project the required front view.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 39 for the introduction.

Fig. 17-2 shows the front view of the cube in the above position, with the corners named in capital letters. Its careful study will show that:

- (a) All the faces of the cube are equally inclined to the V.P. and hence, they are seen as similar and equal rhombuses instead of squares.
- (b) The three lines CB , CD and CG meeting at C and representing the three edges of the solid right-angle are also equally inclined to the V.P. and are therefore, equally foreshortened. They make equal angles of 120° with each other. The line CG being vertical, the other two lines CB and CD make 30° angle each, with the horizontal.
- (c) All the other lines representing the edges of the cube are parallel to one or the other of the above three lines and are also equally foreshortened.
- (d) The diagonal BD of the top face is parallel to the V.P. and hence, retains its true length.

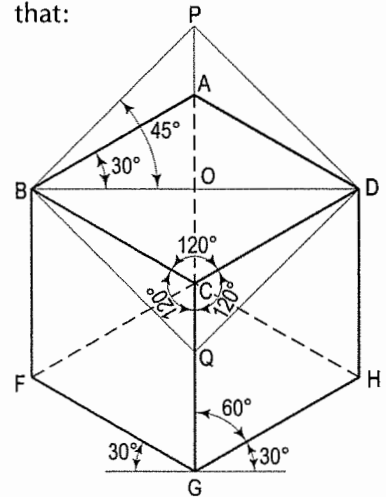


FIG. 17-2

This chapter deals with various topics of isometric projection as shown below:

1. Isometric axes, lines and planes
2. Isometric scale
3. Isometric drawing or isometric view
4. Isometric graph.

17-2. ISOMETRIC AXES, LINES AND PLANES

The three lines CB , CD and CG meeting at the point C and making 120° angles with each other are termed *isometric axes*. The lines parallel to these axes are called *isometric lines*. The planes representing the faces of the cube as well as other planes parallel to these planes are called *isometric planes*.

17-3. ISOMETRIC SCALE

As all the edges of the cube are equally foreshortened, the square faces are seen as rhombuses. The rhombus $ABCD$ (fig. 17-2) shows the isometric projection of the top square face of the cube in which BD is the true length of the diagonal.

Construct a square $BQDP$ around BD as a diagonal. Then BP shows the true length of BA .

$$\text{In triangle } ABO, \frac{BA}{BO} = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}}$$

$$\text{In triangle } PBO, \frac{BP}{BO} = \frac{1}{\cos 45^\circ} = \frac{\sqrt{2}}{1}$$

$$\frac{BA}{BP} = \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{3}} = 0.815$$

$$\text{The ratio, } \frac{\text{isometric length}}{\text{true length}} = \frac{BA}{BP} = \frac{\sqrt{2}}{\sqrt{3}} = 0.815 \text{ or } \frac{9}{11} \text{ (approx.).}$$

Thus, the isometric projection is reduced in the ratio $\sqrt{2} : \sqrt{3}$, i.e. the isometric lengths are 0.815 of the true lengths.

Therefore, while drawing an *isometric projection*, it is necessary to convert true lengths into isometric lengths for measuring and marking the sizes. This is conveniently done by constructing and making use of an isometric scale as shown below.

- (a) Draw a horizontal line BD of any length (fig. 17-3). At the end B , draw lines BA and BP , such that $\angle DBA = 30^\circ$ and $\angle DBP = 45^\circ$. Mark divisions of true length on the line BP and from each division-point, draw verticals to BD meeting BA at respective points. The divisions thus obtained on BA give lengths on isometric scale.

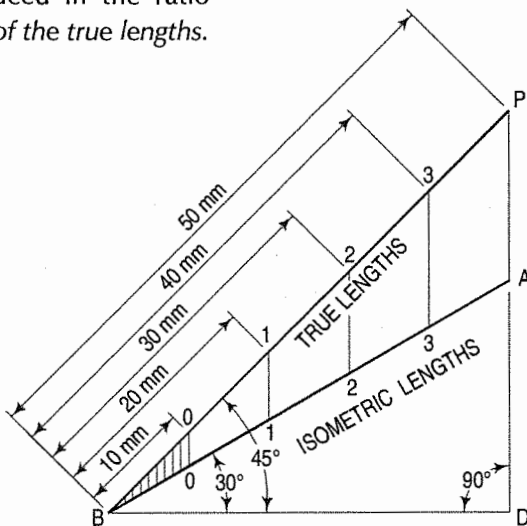


FIG. 17-3

- (b) The same scale may also be drawn with divisions of natural scale on a horizontal line AB (fig. 17-4). At the ends A and B , draw lines AC and BC making 15° and 45° angles with AB respectively, and intersecting each other at C .

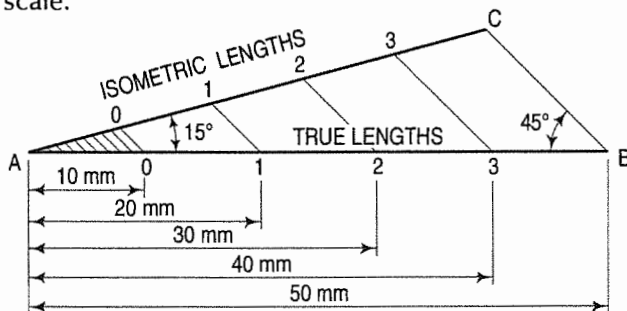


FIG. 17-4

From division-points of true lengths on AB , draw lines parallel to BC and meeting AC at respective points. The divisions along AC give lengths to isometric scale.

The lines BD and AC (fig. 17-2) represent equal diagonals of a square face of the cube, but are not equally shortened in isometric projection. BD retains its true length, while AC is considerably shortened. Thus, it is seen that lines which are not parallel to the isometric axes are not reduced according to any fixed ratio. Such lines are called *non-isometric lines*. The measurements should, therefore, be made on *isometric axes and isometric lines only*. The non-isometric lines are drawn by locating positions of their ends on isometric planes and then joining them.

17-4. ISOMETRIC DRAWING OR ISOMETRIC VIEW

If the foreshortening of the isometric lines in an isometric projection is disregarded and instead, the true lengths are marked, the view obtained [fig. 17-5(iii)] will be exactly of the same shape but larger in proportion (about 22.5%) than that obtained by the use of the isometric scale [fig. 17-5(ii)]. Due to the ease in construction and the advantage of measuring the dimensions directly from the drawing, it has become a general practice to use the true scale instead of the isometric scale.

To avoid confusion, the view drawn with the true scale is called *isometric drawing* or *isometric view*, while that drawn with the use of isometric scale is called *isometric projection*.

Referring again to fig. 17-2, the axes BC and CD represent the sides of a right angle in horizontal position. Each of them together with the vertical axis CG , represents the right angle in vertical position. Hence, in isometric view of any rectangular solid resting on a face on the ground, each horizontal face will have its sides parallel to the two sloping axes; each vertical face will have its vertical sides parallel to the vertical axis and the other sides parallel to one of the sloping axes.

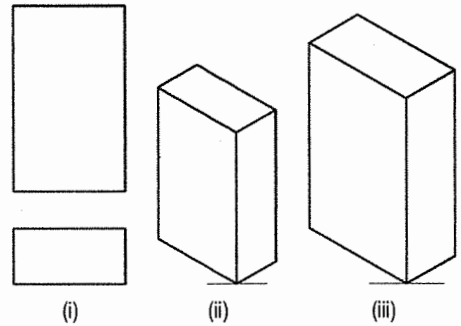


FIG. 17-5

In other words, the vertical edges are shown by vertical lines, while the horizontal edges are represented by lines, making 30° angles with the horizontal. These lines are very conveniently drawn with the T-square and a 30° - 60° set-square or drafter.

17-5. ISOMETRIC GRAPH

An isometric graph as shown in fig. 17-6 facilitates the drawing of isometric view of an object. Students are advised to make practice for drawing of isometric view using such graphs. See fig. 17-55 and fig. 17-56 of problem 17-33.

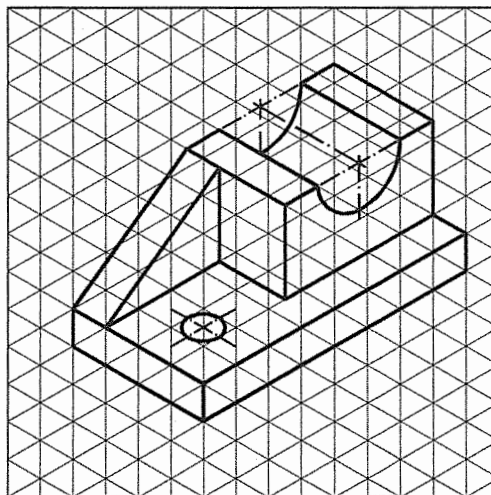


FIG. 17-6

17-6. ILLUSTRATIVE PROBLEMS

The procedure for drawing isometric views of planes, solids and objects of various shapes is explained in stages by means of illustrative problems.

In order that the construction of the view may be clearly understood, construction lines have not been erased. They are, however, drawn fainter than the outlines.

In an isometric view, lines for the hidden edges are generally not shown. In the solutions accompanying the problems, one or two arrows have been shown. They indicate the directions from which if the drawing is viewed, the given orthographic views would be obtained. Students need not show these arrows in their solutions.

17-6-1. ISOMETRIC DRAWING OF PLANES OR PLANE FIGURES



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 40 for the following problem.

Problem 17-1. The front view of a square is given in fig. 17-7(i). Draw its isometric view.

As the front view is a square, the surface of the square is vertical. In isometric view, vertical lines will be drawn vertical, while horizontal lines will be drawn inclined at 30° to the horizontal.

- Through any point d , draw a vertical line $da = DA$ [fig. 17-7(ii)].
- Again through d , draw a line $dc = DC$ inclined at 30° to the horizontal and at 60° to da .
- Complete the rhombus $abcd$ which is the required isometric view. The view can also be drawn in direction of the other sloping axis as shown in fig. 17-7(iii).

Problem 17-2. If fig. 17-7(i) is the top view of a square, draw its isometric view.

As the top view is a square, the surface of the square is horizontal. In isometric view, all the sides will be drawn inclined at 30° to the horizontal.

- From any point d [fig. 17-7(iv)], draw two lines da and dc inclined at 30° to the horizontal and making 120° angle between themselves.
- Complete the rhombus $abcd$ which is the required isometric view.

Problem 17-3. The top view of a rectangle, the surface of which is horizontal is shown in fig. 17-8(i). Draw its isometric view.

Draw the required view as explained in problem 17-2 and as shown in either fig. 17-8(ii) or fig. 17-8(iii).

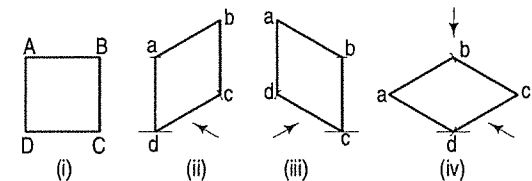


FIG. 17-7

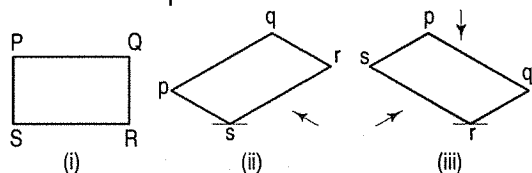


FIG. 17-8

Problem 17-4. The front view of a triangle having its surface parallel to the V.P. is shown in fig. 17-9(i). Draw its isometric view.

The surface of the triangle is vertical and the base ab is horizontal. ab will be drawn parallel to a sloping axis. The two sides of the triangle are inclined.

Hence they will not be drawn parallel to any isometric axis. In an isometric view, angles do not increase or decrease in any fixed proportion. They are drawn after determining the positions of the ends of the arms on isometric lines.

Therefore, enclose the triangle in the rectangle $ABQP$. Draw the isometric view $abqp$ of the rectangle [fig. 17-9(ii)].

Mark a point c in pq such that $pc = PC$. Draw the triangle abc which is the required isometric view. It can also be drawn in the other direction as shown in fig. 17-9(iii).

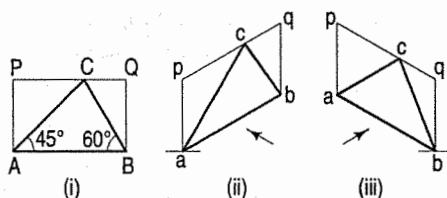


FIG. 17-9

Problem 17-5. The front view of a quadrilateral whose surface is parallel to the V.P. is shown in fig. 17-10(i). Draw its isometric view.

Enclose the quadrilateral in a rectangle $ABEF$.

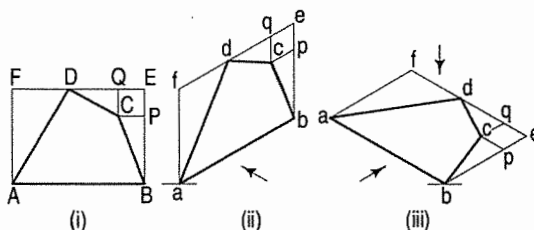


FIG. 17-10

(i) Draw lines CP and CQ parallel to the sides FE and BE respectively.

(ii) Draw the isometric view of the rectangle [fig. 17-10(ii)] and obtain the point d in fe as explained in problem 17-4. Draw the isometric view $cpeq$ of the rectangle $CPEQ$.

(iii) Draw the quadrilateral $abcd$ which is the required isometric view.

If the given view is the top view of a quadrilateral whose surface is horizontal, i.e. parallel to the H.P., its isometric view will be as shown in fig. 17-10(iii).

Problem 17-6. If the view given in fig. 17-11(i) is

- the front view of a hexagon whose surface is parallel to the V.P. or
- the top view of the hexagon whose surface is horizontal, draw its isometric views.

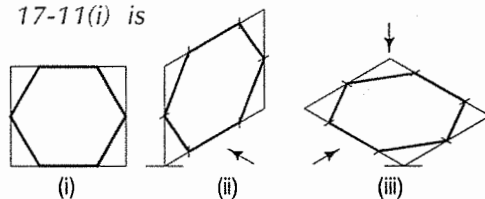


FIG. 17-11

(a) fig 17-11(ii). (b) fig 17-11(iii).

In both cases, the views can be drawn in the other direction also.

Problem 17-7. Fig. 17-12(i) shows the front view of a circle whose surface is parallel to the V.P. Draw the isometric view of the circle.



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 41 for the following problem.

I. Method of points:

- Enclose the circle in a square, touching it in points 1, 2, 3 and 4. Draw the diagonals of the square cutting the circle in points 5, 6, 7 and 8.
- Draw the isometric view of the square [fig. 17-12(ii) and [fig. 17-12(iii)] and on it mark the mid-points 1, 2, 3 and 4 of its sides. Obtain points 5, 6, 7 and 8 on the diagonals as explained in problem 17-5.

Or, after determining the position of one point, draw through it, lines parallel to the sides of the rhombus and obtain the other three points. Draw a neat and smooth curve passing through the eight points viz. 1, 6, 2, 7 etc. The curve is the required isometric view. It is an ellipse.

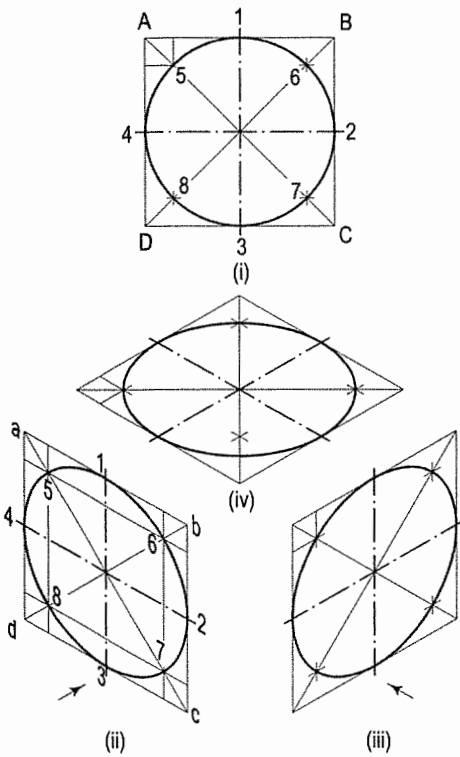


FIG. 17-12

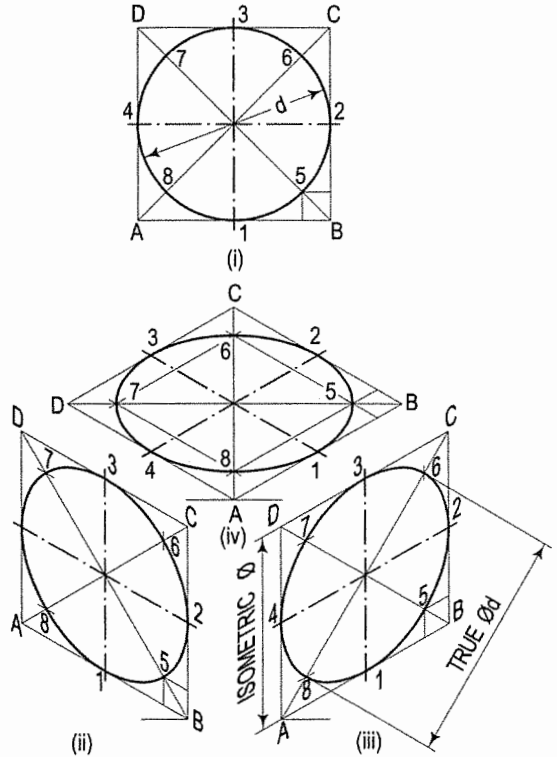


FIG. 17-13

If the view given in fig. 17-12(i) is the top view of a circle whose surface is horizontal, its isometric view will be as shown in fig. 17-12(iv).

As the isometric views have been drawn with the true scale, the major axis of the ellipse is longer than the diameter of the circle.

Fig. 17-13(ii), fig. 17-13(iii) and fig. 17-13(iv) show the isometric projection of the circle drawn with isometric scale. Note that when the length of the side of the rhombus is equal to the isometric diameter of the circle, the length of the major axis of the ellipse is equal to the true diameter of the circle.

II. Four-centre method:

Draw the isometric view of the square [fig. 17-14(i)]. Draw perpendicular bisectors of the sides of the rhombus, intersecting each other on the longer diagonal at points p and q , and which meet at the 120° -angles b and d .

Or, draw lines joining the 120° -angles b and d with the mid-points of the opposite sides and intersecting each other on the longer diagonal at points p and q . Two of these

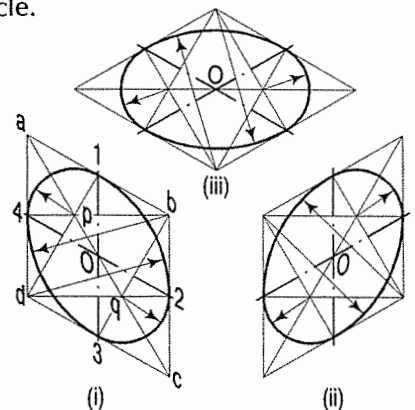


FIG. 17-14

lines will be drawn horizontal, while the other two will make 60° -angles with the horizontal. With centres b and d , draw arcs 3-4 and 1-2 respectively. With centres p and q , draw arcs 1-4 and 2-3 respectively and complete the required ellipse. Fig. 17-14(ii) shows the ellipse obtained in the rhombus drawn in the direction of the other sloping axis. Fig. 17-14(iii) shows the isometric view of the circle when its surface is horizontal.

The ellipse obtained by the four-centre method is not a true ellipse and differs considerably in size and shape from the ellipse plotted through points. But owing to the ease in construction and to avoid the labour of drawing freehand neat curves, this method is generally employed.

Problem 17-8. To draw the isometric view of a circle of a given diameter, around a given point.

Let O be the given point and D the diameter of the circle.

(a) When the surface of the circle is vertical [fig. 17-14(i)].

- (i) Through O , draw a vertical centre line and another centre line inclined at 30° to the horizontal, i.e. parallel to a sloping isometric axis. On these lines, mark points 1, 2, 3 and 4 at a distance equal to $0.5D$ from O .
- (ii) Through these points, draw lines parallel to the centre lines and obtain the rhombus $abcd$ of sides equal to D .
- (iii) Draw the required ellipse in this rhombus by the four-centre method.

By drawing the second centre line parallel to the other sloping axis, the isometric view is obtained in another position as shown in fig. 17-14(ii).

(b) When the surface of the circle is horizontal [fig. 17-14(iii)].

Through O , draw the two centre lines parallel to the two sloping isometric axes, i.e. inclined at 30° to the horizontal. Draw the required ellipse as explained in (a) above.

Note: This construction is very useful in drawing isometric views of circular holes in solids.

Problem 17-9. Fig. 17-15(i) shows the front view of a semi-circle whose surface is parallel to the V.P. Draw its isometric view.

- (i) Enclose the semi-circle in a rectangle. Draw the isometric view of the rectangle [fig. 17-15(ii) and [fig. 17-15(iii)]].
- (ii) Using the four-centre method, draw the half-ellipse in it which is the required view. The centre for the longer arc may be obtained as shown or by completing the rhombus.

If the view given in fig. 17-15(i) is the top view of a horizontal semi-circle, its isometric view would be drawn as shown in fig. 17-16(i) and fig. 17-16(ii).

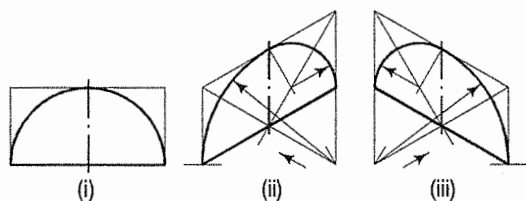


FIG. 17-15

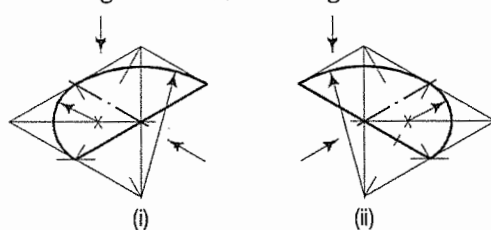


FIG. 17-16

Problem 17-10. Fig. 17-17(i) shows the front view of a semi-circle whose surface is parallel to the V.P. Draw the isometric view of the semi-circle.

See fig. 17-17(ii) and fig. 17-17(iii).

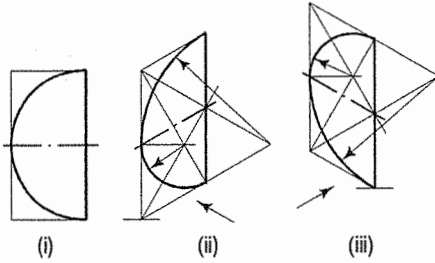


FIG. 17-17

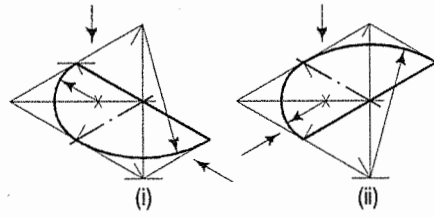


FIG. 17-18

If the view shown in fig. 17-17(i) is the top view of a semi-circle whose surface is horizontal, its isometric view will be as shown in fig. 17-18(i) or fig. 17-18(ii).

Problem 17-11. Fig. 17-19(i) shows the front view of a plane parallel to the V.P. Draw its isometric view.

- (i) The upper two corners of the plane are rounded with quarter circles. Enclose the plane in a rectangle.

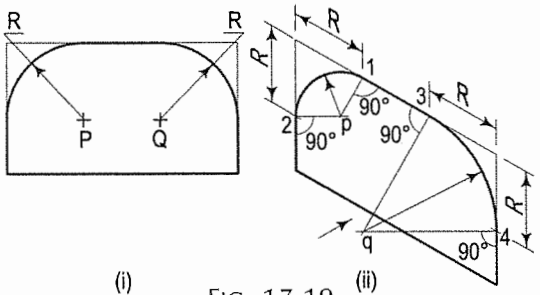


FIG. 17-19

- (ii) Draw the isometric view of the rectangle. From the upper two corners of the parallelogram, mark points on the sides at a distance equal to R , the radius of the arcs. At these points erect perpendiculars to the respective sides to intersect each other at points p and q . With p and q as centres, and radii $p1$ and $q3$, draw the arcs and complete the required view.

It is interesting to note that although the arcs are of the same radius, they are drawn with different radii in their isometric views.

17-6-2. ISOMETRIC DRAWING OF PRISMS AND PYRAMIDS

We have seen that the isometric view of a cube is determined from its orthographic view in a particular position. The three edges of the solid right-angle of the cube are shown by lines parallel to the three isometric axes. A square prism or a rectangular prism also has solid right-angles. Hence, lines for its edges are also drawn parallel to the three isometric axes.

While drawing the isometric view of any solid, the following important points should be carefully noted:

- (i) The isometric view should be drawn according to the given views and in such a way that maximum possible details are visible.
- (ii) At every point for the corner of a solid, at least three lines for the edges must converge. Of these, at least two must be for visible edges.
Lines for the hidden edges need not be shown, but it is advisable to check up every corner so that no line for a visible edge is left out.
- (iii) Two lines (for visible edges) will never cross each other.

Problem 17-12. Draw the isometric view of a square prism, side of the base 20 mm long and the axis 40 mm long, when its axis is (i) vertical and (ii) horizontal.

- (i) When the axis is vertical, the ends of the prism will be horizontal. Draw the isometric view (the rhombus 1-2-3-4) of the top end [fig. 17-20(i)]. Its sides will make 30° -angles with the horizontal. The length of the prism will be drawn in the third direction, i.e. vertical. Hence, from the corners of the rhombus, draw vertical lines 1-5, 2-6 and 3-7 of length equal to the length of the axis. The line 4-8 should not be drawn, as that edge will not be visible. Draw lines 5-6 and 6-7, thus completing the required isometric view. Lines 7-8 and 8-5 also should not be drawn. Beginning may also be made by drawing lines from the point 6 on the horizontal line and then proceeding upwards.
- (ii) When the axis is horizontal, the ends will be vertical. The ends can be drawn in two ways as shown in fig. 17-20(ii) and fig. 17-20(iii). In each case, the length is shown in the direction of the third isometric axis.

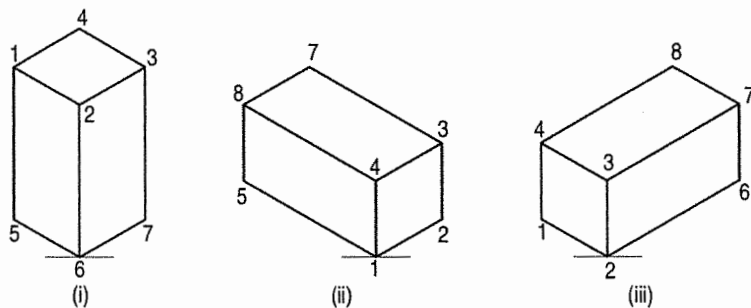


FIG. 17-20

Problem 17-13. Three views of a block are given in fig. 17-21(i). Draw its isometric view.

The block is in the form of a rectangular prism. Its shortest edges are vertical. Lines for these edges will be drawn vertical. Lines for all other edges which are horizontal, will be drawn inclined at 30° to the horizontal in direction of the two sloping axes as shown in fig. 17-21(ii).

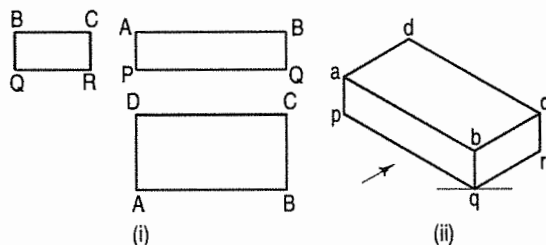


FIG. 17-21

Methods of drawing non-isometric lines.

When an object contains inclined edges which in the isometric view would be shown by non-isometric lines, the view may be drawn by using any one of the following methods:

- (i) box method or
- (ii) co-ordinate or offset method.

(i) **Box method:** This method is used when the non-isometric lines or their ends lie in isometric planes. The object is assumed to be enclosed in a rectangular box. Initially, the box is drawn in isometric. The ends of the lines for the inclined edges are then located by measuring on or from the outlines of the box.

Problem 17-14. Three views of a block are given in fig. 17-22(i). Draw its isometric view.

- Enclose the block in a rectangular box.
- Draw the isometric view of the box [fig. 17-22(ii)].
- Mark points e and f on the line bc such that $be = BE$ and $fc = FC$.
- Complete the required view as shown.

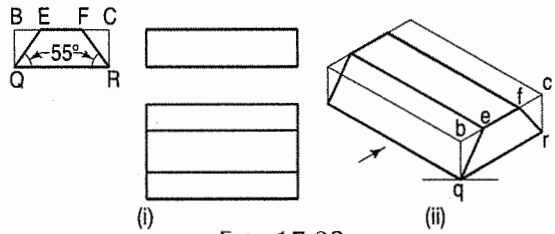


FIG. 17-22

Problem 17-15. Draw the isometric view of the plate shown in three views in fig. 17-23(i).

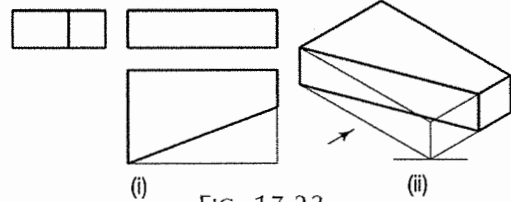


FIG. 17-23

Obtain the required view as explained in problem 17-14 and as shown in fig. 17-23(ii).



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 42 for the following problem.

Problem 17-16. Draw the isometric view of the frustum of the hexagonal pyramid shown in fig. 17-24(i).

- Enclose the front view and the top view in rectangles.
- Draw the isometric view of the rectangular box [fig. 17-24(ii)]. Locate the six points of the base of the frustum on the sides of the bottom of the box. The upper six points on the top surface of the box are located by drawing isometric lines, e.g. $P1$ and $Q1$ intersecting at a point 1.
- Join the corners and complete the isometric view as shown.

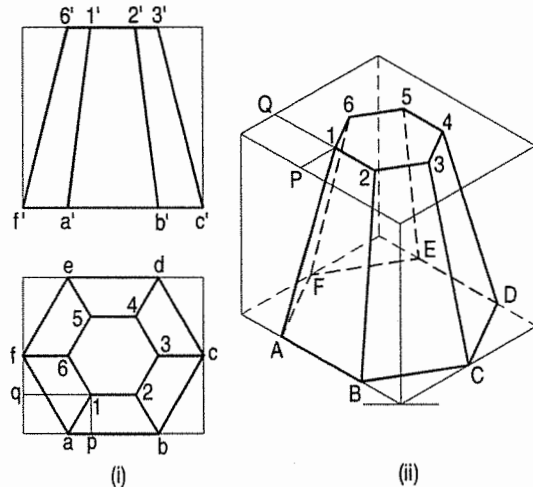


FIG. 17-24

(ii) **Co-ordinate or Offset method:** This method is adopted for objects in which neither non-isometric lines nor their ends lie in isometric planes.

Perpendiculars are dropped from each end of the edge to a horizontal or a vertical reference plane. The points at which the perpendiculars meet the plane, are located by drawing co-ordinates or offsets to the edges of the plane.

This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 43 for the following problem.

Problem 17-17. Draw the isometric view of the pentagonal pyramid, the projections of which are given in fig. 17-25(i).



- (i) Enclose the base (in the top view) in an oblong.
- (ii) Draw an offset oq (i.e. pq) on the line ab .
- (iii) Draw the isometric view of the oblong and locate the corners of the base in it [fig. 17-25(ii)].
- (iv) Mark a point Q on the line AB such that $AQ = aq$. From Q , draw a line QP equal to qo and parallel to $2C$. At P , draw a vertical OP equal to $o'p'$.
- (v) Join O with the corners of the base, thus completing the isometric view of the pyramid.

Fig. 17-25(iii) shows the isometric view of the same pyramid with its axis in horizontal position.

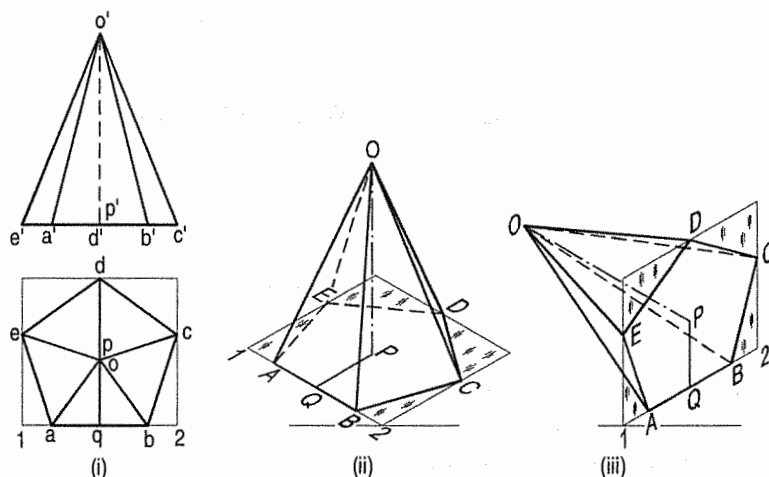


FIG. 17-25

Problem 17-18. Draw the isometric view of the truncated triangular pyramid shown in fig. 17-26(i).

- (i) Draw the perpendiculars $d'x'$, $e'y'$ and $f'z'$ the front view and the offsets dq , er and fc in the top view.
- (ii) Draw the isometric view of the whole pyramid [fig. 17-26(ii)].
- (iii) Transfer the offsets and the verticals to this view and obtain points D , E and F on the lines OA , OB and OC respectively.
- (iv) Draw lines DE , EF and FD and complete the required isometric view.

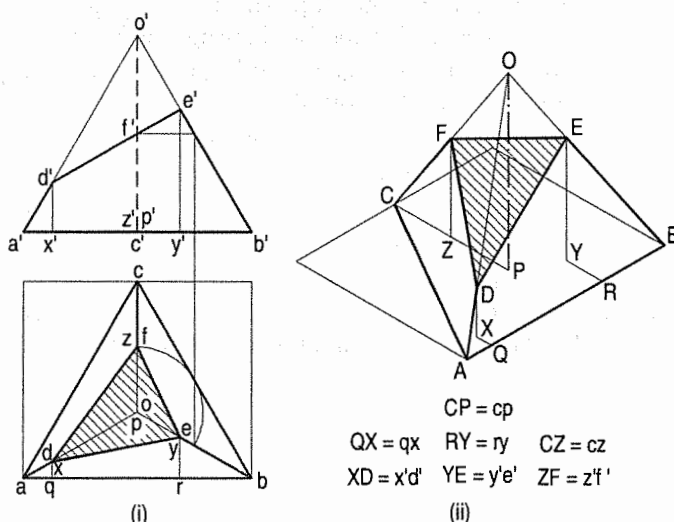


FIG. 17-26

17-6-3. ISOMETRIC DRAWING OF CYLINDERS

Problem 17-19. Draw the isometric view of the cylinder shown in fig. 17-27(i).

The axis of the cylinder is vertical, hence its ends are horizontal. Enclose the cylinder in a square prism.

Method I:

Draw the isometric view of the prism [fig. 17-27(ii)]. In the two rhombuses, draw the ellipses by the four-centre method. Draw two common tangents to the two ellipses. Erase the inner half of the lower ellipse and complete the required view.

Method II:

Draw the rhombus for the upper end of the prism [fig. 17-27(iii)] and in it, draw the ellipse by the four-centre method. From the centres for the arcs, draw vertical lines of length equal to the length of the axis, thus determining the centres for the lower ellipse. Draw the arcs for the half ellipse. Draw common tangents, thus completing the required view.

When the axis of the cylinder is horizontal, its isometric view is drawn by method I as shown in fig. 17-28(i).

Fig. 17-28(ii) shows the view drawn by method II, but the axis is shown sloping in the other direction.

Fig. 17-29 and fig. 17-30 respectively show the isometric views (drawn by method II) of a half-cylindrical disc with its axis in vertical and horizontal positions.

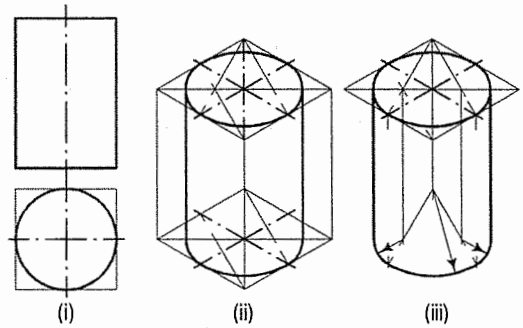


FIG. 17-27

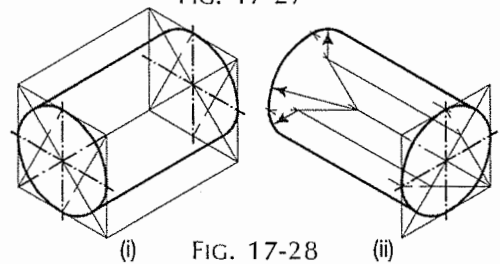


FIG. 17-28

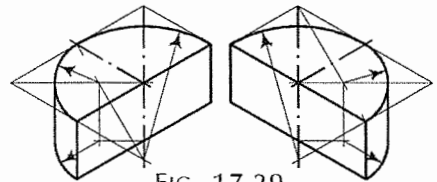


FIG. 17-29

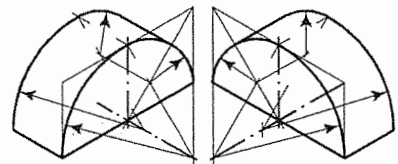


FIG. 17-30

17-6-4. ISOMETRIC DRAWING OF CONES

Problem 17-20. Draw the isometric view of a cone, base 40 mm diameter and axis 55 mm long (i) when its axis is vertical and (ii) when its axis is horizontal.

- (i) Draw the ellipse for the base [fig. 17-31(i)]. Determine the position of the apex by the offset method.
- (ii) Draw tangents to the ellipse from the apex. Erase the part of the ellipse between the tangents and complete the view as shown.

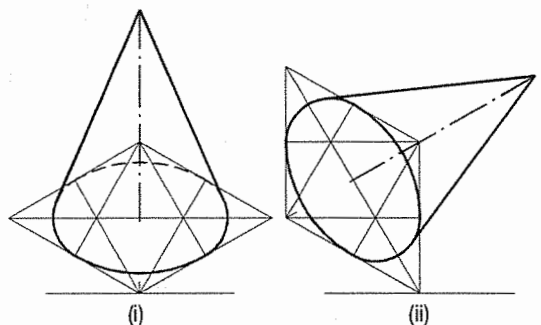


FIG. 17-31

- (iii) See fig. 17-31(ii) which is self-explanatory.

Problem 17-21. Draw the isometric view of the frustum of a cone shown in fig. 17-32(i).

- (i) Draw the ellipse for the base [fig. 17-32(ii)]. Draw the axis.
- (ii) Around the top end of the axis, draw the ellipse for the top.
- (iii) Draw common tangents, erase the unwanted part of the ellipse and complete the view as shown.

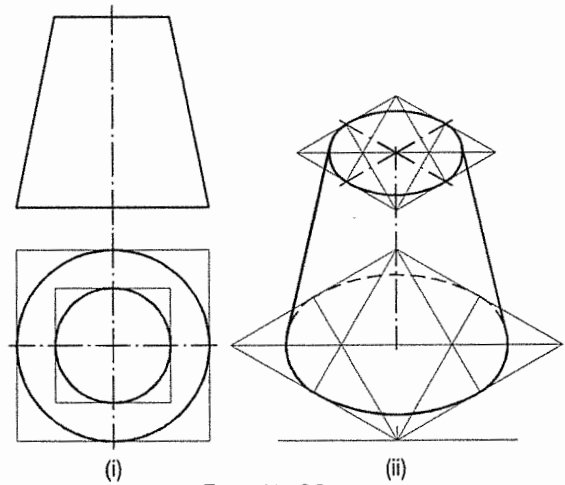


FIG. 17-32

17-6-5. ISOMETRIC DRAWING OF SPHERE

The orthographic view of a sphere seen from any direction is a circle of diameter equal to the diameter of the sphere. Hence, the isometric projection of a sphere is also a circle of the same diameter as explained below.

The front view and the top view of a sphere resting on the ground are shown in fig. 17-33(i). C is its centre, D is the diameter and P is the point of its contact with the ground.

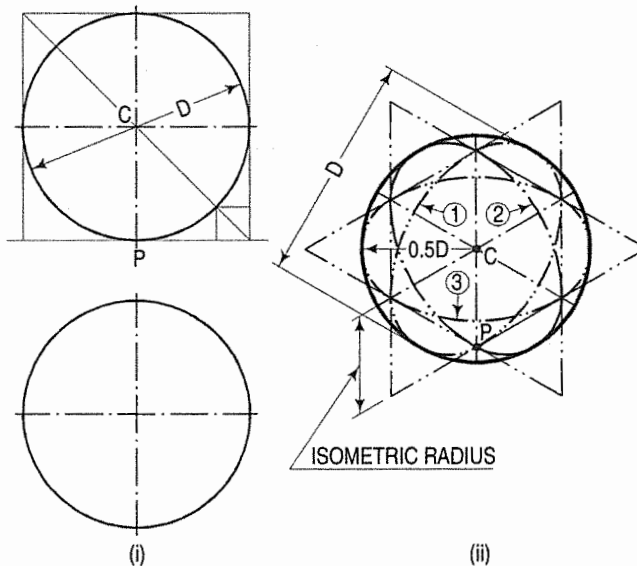


FIG. 17-33

Assume a vertical section through the centre of the sphere. Its shape will be a circle of diameter D . The isometric projection of this circle is shown in fig. 17-33(ii) by ellipses 1 and 2, drawn in two different vertical positions around the same centre C . The length of the major axis in each case is equal to D . The distance of the point P from the centre C is equal to the isometric radius of the sphere.

Again, assume a horizontal section through the centre of the sphere. The isometric projection of this circle is shown by the ellipse 3, drawn in a horizontal position around the same centre C . In this case also, the distance of the outermost points on the ellipse from the centre C is equal to $0.5D$.

Thus, it can be seen that in an isometric projection, the distances of all the points on the surface of a sphere from its centre, are equal to the radius of the sphere.

Hence, the isometric projection of a sphere is a circle whose diameter is equal to the true diameter of the sphere.

Also, the distance of the centre of the sphere from its point of contact with the ground is equal to the isometric radius of the sphere, viz. CP .

It is, therefore, of the utmost importance to note that, *isometric scale must invariably be used, while drawing isometric projections of solids in conjunction with spheres or having spherical parts.*

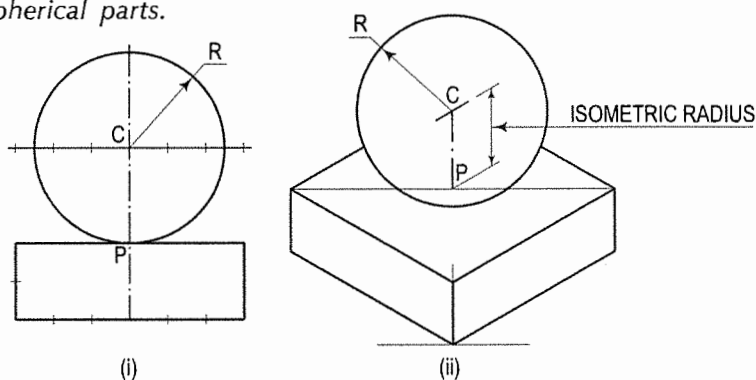


FIG. 17-34

Problem 17-22. Draw the isometric projection of a sphere resting centrally on the top of a square prism, the front view of which is shown in fig. 17-34(i).

- (i) Draw the isometric projection (using isometric scale) of the square prism and locate the centre P of its top surface [fig. 17-34(ii)].
- (ii) Draw a vertical at P and mark a point C on it, such that PC = the isometric radius of the sphere.
- (iii) With C as centre and radius equal to the radius of the sphere, draw a circle which will be the isometric projection of the sphere.

17-7. TYPICAL PROBLEMS OF ISOMETRIC DRAWING

The solutions given in the following typical problems are mostly self-explanatory. Explanations are however given where deemed necessary. Construction lines are left intact for guidance. Dotted lines for hidden edges have been shown in some views to make the construction more clear. Unless otherwise stated, all dimensions are given in millimetres.

Problem 17-23. A hexagonal prism having the side of base 26 mm and the height of 60 mm is resting on one of the corner of the base and its axis is inclined to 30° to the H.P. Draw its projections and also prepare the isometric view of the prism in the above stated condition.

- (i) Draw the projections of the prism as shown in figure 17-35.
- (ii) Construct the isometric view as shown in fig. 17-36.

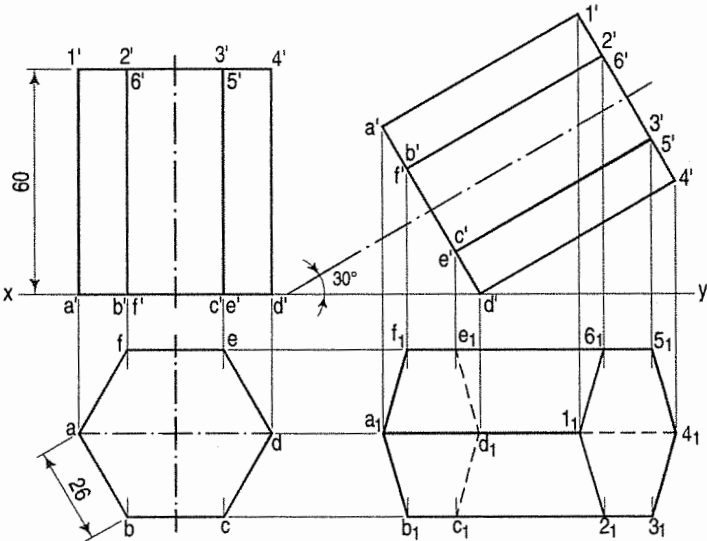


FIG. 17-35

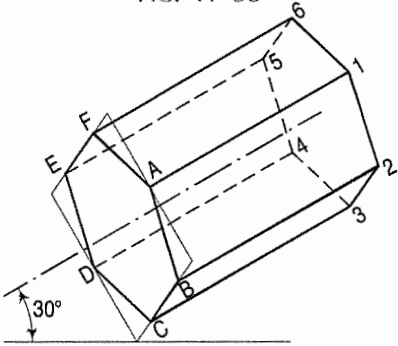


FIG. 17-36

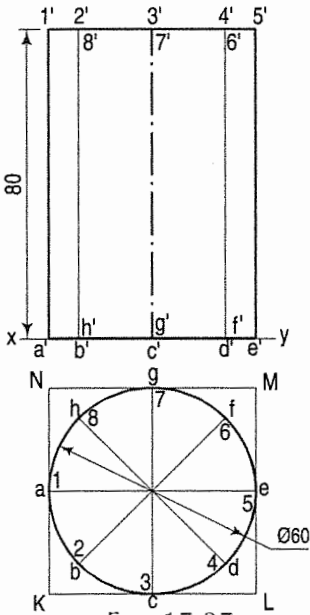


FIG. 17-37

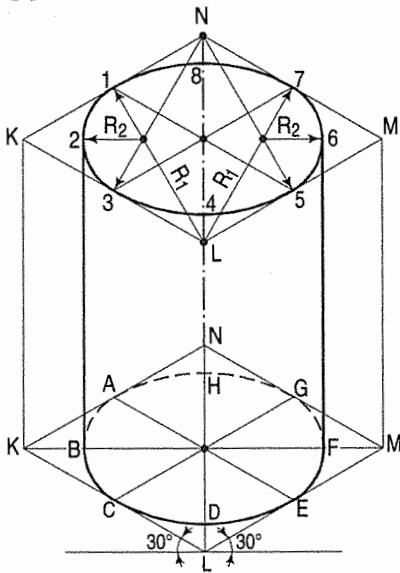


FIG. 17-38

Problem 17-24. (fig. 17-37): A cylindrical block of base, 60 mm diameter and height 80 mm, standing on the H.P. with its axis perpendicular to the H.P. Draw its isometric view.

The method shown in fig. 17-38 is self-explanatory.

Problem 17-25. The projection of pentagonal pyramid is shown in fig. 17-39. Draw its isometric view.

See fig. 17-40.

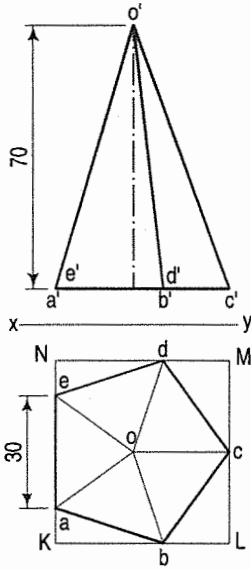


FIG. 17-39

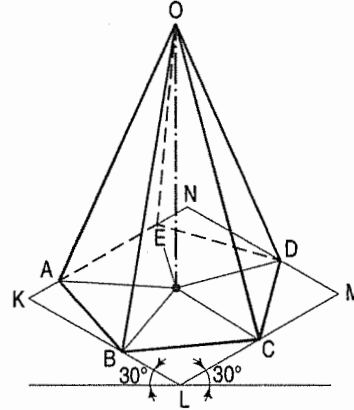


FIG. 17-40

Problem 17-26. The projection of the frustum of the cone is shown in fig. 17-41. Draw its isometric view.

See fig. 17-42.

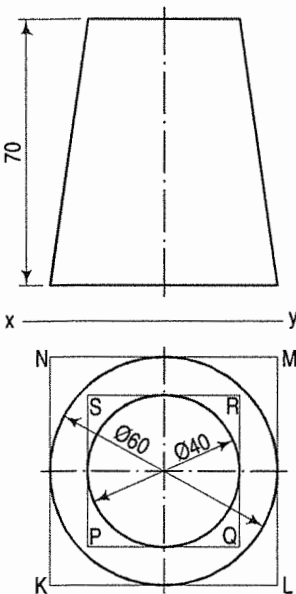


FIG. 17-41

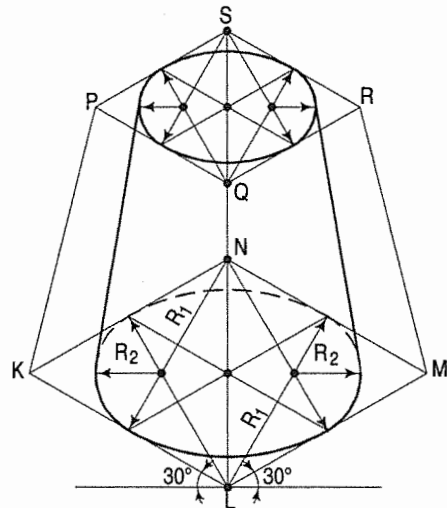


FIG. 17-42

Problem 17-27. The orthographic projections of the object is shown in fig. 17-43. Draw the isometric view of the object.

See fig. 17-44.

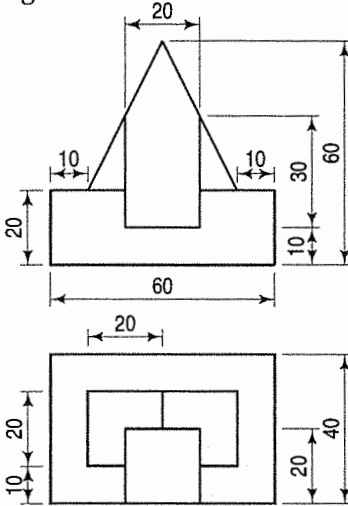


FIG. 17-43

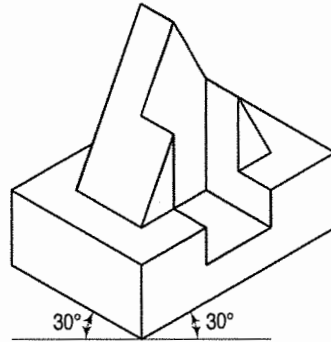


FIG. 17-44

Problem 17-28. Draw the isometric view of the casting shown in two views in fig. 17-45.

See fig. 17-46. Lines for the visible lower edges of the rectangular hole should be shown.

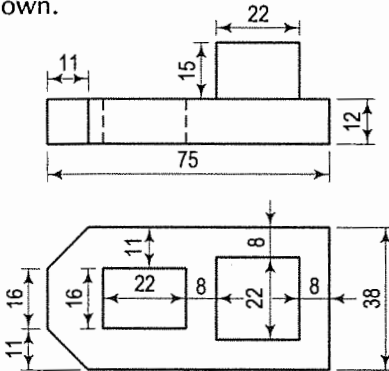


FIG. 17-45

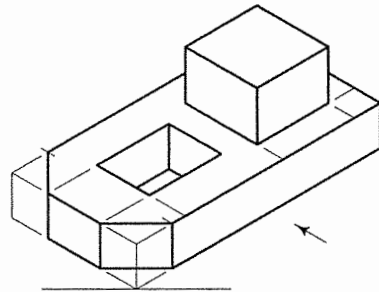


FIG. 17-46

Problem 17-29. Draw the isometric view of the block, two views of which are shown in fig. 17-47.

See fig. 17-48. Centres for the arcs for lower circular edges are obtained by drawing vertical lines from the centres for the upper arcs.

Problem 17-30. Draw the isometric view of the casting, shown in three views in fig. 17-49.

See fig. 17-50.

Problem 17-31. Draw the isometric view of the casting, two views of which are shown in fig. 17-51.

See fig. 17-52.

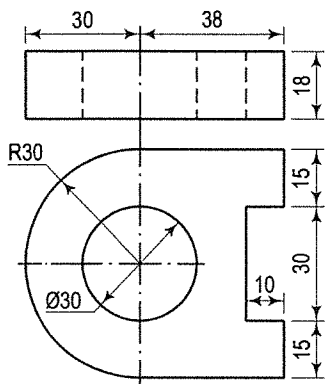


FIG. 17-47

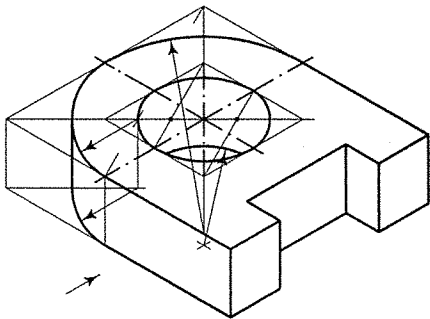
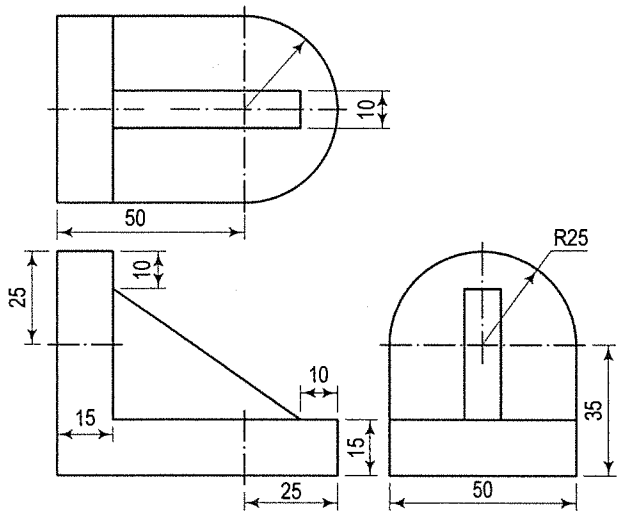


FIG. 17-48



(Third-angle projection)

FIG. 17-49

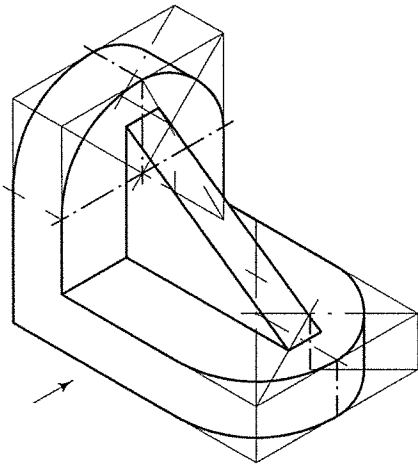
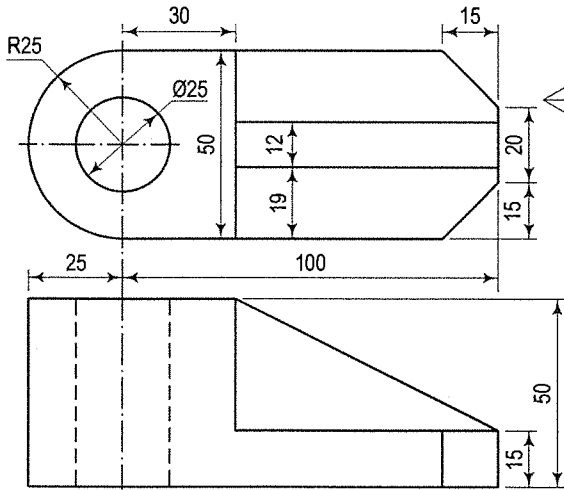


FIG. 17-50



(Third-angle projection)

FIG. 17-51

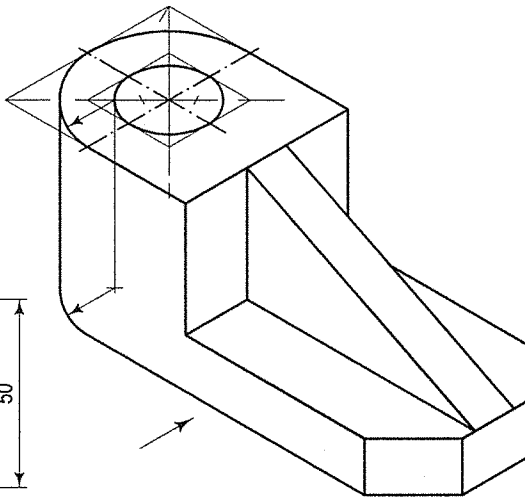
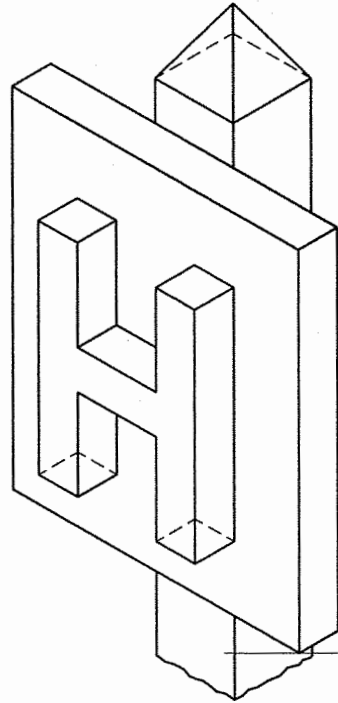
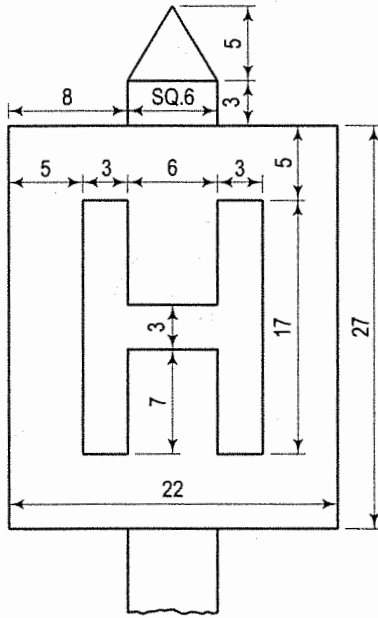


FIG. 17-52

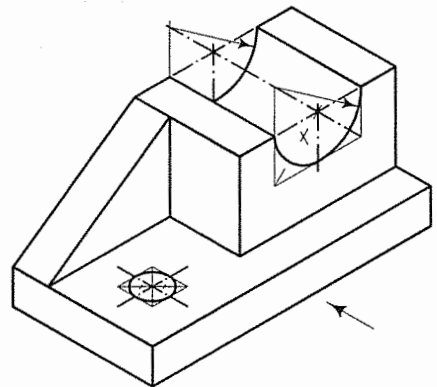
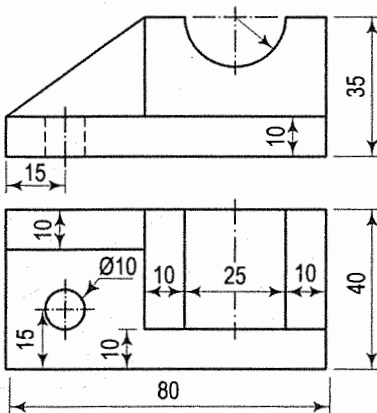
Problem 17-32. The front view of a board fitted with a letter H and mounted on a wooden post is given in fig. 17-53. Draw its isometric view, assuming the thickness of the board and of the letter to be equal to 3 cm. Scale, half full size. (All dimensions are given in centimeters.)

See fig. 17-54.



Problem 17-33. Draw the isometric view of the casting shown in two views in fig. 17-55.

See fig. 17-56.



Problem 17-34. Draw the isometric view of the model of steps, two views of which are shown in fig. 17-57.

See fig. 17-58.

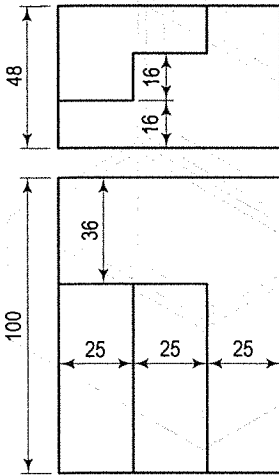


FIG. 17-57

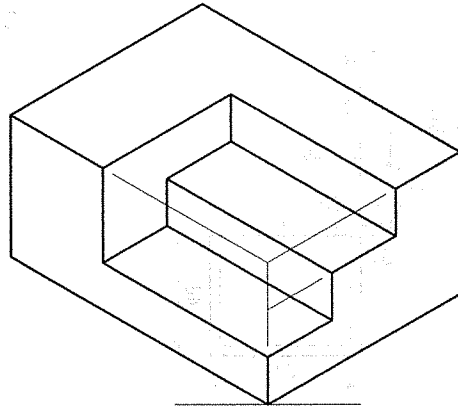


FIG. 17-58

Problem 17-35. Two pieces of wood joined together by a dovetail joint are shown in two views in fig. 17-59. Draw the isometric view of the two pieces separated but in a position ready for fitting.

See fig. 17-60.

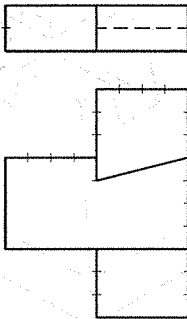


FIG. 17-59

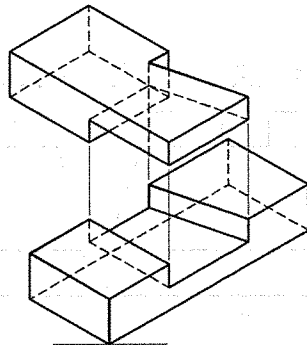


FIG. 17-60

Problem 17-36. The outside dimensions of a box made of 4 cm thick planks are 90 cm $\times$ 60 cm $\times$ 60 cm height. The depth of the lid on the outside is 12 cm. Draw the isometric view of the box when the lid is (a) 90° open and (b) 120° open.

Draw the orthographic view of the box with the lid in required positions as shown in fig. 17-61.

- (a) This position is simple to draw in isometric view. Care must, however, be taken to deduct the thickness of the wood for the bottom and the top, when showing in the lines for the inside of the box and the lid (fig. 17-62).

- (b) In this position, points P , Q , R etc. for the lid are located by enclosing the lid in the oblong and transferring the same on the isometric view as shown in fig. 17-62. The view is left incomplete to avoid congestion.

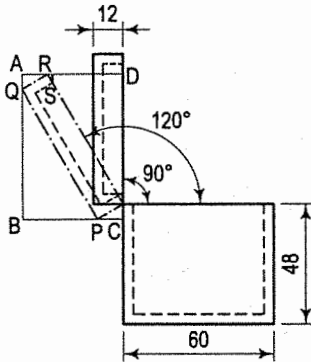


FIG. 17-61

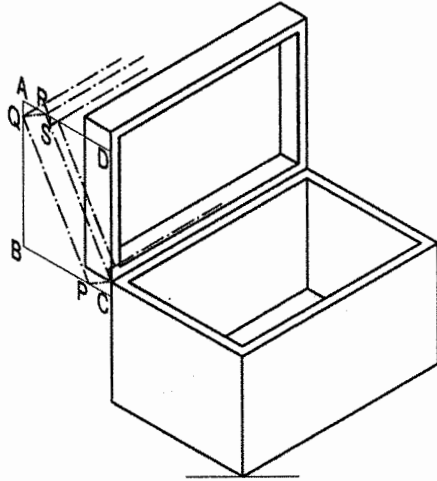


FIG. 17-62

Problem 17-37. Two views of a cast-iron block are shown in fig. 17-63. Draw its isometric view.

See fig. 17-64.

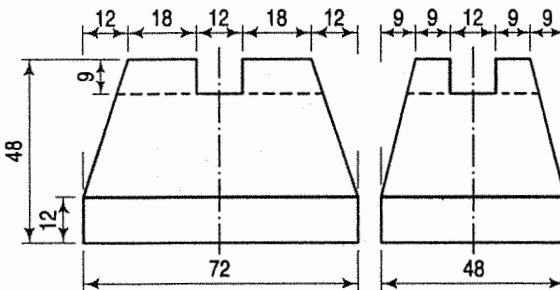


FIG. 17-63

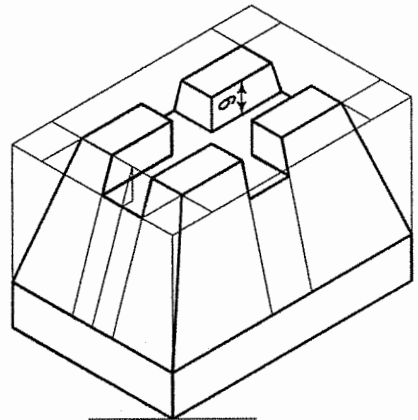


FIG. 17-64

The slope of the lines for the grooves on the outer surface on all the four sides is different and is obtained as shown by construction lines. The depth is measured along vertical lines.

Problem 17-38. Draw the isometric view of the casting shown in two views in fig. 17-65.

See fig. 17-66.

Problem 17-39. Draw the isometric view of the simple moulding shown in fig. 17-67.

See fig. 17-68.

The points on the curve are located by co-ordinate method. The parallel curve is obtained by drawing lines in the third direction and equal to the thickness of the moulding.

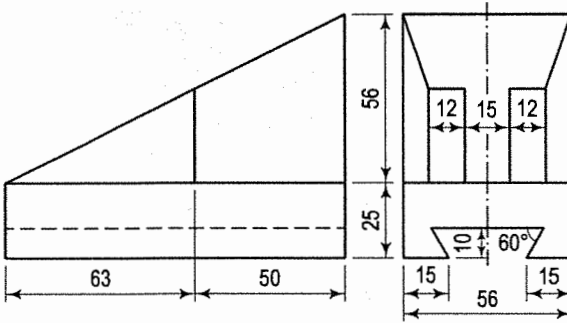


FIG. 17-65

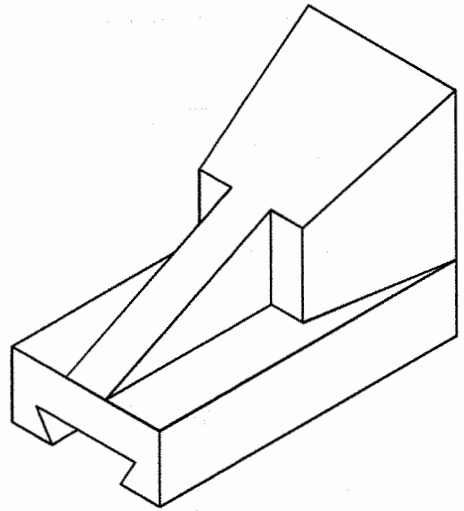


FIG. 17-66

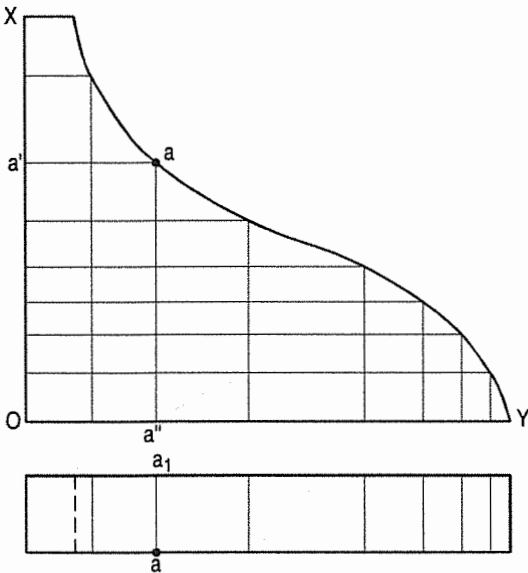


FIG. 17-67

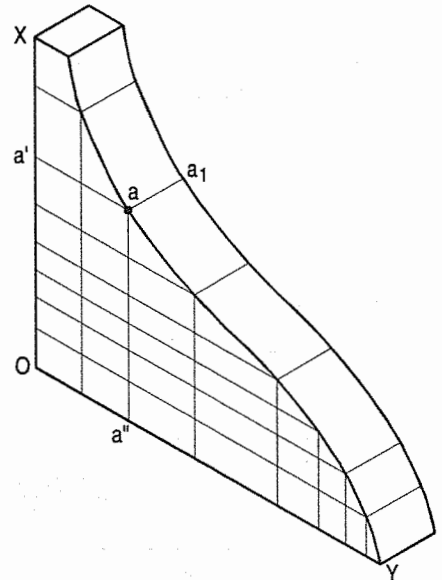


FIG. 17-68

Problem 17-40. The front view of three solids placed one above the other, with their axes in a straight line is shown in fig. 17-69. Draw the isometric view of the arrangement.

See fig. 17-70.

In this problem, isometric lengths must be taken for all dimensions except for the radius of the circle for the sphere.

The centre C of the sphere is at a distance equal to the isometric radius from the centre P of the top face of the cone frustum. The circle for the sphere is drawn with the true radius.

The ellipse for the section of the sphere is drawn within the rhombus constructed around the point Q on the axis.

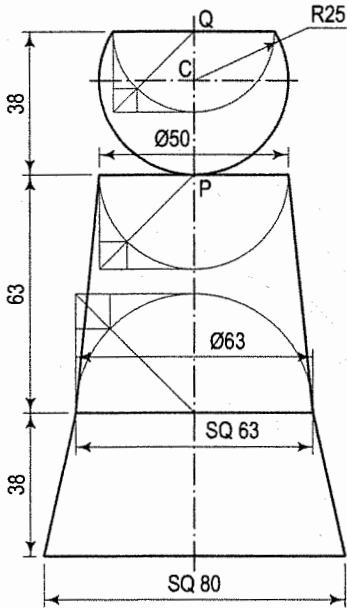


FIG. 17-69

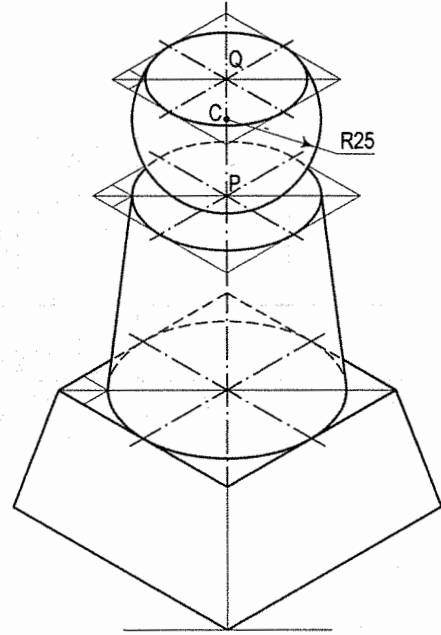


FIG. 17-70

Problem 17-41. Draw the isometric view of the clamping piece shown in fig. 17-71.

See fig. 17-72.

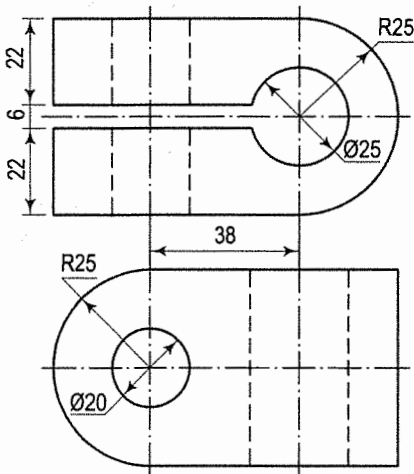


FIG. 17-71

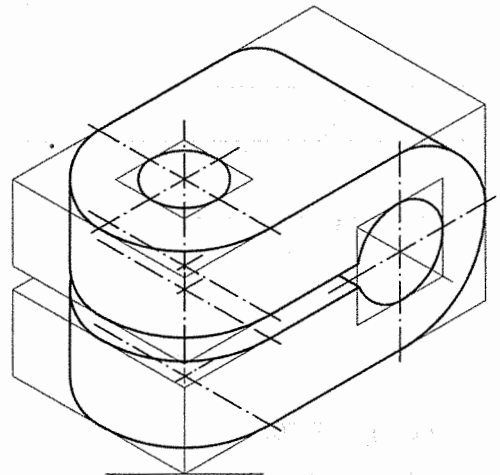


FIG. 17-72

Problem 17-42. (fig. 17-73): Draw the isometric view of a hexagonal nut for a 24 mm diameter bolt, assuming approximate dimensions. The threads may be neglected but chamfer must be shown.

See fig. 17-74.

Problem 17-43. Draw the isometric view of the paper-weight with spherical knob shown in fig. 17-75.

See fig. 17-76. Isometric scale should be used for all dimensions except for the radius of the circle for the sphere.

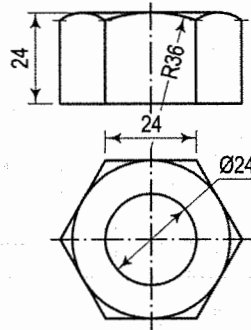


FIG. 17-73

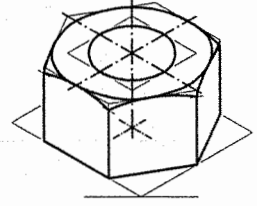


FIG. 17-74

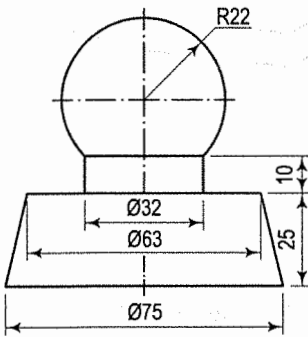


FIG. 17-75

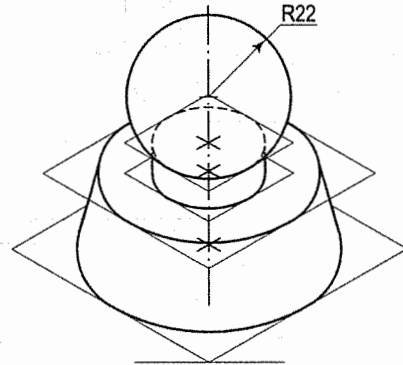


FIG. 17-76

Problem 17-44. (fig. 17-77): Draw the isometric view of a square-headed bolt 24 mm diameter and 70 mm long, with a square neck 18 mm thick and a head, 40 mm square and 18 mm thick.

See fig. 17-78.

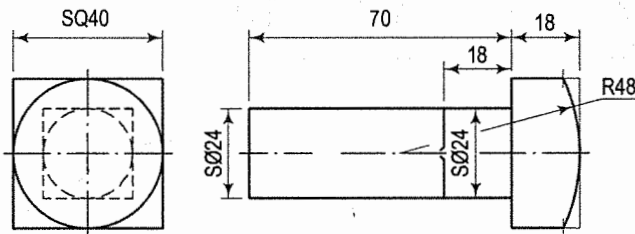


FIG. 17-77

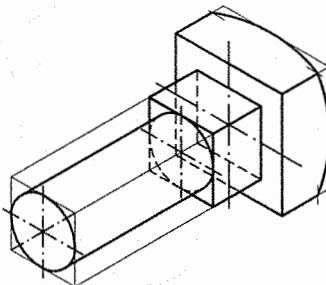


FIG. 17-78

Problem 17-45. Draw the isometric view of the casting shown in fig. 17-79.
See fig. 17-80.

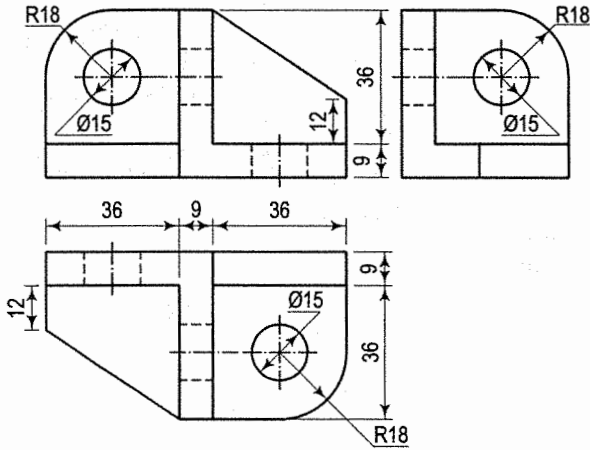


FIG. 17-79

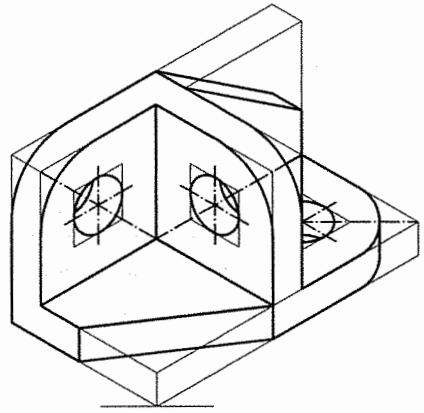


FIG. 17-80

Problem 17-46. The projections of a casting are shown in fig. 17-81.
Draw its isometric view.

See fig. 17-82.

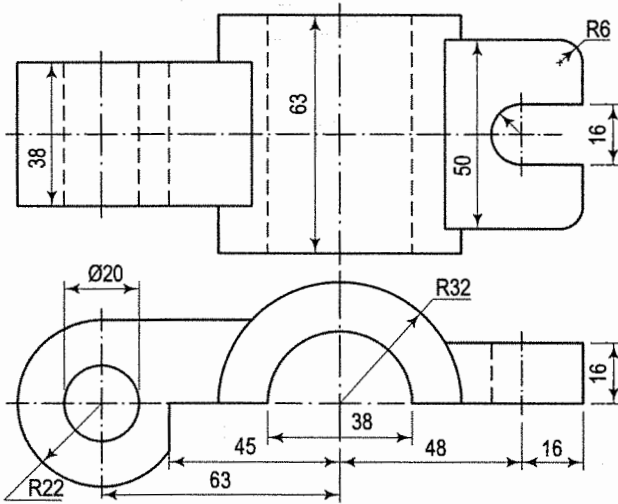


FIG. 17-81

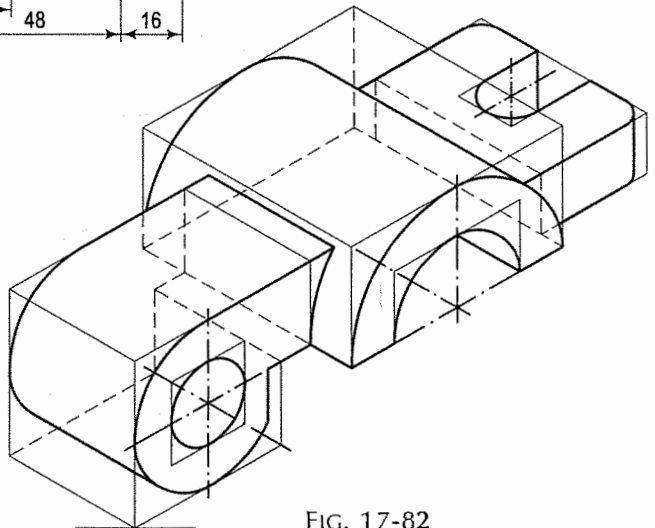


FIG. 17-82

Problem 17-47. Draw the isometric view of the bracket shown in two views in fig. 17-83.

See fig. 17-84.

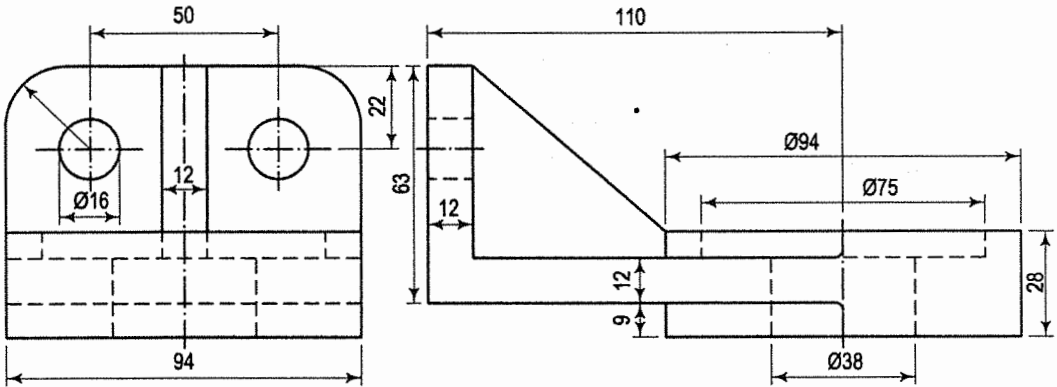


FIG. 17-83

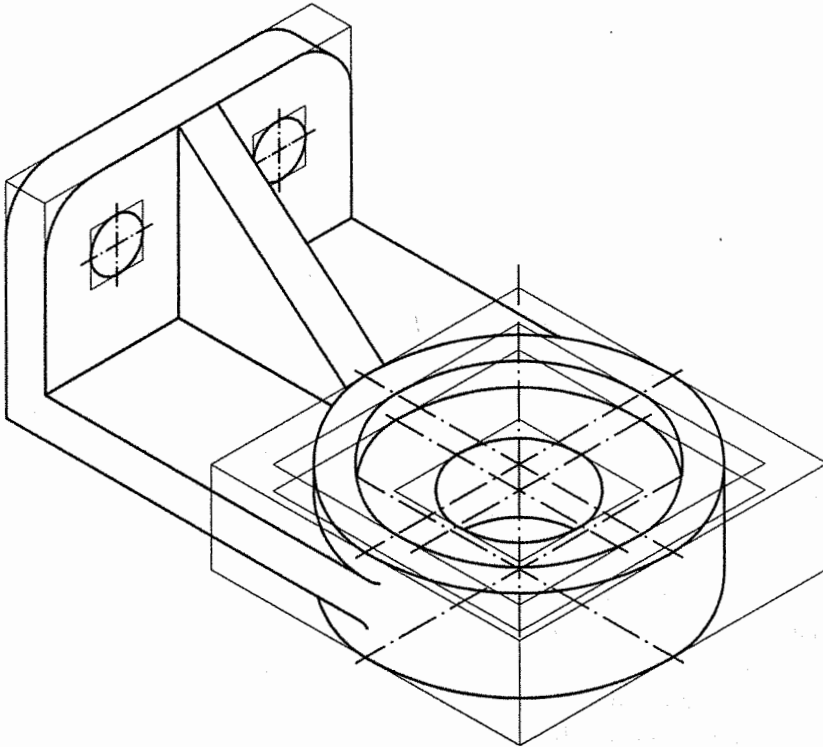


FIG. 17-84

Problem 17-48. Draw the isometric view of the machine-handle shown in fig. 17-85.

See fig. 17-86.

All measurements must be in isometric lengths except those for the diameters of spherical parts.

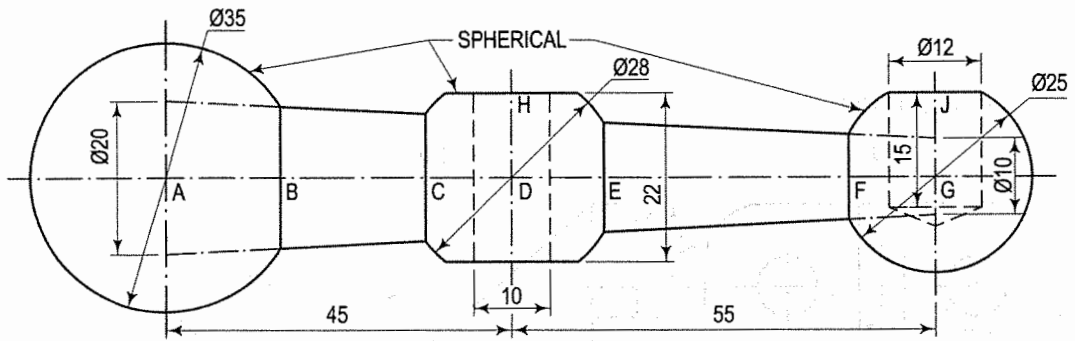


FIG. 17-85

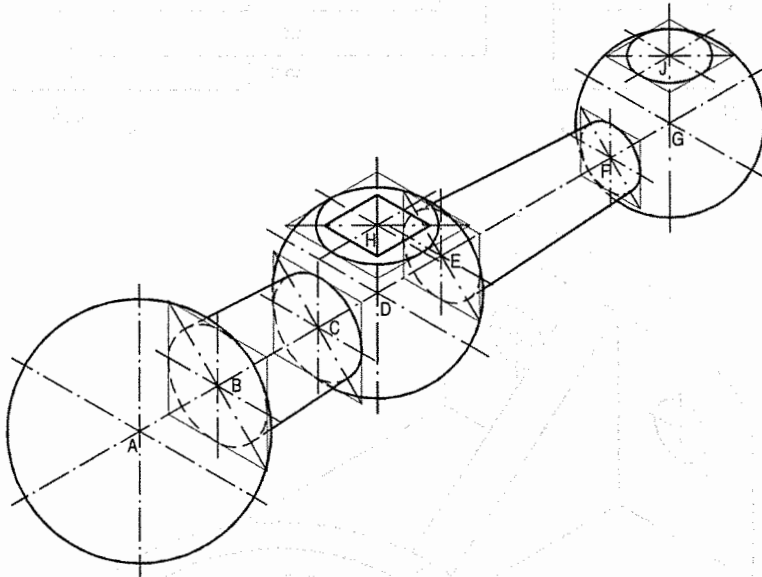


FIG. 17-86

- (i) Draw an axis and mark on it the positions of points *A*, *B* etc.
- (ii) At points *B*, *C*, *E* and *F*, draw ellipses for circular sections of the conical handle. Ellipses at *B* and *E* will be completely hidden.
- (iii) With points *A*, *D* and *G* as centres, draw circles for the spheres, with their respective true radii.
- (iv) Mark points *H* and *J* on the vertical axes through *D* and *G* respectively and draw ellipses for the respective sections of the spheres.
- (v) Around *H*, draw a rhombus for the square hole.
- (vi) The dotted lines for the depth of the holes are omitted.

Problem 17-49. The front view of a stool having a square top and four legs is shown in fig. 17-87. Draw its isometric view.

The legs lie along the slant edges of a frustum of a square pyramid (fig. 17-88).

Positions of the connecting horizontal strips between the legs at the top and at the bottom are determined by marking the heights along the axis and then drawing isometric lines upto the line *AB*, which shows the slope of the face of the frustum.

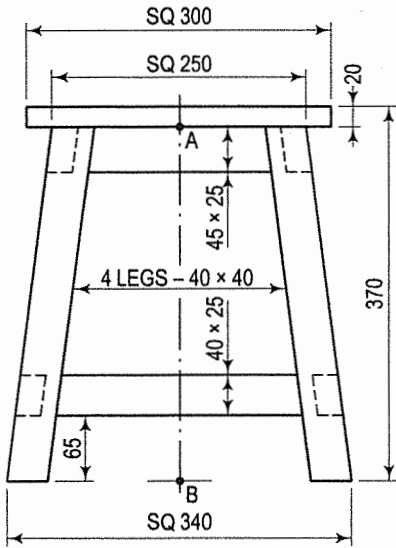


FIG. 17-87

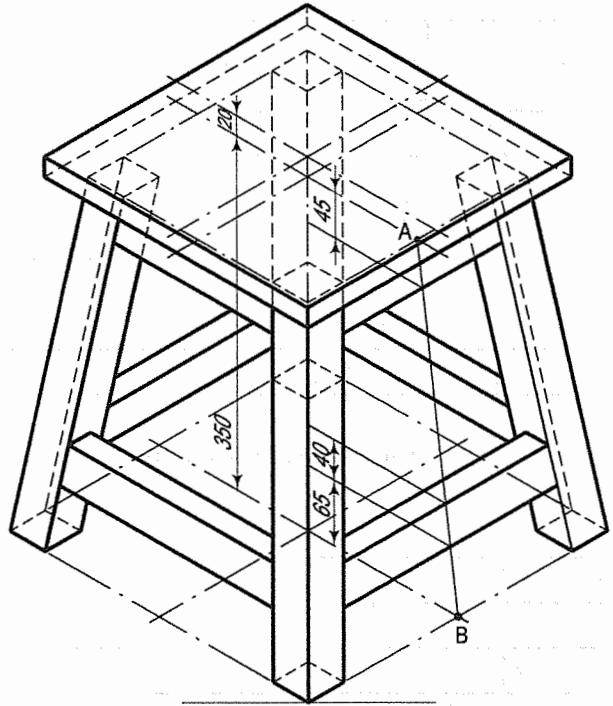


FIG. 17-88

EXERCISES 17

- Projections of castings of various shapes are given in figs. 17-89 to 17-115. Draw the isometric view of each casting.

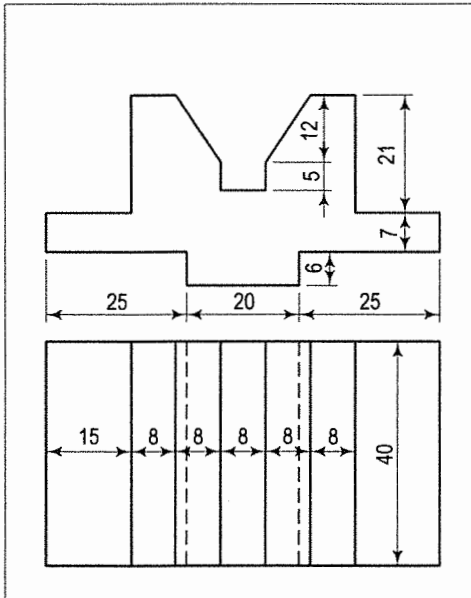


FIG. 17-89

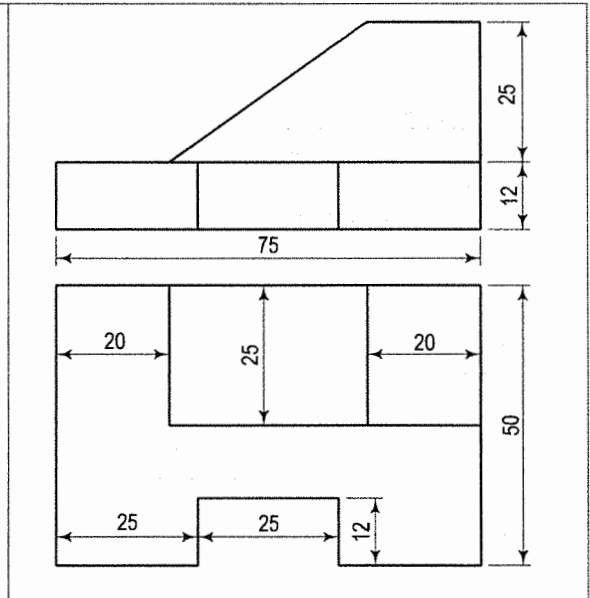
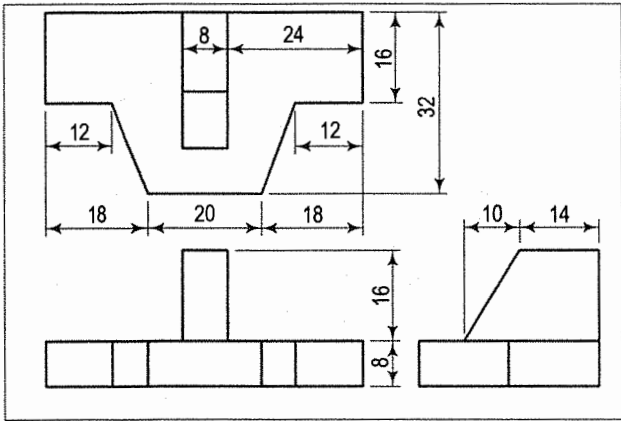


FIG. 17-90



(Third-angle projection)
FIG. 17-91

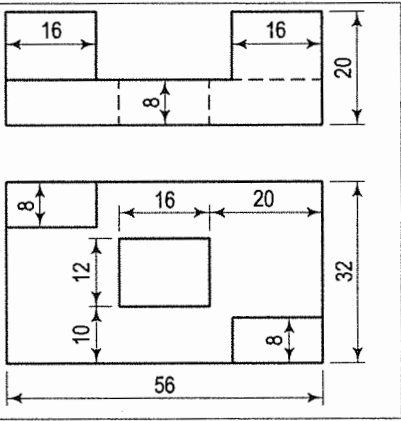


FIG. 17-92

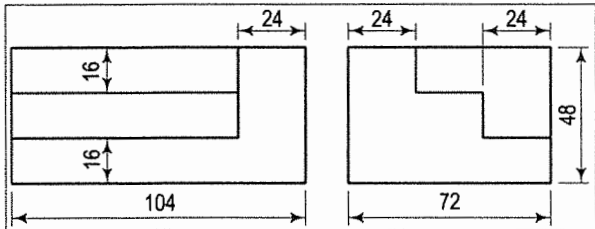


FIG. 17-93

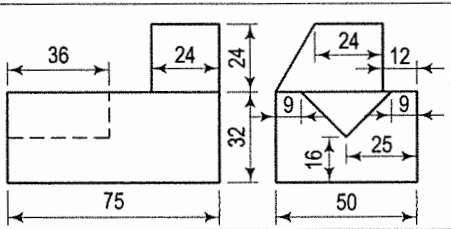
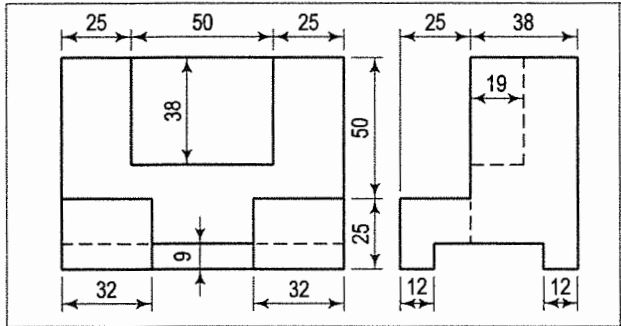
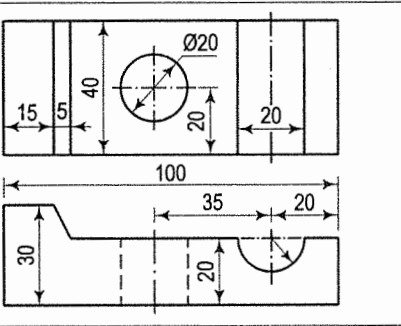


FIG. 17-94



(Third-angle projection)
FIG. 17-95



(Third-angle projection)
FIG. 17-96

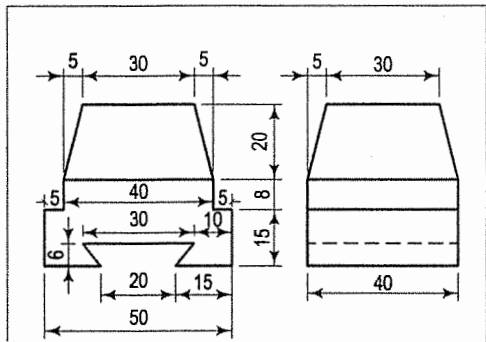


FIG. 17-97

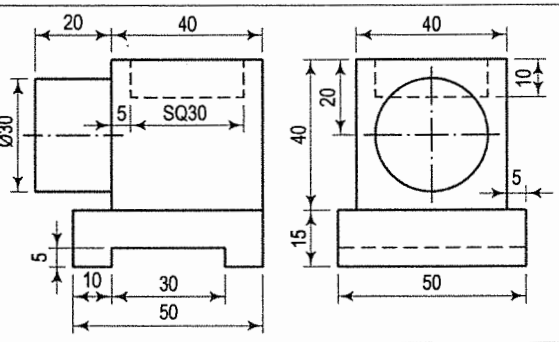


FIG. 17-98

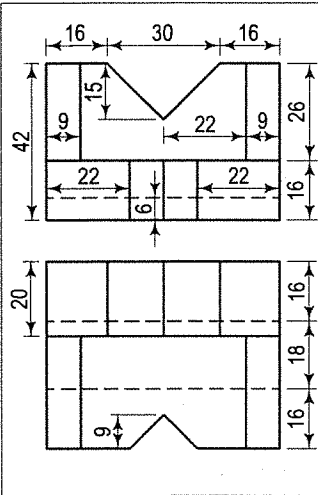


FIG. 17-99

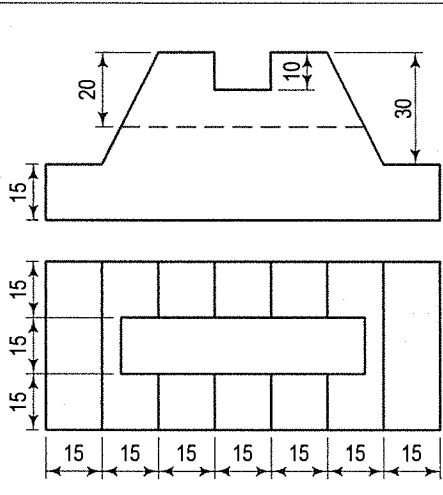


FIG. 17-100

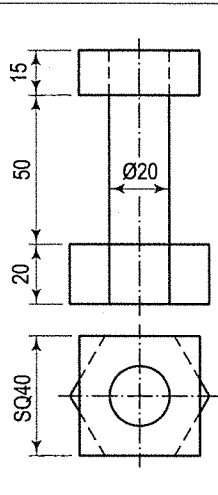


FIG. 17-101

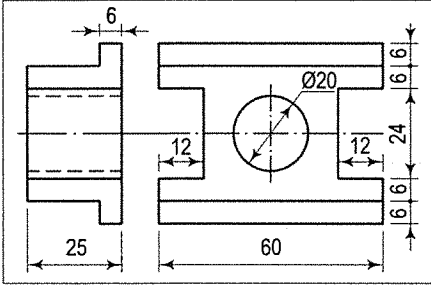


FIG. 17-102

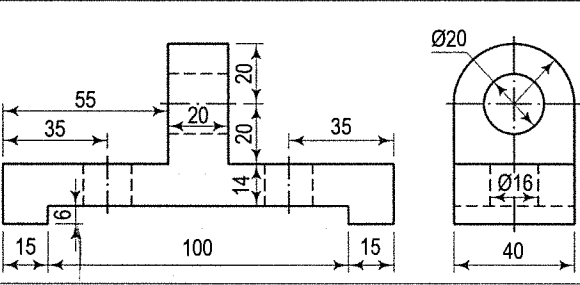


FIG. 17-103

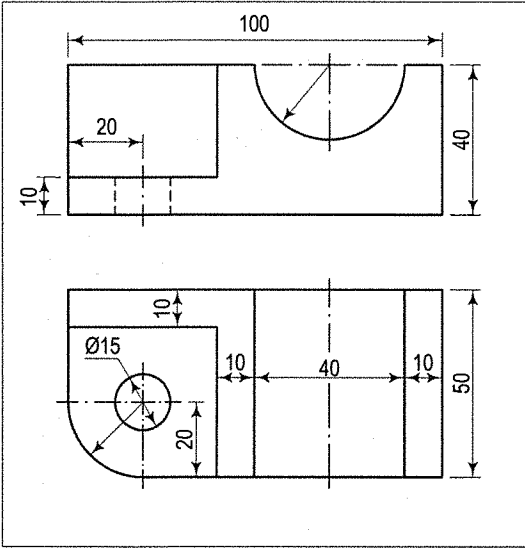
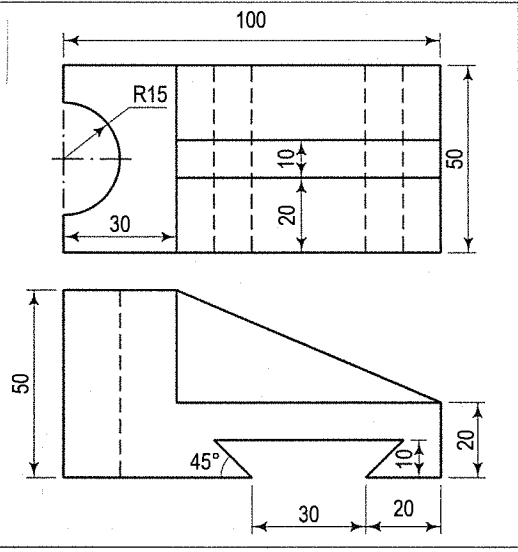


FIG. 17-104



(Third-angle projection)
FIG. 17-105

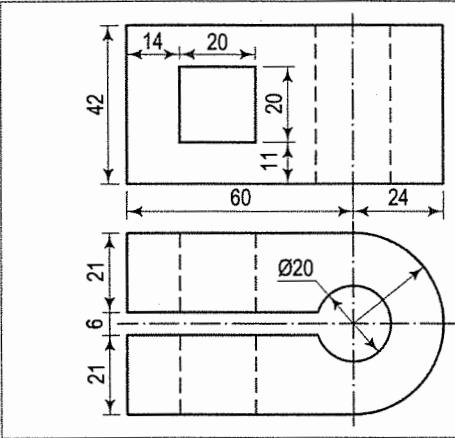
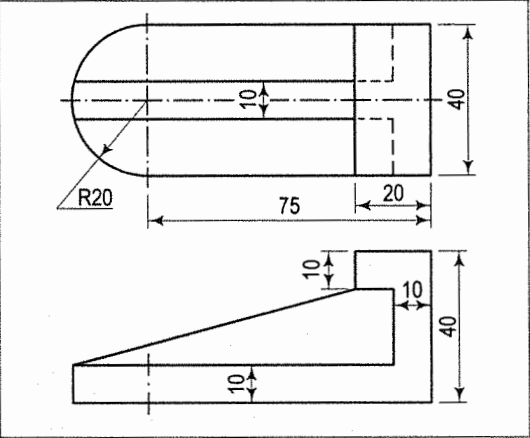


FIG. 17-106



(Third-angle projection)
FIG. 17-107

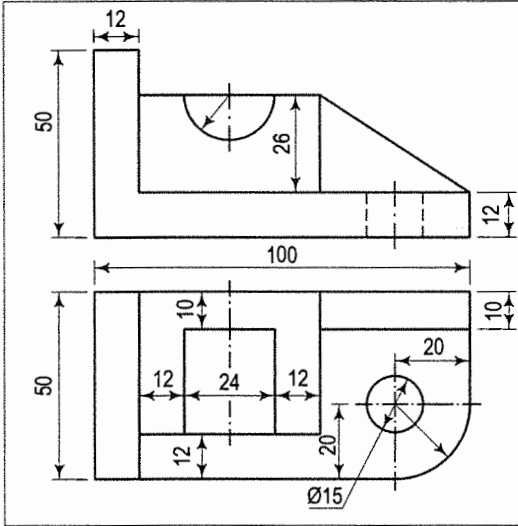
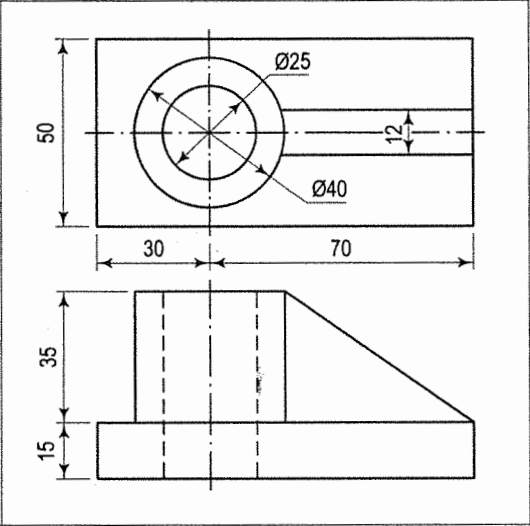


FIG. 17-108



(Third-angle projection)
FIG. 17-109

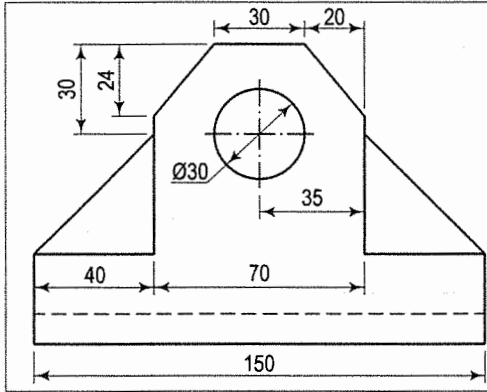


FIG. 17-110

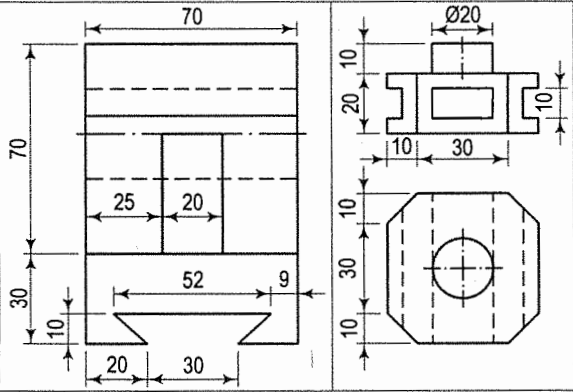
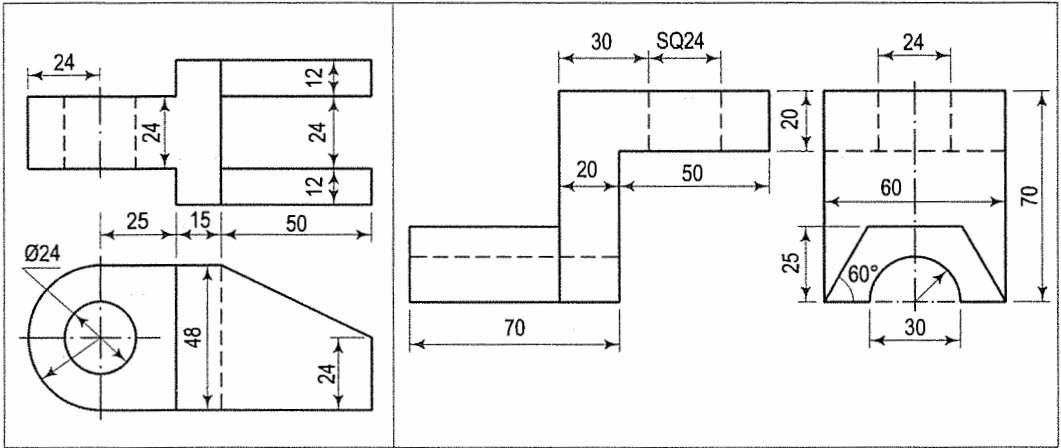


FIG. 17-111



(Third-angle projection)
FIG. 17-112

FIG. 17-113

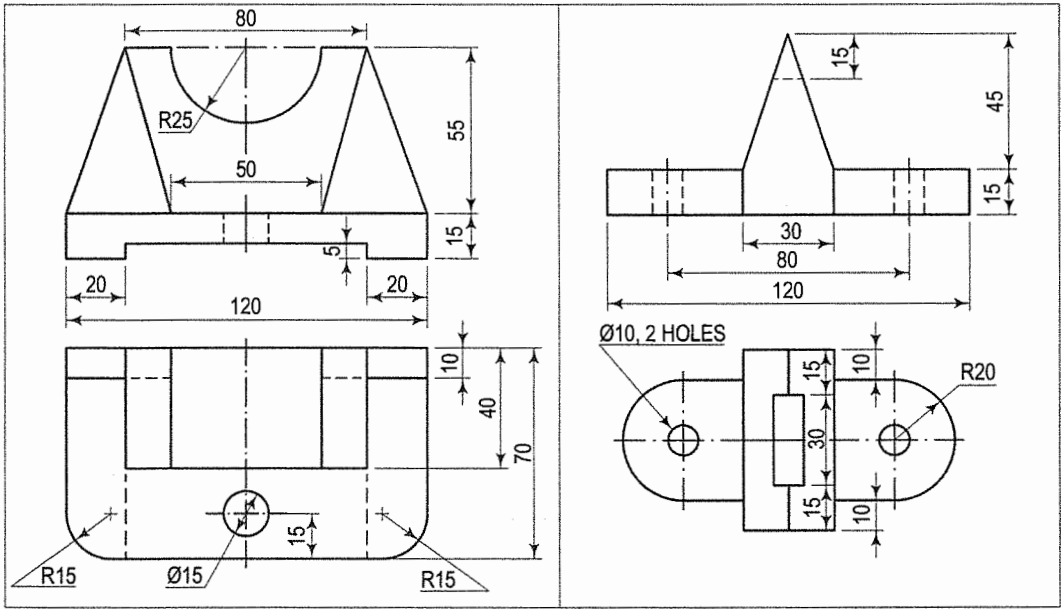
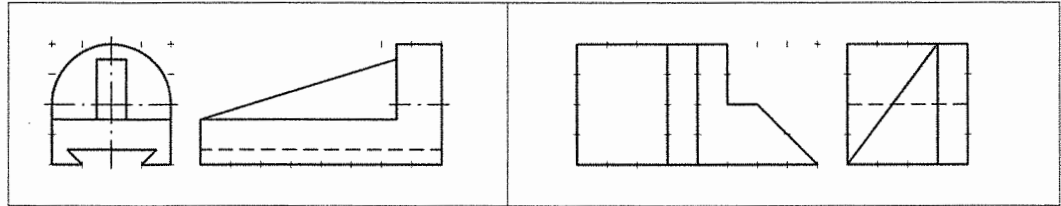


FIG. 17-114

FIG. 17-115

2. Assuming unit length to be equal to 10 mm, draw the isometric views of objects shown in figs. 17-116 to 17-125.



(Third-angle projection)
FIG. 17-116

FIG. 17-117

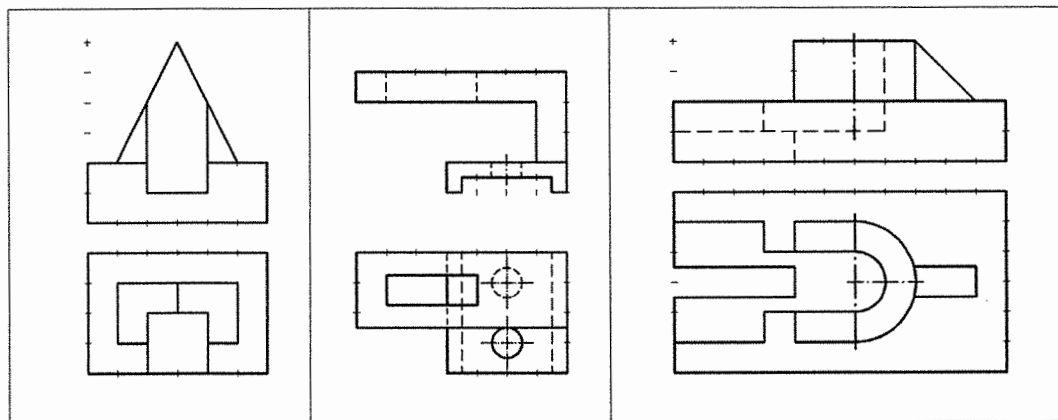
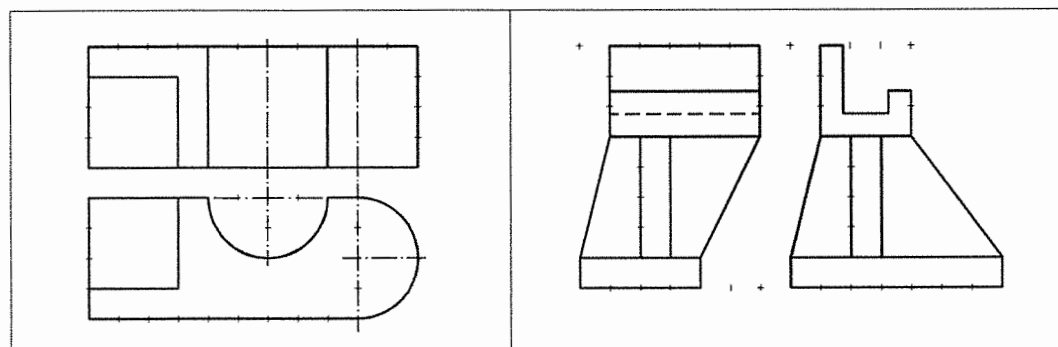


FIG. 17-118

FIG. 17-119

FIG. 17-120



(Third-angle projection)

FIG. 17-121

FIG. 17-122

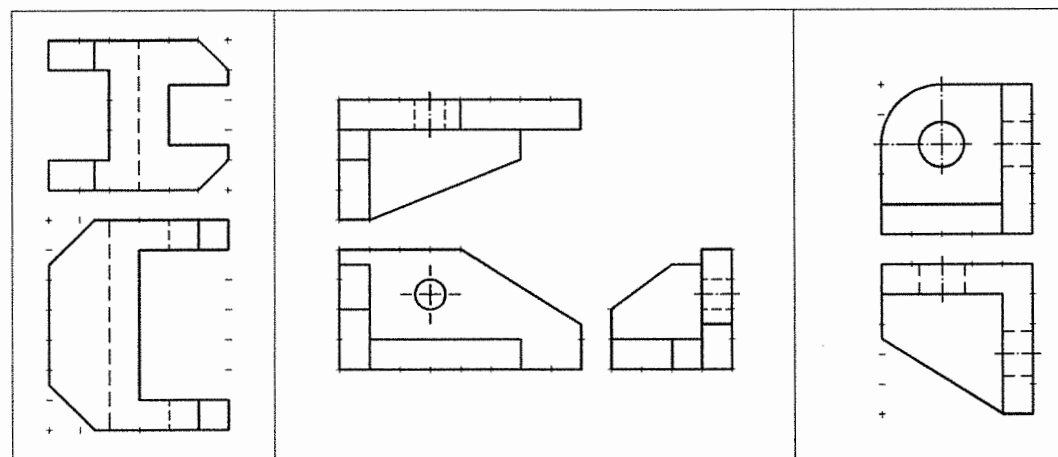


FIG. 17-123

(Third-angle projection)

FIG. 17-124

FIG. 17-125

3. Assuming simple graph of 10 mm grid, draw the isometric views of objects shown in fig. 17-126 and fig. 17-127.

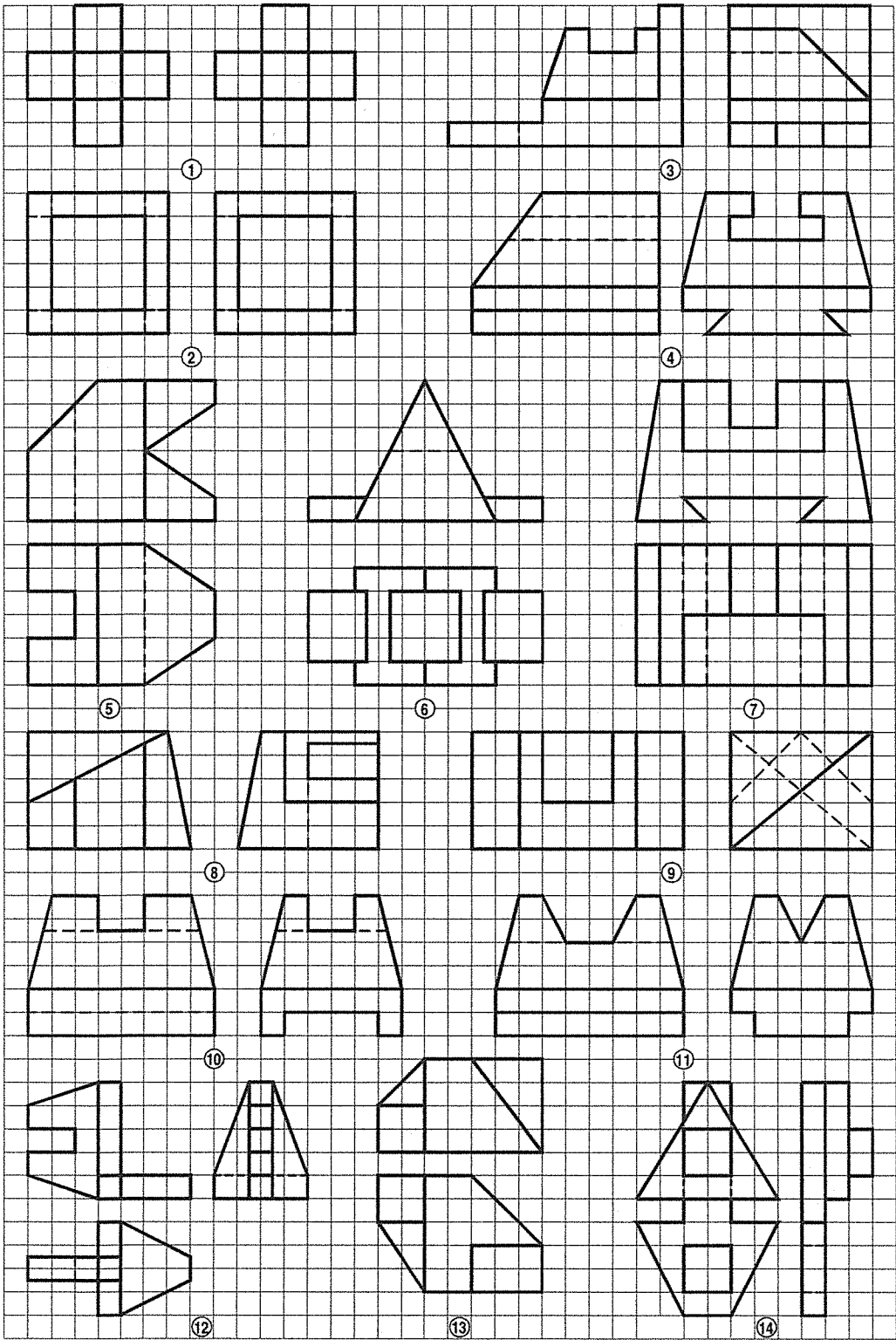


FIG. 17-126

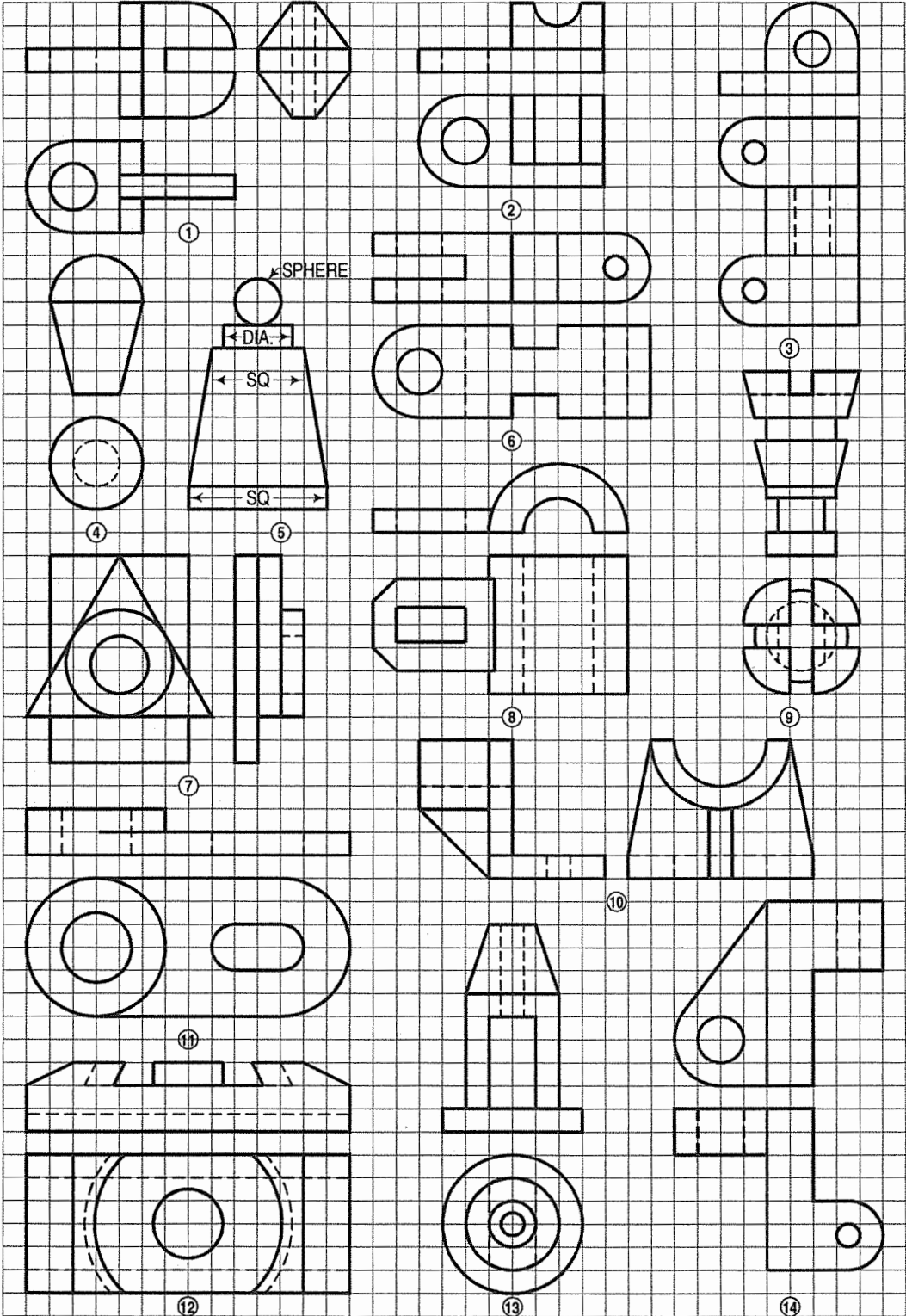


FIG. 17-127

4. Orthographic views of 35 objects with either (i) a line or (ii) lines or (iii) a view missing are given in fig. 17-128. Complete the given views. Also draw freehand, the isometric view of each object.

[For answer see fig. 17-194.]

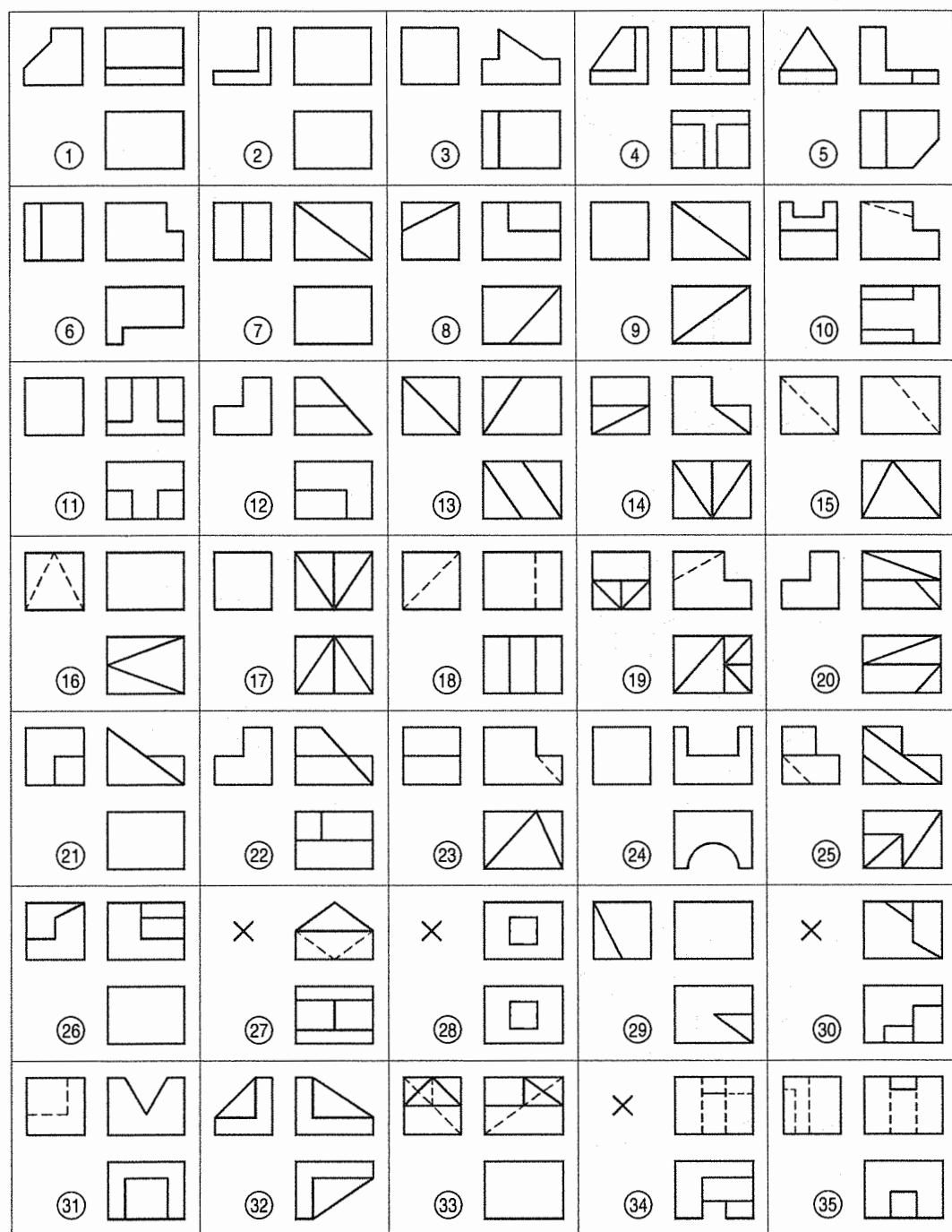
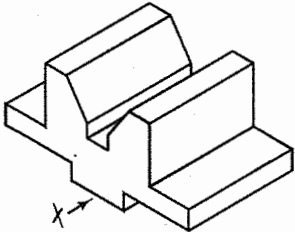
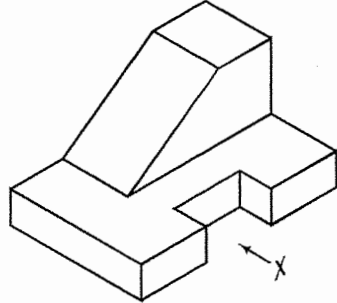


FIG. 17-128

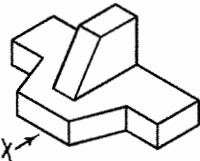
SOLUTIONS TO EXERCISES 17



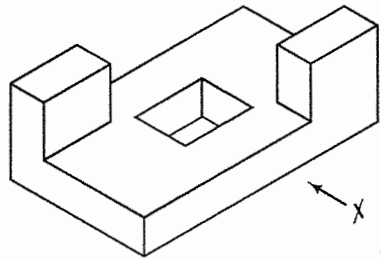
(Ex. 1. FIG. 17-89)
FIG. 17-129



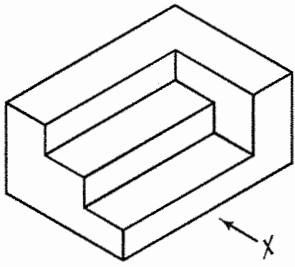
(Ex. 1. FIG. 17-90)
FIG. 17-130



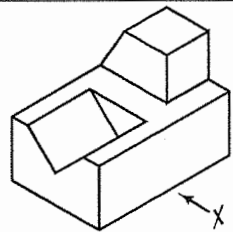
(Ex. 1. FIG. 17-91)
FIG. 17-131



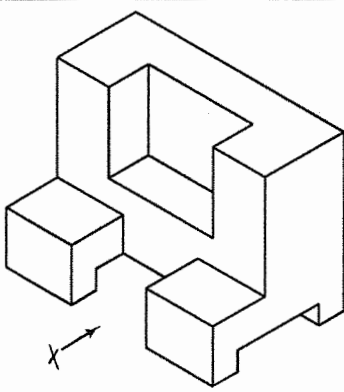
(Ex. 1. FIG. 17-92)
FIG. 17-132



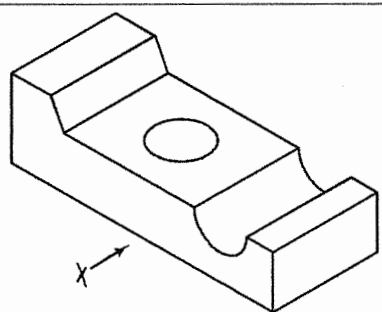
(Ex. 1. FIG. 17-93)
FIG. 17-133



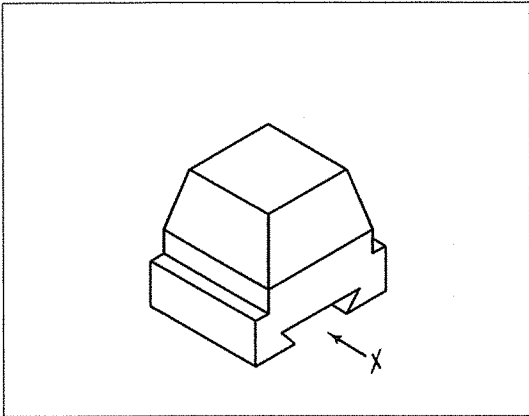
(Ex. 1. FIG. 17-94)
FIG. 17-134



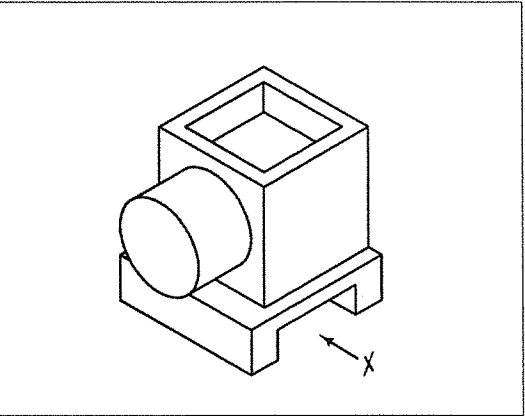
(Ex. 1. FIG. 17-95)
FIG. 17-135



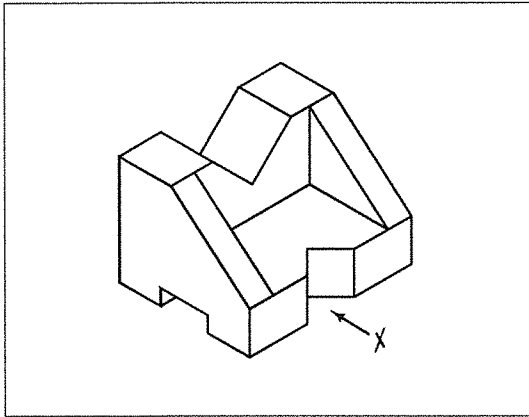
(Ex. 1. FIG. 17-96)
FIG. 17-136



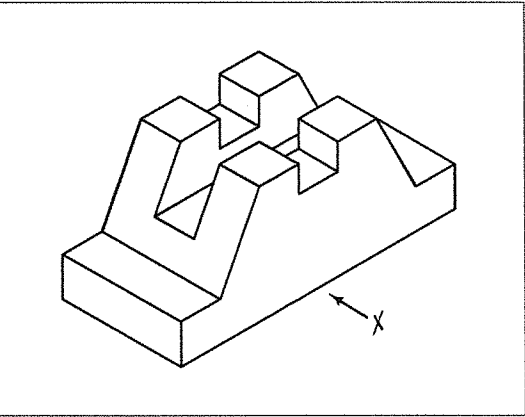
(Ex. 1. FIG. 17-97)
FIG. 17-137



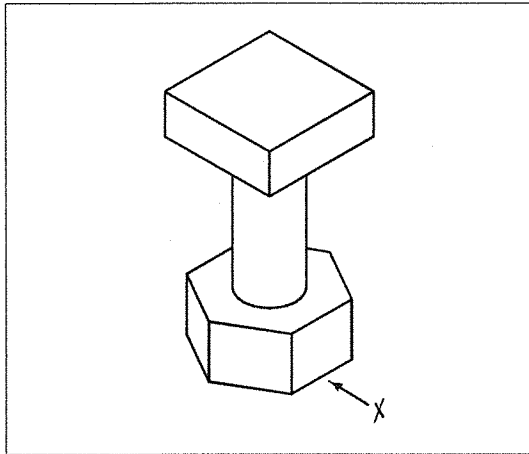
(Ex. 1. FIG. 17-98)
FIG. 17-138



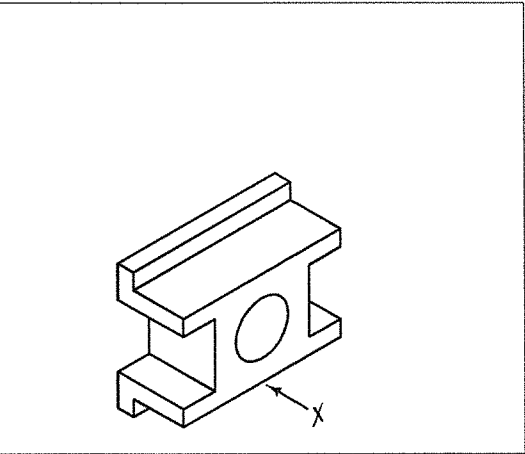
(Ex. 1. FIG. 17-99)
FIG. 17-139



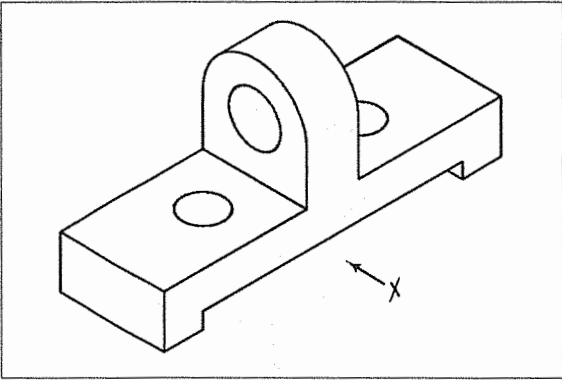
(Ex. 1. FIG. 17-100)
FIG. 17-140



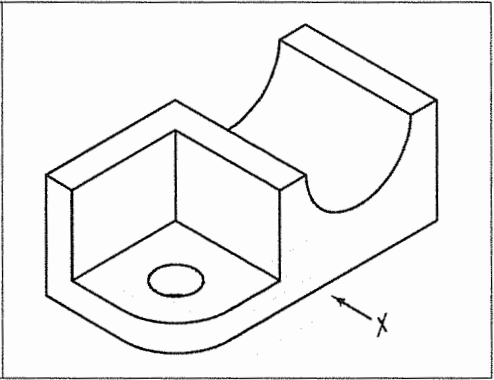
(Ex. 1. FIG. 17-101)
FIG. 17-141



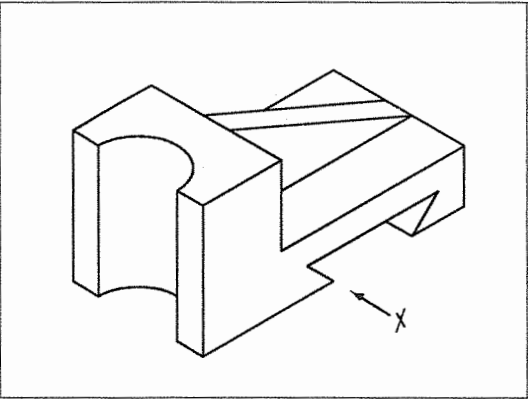
(Ex. 1. FIG. 17-102)
FIG. 17-142



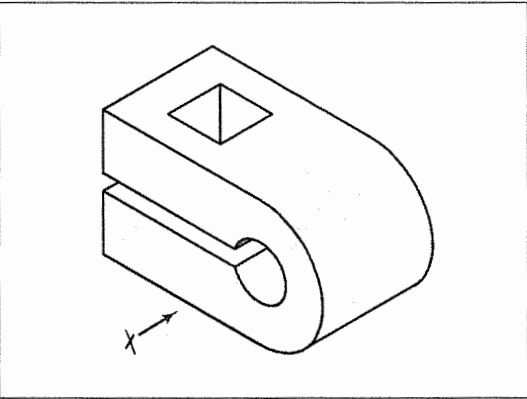
(EX. 1. FIG. 17-103)
FIG. 17-143



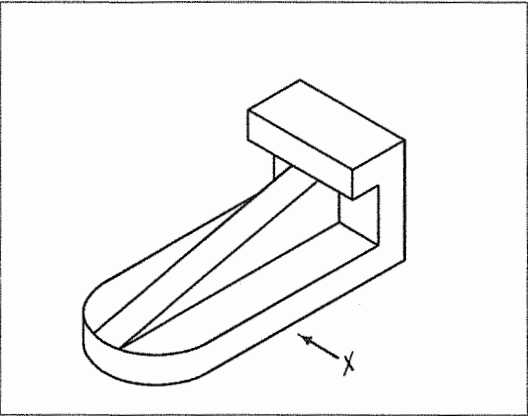
(EX. 1. FIG. 17-104)
FIG. 17-144



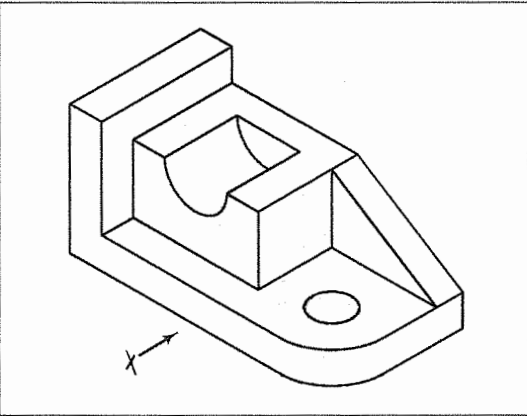
(EX. 1. FIG. 17-105)
FIG. 17-145



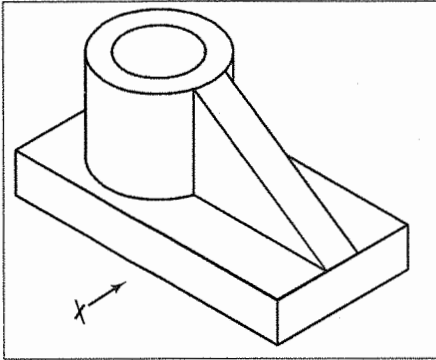
(EX. 1. FIG. 17-106)
FIG. 17-146



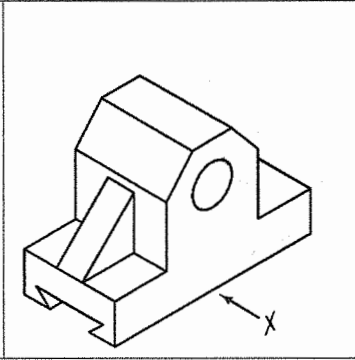
(EX. 1. FIG. 17-107)
FIG. 17-147



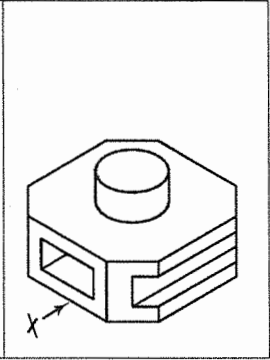
(EX. 1. FIG. 17-108)
FIG. 17-148



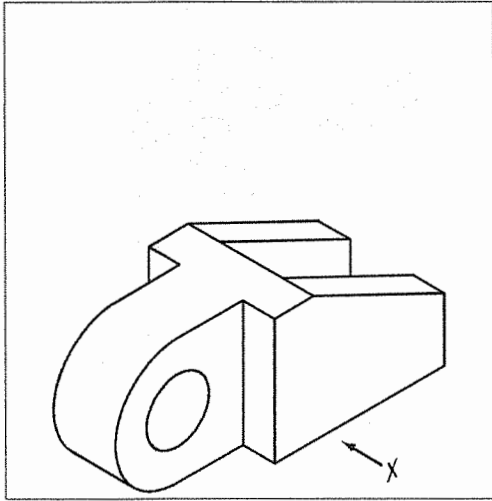
(EX. 1. FIG. 17-109)
FIG. 17-149



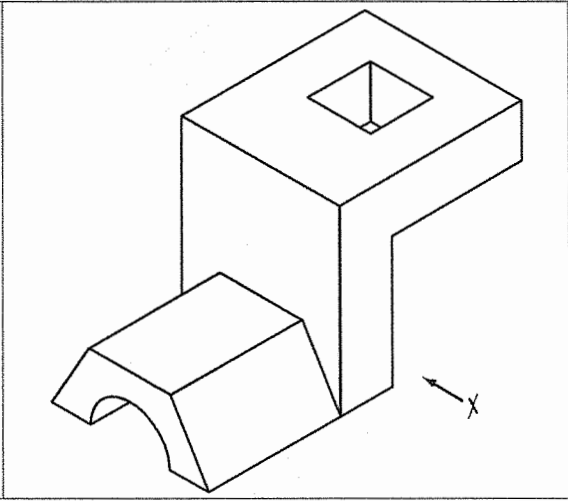
(EX. 1. FIG. 17-110)
FIG. 17-150



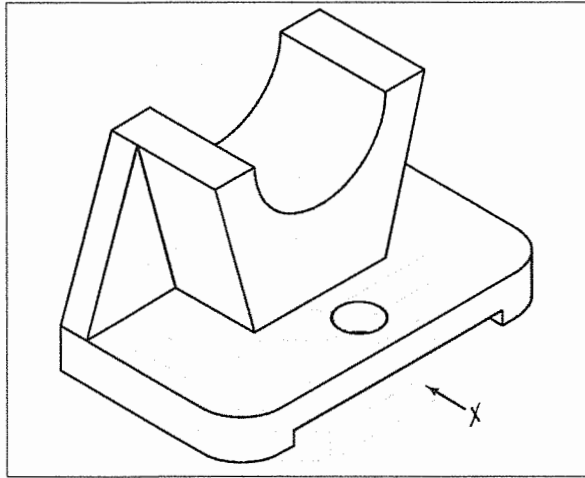
(EX. 1. FIG. 17-111)
FIG. 17-151



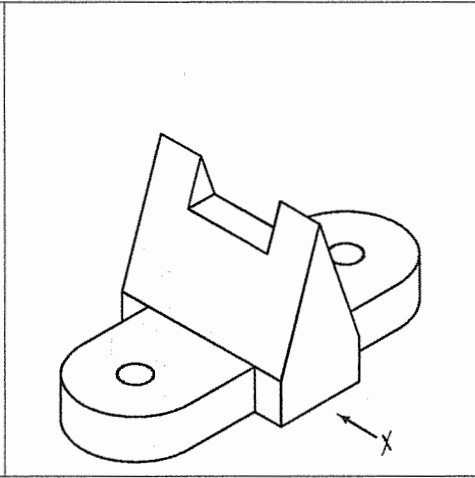
(EX. 1. FIG. 17-112)
FIG. 17-152



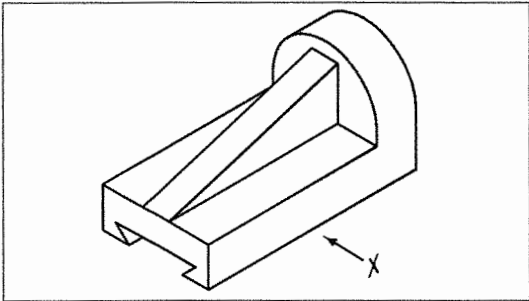
(EX. 1. FIG. 17-113)
FIG. 17-153



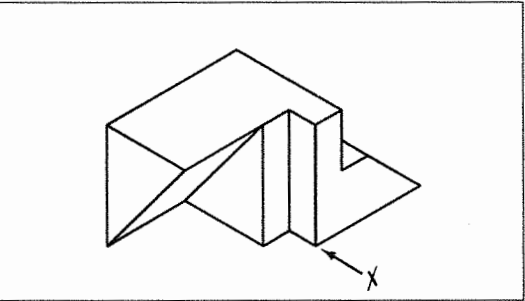
(EX. 1. FIG. 17-114)
FIG. 17-154



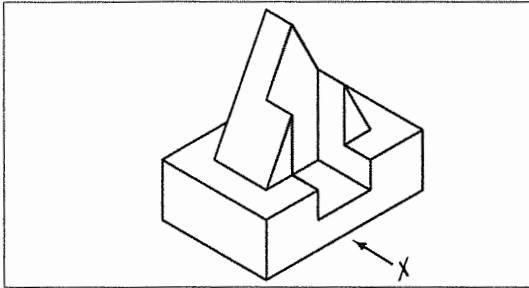
(EX. 1. FIG. 17-115)
FIG. 17-155



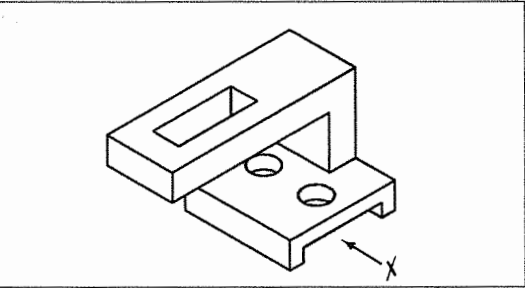
(EX. 2. FIG. 17-116)
FIG. 17-156



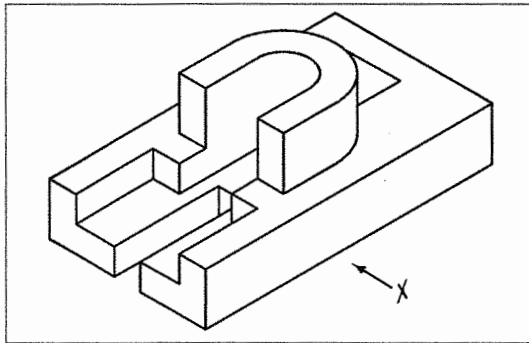
(EX. 2. FIG. 17-117)
FIG. 17-157



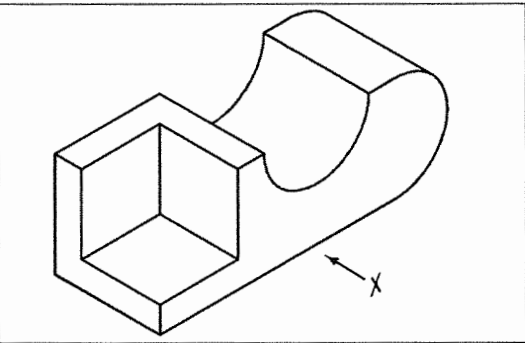
(EX. 2. FIG. 17-118)
FIG. 17-158



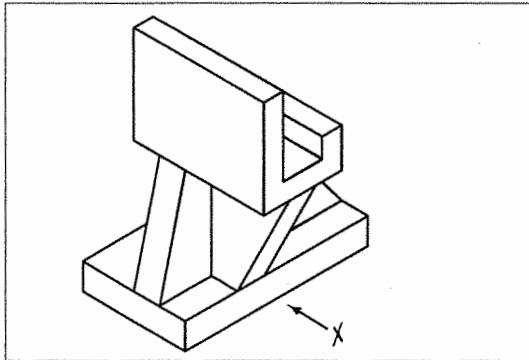
(EX. 2. FIG. 17-119)
FIG. 17-159



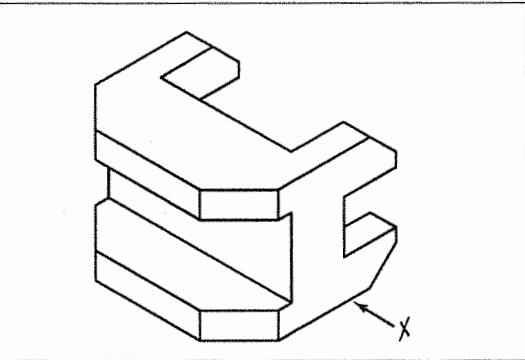
(EX. 2. FIG. 17-120)
FIG. 17-160



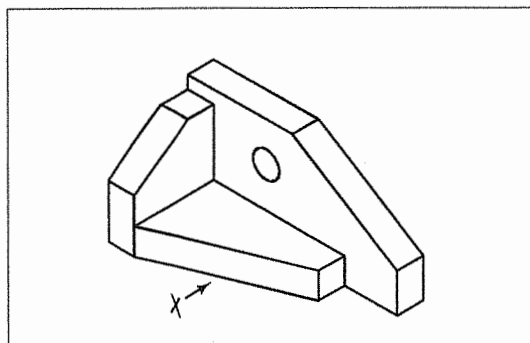
(EX. 2. FIG. 17-121)
FIG. 17-161



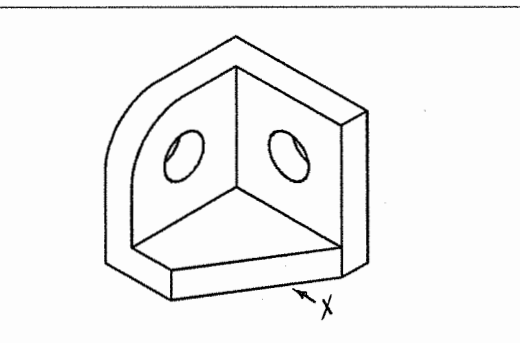
(EX. 2. FIG. 17-122)
FIG. 17-162



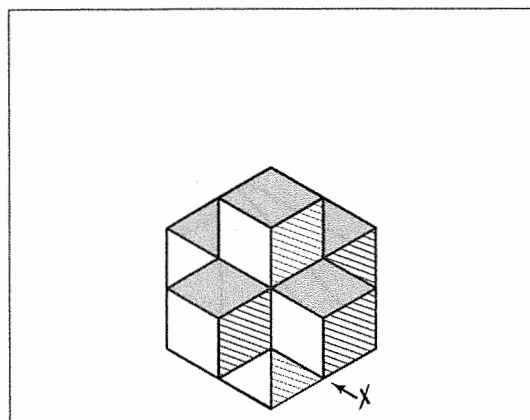
(EX. 2. FIG. 17-123)
FIG. 17-163



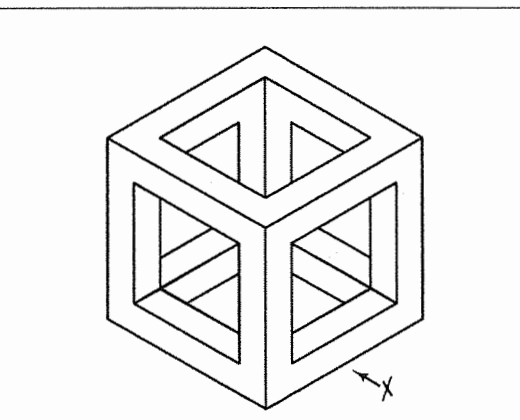
(Ex. 2. FIG. 17-124)
FIG. 17-164



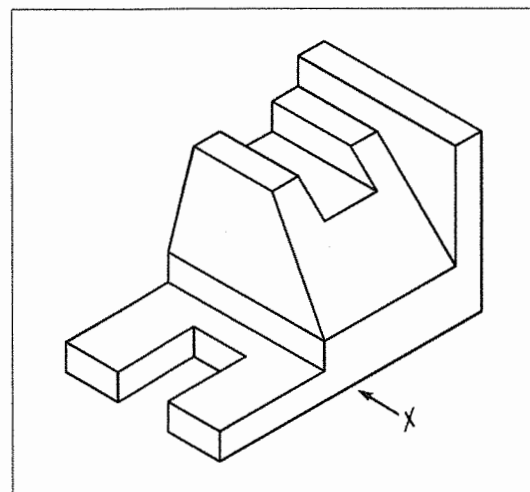
(Ex. 2. FIG. 17-125)
FIG. 17-165



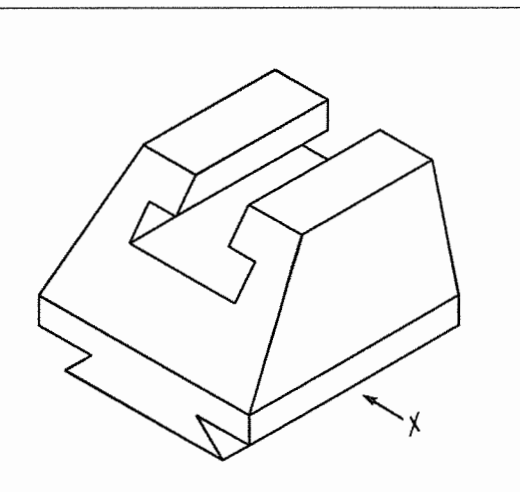
[Ex. 3. FIG. 17-126(1)]
FIG. 17-166



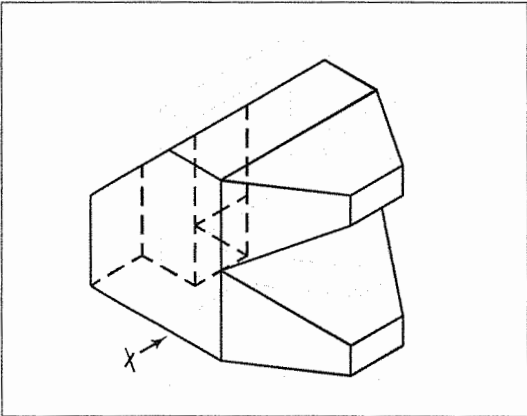
[Ex. 3. FIG. 17-126(2)]
FIG. 17-167



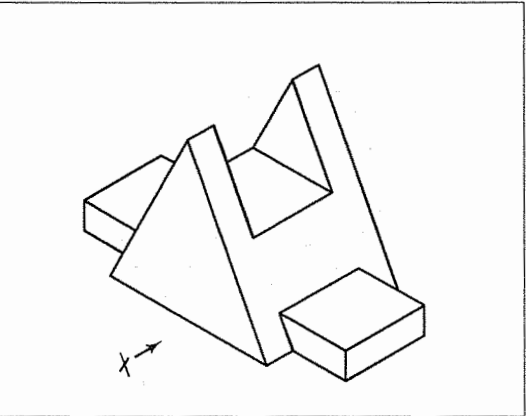
[Ex. 3. FIG. 17-126(3)]
FIG. 17-168



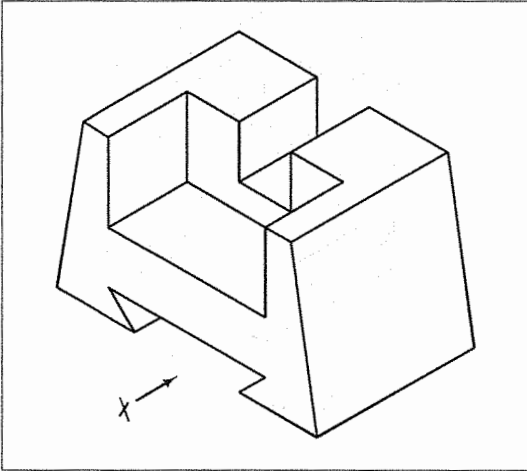
[Ex. 3. FIG. 17-126(4)]
FIG. 17-169



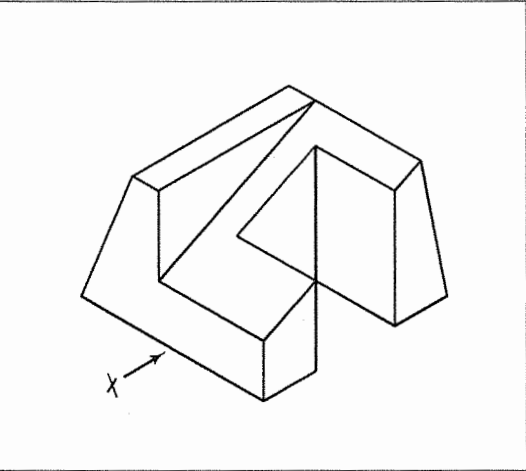
[EX. 3. FIG. 17-126(5)]
FIG. 17-170



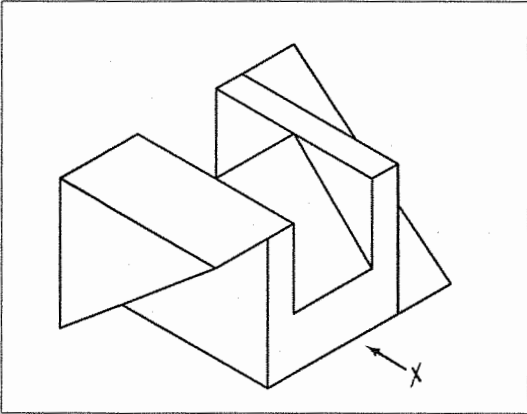
[EX. 3. FIG. 17-126(6)]
FIG. 17-171



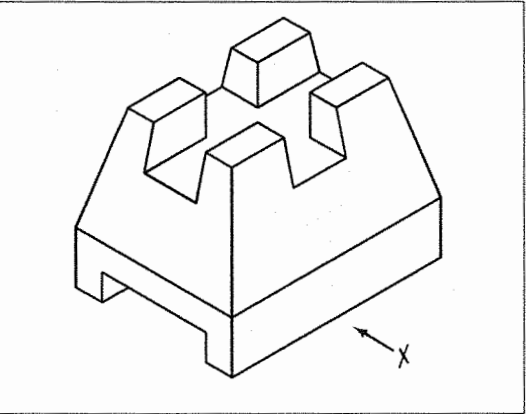
[EX. 3. FIG. 17-126(7)]
FIG. 17-172



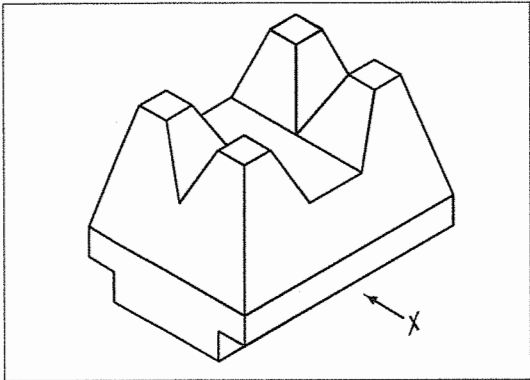
[EX. 3. FIG. 17-126(8)]
FIG. 17-173



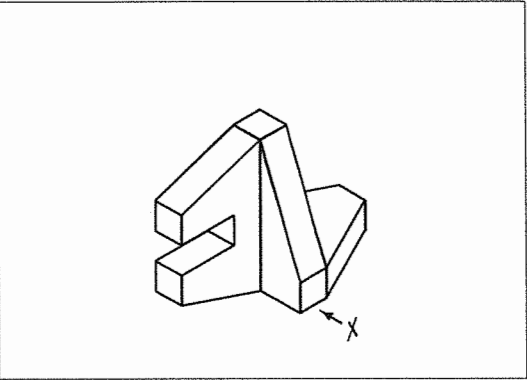
[EX. 3. FIG. 17-126(9)]
FIG. 17-174



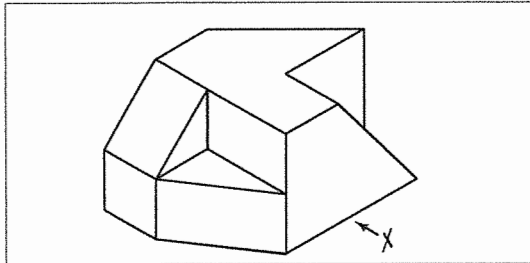
[EX. 3. FIG. 17-126(10)]
FIG. 17-175



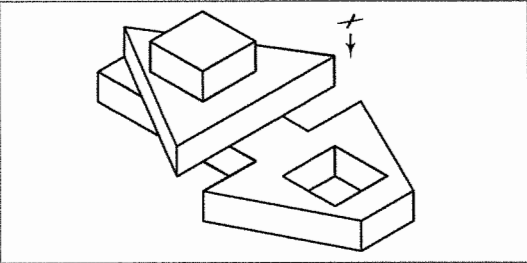
[Ex. 3. FIG. 17-126(11)]
FIG. 17-176



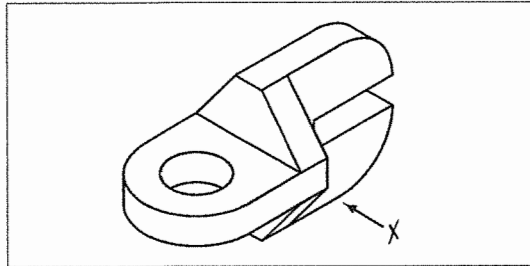
[Ex. 3. FIG. 17-126(12)]
FIG. 17-177



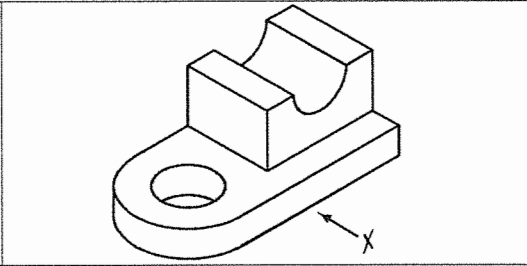
[Ex. 3. FIG. 17-126(13)]
FIG. 17-178



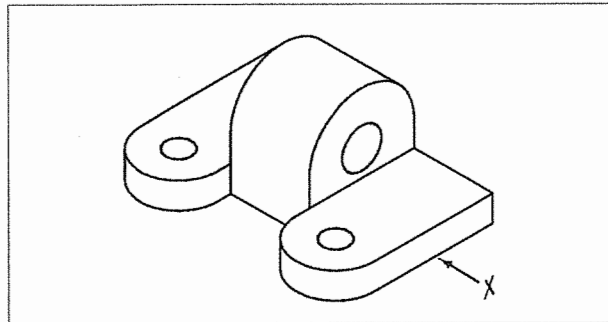
[Ex. 3. FIG. 17-126(14)]
FIG. 17-179



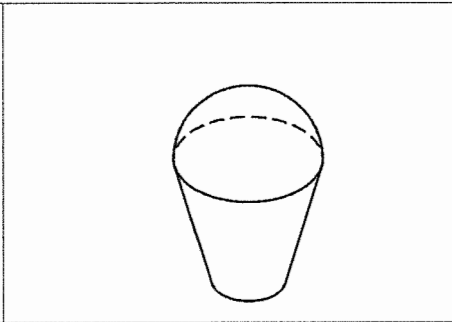
[Ex. 3. FIG. 17-127(1)]
FIG. 17-180



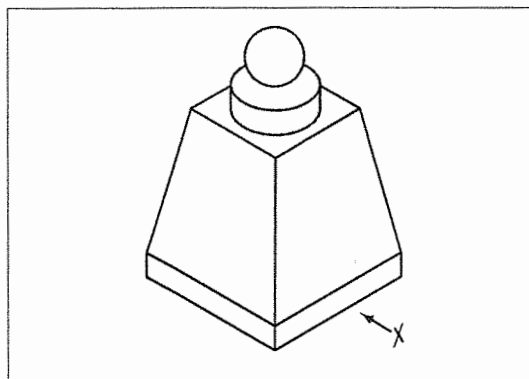
[Ex. 3. FIG. 17-127(2)]
FIG. 17-181



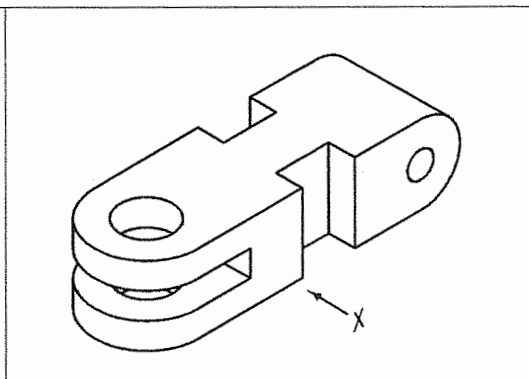
[Ex. 3. FIG. 17-127(3)]
FIG. 17-182



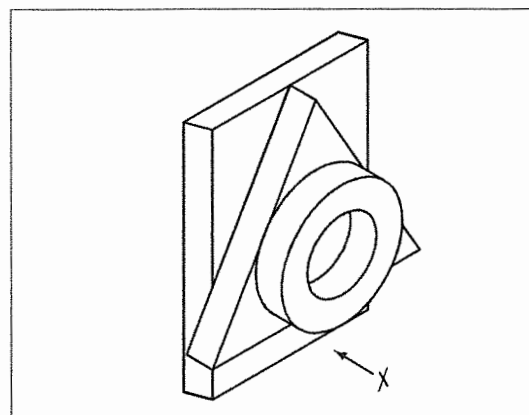
[Ex. 3. FIG. 17-127(4)]
FIG. 17-183



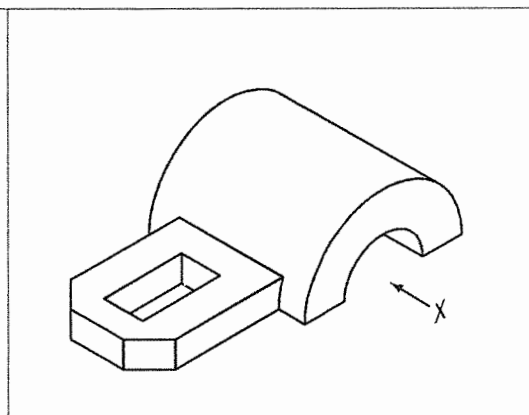
[Ex. 3. FIG. 17-127(5)]
FIG. 17-184



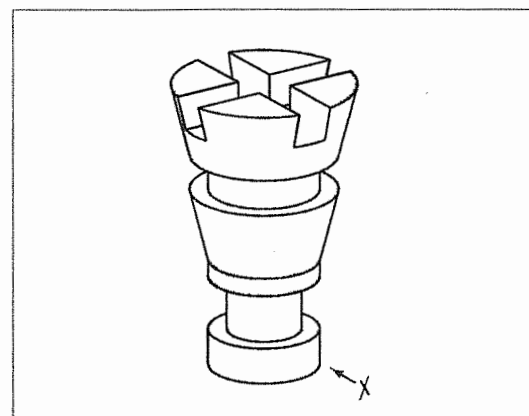
[Ex. 3. FIG. 17-127(6)]
FIG. 17-185



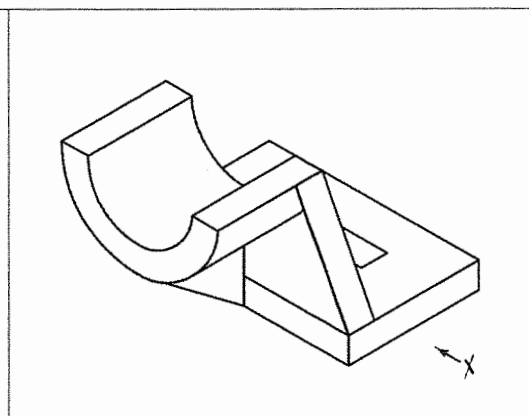
[Ex. 3. FIG. 17-127(7)]
FIG. 17-186



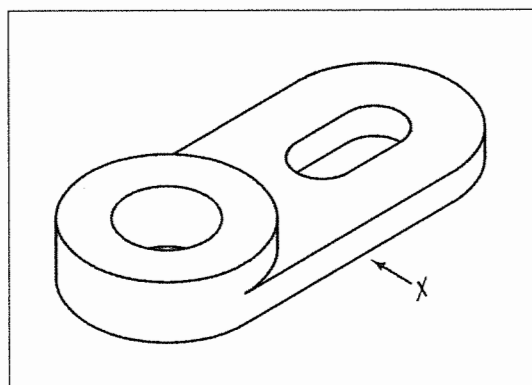
[Ex. 3. FIG. 17-127(8)]
FIG. 17-187



[Ex. 3. FIG. 17-127(9)]
FIG. 17-188

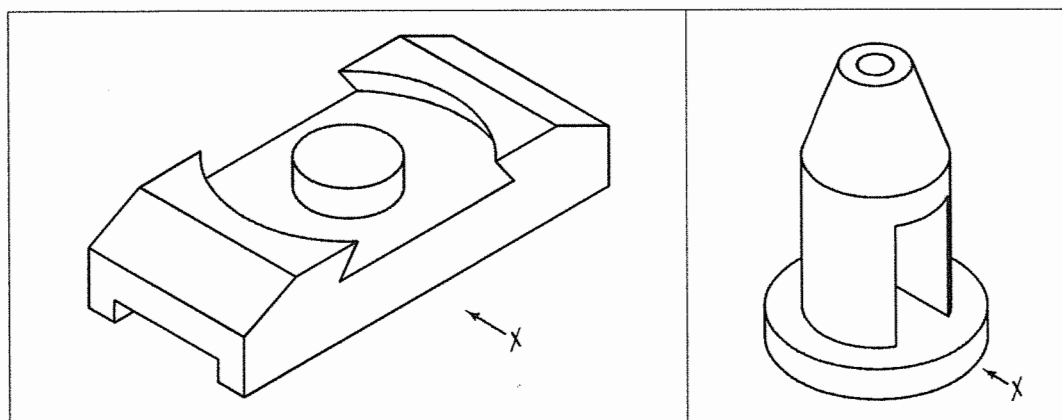


[Ex. 3. FIG. 17-127(10)]
FIG. 17-189



[Ex. 3. FIG. 17-127(11)]

FIG. 17-190

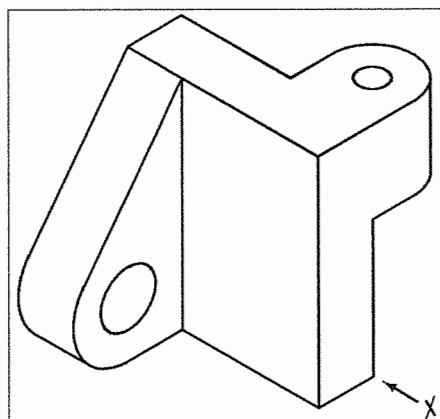


[Ex. 3. FIG. 17-127(12)]

FIG. 17-191

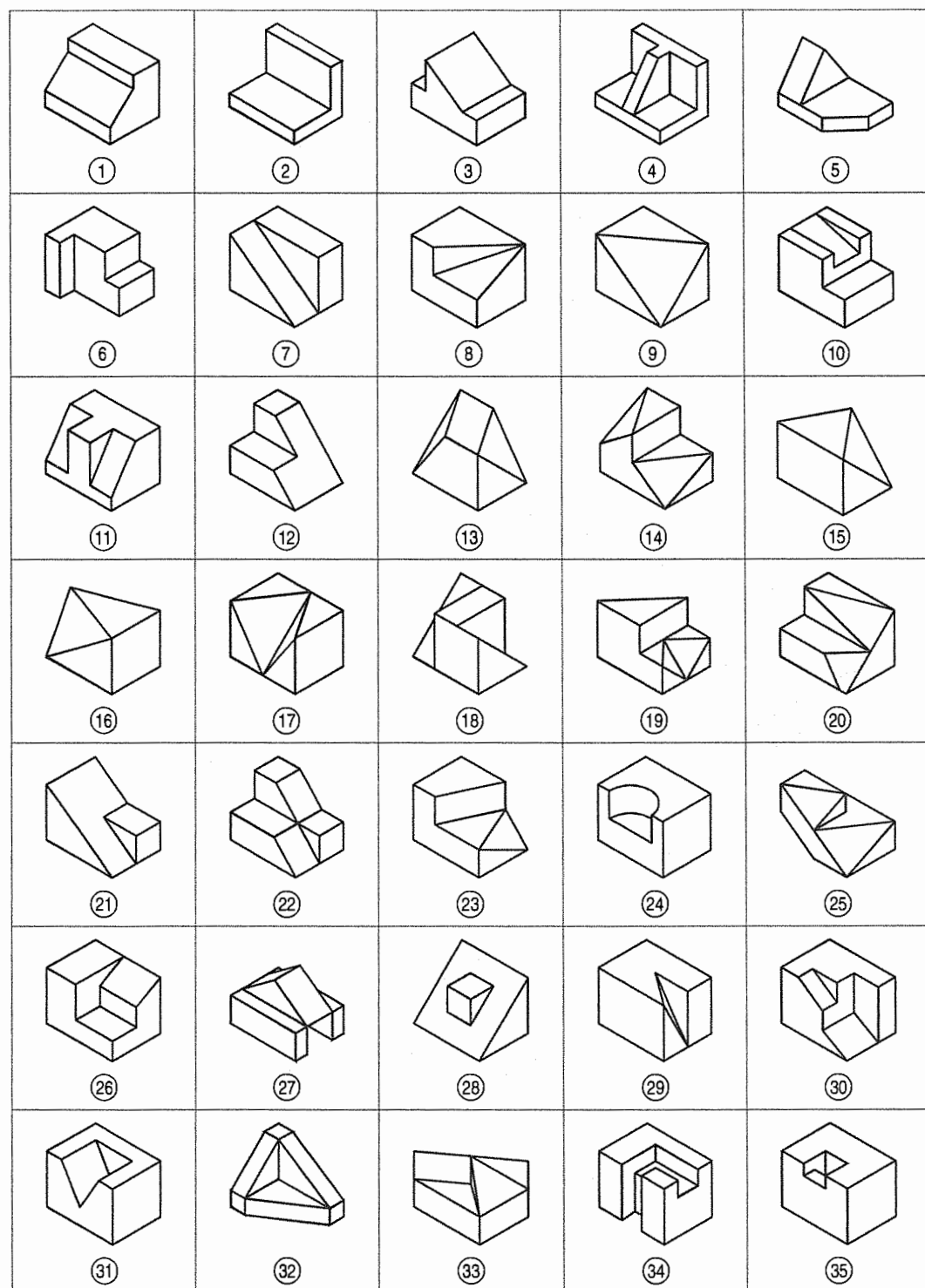
[Ex. 3. FIG. 17-127(13)]

FIG. 17-192



[Ex. 3. FIG. 17-127(14)]

FIG. 17-193



(Ex. 4. FIG. 17-128)

FIG. 17-194

Chapter 18



OBLIQUE PROJECTION

18-1. INTRODUCTION



Pictorial projections are becoming popular due to the use of a computer in a modern drawing office. An oblique projection like isometric projection is another method of pictorial projection. The oblique projection represents three dimensional object on the projection plane by one view only. This type of drawing is useful for making an assembly of an object and provides directly a production drawing (working drawing) of the object for the manufacturing purpose.

This chapter deals with the following topics of oblique projection:

1. Principle of the oblique projection
2. The oblique projection and the isometric projection
3. Receding lines and receding angles
4. Types of the oblique projection
5. Rules for the choice of position of an object
6. Steps for drawing the oblique projection
7. Oblique projection of pyramid
8. Oblique projection of circle
9. Oblique projection of cylinder
10. Oblique projection of prism.

18-2. PRINCIPLE OF THE OBLIQUE PROJECTION



When an observer looks towards an object from infinity, the lines of sights (projectors) will be parallel to each other and inclined to the plane of projection, the resulting projection is known as the *oblique projection*.

Imagine that a cuboid is placed in front of a vertical plane as shown in fig. 18-1. In orthographic projection, the visual rays (edges) are perpendicular to the plane of projection and in oblique projection, the visual rays (edges) makes an angle of 60° with the plane of projection.

The face *PQRS* which is parallel to the plane of projection, is projected in its true shape and size. The faces which are perpendicular to the plane of projection, are distorted in shape. The edges of perpendicular faces are inclined with the horizontal and projected to their true length.

The principal difference between the *oblique projection* and the *isometric projection* is that in case of the isometric projection all the three axes are inclined at an angle of 120° to each other whereas in the oblique projection third axis is inclined at an angle of 30° or 45° or 60° with respect to two perpendicular axes.

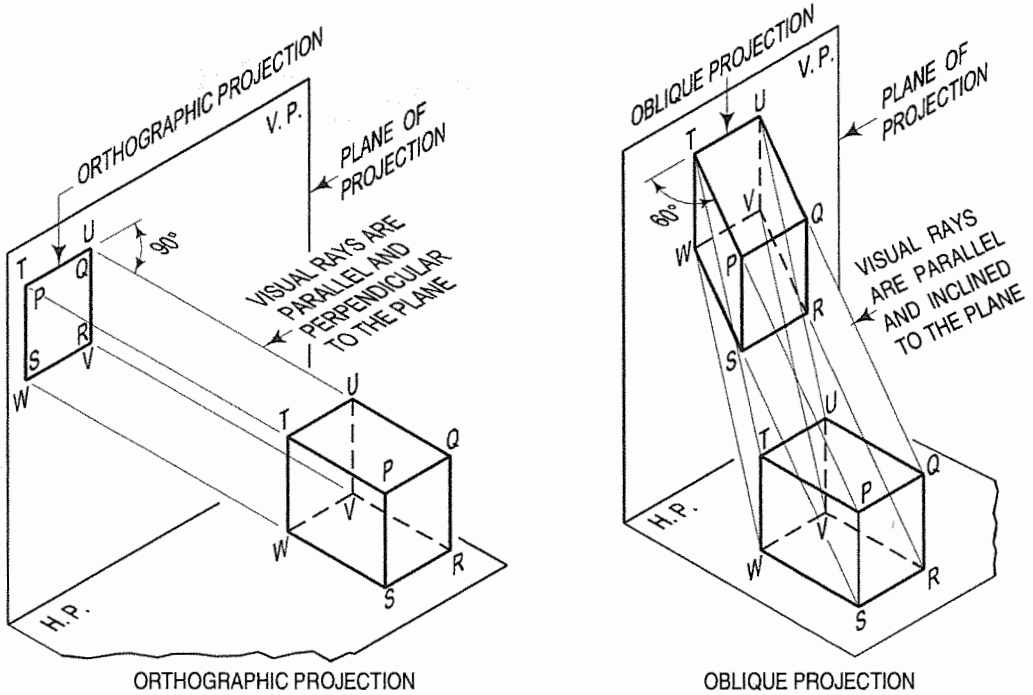


FIG. 18-1

18-3. THE OBLIQUE PROJECTION AND THE ISOMETRIC PROJECTION

The difference between the oblique projection and the isometric projection is given in table 18-1.

TABLE 18-1

Oblique projection	Isometric projection
(i) Projectors from an object are parallel to each other and inclined to the plane of projection.	(i) Projectors from an object are parallel to each other and perpendicular to the plane of picture.
(ii) The object is placed in such a way that one of its prominent faces remains parallel to the plane of projection. The object rests on one of its faces.	(ii) The object is kept in such a way that its three mutual perpendicular edges (axes) make equal angles with the plane of projection. The object stands on one of its corners.
(iii) The object is drawn with the actual dimensions.	(iii) The object is drawn with the reduced (about 82%) dimensions.
(iv) The faces of object which are perpendicular to the plane of projection will be distorted in the shape and size.	(iv) All the faces of the object are distorted in the shape and size.
(v) The choice of position of the object depends upon the shape and size.	(v) The choice of position of the object depends upon the shape and size.

18-4. RECEDING LINES AND RECEDING ANGLES

In oblique projection, the faces of object which are perpendicular to the plane of projection are projected in the distorted shapes. The perpendicular edges of such planes are drawn at an angle of 30° or 45° or 60° with the horizontal. These inclined lines are known as the *receding lines* and their inclinations to the horizontal is known as *receding angles*.

The appearance of distortion of an object can be improved by shortening the length of the receding lines. Refer to fig. 18-3.

The receding lines may be inclined either upwards or downwards, or to the left or right depending upon the necessity to show the details of an object effectively. Refer fig. 18-2.

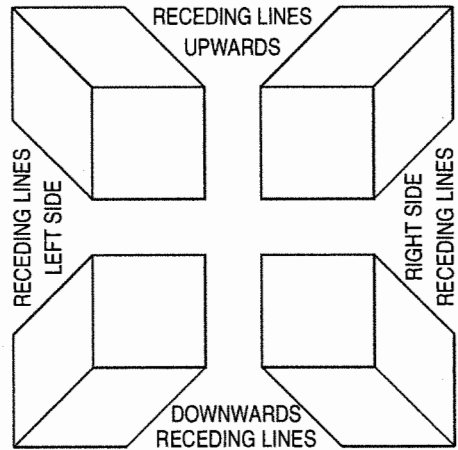


FIG. 18-2

18-5. TYPES OF THE OBLIQUE PROJECTION

The oblique projection is based on scales by which the receding lines are drawn.

(1) **Cavalier projection:** When the receding lines are drawn to full size scale and the projectors inclined at an angle of 30° or 45° or 60° to the plane of projection, such oblique projection is known as *cavalier projection*. Refer to fig. 18-3(i).

(2) **Cabinet projection:** If the receding lines are drawn to half size scale such oblique projection is known as the *cabinet projection*. Refer to fig. 18-3(ii).

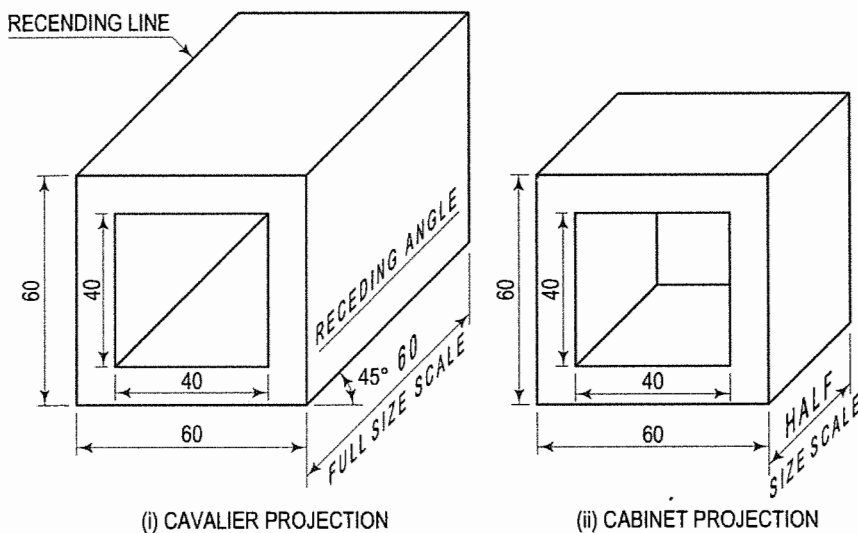


FIG. 18-3

18-7. STEPS FOR DRAWING THE OBLIQUE PROJECTION



The steps for preparing an oblique projection from orthographic projections are illustrated in the following problem.

Problem 18-1. (fig. 18-8 and fig. 18-9): *The orthographic projections of a casting are given in fig. 18-8. Draw the oblique projection when the receding axis is inclined at an angle of 30° to the horizontal.*

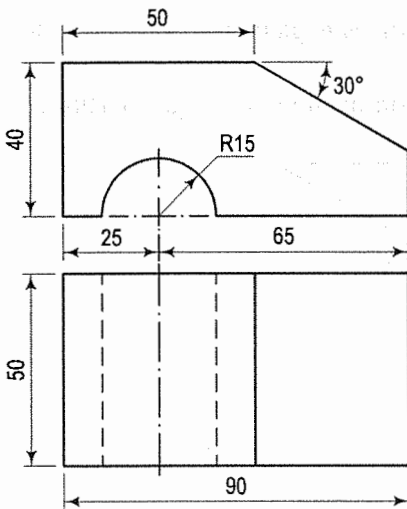


FIG. 18-8

- (i) Mark the point o and draw ox and oy mutually perpendicular axes. Draw receding axis oz at an angle of 30° with the axis ox (horizontal). Fig. 18-9(i).
- (ii) Construct a box by marking distances of 90 mm, 40 mm and 50 mm along the axes ox , oy and oz respectively as shown in fig. 18-9(ii).
- (iii) Mark a centre c of semi-circle along the axis ox at a distance of 65 mm from the point o . With the point c as centre and radius equal to 15 mm, draw a semi-circle. Mark a distance of 50 mm on the line parallel to ox and draw an angle of 30° as shown in fig. 18-9(iii).
- (iv) Complete the oblique view as shown in fig. 18-9(iv).

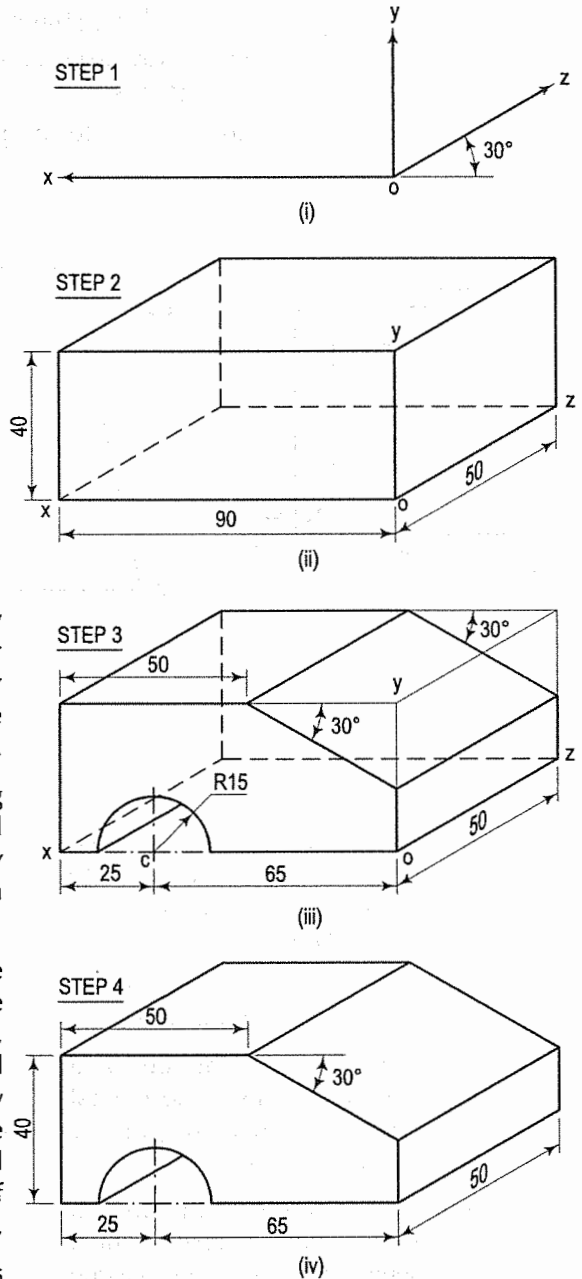
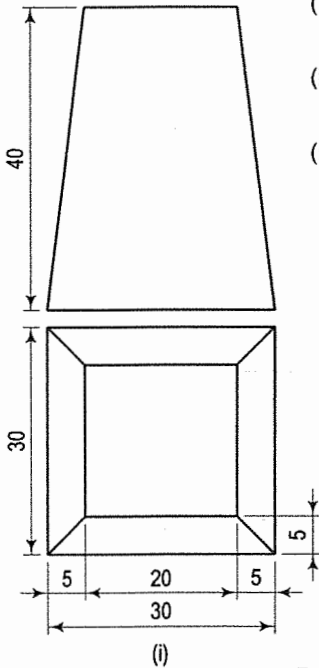


FIG. 18-9

18-8. OBLIQUE DRAWING OF PYRAMID

Problem 18-2. (fig. 18-10): A frustum of a square pyramid has its base 30 mm, top 20 mm, side and height 40 mm. Draw the oblique projection of the pyramid when it rests on its base on the H.P. with one of the sides of the base perpendicular to the V.P.



- (i) Draw the front view and the top view of the frustum of the pyramid as per given conditions.
- (ii) Draw axes ox , oy and oz as explained in problem 18-1 [steps (i) and (ii)].
- (iii) Complete the projection as shown in fig. 18-10(ii).

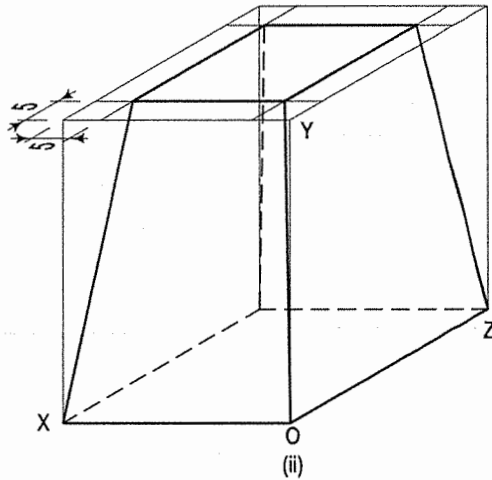


FIG. 18-10

18-9. OBLIQUE DRAWING OF CIRCLE

(1) Offset method (fig. 18-11):

- (i) As shown in fig. 18-11, draw square to enclose circle in the front plane and divide the circle into 12 divisions.
- (ii) Through division points draw vertical and horizontal lines.
- (iii) Create rhombus in the side and top plane and transfer horizontal and vertical lines to get 12 intersection points.
- (iv) Join these points with smooth curve to get ellipse which represents the circle in the oblique projection.

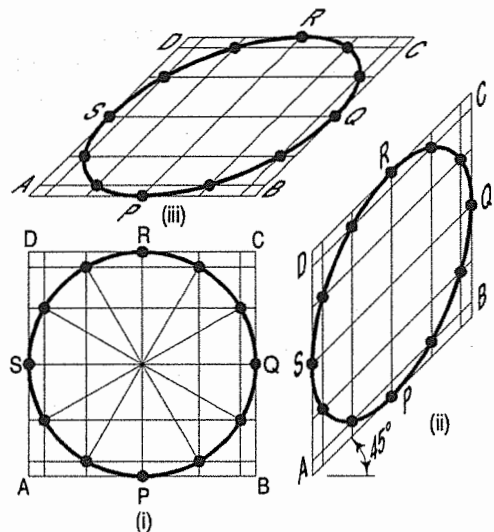


FIG. 18-11

(2) Four centre approximate method (fig. 18-12):

- (i) Draw the rhombus $ABCD$ in top and side plane with the length of side equal to the diameter of a circle.
- (ii) Mark P , Q , R and S as the mid points of the respective sides.
- (iii) From P and Q , draw line perpendicular to AB and BC respectively, such that they intersect at C_1 .
- (iv) Similarly draw perpendiculars from R and S to get C_2 .
- (v) Also mark C_3 and C_4 as the intersection points of perpendicular lines.
- (vi) With C_1 and C_2 as centre and C_1P and C_2R as radius draw arc PQ and arc RS respectively.

(vii) Similarly with C_3 and C_4 as centre and C_3P and C_4R as radius draw arc PS and arc RQ respectively.

(viii) These four arcs forms the ellipse, which represents the circle in the oblique projection.

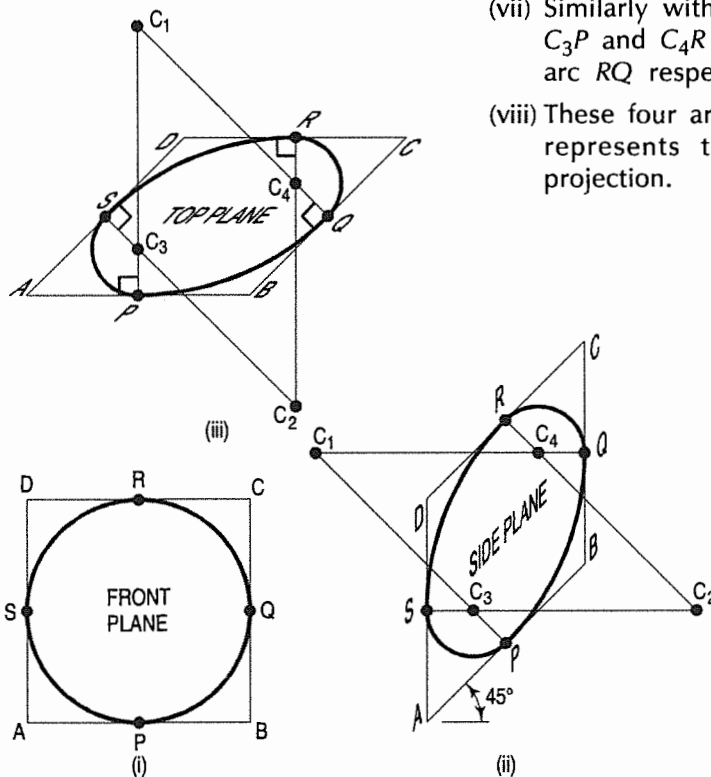


FIG. 18-12

18-10. OBLIQUE DRAWING OF CYLINDER

Problem 18-3. (fig. 18-13): A cylinder of diameter of base 40 mm and height 50 mm rests with its base on the H.P. Draw the oblique projection using the cavalier method.

- (i) Construct the box of 40 mm $\times$ 40 mm and height 50 mm.
- (ii) Using four centre approximate method, construct oblique circle on top and bottom face of box.
- (iii) Complete the projection as shown in fig. 18-13(ii).

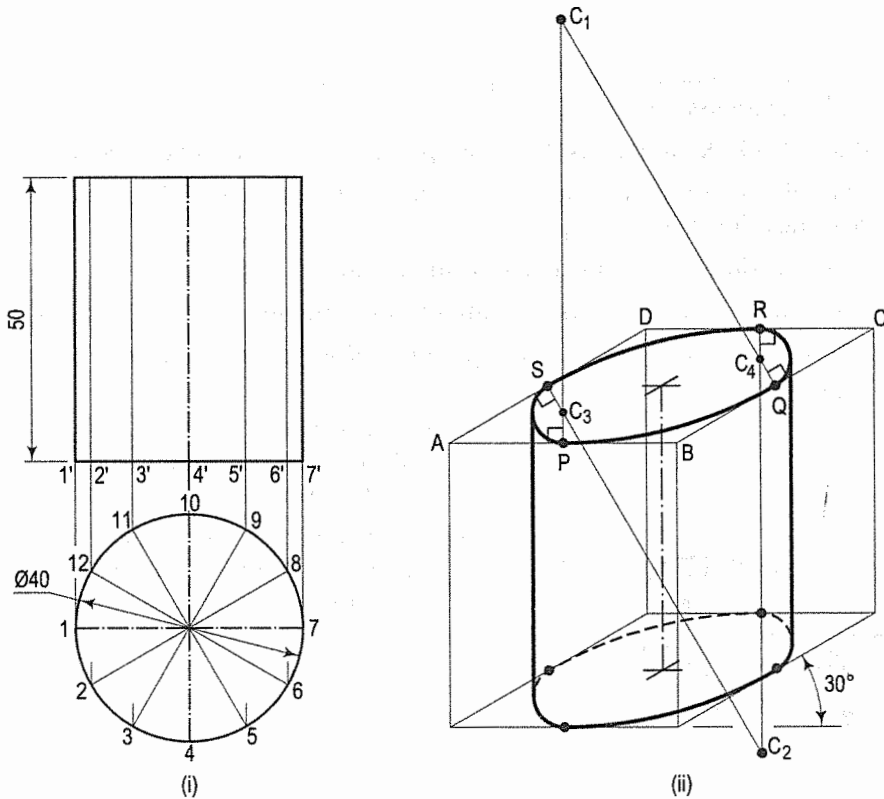


FIG. 18-13

18-11. OBLIQUE DRAWING OF PRISM



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 44 for the following problem.

Problem 18-4. A hexagonal prism, base 50 mm side and axis 100 mm long, resting on its base on H.P. is cut by a section plane perpendicular to V.P. and makes the angle of 45° with the H.P., passes through a point on the axis 20 mm from its top. Draw the oblique projection of the prism by the cavalier method with receding axis inclined at 45° with the horizontal when, (a) Two rectangular faces of the prism are equally inclined with V.P. (b) One rectangular face of the prism is parallel to V.P.

- (i) Draw the front view and top view of the hexagonal prism as shown in fig. 18-14 and fig. 18-17.
- (ii) Enclose the top views in rectangular box.
- (iii) Draw oblique projection as shown in fig. 18-15 and fig. 18-18 using box method as explained in problem 17-14, problem 17-15 and problem 17-16.
- (iv) Similarly using the Box method, draw sectional oblique projection as shown in fig. 18-16 and fig. 18-19.

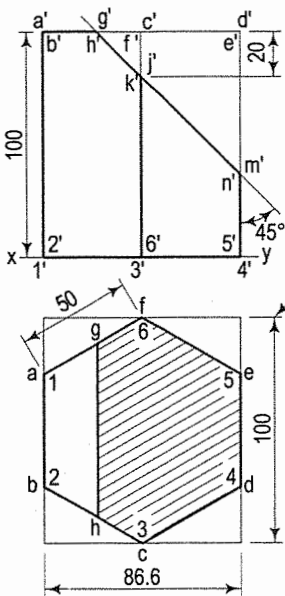


FIG. 18-14

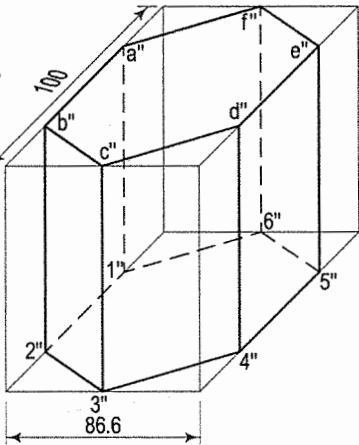


FIG. 18-15

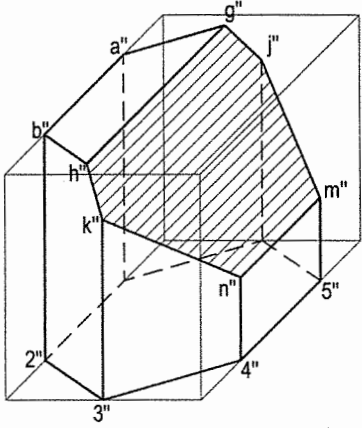


FIG. 18-16

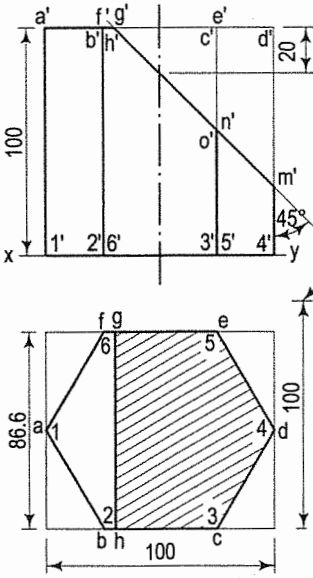


FIG. 18-17

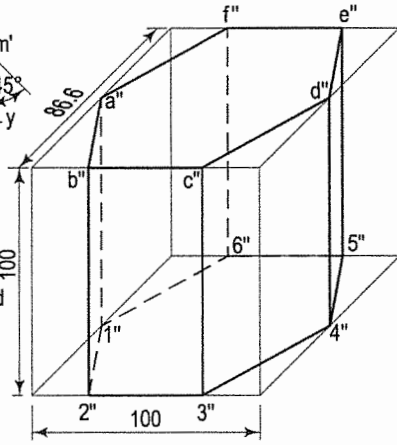


FIG. 18-18

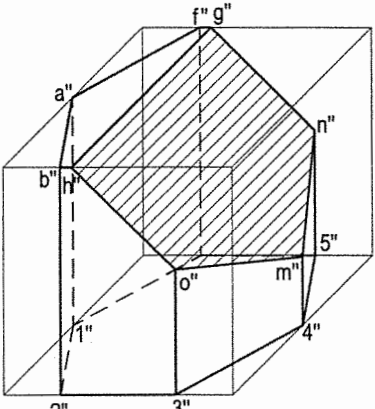


FIG. 18-19

Problem 18-5. (fig. 18-20): Draw the development of an oblique square prism whose cross-section is a regular square of 20 mm side and the top and the bottom bases are parallel to each other. The axis is 50 mm long. It is inclined at an angle of 60° to the base. The edges of base are equally inclined with the V.P.

- (i) Draw the top view and the front view of the oblique prism as shown in fig. 18-20(i).
- (ii) From the points a' , b' , c' and d' and $1'$, $2'$, $3'$ and $4'$, draw the projectors. Mark 80 mm (the perimeter length of the square) along the projector of the point a' . Divide this length (80 mm) into four equal parts. Draw the edges of prism a_1 , b_2 , c_3 and d_4 .

(iii) Complete the development as shown in fig. 18-20(ii).

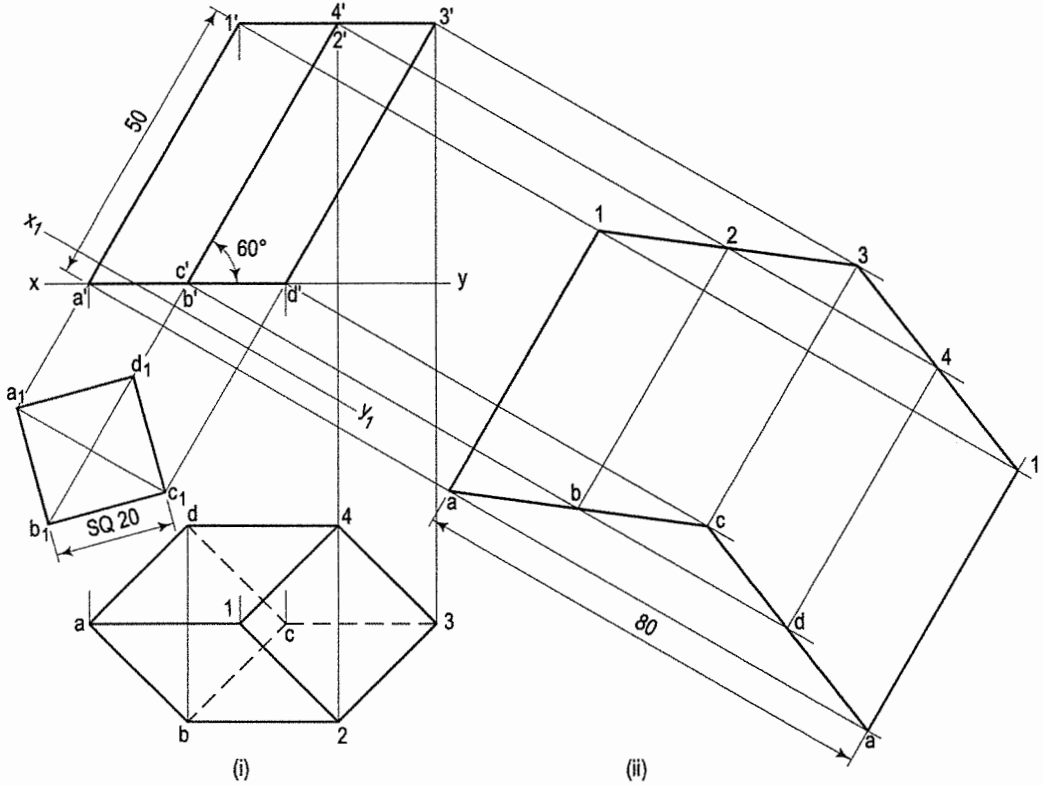


FIG. 18-20

18-12. TYPICAL PROBLEMS OF OBLIQUE PROJECTION

Problem 18-6. (fig. 18-21 and fig. 18-22): The orthographic projections of a casting are shown in fig. 18-21. Draw the oblique projection when the receding axis is inclined at an angle of 30° with the horizontal.

See fig. 18-22.

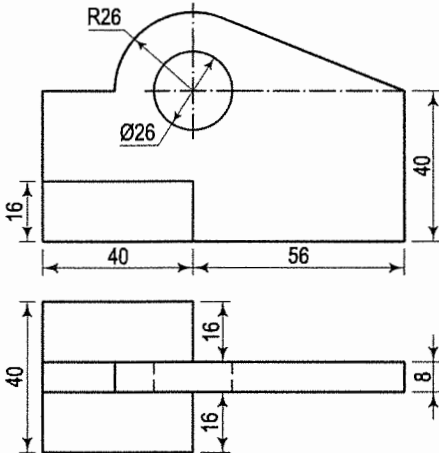


FIG. 18-21

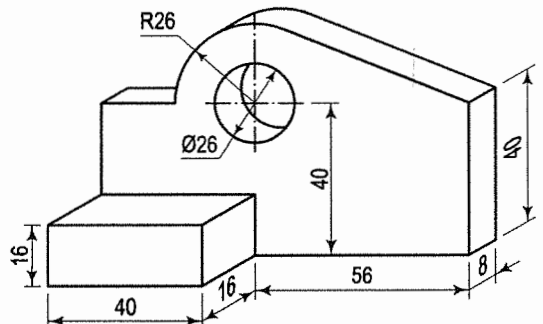


FIG. 18-22

Problem 18-7. (fig. 18-23 and fig. 18-24): The front view and sectional side view of a casting is shown in fig. 18-23. Draw the oblique projection when the receding axis is inclined at an angle of 45° with the horizontal.

See fig. 18-24.

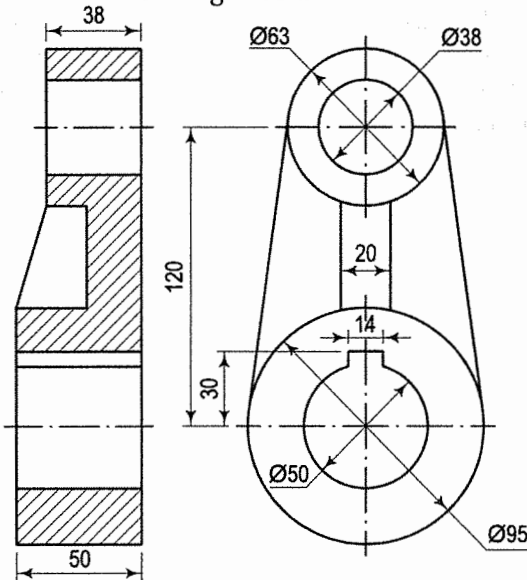


FIG. 18-23

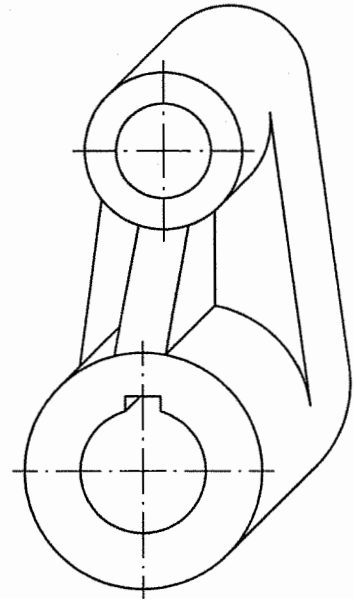


FIG. 18-24

EXERCISES 18

1. Differentiate clearly between the oblique projection and the isometric projection.
2. Explain the terms receding lines and receding angles.
3. Explain the rules pertaining to the choice of position for drawing oblique projection with minimum distortions.
4. Draw the oblique projection of a regular hexagonal prism, 25 mm side of the base and height 60 mm. When it rests on its base in the horizontal plane, the receding axis is inclined at an angle of 45° .
5. A right cylinder of 60 mm diameter and an axis 70 mm long has a co-axial through square hole of 20 mm side. Draw the oblique view when the cylinder rests on its base on the horizontal plane.
6. A cone of base 50 mm diameter and 60 mm long rests on its base on the H.P. A section plane perpendicular to the V.P. and inclined at 75° to the H.P., cuts the axis at a point 20 mm above the base. Draw the oblique projection of the cone. Assume that the receding axis is inclined at an angle of 30° with the horizontal.
7. Draw the oblique projection of two pieces of a mitre faced bridle joint made from the two wooden pieces of 160 mm $\times$ 65 mm $\times$ 75 mm by the cabinet method. The thickness of grooves is 18 mm.
8. Two pieces of wood joined together by a dovetail joint are shown in two views in fig. 17-48. Draw the oblique projection of the two pieces separated but in a position ready for fitting.

9. Draw the oblique view of a square with a square pyramid placed on its top.
10. A frustum of a hexagonal pyramid of 25 mm side of the top and 30 mm side of the base, and height 60 mm rests with one of its base corners on the H.P. such that two of the base edges passing through the corner on which it rests make equal angle with the H.P. The axis of the pyramid is parallel to the H.P. and perpendicular to V.P. Draw the oblique projection by the cabinet method.
11. The orthographic projections of few objects are shown in fig. 18-25, fig. 18-26 and fig. 18-27. Draw the oblique projections.

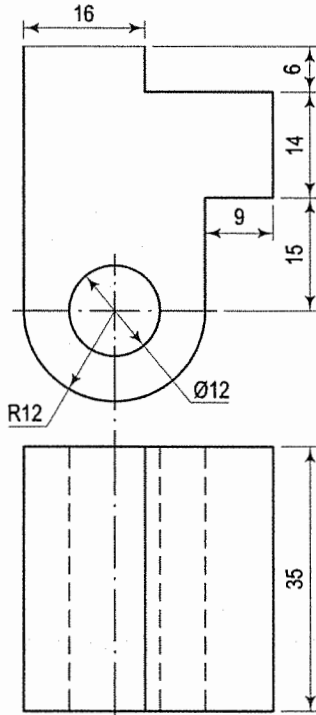


FIG. 18-25

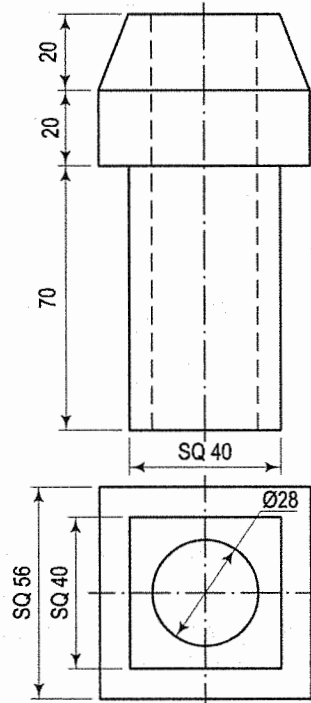


FIG. 18-26

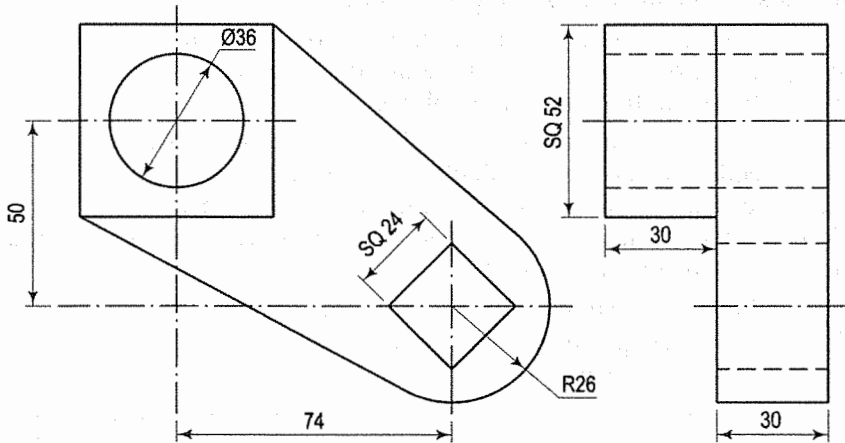


FIG. 18-27



19-1. INTRODUCTION

Perspective projection or perspective drawing is the representation of an object on a plane surface, called the *picture plane*, as it would appear to the eye, when viewed from a fixed position.

It may also be defined as the figure formed on the picture plane when visual rays from the eye to the object cut the picture plane. Perspective is mainly used in architecture. By means of perspective, the architect is able to show how an object would appear when constructed.

It is essential to have full knowledge of the principles of orthographic projection (third-angle method) before the theory of perspective drawing can be studied.

In this chapter, we shall deal with the topics of perspective drawing as follows:

- | | |
|--|---|
| 1. Principle of perspective projection | 6. Methods of drawing perspective view |
| 2. Definitions of perspective elements | 7. Types of perspective |
| 3. Station point | 8. Distance points |
| 4. Angle of vision | 9. Measuring line or line of heights |
| 5. Picture plane | 10. Perspectives of circles and solids. |

19-2. PRINCIPLE OF PERSPECTIVE PROJECTION

In perspective projection, the eye is assumed to be situated at a definite position relative to the object. The vertical plane, which (in perspective) is called the picture plane, is placed between the object and the eye.

Visual rays from the eye to the object converge to a point in the eye and are, therefore, inclined to the picture plane. The rays pierce the picture plane and form an image on it. This image is the perspective of the object.

19-3. DEFINITIONS OF PERSPECTIVE ELEMENTS

This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 45 for the definitions of perspective elements.



Various elements used in obtaining the perspective view are defined below. Refer fig. 19-1.

(1) **Ground plane (GP):** It is a horizontal plane on which the object is assumed to be situated.

(2) **Station point (S):** It is the point where the eye of the observer is located while viewing the object.

(3) **Picture plane (PP):** It is a vertical transparent plane located between the station point and the object which is to be viewed. It is the plane on which the perspective is formed. The front view of perspective elements and of the object (if necessary) is also projected on this plane.

(4) **Horizontal plane (HP):** This imaginary plane is at the level of the eye, i.e. the station point. It is a horizontal plane, above the ground plane and at right angles to the picture plane.

(5) **Auxiliary ground plane (AGP):** It is a horizontal plane placed above the horizon plane. The top view of the object and of the perspective elements is projected on this plane.

(6) **Ground line (GL):** The line of intersection of the picture plane with the ground plane is called the ground line.

(7) **Horizon line (HL):** It is the line in which the horizon plane intersects the picture plane. It is parallel to the ground line.

(8) **Perpendicular axis (PA):** It is the line drawn through the station point, perpendicular to the picture plane. It is, sometimes called the *Line of sight* or *Axis of vision*.

(9) **Centre of vision (C):** The point in which the perpendicular axis pierces the picture plane is called the centre of vision. It lies on the horizon line.

(10) **Central plane (CP):** It is an imaginary vertical plane, which passes through the station point and the centre of vision. It contains the perpendicular axis. It is perpendicular to both, the picture plane and the ground plane.

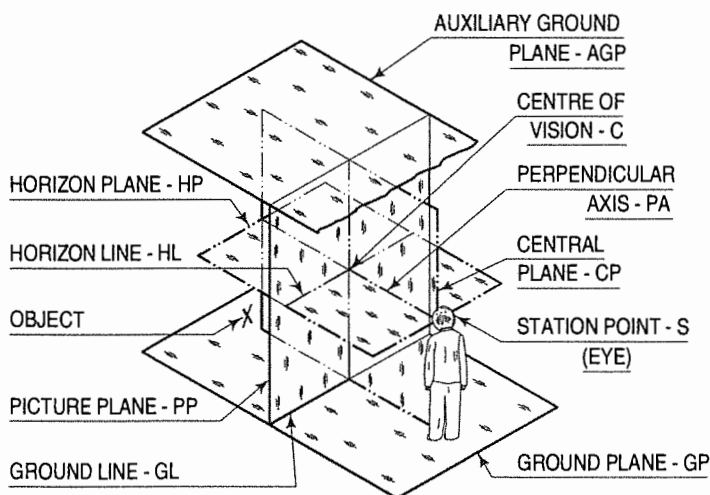


FIG. 19-1

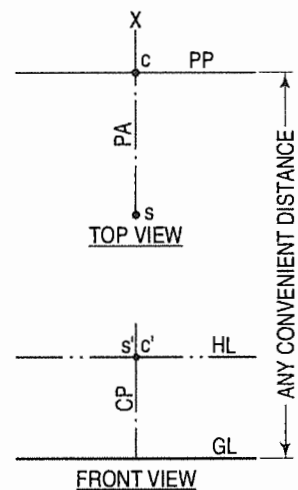


FIG. 19-2

Fig. 19-2 shows the projections of the perspective elements. In the top view, the picture plane PP is seen as a horizontal line. The object X is above PP, while S the station point is below PP. The line SC is the perpendicular axis and represents the central plane also. The ground plane, the horizon plane and the auxiliary ground plane will be seen as rectangles, but are not shown. In the front view, lines GL (ground line) and HL (horizon line) represent respectively the ground plane and the horizon plane. The station point s' and the centre of vision c' coincide with each other on HL. The central plane CP is seen as a vertical line through s' . The picture plane will be seen as a rectangle, but is not shown. The perspective view (when drawn) will be seen above or around GL.

19-4. STATION POINT

The position of the station point is of great importance. Upon its position, the general appearance of the perspective depends. Hence, it should be so located as to view the object in the best manner.

For large objects such as buildings, the station point is usually taken at the eye level of a person of normal height as shown in fig. 19-1 i.e. about 1.8 metres. For small objects, the station point should be fixed at such a height as would give a good view of the top surface as well as side surfaces.

The distance of the station point from the picture plane, when taken equal to about twice the greatest dimension of the object, usually gives good view in the perspective.

For objects having heights and widths more or less equal, the location of the station point may be so fixed that the angle between the visual rays from the station point to the outer-most boundaries of the object is approximately 30° .

The station point should be so situated in front of the object that the central plane passes through the centre of interest of the object. It may not, necessarily, be placed in front of the exact middle of the object. Refer fig. 19-1.

19-5. ANGLE OF VISION

Angle of vision is angle subtended by eye in horizontal or vertical direction in which one can visualize the things clearly. Horizontal and vertical angle of vision is generally 60° and 45° respectively. Refer fig. 19-3.

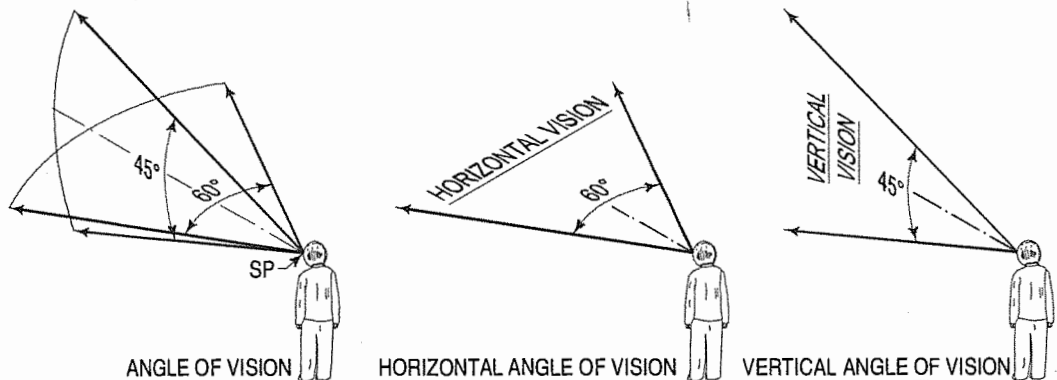


FIG. 19-3

19-6. PICTURE PLANE

The position of the picture plane relative to the object, determines the size of the perspective view. The perspective will show the object reduced in size when it is placed behind the picture plane. If the object is moved nearer the picture plane, the size of the perspective will increase. When the picture plane coincides with the object, the perspective of the object will be of its exact size. When the object is placed in front of the picture plane, its perspective, when projected back, will show the object enlarged in size.

In fig. 19-4(i), the line AB is behind the picture plane. Its perspective $A'B'$ is shorter than AB . In fig. 19-4(ii), AB is in the picture plane; its perspective $A'B'$ is equal to AB and coincides with it. Fig. 19-4(iii) shows the line AB placed in front of the picture plane; when projected back on the picture plane, its perspective $A'B'$ is longer than AB .

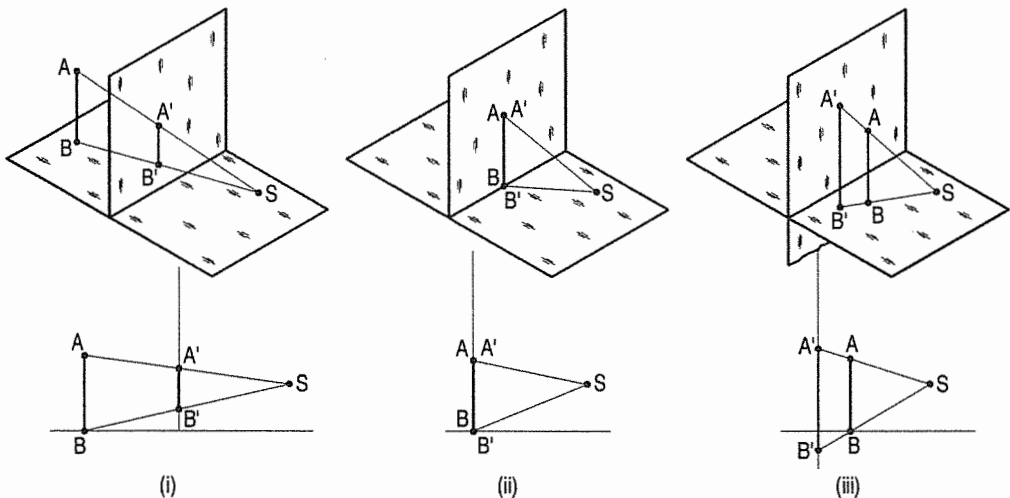


FIG. 19-4

19-7. METHODS OF DRAWING PERSPECTIVE VIEW

The perspective view of an object may be obtained by either

- (1) Visual-ray method
- (2) Vanishing-point method.

In the visual-ray method, points on the perspective are obtained by projecting

- (i) the top view and
- (ii) either the front view or the side view of the visual rays.

Vanishing-point method is comparatively simple. In addition to the top view of the visual rays, use of vanishing points of straight lines is made in this method. A front view or a side view of the object is also required to be drawn, for determining the heights.

19-7-1. VISUAL-RAY METHOD

This method is explained by means of the following three illustrative problems:

Problem 19-1. (fig. 19-5): A point A is situated 15 mm behind the picture plane and 10 mm above the ground plane. The station point S is 25 mm in front of the picture plane, 20 mm above the ground plane and lies in a central plane 10 mm to the right of the point. Draw the perspective view of the point A .

The pictorial view of the ground plane, the picture plane, the given point and the station point in their respective positions is given in fig. 19-5(i). The visual ray AS from the station point S to the point A is also shown. It passes through the picture plane. To mark the perspective of A , the point A' at which AS pierces the picture plane should be located.

In fig. 19-5(ii), an auxiliary ground plane (AGP) is shown placed above the point A , and the visual ray AS is shown projected on it, as is the top view of AS and a_1 is the top view of the point A' at which AS pierces the picture plane. a_1 shows the position of the point A' along the length of the picture plane. When the auxiliary ground plane is revolved and brought in the same plane as that of the picture plane, the view will be as shown in fig. 19-5(iii).

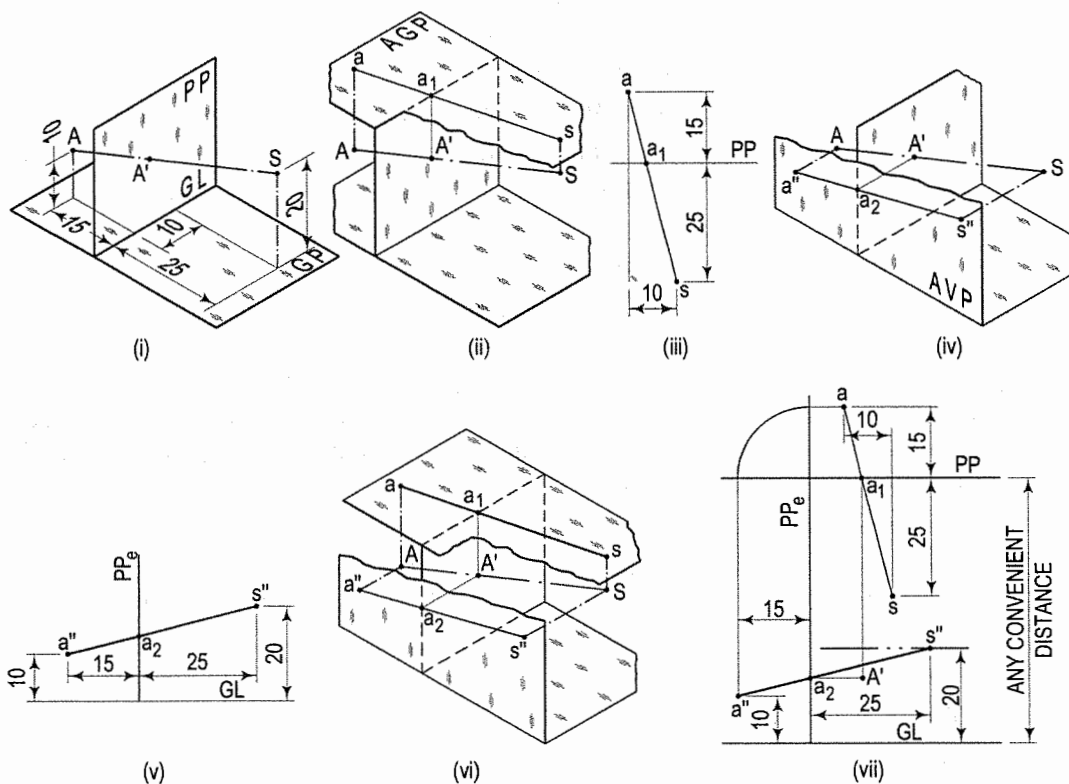


FIG. 19-5

To obtain the height of A' above the ground plane, an auxiliary vertical plane (AVP) perpendicular to both the picture plane and the ground plane is placed to the left of A and the side view of AS is projected on it [fig. 19-5(iv)]. $a''s''$ is the side

view of AS and a_2 is the side view of A' . It shows the height of A' above the ground plane. Fig. 19-5(v) shows the orthographic view (side view) when AVP is revolved and brought in the same plane as that of the picture plane.

Fig. 19-5(vii) shows the top view [fig. 19-5(iii)] and the side view [fig. 19-5(v)] combined together. A horizontal line drawn through a_2 and intersecting the vertical line through a_1 gives the point A' which is the perspective view of the point A. It is quite clear from the pictorial view [fig. 19-5(vi)] that A' lies in the picture plane on the line AS.

Steps in drawing the perspective view of the point A [fig. 19-5(vii)]:

- (i) Draw a horizontal line PP representing the picture plane in the top view.
- (ii) Mark a , the top view of A, 15 mm above PP.
- (iii) Draw a line (representing the central plane) perpendicular to PP and 10 mm to the right of a . On this line, mark s , the top view of the station point, 25 mm below PP.
- (iv) Draw a line joining a with s and intersecting PP at a point a_1 .
- (v) At any convenient distance below PP, draw a horizontal line GL. It is the ground line and also represents the ground plane in the front view.
- (vi) Draw a line HL parallel to and 20 mm above GL. It is the horizon line and also represents the horizon plane in the front view.
- (vii) At any point on GL and to the left of a , draw a vertical line PP_e (representing the picture plane in the side view).
- (viii) Mark a'' , the side view of A, 10 mm above GL and 15 mm to the left of PP_e .
- (ix) Mark s'' , the side view of the station point, on HL and 25 mm to the right of PP_e .
- (x) Draw a line joining a'' with s'' and intersecting PP_e at a point a_2 .
- (xi) Through a_1 , draw a vertical line. Through a_2 , draw a horizontal line intersecting the vertical line at a point A' .

Then A' is the perspective view of the point A.

Alternative method: Instead of the side view of AS, its front view $a's'$ may be projected on the picture plane (considering it as a vertical plane of projection) as shown in fig 19-6(i). The point A' must lie on this line $a's'$. It can be located by combining the top view and the front view as shown in fig. 19-6(ii) and fig. 19-6(iii), and as described below.

- (i) Draw the line as in the top view [steps (i) to (iv)].
- (ii) Draw the ground line GL at any convenient distance below PP and mark a' , the front view of A, 10 mm above GL and in projection with a .
- (iii) Draw the horizon line HL, 20 mm above GL and on it, mark s' , the front view of S, in projection with s .
- (iv) Draw a line joining a' with s' .
- (v) Draw a vertical line through a_1 intersecting $a's'$ at a point A' .

Then A' is the perspective view of the point A.

This method is comparatively simple and is generally adopted. In case of large objects, the perspective view often partly overlaps the front view. This, sometimes, causes confusion.

Perspective view of a straight line by the visual-ray method is drawn by first marking the perspectives of its ends which are points and then joining them.

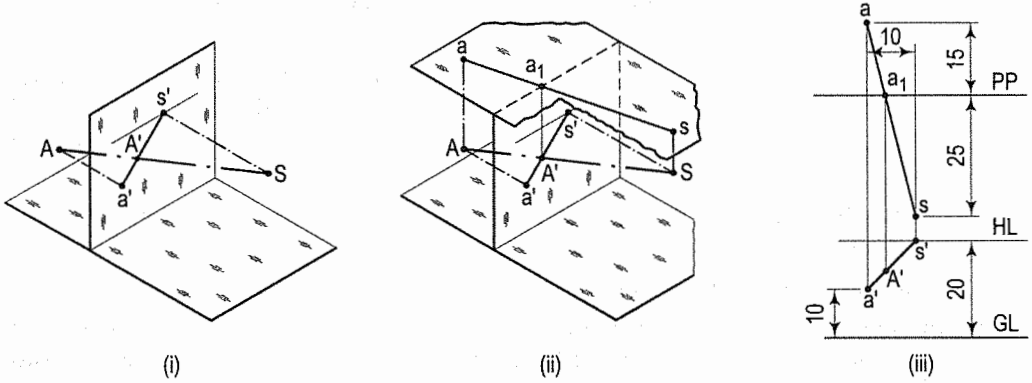


FIG. 19-6

Problem 19-2. (fig. 19-7): A straight line AB , 40 mm long, is parallel to and 15 mm above the ground plane, and inclined at 30° to the picture plane. The end A is 20 mm behind the picture plane. The station point is 40 mm above the ground plane, 50 mm in front of the picture plane and lies in a central plane which passes through the mid-point of AB . Draw its perspective view.

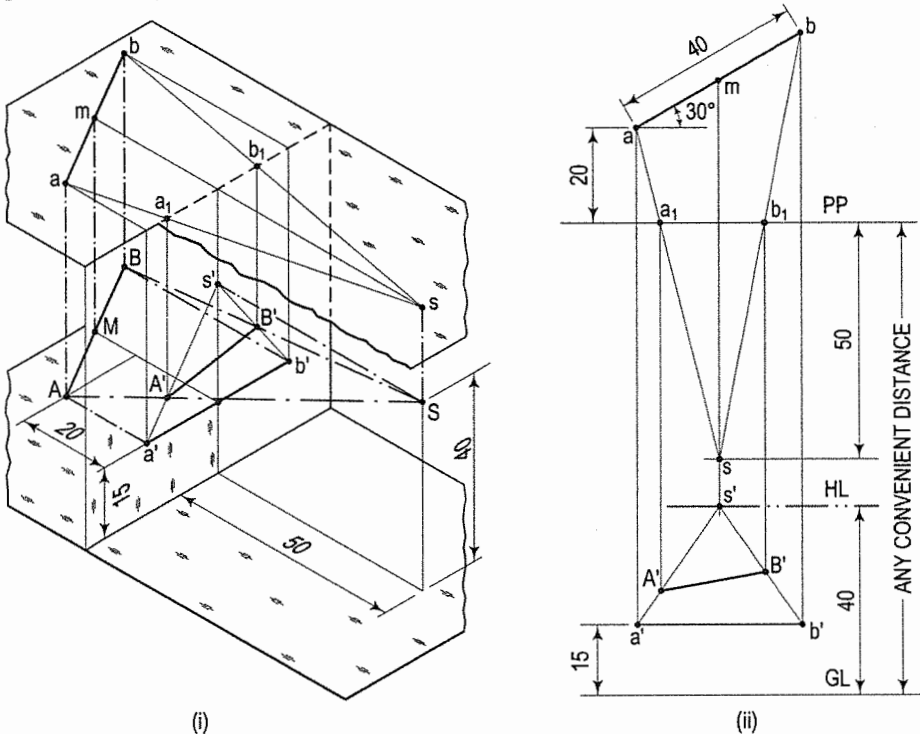


FIG. 19-7

- (i) Draw a horizontal line PP . As AB is parallel to the ground plane, its top view will show its true length. Therefore, draw a line ab , 40 mm long, inclined at 30° to PP and the end a , 20 mm above PP .

- (ii) Draw a vertical line through m , the mid-point of ab and on it mark s , the top view of the station point, 50 mm below PP.
- (iii) Draw lines joining s with a and b , and intersecting PP at points a_1 and b_1 respectively.
- (iv) Draw the ground line GL at any convenient distance below PP. Draw the horizon line HL parallel to and 40 mm above GL. Project s' , the front view on HL.
- (v) From ab , project the front view $a'b'$, parallel to and 15 mm above GL. Draw lines joining s' with a' and b' .
- (vi) Through a_1 and b_1 , draw verticals to intersect $a's'$ and $b's'$ at points A' and B' respectively.
- (vii) Join A' with B' . Then $A'B'$ is the required perspective view of AB .

The perspective can also be obtained with the aid of the side view instead of the front view. Perspective view of any solid (by visual-ray method) can similarly be drawn by first obtaining the perspective of each corner and then joining them in correct sequence, taking care to show the hidden edges by dashed lines.

Problem 19-3. (fig. 19-8 and fig. 19-9): A rectangular pyramid, base $30\text{ mm} \times 20\text{ mm}$ and axis 35 mm long, is placed on the ground plane on its base, with the longer edge of the base parallel to and 30 mm behind the picture plane. The central plane is 30 mm to the left of the apex and the station point is 50 mm in front of the picture plane and 25 mm above the ground plane. Draw the perspective view of the pyramid.

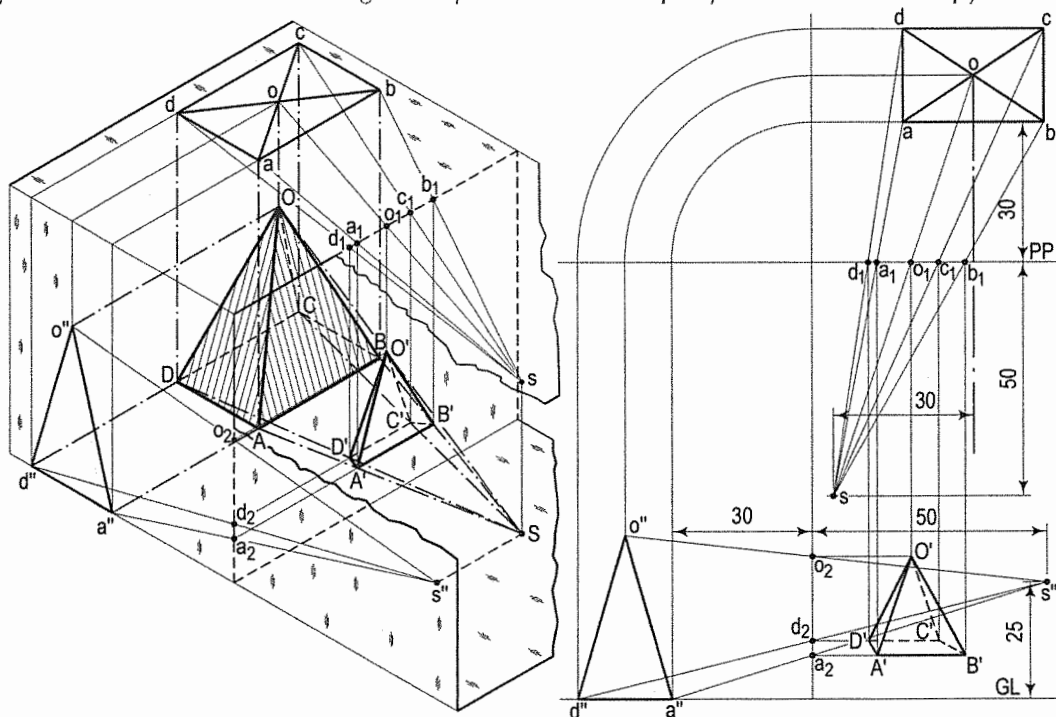


FIG. 19-8

Fig. 19-8 shows the perspective view of the pyramid obtained by means of its top view and the side view. The pictorial view shows clearly that points on the perspective lie in the picture plane on respective visual rays.

In fig. 19-9, the perspective view is drawn by means of its top view and the front view. It partly overlaps the front view.

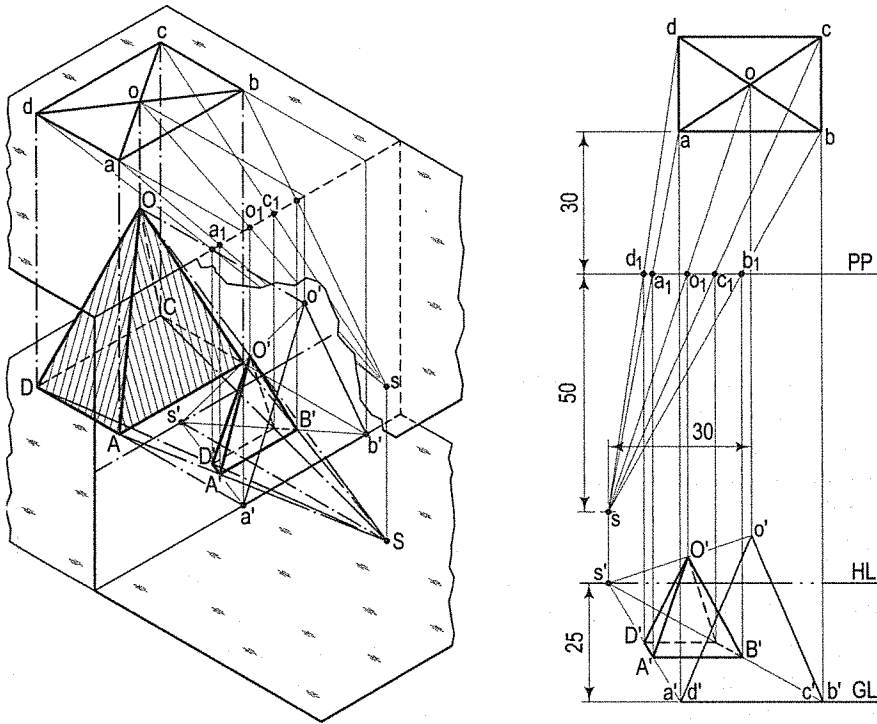


FIG. 19-9

19-7-2. VANISHING-POINT METHOD



Vanishing points: These are imaginary points infinite distance away from the station point. In practice, the point at which the visual ray from the eye to that infinitely distant vanishing point pierces the picture plane is referred to as the vanishing point.

If we stand between the rails of a long straight stretch of a railway track, it would appear as if the rails meet very far away at a point just at the level of the eye, i.e. on the horizon line. Even the telegraph and telephone wires running along the track at the sides of the track appear to meet at the same point. This point is a vanishing point.

In fig. 19-10, ab is the top view of a line AB lying on the ground plane and inclined at angle θ to the picture plane. When viewed from the station point s , its intercept on PP is a_1b_1 .

If the line is moved along the ground to the right, keeping the same inclination θ with the picture plane, its intercept will go on decreasing. The intercept becomes zero, or the line vanishes in a point at v when ab and the visual ray fall in a straight line. The point v is the top view position of the vanishing point for the horizontal line AB and for all lines parallel to AB , irrespective of their positions. The front-view position V of the vanishing point is obtained by projecting v , vertically on the horizon line.

Therefore, the vanishing point for any horizontal line is found by drawing a line parallel to the top view of that line from the top view of the station point. The point at which this line intersects the top view of the picture plane is then projected

on the horizon line. This point on the horizon line is the front-view position of the vanishing point.

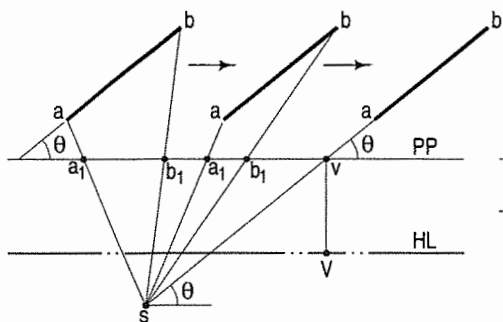


FIG. 19-10

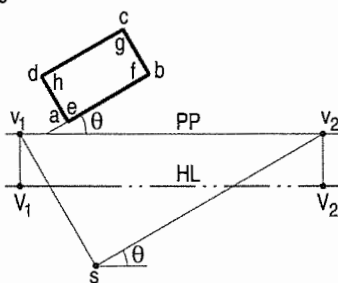


FIG. 19-11

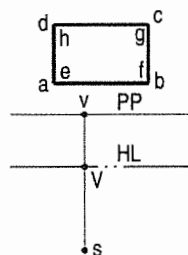


FIG. 19-12

In fig. 19-11, $abcd$ is the top view of a rectangular block placed on the ground plane so that a vertical face is inclined at angle θ to the picture plane.

The vanishing point for the line ab and for lines cd , ef and gh (which are parallel to ab) is obtained by drawing a line through s , parallel to ab and intersecting PP at a point v_2 . Through v_2 , a vertical line is drawn to meet HL at a point V_2 .

Then V_2 is the front-view position of the vanishing point. In perspective view of the block, edges AB , CD , EF and GH will converge to this point V_2 .

Similarly, V_1 is the vanishing point to which edges AD , BC , EH and FG will converge.

Thus, perspectives of all horizontal lines, if produced, pass through their respective vanishing points on the horizon line. Perspectives of all horizontal parallel lines converge to a vanishing point on the horizon line.

Vanishing point for lines perpendicular to the picture plane is obtained by drawing a line through the top view of the station point, and perpendicular to the picture plane. It lies on the horizon line and coincides with the centre of vision. It is the front-view position of the station point.

In fig. 19-12, V is the front-view position of the station point and the vanishing point, at which perspectives of lines AD , BC , EH and FG will converge. Thus, perspectives of all lines perpendicular to the picture plane converge to the centre of vision on the horizon line.

Lines which are parallel to the picture plane will have no vanishing points. They vanish at infinity. Therefore, perspectives of vertical lines are vertical; perspectives of horizontal lines which are parallel to the picture plane, remain horizontal; and perspectives of lines inclined to the ground plane and parallel to the picture plane will be inclined in the same direction (see fig. 19-13).

19-8. TYPES OF PERSPECTIVE



This book is accompanied by a computer CD, which contains an audiovisual animation presented for better visualization and understanding of the subject. Readers are requested to refer Presentation module 46 for the parallel or one point perspective.

(1) **Parallel perspective or one point perspective:** When an object has its one or more faces parallel to the picture plane, its perspective is called *parallel perspective* also called *one point perspective* as the edges converge to a single vanishing point of the parallel faces.